FHIR-STU3 — Clinical Summary Module (sections 9 – 9.11.11)

Source: HL7 FHIR STU3 (v3.0.2). Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

9 Clinical Summary Module

URL: https://hl7.org/fhir/STU3/clinicalsummary-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group Patient Care | Ballot Status: Informative

9.0 Clinical Summary Module

9.0.1 Introduction

This Clinical Summary Module focuses on the FHIR Resources that represent core clinical information for a patient. The information contained in these Resources are those frequently documented, created or retrieved by healthcare providers during the course of clinical care. Resources generated during the course of diagnostic studies can be found in the Diagnostics Module, whereas the Resources related to medication ordering and administration process can be found in the Medications Module.

As an introduction to FHIR APIs and Resources, please see the Developer's Introduction or Clinical Introduction in the Overview section of the Foundation Module.

9.0.2 Index

The Clinical Summary Module covers the following resources:

AllergyIntolerance Condition (Problem) Procedure FamilyMemberHistory | CarePlan Goal CareTeam ClinicalImpression | AdverseEvent DetectedIssue RiskAssessment

FHIR Resources have a low, moderate or high levels of complexity with respect to the number of primary and child elements as well as the number of referenced Resources, found in this module and others. To better understand the relationships between Resources, we recommend beginning with the lower complexity, core Resources such as Patient, Condition, and FamilyMemberHistory before addressing a high complexity Resource such as CarePlan.

9.0.3 Security and Privacy

The diagnostic resources often represent patient-related data, and as such are susceptible to data breaching. Necessary privacy and security provision must be in place for searching and fetching this information. For more general considerations, see the Security and Privacy module.

9.0.4 Common use Cases

* Documenting a patient's condition - The Condition Resource is used extensively throughout FHIR Resources to associate information and activities with specific conditions. The Condition Resource is broadly defined to include problems, diagnoses and health concerns.

* Retrieving a list of the patient's Problem List

* Documenting and retrieving the patient's allergies - The AllergyIntolerance Resource is used to represent the patient's allergy or intolerance to a substance. There is vibrant debate within clinical community regarding what is appropriate to document as an allergy or intolerance. These terms are used both formally by the Allergy and Immunology community as well as informally by patients leading to confusion. Readers are referred to the Resource definition for the Scope and Usage of this resource. The AllergyIntolerance Resource also supports the documenting of the absence of an allergy.

* Family History - The FamilyMemberHistory Resource can be used to document know conditions of family members and support the creation of pedigrees.

* Care Plans - The CarePlan resource supports a problem based care plan with references to other Resources including CareTeam, Condition, Goal, and activities such as ReferralRequest

9.0.5 Developmental Roadmap

Over the next 18 months, we will continue to advance the resources through the Maturity Levels through the process of development and testing of the Resources. We anticipate more widespread implementation of core Resources such as Condition. Complex Resources such as CarePlan are dependent on the maturation of its referred Resources and are expected to mature more gradually. The clinical community will need to develop use cases to test further mature the ReferralRequest and ProcedureRequest Resources at opportunities such as the Clinicians on FHIR sessions at the HL7 Working Group Meetings.

9.1 Resource AllergyIntolerance - Content

URL: https://hl7.org/fhir/STU3/allergyintolerance.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Risk of harmful or undesirable, physiological response which is unique to an individual and associated with exposure to a substance.

9.1.1 Scope and Usage

A record of a clinical assessment of an allergy or intolerance; a propensity, or a potential risk to an individual, to have an adverse reaction on future exposure to the specified substance, or class of substance.

Where a propensity is identified, to record information or evidence about a reaction event that is characterized by any harmful or undesirable physiological response that is specific to the individual and triggered by exposure of an individual to the identified substance or class of substance.

Substances include, but are not limited to: a therapeutic substance administered correctly at an appropriate dosage for the individual; food; material derived from plants or animals; or venom from insect stings.

Note for Reviewers

Presently open issues for this resource:

* This resource represents a condition of susceptibility to a substance, with a list of supporting events and/or symptoms, and has no direct relationship to an event reporting framework; this will be re-assessed when adverse event reporting resource(s) and/or profiles are added to FHIR

* Other HL7 models and the openEHR archetype have "exposure date" but this is not found in any surveyed systems, so this is left as an extension (more appropriate for adverse event reporting)

9.1.2 Boundaries and Relationships

This resource is used to provide a single place within the health record to document a range of clinical statements about adverse reactions to substances/products, including:

* record a clinical assessment of the individual's propensity to a potential future reaction upon re-exposure

* record cumulative information about the reaction to each exposure, including 'no reaction' if appropriate

Use to record information about the positive presence of the risk of an adverse reaction:

* to support direct clinical care of an individual

* as part of a managed adverse reaction or allergy/intolerance list

* to support exchange of information about the propensity and events related to adverse reactions

* to inform adverse reaction reporting

* to assist computerized knowledge-based activities such as clinical decision support and alerts

Use to record information about adverse reactions to a broad range of substances, including: biological & blood products; incipients and excipients in medicinal preparations; foods; metal salts; and organic chemical compounds.

Adverse reactions may be:

* an allergy (typically type I hypersensitivity, plus other "allergy-like" reactions, including pseudoallergy)

* an intolerance (typically non-immune adverse reactions that are not determined or perceived to be allergic or "allergy-like", and are to some degree idiosyncratic and/or individually specific [i.e. are not a reaction that is expected to occur with most or all patients given similar circumstances])

In clinical practice distinguishing between allergy and intolerance is difficult and may not be practical. Often the term "allergy" is used rather generically and may overlap with "intolerance", and the boundaries between these concepts may not be well-defined or understood. As noted above, the term "intolerance" should generally be applied to a propensity for adverse reactions which is either determined (to the extent that is possible) or perceived to not be allergic or "allergy-like". If it is not possible to determine whether a particular propensity condition is an allergy or an intolerance, then the type element should be omitted from the resource. Identification of the type of reaction is not a proxy for seriousness or risk of harm to the patient, which is better expressed in the documentation of the clinical manifestation and the assessment of criticality.

The sensitivity in the case of either an allergy or intolerance is unique to the individual, and is distinguished from those reactions that are a property of the circumstance, such as toxicity of a food or drug, overdose, drug-drug, drug-food, or drug-disease interaction (which are reactions that would be expected to occur for any individual given the same circumstances).

The risk of an adverse reaction event or manifestation should not be recorded without identifying a proposed causative substance (including pharmaceutical products) or class of substance. If there is uncertainty that a specific substance is the cause, this uncertainty can be recorded using the 'verificationStatus' data element. If there are multiple possible substances that may have caused a reaction/manifestation, each substance should be recorded using a separate instance of this resource with the 'verificationStatus' set to an initial state of 'unconfirmed' so that adverse reaction checking can be supported in clinical systems. If a substance, agent or class is later proven not to be the cause for a given reaction then the 'verificationStatus' can be modified to 'refuted'.

This resource has been designed to allow recording of information about a specific substance (e.g., amoxicillin, oysters, or bee sting venom) or pharmaceutical product or, alternatively, a class of substance (e.g., penicillins). If a class of substance is recorded, then identification of the exact substance can be recorded on a per exposure basis.

The scope of this FHIR resource has deliberately focused on identifying a pragmatic data set that is used in most clinical systems or will be suitable for most common clinical scenarios; extensions can be used to add additional detail if required. Examples of clinical situations where the extension may be required include: a detailed allergist/immunologist assessment, for reporting to regulatory bodies or use in a clinical trial.

The act of recording any adverse reaction in a health record involves the clinical assessment that a potential hazard exists for an individual if they are exposed to the same substance/product/class in the future - that is, a relative contraindication - and, in the absence of additional information indicating a higher level of potential risk, the default 'criticality' value should be set to 'Low Risk'. If a clinician considers that it is not safe for the individual to be deliberately re-exposed to the substance/product again, for example, following a manifestation of a life-threatening anaphylaxis, then the 'criticality' data element should be amended to 'High Risk'.

A formal adverse event report to regulatory bodies is a document that will contain a broad range of information in addition to the specific details about the adverse reaction. The report could utilize parts of this resource plus include additional data as required per jurisdiction.

An adverse reaction or allergy/intolerance list is a record of all identified propensities for an adverse reaction for the individual upon future exposure to the substance/product or class, plus provides potential access to the evidence provided by details about each reaction event, such as manifestation.

Valuable first-level information that could be presented to the clinician when they need to assess propensity for future reactions are:

* statements about previous clinical manifestations following exposure

* source of the information/reporter

* the 'criticality' flag

Second-level information can be drawn from each exposure event and links to additional detailed information such as history, examination and diagnoses stored elsewhere in the record, if it is available.

AllergyIntolerance and RiskAssessment

AllergyIntolerance describes a specific type of risk - propensity to reaction to a substance/product while RiskAssessment describes general risks to a subject, not generally based on a reaction.

AllergyIntolerance and Immunization.reaction

Immunization.reaction may be an indication of an allergy or intolerance. If this is deemed to be the case, a separate AllergyIntolerance record should be created to indicate it, as most systems will not query against past immunization.reactions.

Misuse

* The allergy/intolerance list exists as a patient safety tool to inform decision support around ordering of medications and nutrition and to guide clinical treatment. Other reactions triggered by physical stimuli -- light, heat, cold, pressure, vibration, which may mimic allergic or intolerance reactions, should be recorded as Condition on the problem list, not using AllergyIntolerance.

* Not to be used to record adverse events, including failures of clinical process, interventions or products. For example, abnormal use or mistakes/errors made in maladministration of an agent or substance, incorrect dosage, mislabeling, harm or injury caused by an intervention or procedure, overdose/poisoning, etc.

* Not to be used as a proxy for an adverse event report. See above for how it may be used as one component of an adverse event report.

* Not to be used for recording alerts. Alerts are handled using Flag or - where event-specific, DetectedIssue.

* Not to be used for recording failed therapy.

STU Note:Requests have been received (GF#10369) to add codes to the extension-allergyintolerance-certainty value set (reaction-event-certainty), which is a required binding. The requested codes include "unknown", "ruled out" and "possible". The Patient Care WG has voted to add "unknown" to the value set, but recommends that if other terms (including "ruled out" and "possible") are desired for use in a specific setting, an extension or profile should be used. During the STU period feedback is solicited regarding (1) the need and desirability of adding the code "unknown" to the value set, vs. omitting the element if the certainty is not known; and (2) whether or not additional codes besides "unknown" should be added to the value set for the core specification, or whether additional codes, if needed, should be added in an extension/profile.

Feedback is welcome here .

This resource is referenced by AdverseEvent, ClinicalImpression, FamilyMemberHistory, ImmunizationRecommendation and NutritionOrder

9.1.3 Resource Content

* Structure

* UML

* XML

* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
AllergyIntolerance | I | | DomainResource | Allergy or Intolerance (generally: Risk of adverse reaction to a substance) + AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. + AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External ids for this item
clinicalStatus | ?!?I | 0..1 | code | active | inactive | resolved AllergyIntoleranceClinicalStatus (Required)
verificationStatus | ?!?I | 1..1 | code | unconfirmed | confirmed | refuted | entered-in-error AllergyIntoleranceVerificationStatus (Required)
type | ? | 0..1 | code | allergy | intolerance - Underlying mechanism (if known) AllergyIntoleranceType (Required)
category | ? | 0..* | code | food | medication | environment | biologic AllergyIntoleranceCategory (Required)
criticality | ? | 0..1 | code | low | high | unable-to-assess AllergyIntoleranceCriticality (Required)
code | ? | 0..1 | CodeableConcept | Code that identifies the allergy or intolerance AllergyIntolerance Substance/Product, Condition and Negation Codes (Example)
patient | ? | 1..1 | Reference(Patient) | Who the sensitivity is for
onset[x] | | 0..1 | | When allergy or intolerance was identified
onsetDateTime | | | dateTime
onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
assertedDate | | 0..1 | dateTime | Date record was believed accurate
recorder | | 0..1 | Reference(Practitioner | Patient) | Who recorded the sensitivity
asserter | ? | 0..1 | Reference(Patient | RelatedPerson | Practitioner) | Source of the information about the allergy
lastOccurrence | | 0..1 | dateTime | Date(/time) of last known occurrence of a reaction
note | | 0..* | Annotation | Additional text not captured in other fields
reaction | | 0..* | BackboneElement | Adverse Reaction Events linked to exposure to substance
substance | | 0..1 | CodeableConcept | Specific substance or pharmaceutical product considered to be responsible for event Substance Code (Example)
manifestation | | 1..* | CodeableConcept | Clinical symptoms/signs associated with the Event SNOMED CT Clinical Findings (Example)
description | | 0..1 | string | Description of the event as a whole
onset | | 0..1 | dateTime | Date(/time) when manifestations showed
severity | | 0..1 | code | mild | moderate | severe (of event as a whole) AllergyIntoleranceSeverity (Required)
exposureRoute | | 0..1 | CodeableConcept | How the subject was exposed to the substance SNOMED CT Route Codes (Example)
note | | 0..* | Annotation | Text about event not captured in other fields
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AllergyIntolerance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this item --></identifier>
 <clinicalStatus value="[code]"/><!--  0..1 active | inactive | resolved -->
 <verificationStatus value="[code]"/><!--  1..1 unconfirmed | confirmed | refuted | entered-in-error -->
 <type value="[code]"/><!-- 0..1 allergy | intolerance - Underlying mechanism (if known) -->
 <category value="[code]"/><!-- 0..* food | medication | environment | biologic -->
 <criticality value="[code]"/><!-- 0..1 low | high | unable-to-assess -->
 <code><!-- 0..1 CodeableConcept Code that identifies the allergy or intolerance --></code>
 <patient><!-- 1..1 Reference(Patient) Who the sensitivity is for --></patient>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string When allergy or intolerance was identified --></onset[x]>
 <assertedDate value="[dateTime]"/><!-- 0..1 Date record was believed accurate -->
 <recorder><!-- 0..1 Reference(Practitioner|Patient) Who recorded the sensitivity --></recorder>
 <asserter><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner) Source of the information about the allergy
    --></asserter>
 <lastOccurrence value="[dateTime]"/><!-- 0..1 Date(/time) of last known occurrence of a reaction -->
 <note><!-- 0..* Annotation Additional text not captured in other fields --></note>
 <reaction>  <!-- 0..* Adverse Reaction Events linked to exposure to substance -->
  <substance><!-- 0..1 CodeableConcept Specific substance or pharmaceutical product considered to be responsible for
    event --></substance>
  <manifestation><!-- 1..* CodeableConcept Clinical symptoms/signs associated with the Event --></manifestation>
  <description value="[string]"/><!-- 0..1 Description of the event as a whole -->
  <onset value="[dateTime]"/><!-- 0..1 Date(/time) when manifestations showed -->
  <severity value="[code]"/><!-- 0..1 mild | moderate | severe (of event as a whole) -->
  <exposureRoute><!-- 0..1 CodeableConcept How the subject was exposed to the substance --></exposureRoute>
  <note><!-- 0..* Annotation Text about event not captured in other fields --></note>
 </reaction>
</AllergyIntolerance>
```

JSON Template

```
{
  "resourceType" : "AllergyIntolerance",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this item
  "clinicalStatus" : "<code>", // C? active | inactive | resolved
  "verificationStatus" : "<code>", // C? R!  unconfirmed | confirmed | refuted | entered-in-error
  "type" : "<code>", // allergy | intolerance - Underlying mechanism (if known)
  "category" : ["<code>"], // food | medication | environment | biologic
  "criticality" : "<code>", // low | high | unable-to-assess
  "code" : { CodeableConcept }, // Code that identifies the allergy or intolerance
  "patient" : { Reference(Patient) }, // R!  Who the sensitivity is for
  // onset[x]: When allergy or intolerance was identified. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  "assertedDate" : "<dateTime>", // Date record was believed accurate
  "recorder" : { Reference(Practitioner|Patient) }, // Who recorded the sensitivity
  "asserter" : { Reference(Patient|RelatedPerson|Practitioner) }, // Source of the information about the allergy
  "lastOccurrence" : "<dateTime>", // Date(/time) of last known occurrence of a reaction
  "note" : [{ Annotation }], // Additional text not captured in other fields
  "reaction" : [{ // Adverse Reaction Events linked to exposure to substance
    "substance" : { CodeableConcept }, // Specific substance or pharmaceutical product considered to be responsible for
    event
    "manifestation" : [{ CodeableConcept }], // R!  Clinical symptoms/signs associated with the Event
    "description" : "<string>", // Description of the event as a whole
    "onset" : "<dateTime>", // Date(/time) when manifestations showed
    "severity" : "<code>", // mild | moderate | severe (of event as a whole)
    "exposureRoute" : { CodeableConcept }, // How the subject was exposed to the substance
    "note" : [{ Annotation }] // Text about event not captured in other fields
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AllergyIntolerance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AllergyIntolerance.identifier [ Identifier ], ... ; # 0..* External ids for this item
  fhir:AllergyIntolerance.clinicalStatus [ code ]; # 0..1 active | inactive | resolved
  fhir:AllergyIntolerance.verificationStatus [ code ]; # 1..1 unconfirmed | confirmed | refuted | entered-in-error
  fhir:AllergyIntolerance.type [ code ]; # 0..1 allergy | intolerance - Underlying mechanism (if known)
  fhir:AllergyIntolerance.category [ code ], ... ; # 0..* food | medication | environment | biologic
  fhir:AllergyIntolerance.criticality [ code ]; # 0..1 low | high | unable-to-assess
  fhir:AllergyIntolerance.code [ CodeableConcept ]; # 0..1 Code that identifies the allergy or intolerance
  fhir:AllergyIntolerance.patient [ Reference(Patient) ]; # 1..1 Who the sensitivity is for
  # AllergyIntolerance.onset[x] : 0..1 When allergy or intolerance was identified. One of these 5
    fhir:AllergyIntolerance.onsetDateTime [ dateTime ]
    fhir:AllergyIntolerance.onsetAge [ Age ]
    fhir:AllergyIntolerance.onsetPeriod [ Period ]
    fhir:AllergyIntolerance.onsetRange [ Range ]
    fhir:AllergyIntolerance.onsetString [ string ]
  fhir:AllergyIntolerance.assertedDate [ dateTime ]; # 0..1 Date record was believed accurate
  fhir:AllergyIntolerance.recorder [ Reference(Practitioner|Patient) ]; # 0..1 Who recorded the sensitivity
  fhir:AllergyIntolerance.asserter [ Reference(Patient|RelatedPerson|Practitioner) ]; # 0..1 Source of the information
    about the allergy
  fhir:AllergyIntolerance.lastOccurrence [ dateTime ]; # 0..1 Date(/time) of last known occurrence of a reaction
  fhir:AllergyIntolerance.note [ Annotation ], ... ; # 0..* Additional text not captured in other fields
  fhir:AllergyIntolerance.reaction [ # 0..* Adverse Reaction Events linked to exposure to substance
    fhir:AllergyIntolerance.reaction.substance [ CodeableConcept ]; # 0..1 Specific substance or pharmaceutical product
    considered to be responsible for event
    fhir:AllergyIntolerance.reaction.manifestation [ CodeableConcept ], ... ; # 1..* Clinical symptoms/signs associated
    with the Event
    fhir:AllergyIntolerance.reaction.description [ string ]; # 0..1 Description of the event as a whole
    fhir:AllergyIntolerance.reaction.onset [ dateTime ]; # 0..1 Date(/time) when manifestations showed
    fhir:AllergyIntolerance.reaction.severity [ code ]; # 0..1 mild | moderate | severe (of event as a whole)
    fhir:AllergyIntolerance.reaction.exposureRoute [ CodeableConcept ]; # 0..1 How the subject was exposed to the
    substance
    fhir:AllergyIntolerance.reaction.note [ Annotation ], ... ; # 0..* Text about event not captured in other fields
  ], ...;
]
```

Changes since DSTU2

AllergyIntolerance

AllergyIntolerance.clinicalStatus | Added Element

AllergyIntolerance.verificationStatus | Added Element

AllergyIntolerance.category | Max Cardinality changed from 1 to *

AllergyIntolerance.code | Renamed from substance to code Min Cardinality changed from 1 to 0

AllergyIntolerance.onset[x] | Renamed from onset to onset[x] Add Age, Add Period, Add Range, Add string

AllergyIntolerance.assertedDate | Added Element

AllergyIntolerance.asserter | Renamed from reporter to asserter

AllergyIntolerance.lastOccurrence | Renamed from lastOccurence to lastOccurrence

AllergyIntolerance.note | Max Cardinality changed from 1 to *

AllergyIntolerance.reaction.note | Max Cardinality changed from 1 to *
AllergyIntolerance.recordedDate | deleted
AllergyIntolerance.status | deleted
AllergyIntolerance.reaction.certainty | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 5 tests that all execute ok. 2 fail round-trip testing and 4 r3 resources are invalid (6 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
AllergyIntolerance | I | | DomainResource | Allergy or Intolerance (generally: Risk of adverse reaction to a substance) + AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. + AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External ids for this item
clinicalStatus | ?!?I | 0..1 | code | active | inactive | resolved AllergyIntoleranceClinicalStatus (Required)
verificationStatus | ?!?I | 1..1 | code | unconfirmed | confirmed | refuted | entered-in-error AllergyIntoleranceVerificationStatus (Required)
type | ? | 0..1 | code | allergy | intolerance - Underlying mechanism (if known) AllergyIntoleranceType (Required)
category | ? | 0..* | code | food | medication | environment | biologic AllergyIntoleranceCategory (Required)
criticality | ? | 0..1 | code | low | high | unable-to-assess AllergyIntoleranceCriticality (Required)
code | ? | 0..1 | CodeableConcept | Code that identifies the allergy or intolerance AllergyIntolerance Substance/Product, Condition and Negation Codes (Example)
patient | ? | 1..1 | Reference(Patient) | Who the sensitivity is for
onset[x] | | 0..1 | | When allergy or intolerance was identified
onsetDateTime | | | dateTime
onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
assertedDate | | 0..1 | dateTime | Date record was believed accurate
recorder | | 0..1 | Reference(Practitioner | Patient) | Who recorded the sensitivity
asserter | ? | 0..1 | Reference(Patient | RelatedPerson | Practitioner) | Source of the information about the allergy
lastOccurrence | | 0..1 | dateTime | Date(/time) of last known occurrence of a reaction
note | | 0..* | Annotation | Additional text not captured in other fields
reaction | | 0..* | BackboneElement | Adverse Reaction Events linked to exposure to substance
substance | | 0..1 | CodeableConcept | Specific substance or pharmaceutical product considered to be responsible for event Substance Code (Example)
manifestation | | 1..* | CodeableConcept | Clinical symptoms/signs associated with the Event SNOMED CT Clinical Findings (Example)
description | | 0..1 | string | Description of the event as a whole
onset | | 0..1 | dateTime | Date(/time) when manifestations showed
severity | | 0..1 | code | mild | moderate | severe (of event as a whole) AllergyIntoleranceSeverity (Required)
exposureRoute | | 0..1 | CodeableConcept | How the subject was exposed to the substance SNOMED CT Route Codes (Example)
note | | 0..* | Annotation | Text about event not captured in other fields
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AllergyIntolerance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this item --></identifier>
 <clinicalStatus value="[code]"/><!--  0..1 active | inactive | resolved -->
 <verificationStatus value="[code]"/><!--  1..1 unconfirmed | confirmed | refuted | entered-in-error -->
 <type value="[code]"/><!-- 0..1 allergy | intolerance - Underlying mechanism (if known) -->
 <category value="[code]"/><!-- 0..* food | medication | environment | biologic -->
 <criticality value="[code]"/><!-- 0..1 low | high | unable-to-assess -->
 <code><!-- 0..1 CodeableConcept Code that identifies the allergy or intolerance --></code>
 <patient><!-- 1..1 Reference(Patient) Who the sensitivity is for --></patient>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string When allergy or intolerance was identified --></onset[x]>
 <assertedDate value="[dateTime]"/><!-- 0..1 Date record was believed accurate -->
 <recorder><!-- 0..1 Reference(Practitioner|Patient) Who recorded the sensitivity --></recorder>
 <asserter><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner) Source of the information about the allergy
    --></asserter>
 <lastOccurrence value="[dateTime]"/><!-- 0..1 Date(/time) of last known occurrence of a reaction -->
 <note><!-- 0..* Annotation Additional text not captured in other fields --></note>
 <reaction>  <!-- 0..* Adverse Reaction Events linked to exposure to substance -->
  <substance><!-- 0..1 CodeableConcept Specific substance or pharmaceutical product considered to be responsible for
    event --></substance>
  <manifestation><!-- 1..* CodeableConcept Clinical symptoms/signs associated with the Event --></manifestation>
  <description value="[string]"/><!-- 0..1 Description of the event as a whole -->
  <onset value="[dateTime]"/><!-- 0..1 Date(/time) when manifestations showed -->
  <severity value="[code]"/><!-- 0..1 mild | moderate | severe (of event as a whole) -->
  <exposureRoute><!-- 0..1 CodeableConcept How the subject was exposed to the substance --></exposureRoute>
  <note><!-- 0..* Annotation Text about event not captured in other fields --></note>
 </reaction>
</AllergyIntolerance>
```

JSON Template

```
{
  "resourceType" : "AllergyIntolerance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this item
  "clinicalStatus" : "<code>", // C? active | inactive | resolved
  "verificationStatus" : "<code>", // C? R!  unconfirmed | confirmed | refuted | entered-in-error
  "type" : "<code>", // allergy | intolerance - Underlying mechanism (if known)
  "category" : ["<code>"], // food | medication | environment | biologic
  "criticality" : "<code>", // low | high | unable-to-assess
  "code" : { CodeableConcept }, // Code that identifies the allergy or intolerance
  "patient" : { Reference(Patient) }, // R!  Who the sensitivity is for
  // onset[x]: When allergy or intolerance was identified. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  "assertedDate" : "<dateTime>", // Date record was believed accurate
  "recorder" : { Reference(Practitioner|Patient) }, // Who recorded the sensitivity
  "asserter" : { Reference(Patient|RelatedPerson|Practitioner) }, // Source of the information about the allergy
  "lastOccurrence" : "<dateTime>", // Date(/time) of last known occurrence of a reaction
  "note" : [{ Annotation }], // Additional text not captured in other fields
  "reaction" : [{ // Adverse Reaction Events linked to exposure to substance
    "substance" : { CodeableConcept }, // Specific substance or pharmaceutical product considered to be responsible for
    event
    "manifestation" : [{ CodeableConcept }], // R!  Clinical symptoms/signs associated with the Event
    "description" : "<string>", // Description of the event as a whole
    "onset" : "<dateTime>", // Date(/time) when manifestations showed
    "severity" : "<code>", // mild | moderate | severe (of event as a whole)
    "exposureRoute" : { CodeableConcept }, // How the subject was exposed to the substance
    "note" : [{ Annotation }] // Text about event not captured in other fields
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AllergyIntolerance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AllergyIntolerance.identifier [ Identifier ], ... ; # 0..* External ids for this item
  fhir:AllergyIntolerance.clinicalStatus [ code ]; # 0..1 active | inactive | resolved
  fhir:AllergyIntolerance.verificationStatus [ code ]; # 1..1 unconfirmed | confirmed | refuted | entered-in-error
  fhir:AllergyIntolerance.type [ code ]; # 0..1 allergy | intolerance - Underlying mechanism (if known)
  fhir:AllergyIntolerance.category [ code ], ... ; # 0..* food | medication | environment | biologic
  fhir:AllergyIntolerance.criticality [ code ]; # 0..1 low | high | unable-to-assess
  fhir:AllergyIntolerance.code [ CodeableConcept ]; # 0..1 Code that identifies the allergy or intolerance
  fhir:AllergyIntolerance.patient [ Reference(Patient) ]; # 1..1 Who the sensitivity is for
  # AllergyIntolerance.onset[x] : 0..1 When allergy or intolerance was identified. One of these 5
    fhir:AllergyIntolerance.onsetDateTime [ dateTime ]
    fhir:AllergyIntolerance.onsetAge [ Age ]
    fhir:AllergyIntolerance.onsetPeriod [ Period ]
    fhir:AllergyIntolerance.onsetRange [ Range ]
    fhir:AllergyIntolerance.onsetString [ string ]
  fhir:AllergyIntolerance.assertedDate [ dateTime ]; # 0..1 Date record was believed accurate
  fhir:AllergyIntolerance.recorder [ Reference(Practitioner|Patient) ]; # 0..1 Who recorded the sensitivity
  fhir:AllergyIntolerance.asserter [ Reference(Patient|RelatedPerson|Practitioner) ]; # 0..1 Source of the information
    about the allergy
  fhir:AllergyIntolerance.lastOccurrence [ dateTime ]; # 0..1 Date(/time) of last known occurrence of a reaction
  fhir:AllergyIntolerance.note [ Annotation ], ... ; # 0..* Additional text not captured in other fields
  fhir:AllergyIntolerance.reaction [ # 0..* Adverse Reaction Events linked to exposure to substance
    fhir:AllergyIntolerance.reaction.substance [ CodeableConcept ]; # 0..1 Specific substance or pharmaceutical product
    considered to be responsible for event
    fhir:AllergyIntolerance.reaction.manifestation [ CodeableConcept ], ... ; # 1..* Clinical symptoms/signs associated
    with the Event
    fhir:AllergyIntolerance.reaction.description [ string ]; # 0..1 Description of the event as a whole
    fhir:AllergyIntolerance.reaction.onset [ dateTime ]; # 0..1 Date(/time) when manifestations showed
    fhir:AllergyIntolerance.reaction.severity [ code ]; # 0..1 mild | moderate | severe (of event as a whole)
    fhir:AllergyIntolerance.reaction.exposureRoute [ CodeableConcept ]; # 0..1 How the subject was exposed to the
    substance
    fhir:AllergyIntolerance.reaction.note [ Annotation ], ... ; # 0..* Text about event not captured in other fields
  ], ...;
]
```

Changes since DSTU2

AllergyIntolerance

AllergyIntolerance.clinicalStatus | Added Element

AllergyIntolerance.verificationStatus | Added Element

AllergyIntolerance.category | Max Cardinality changed from 1 to *

AllergyIntolerance.code | Renamed from substance to code Min Cardinality changed from 1 to 0

AllergyIntolerance.onset[x] | Renamed from onset to onset[x] Add Age, Add Period, Add Range, Add string

AllergyIntolerance.assertedDate | Added Element

AllergyIntolerance.asserter | Renamed from reporter to asserter
AllergyIntolerance.lastOccurrence | Renamed from lastOccurence to lastOccurrence
AllergyIntolerance.note | Max Cardinality changed from 1 to *
AllergyIntolerance.reaction.note | Max Cardinality changed from 1 to *
AllergyIntolerance.recordedDate | deleted
AllergyIntolerance.status | deleted
AllergyIntolerance.reaction.certainty | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 5 tests that all execute ok. 2 fail round-trip testing and 4 r3 resources are invalid (6 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.1.3.1 Terminology Bindings

Path | Definition | Type | Reference
AllergyIntolerance.clinicalStatus | The clinical status of the allergy or intolerance. | Required | AllergyIntoleranceClinicalStatus
AllergyIntolerance.verificationStatus | Assertion about certainty associated with a propensity, or potential risk, of a reaction to the identified substance. | Required | AllergyIntoleranceVerificationStatus
AllergyIntolerance.type | Identification of the underlying physiological mechanism for a Reaction Risk. | Required | AllergyIntoleranceType
AllergyIntolerance.category | Category of an identified substance. | Required | AllergyIntoleranceCategory
AllergyIntolerance.criticality | Estimate of the potential clinical harm, or seriousness, of a reaction to an identified substance. | Required | AllergyIntoleranceCriticality
AllergyIntolerance.code | Type of the substance/product, allergy or intolerance condition, or negation/exclusion codes for reporting no known allergies. | Example | AllergyIntolerance Substance/Product, Condition and Negation Codes
AllergyIntolerance.reaction.substance | Codes defining the type of the substance (including pharmaceutical products). | Example | Substance Code
AllergyIntolerance.reaction.manifestation | Clinical symptoms and/or signs that are observed or associated with an Adverse Reaction Event. | Example | SNOMED CT Clinical Findings
AllergyIntolerance.reaction.severity | Clinical assessment of the severity of a reaction event as a whole, potentially considering multiple different manifestations. | Required | AllergyIntoleranceSeverity
AllergyIntolerance.reaction.exposureRoute | A coded concept describing the route or physiological path of administration of a therapeutic agent into or onto the body of a subject. | Example | SNOMED CT Route Codes

9.1.3.2 Constraints

* ait-1: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. (expression : verificationStatus='entered-in-error' or clinicalStatus.exists())

* ait-2: AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error (expression : verificationStatus!='entered-in-error' or clinicalStatus.empty())

9.1.3.3 Negated Allergies and Intolerances

It is important to differentiate between affirmatively stating that a patient has no known allergies versus either not including allergies in the record (for example an episodic document where the allergies are not considered relevant to the document); or asserting that allergies were not reviewed and are unknown.

Allergies with the verificationStatus "entered-in-error" indicates that the allergy or intolerance statement is entered by mistake and hence invalid.

Allergies with the verificationStatus "refuted" must be displayed to indicate that a reaction to a substance has been ruled out with the high level of clinical certainty (e.g. additional testing, rechallenging).

Prior to adding a new allergy/intolerance, a list of existing negated and refuted reactions should be reviewed and reconciled.

Allergies Not Reviewed, Not Asked

When a sending system does not have any information about allergies being reviewed or the statement is about allergies not being asked yet, then the List resource should be used to indicate the List.emptyReason.code="notasked".

Allergies Reviewed, None Identified

Systems may use the List.emptyReason when a statement is about the full scope of the list (i.e. the patient has no known allergies or intolerances of any type). However, it is generally preferred to use a code for "No known allergies" (e.g., SNOMED CT: 716186003 |No known allergy (situation)|), so that all allergy data will be available and queryable from AllergyIntolerance resource instances. Negated AllergyIntolerance instances are also typically used when the record is more fine-grained (e.g. no drug allergies, no food allergies, no nut allergies, etc.).

However, it is possible to include negation statements that apply at the level of the whole list and it is also possible to have separate lists for things like medication allergies vs. food allergies, where that is appropriate to the architecture. Also note that care should be used when adding new AllergyIntolerances to a list to ensure that any negation statements that are voided by the addition of a new record are removed from the list. E.g. If the list contains a "no known food allergies" record and you add an "intolerance to grape flavor" record, then be sure you remove the "no known food allergies" record.

The substanceExposureRisk extension is also available for use as a more completely structured and flexible alternative to the 'code' element for representing positive and negative allergy and intolerance statements (either the 'code' element or the substanceExposureRisk extension may be used, but not both).

STU Note:There are two primary ways of reporting "no known allergies" in the current specification: using the CodeableConcept, as described above, or using the List resource with emptyReason. The third available option is using the substanceExposureRisk extension. During the STU period, it is not recommended to use the List resource for "no known allergies" reporting purposes. The principal reason for this is to allow all allergy or intolerance data to be found and to be consistently queryable from the single location of the AllergyIntolerance resource.

Provide feedback here .

No Known Allergies, using List.emptyReason (discouraged)

```
  <List xmlns="http://hl7.org/fhir" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://hl7.org/fhir ..\..\schema\list.xsd">
 <id value="example-empty-allergy"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p> The patient is not aware of any allergies.</p>
    </div>
  </text>
<code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="52472-8"/>
      <display value="Allergies and Adverse Drug Reactions"/>
    </coding>
    <text value="Current Allergy List"/>
  </code>
  <source>
    <reference value="Patient/example"/>
  </source>
  <status value="current"/>
  <date value="2012-11-26T07:30:23+11:00"/>
  <mode value="snapshot"/>
  <emptyReason>
    <coding>
      <system value="http://hl7.org/fhir/special-values"/>
      <code value="nil-known"/>
      <display value="Nil Known"/>
    </coding>
    <text value="The patient is not aware of any allergies."/>
  </emptyReason>
</List>
```

No Known Food Allergies and Medication Allergy List

```
  <?xml version="1.0" encoding="UTF-8"?>
<List xmlns="http://hl7.org/fhir" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://hl7.org/fhir ..\..\schema\list.xsd">
  <id value="current-allergies"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Patient Peter Chalmers, DOB = Dec 25, 1974, MRN = 12345 (Acme Healthcare) has the following allergies</p>
      <ul>
        <li>No known food allergies</li>
        <li>Allergenic extract, penicillin (high)</li>
      </ul>
    </div>
  </text>
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="52472-8"/>
      <display value="Allergies and Adverse Drug Reactions"/>
    </coding>
    <text value="Current Allergy List"/>
  </code>
  <source>
    <reference value="Patient/example"/>
  </source>
  <status value="current"/>
  <date value="2015-07-14T23:10:23+11:00"/>
  <mode value="snapshot"/>
  <entry>
    <item>
      <reference value="AllergyIntolerance/nofoodallergies"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="AllergyIntolerance/penicillin"/>
    </item>
  </entry>
</List>
```

If a new allergy is discovered, the negated allergy record must be updated with the "refuted" verificationStatus - to ensure that systems referring to this record are aware that this is no longer true.

9.1.3.4 Use of AllergyIntolerance.criticality

Systems that only support one notion will have to determine whether what they're capturing is criticality or severity and map to the appropriate place. Criticality refers to the likelihood the allergy/intolerance could result in significant harm. Severity refers to the degree of manifestation of the reaction symptom. Moderate breathing difficulty would have high criticality while a severe rash would have low criticality. Severity is specific to a particular reaction occurrence.

For systems that only track generic reaction characteristics rather than a specific reaction will provide guidance to use the "reaction" structure and simply provide no date.

9.1.3.5 References

* Adverse Reaction, draft archetype, National eHealth Transition Authority [Internet]. NEHTA Clinical Knowledge Manager. Authored: 08 Nov 2010. Available at: http://dcm.nehta.org.au/ckm/OKM.html#showarchetype_1013.1.868_7 (accessed Jan 16, 2012).

* Allergy and Intolerance Domain Analysis Model, Release 1, HL7 [Internet]. Publication pending, expected August 2014; Available at http://wiki.hl7.org/images/1/1b/Allergy_and_Intolerance_INFORM_2013_MAY.pdf (accessed 06 July 2014).

* Allergy, clinical element model, GE/Intermountain Healthcare. Clinical Element Model Search. Available at: http://intermountainhealthcare.org/cem/Pages/Detail.aspx?NCID=520861661&k=allergy (accessed Jan 16, 2012).

* Edwards IR, Aronson JK. Adverse drug reactions: definitions, diagnosis, and management. Lancet. 2000 Oct 7;356(9237):1255-9. PubMed PMID: 11072960.

* Horsfield P, Sibeko S. Representation in Electronic Patient Records of Allergic Reactions, Adverse Reactions, and Intolerance of Pharmaceutical Products [Internet]. London, UK: National Health Service; 2006 Sep 07 [cited 2011 Jun 21]; Available at https://svn.connectingforhealth.nhs.uk/svn/public/nhscontentmodels/TRUNK/ref/NPfIT/Allergy_ADR_Intolerance%20v%201.2Final.doc .

* Long R, Bentley S. SCG Guidance on the Representation of Allergies and Adverse Reaction Information Using NHS Message Templates [Internet]. London, UK: National Health Service; 2008 Apr 30 [cited 2011 Jun 21]; Available at http://www.connectingforhealth.nhs.uk/systemsandservices/data/scg/scg0001.pdf .

* Microsoft. Design Guidance: Displaying Adverse Drug Reaction Risks [Internet]. 2009 January 28 [cited 2011 Jun 21]; Available at www.mscui.net/DesignGuide/DisplayingAllergies.aspx .

* Microsoft. Design Guidance: Recording Adverse Drug Reaction Risks [Internet]. 2009 March 27 [cited 2011 Jun 21]; Available at www.mscui.net/DesignGuide/RecordingAllergies.aspx .

* Mosby. Mosby's Pocket Dictionary of Medicine, Nursing and Health Professions. 6th Edition. USA: Mosby Elsevier; 2010

* National E-Health Transition Authority. Adverse Reactions (Data Specifications) Version 1.1 [Internet]. Sydney, Australia: NEHTA; 2008 Feb 29 [cited 2011 Jun 21]; Available at http://www.nehta.gov.au/component/docman/doc_download/453-adverse-reaction-data-specification-v11

* Riedl MA, Casillas AM. Adverse drug reactions: types and treatment options. Am Fam Physician. 2003 Nov 1;68(9):1781-90. Review. PubMed PMID: 14620598.

* Royal Australian College of General Practitioners. Fact Sheet: Allergies & Adverse Reactions (Draft). 2010.

* Thien FC. Drug hypersensitivity. Med J Aust. 2006 Sep 18;185(6):333-8. Review. PubMed PMID: 16999678.

* Uppsala Monitoring Centre (WHO): http://www.who-umc.org/

* European Medicines Agency: http://www.ema.europa.eu/ema/

* DIRECTIVE 2010/84/EU OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL, of 15 December 2010, amending, as regards pharmacovigilance, Directive 2001/83/EC on the Community code relating to medicinal products for human use: http://ec.europa.eu/health/files/eudralex/vol-1/dir_2010_84/dir_2010_84_en.pdf

9.1.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
asserter	reference	Source of the information about the allergy	AllergyIntolerance.asserter (Practitioner, Patient, RelatedPerson)	
category	token	food | medication | environment | biologic	AllergyIntolerance.category	
clinical-status	token	active | inactive | resolved	AllergyIntolerance.clinicalStatus	
code	token	Code that identifies the allergy or intolerance	AllergyIntolerance.code | AllergyIntolerance.reaction.substance	8 Resources
criticality	token	low | high | unable-to-assess	AllergyIntolerance.criticality	
date	date	Date record was believed accurate	AllergyIntolerance.assertedDate	18 Resources
identifier	token	External ids for this item	AllergyIntolerance.identifier	26 Resources
last-date	date	Date(/time) of last known occurrence of a reaction	AllergyIntolerance.lastOccurrence	
manifestation	token	Clinical symptoms/signs associated with the Event	AllergyIntolerance.reaction.manifestation	
onset	date	Date(/time) when manifestations showed	AllergyIntolerance.reaction.onset	
patient	reference	Who the sensitivity is for	AllergyIntolerance.patient (Patient)	31 Resources
recorder	reference	Who recorded the sensitivity	AllergyIntolerance.recorder (Practitioner, Patient)	
route	token	How the subject was exposed to the substance	AllergyIntolerance.reaction.exposureRoute	
severity	token	mild | moderate | severe (of event as a whole)	AllergyIntolerance.reaction.severity	
type	token	allergy | intolerance - Underlying mechanism (if known)	AllergyIntolerance.type	6 Resources
verification-status	token	unconfirmed | confirmed | refuted | entered-in-error	AllergyIntolerance.verificationStatus	

9.1.5 Resource AllergyIntolerance - Examples

URL: https://hl7.org/fhir/STU3/allergyintolerance-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.5 Resource AllergyIntolerance - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Example List:

A typical Food Allergy | example | XML | JSON | Turtle

A typical Medication allergy from a clinical system | medication | XML | JSON | Turtle

An allergy to a substance with additional details in notes | fishallergy | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.1.6 Resource AllergyIntolerance - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/allergyintolerance-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.6 Resource AllergyIntolerance - Detailed Descriptions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the AllergyIntolerance resource.

AllergyIntolerance

Definition | Risk of harmful or undesirable, physiological response which is unique to an individual and associated with exposure to a substance.

Control | 1..1

Requirements | To record a clinical assessment of a propensity, or potential risk to an individual, of an adverse reaction upon future exposure to the specified substance, or class of substance.

Alternate Names | Allergy; Intolerance; Adverse Reaction

Comments | Substances include, but are not limited to: a therapeutic substance administered correctly at an appropriate dosage for the individual; food; material derived from plants or animals; or venom from insect stings.

Invariants | Defined on this element ait-1: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. (expression : verificationStatus='entered-in-error' or clinicalStatus.exists(), xpath: f:verificationStatus/@value='entered-in-error' or exists(f:clinicalStatus)) ait-2: AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error (expression : verificationStatus!='entered-in-error' or clinicalStatus.empty(), xpath: f:verificationStatus/@value!='entered-in-error' or not(exists(f:clinicalStatus)))

AllergyIntolerance.identifier

Definition | This records identifiers associated with this allergy/intolerance concern that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).

Note | This is a business identifer, not a resource identifier (see discussion)

Control | 0..*

Type | Identifier

Summary | true

AllergyIntolerance.clinicalStatus

Definition | The clinical status of the allergy or intolerance.

Control | 0..1

Terminology Binding | AllergyIntoleranceClinicalStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | This element is labeled as a modifier because the status contains the codes inactive and resolved that mark the AllergyIntolerance as not currently valid.

Invariants | Affect this element ait-1: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. (expression : verificationStatus='entered-in-error' or clinicalStatus.exists(), xpath: f:verificationStatus/@value='entered-in-error' or exists(f:clinicalStatus)) ait-2: AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error (expression : verificationStatus!='entered-in-error' or clinicalStatus.empty(), xpath: f:verificationStatus/@value!='entered-in-error' or not(exists(f:clinicalStatus)))

AllergyIntolerance.verificationStatus

Definition | Assertion about certainty associated with the propensity, or potential risk, of a reaction to the identified substance (including pharmaceutical product).

Control | 1..1

Terminology Binding | AllergyIntoleranceVerificationStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | This element is labeled as a modifier because the status contains the codes refuted and entered-in-error that mark the AllergyIntolerance as not currently valid.

Invariants | Affect this element ait-1: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. (expression : verificationStatus='entered-in-error' or clinicalStatus.exists(), xpath: f:verificationStatus/@value='entered-in-error' or exists(f:clinicalStatus)) ait-2: AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error (expression : verificationStatus!='entered-in-error' or clinicalStatus.empty(), xpath: f:verificationStatus/@value!='entered-in-error' or not(exists(f:clinicalStatus)))

AllergyIntolerance.type
Definition | Identification of the underlying physiological mechanism for the reaction risk.
Control | 0..1
Terminology Binding | AllergyIntoleranceType (Required)
Type | code
Alternate Names | Category; Class
Summary | true
Comments | Allergic (typically immune-mediated) reactions have been traditionally regarded as an indicator for potential escalation to significant future risk. Contemporary knowledge suggests that some reactions previously thought to be immune-mediated are, in fact, non-immune, but in some cases can still pose a life threatening risk. It is acknowledged that many clinicians may not be in a position to distinguish the mechanism of a particular reaction. Often the term "allergy" is used rather generically and may overlap with the use of "intolerance" - in practice the boundaries between these two concepts may not be well-defined or understood. This data element is included nevertheless, because many legacy systems have captured this attribute. Immunologic testing may provide supporting evidence for the basis of the reaction and the causative substance, but no tests are 100% sensitive or specific for sensitivity to a particular substance. If, as is commonly the case, it is unclear whether the reaction is due to an allergy or an intolerance, then the type element should be omitted from the resource.
AllergyIntolerance.category
Definition | Category of the identified substance.
Control | 0..*
Terminology Binding | AllergyIntoleranceCategory (Required)
Type | code
Alternate Names | Category; Type; Reaction Type; Class
Summary | true
Comments | This data element has been included because it is currently being captured in some clinical systems. This data can be derived from the substance where coding systems are used, and is effectively redundant in that situation. When searching on category, consider the implications of AllergyIntolerance resources without a category. For example, when searching on category = medication, medication allergies that don't have a category valued will not be returned. Refer to search for more information on how to search category with a :missing modifier to get allergies that don't have a category. Additionally, category should be used with caution because category can be subjective based on the sender.
AllergyIntolerance.criticality
Definition | Estimate of the potential clinical harm, or seriousness, of the reaction to the identified substance.
Control | 0..1
Terminology Binding | AllergyIntoleranceCriticality (Required)
Type | code
Alternate Names | Severity; Seriousness; Contra-indication; Risk
Summary | true
Comments | The default criticality value for any propensity to an adverse reaction should be 'Low Risk', indicating at the very least a relative contraindication to deliberate or voluntary exposure to the substance. 'High Risk' is flagged if the clinician has identified a propensity for a more serious or potentially life-threatening reaction, such as anaphylaxis, and implies an absolute contraindication to deliberate or voluntary exposure to the substance. If this element is missing, the criticality is unknown (though it may be known elsewhere). Systems that capture a severity at the condition level are actually representing the concept of criticality whereas the severity documented at the reaction level is representing the true reaction severity. Existing systems that are capturing both condition criticality and reaction severity may use the term "severity" to represent both. Criticality is the worst it could be in the future (i.e. situation-agnostic) whereas severity is situation-dependent.
AllergyIntolerance.code
Definition | Code for an allergy or intolerance statement (either a positive or a negated/excluded statement). This may be a code for a substance or pharmaceutical product that is considered to be responsible for the adverse reaction risk (e.g., "Latex"), an allergy or intolerance condition (e.g., "Latex allergy"), or a negated/excluded code for a specific substance or class (e.g., "No latex allergy") or a general or categorical negated statement (e.g., "No known allergy", "No known drug allergies").
Control | 0..1
Terminology Binding | AllergyIntolerance Substance/Product, Condition and Negation Codes (Example)
Type | CodeableConcept
Alternate Names | Code
Summary | true
Comments | It is strongly recommended that this element be populated using a terminology, where possible. For example, some terminologies used include RxNorm, SNOMED CT, DM+D, NDFRT, ICD-9, IDC-10, UNI, ATC and CPT. Plain text should only be used if there is no appropriate terminology available. Additional details can be specified in the text. When a substance or product code is specified for the 'code' element, the "default" semantic context is that this is a positive statement of an allergy or intolerance (depending on the value of the 'type' element, if present) condition to the specified substance/product. In the corresponding SNOMED CT allergy model, the specified substance/product is the target (destination) of the "Causative agent" relationship. The 'substanceExposureRisk' extension is available as a structured and more flexible alternative to the 'code' element for making positive or negative allergy or intolerance statements. This extension provides the capability to make "no known allergy" (or "no risk of adverse reaction") statements regarding any coded substance/product (including cases when a pre-coordinated "no allergy to x" concept for that substance/product does not exist). If the 'substanceExposureRisk' extension is present, the AllergyIntolerance.code element SHALL be omitted.
AllergyIntolerance.patient
Definition | The patient who has the allergy or intolerance.
Control | 1..1
Type | Reference(Patient)
Alternate Names | Patient
Summary | true
AllergyIntolerance.onset[x]
Definition | Estimated or actual date, date-time, or age when allergy or intolerance was identified.
Control | 0..1
Type | dateTime|Age|Period|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
AllergyIntolerance.assertedDate
Definition | The date on which the existance of the AllergyIntolerance was first asserted or acknowledged.
Control | 0..1

Type | dateTime
AllergyIntolerance.recorder
Definition | Individual who recorded the record and takes responsibility for its content.
Control | 0..1
Type | Reference(Practitioner | Patient)
Alternate Names | Author
AllergyIntolerance.asserter
Definition | The source of the information about the allergy that is recorded.
Control | 0..1
Type | Reference(Patient | RelatedPerson | Practitioner)
Alternate Names | Source; Informant
Summary | true
Comments | The recorder takes repsonsibility for the content, but can reference the source from where they got it.
AllergyIntolerance.lastOccurrence
Definition | Represents the date and/or time of the last known occurrence of a reaction event.
Control | 0..1
Type | dateTime
Comments | This date may be replicated by one of the Onset of Reaction dates. Where a textual representation of the date of last occurrence is required e.g. 'In Childhood, '10 years ago' the Comment element should be used.
AllergyIntolerance.note
Definition | Additional narrative about the propensity for the Adverse Reaction, not captured in other fields.
Control | 0..*
Type | Annotation
Comments | For example: including reason for flagging a seriousness of 'High Risk'; and instructions related to future exposure or administration of the substance, such as administration within an Intensive Care Unit or under corticosteroid cover. The notes should be related to an allergy or intolerance as a condition in general and not related to any particular episode of it. For episode notes and descriptions, use AllergyIntolerance.event.description and AllergyIntolerance.event.notes.
AllergyIntolerance.reaction
Definition | Details about each adverse reaction event linked to exposure to the identified substance.
Control | 0..*
AllergyIntolerance.reaction.substance
Definition | Identification of the specific substance (or pharmaceutical product) considered to be responsible for the Adverse Reaction event. Note: the substance for a specific reaction may be different from the substance identified as the cause of the risk, but it must be consistent with it. For instance, it may be a more specific substance (e.g. a brand medication) or a composite product that includes the identified substance. It must be clinically safe to only process the 'code' and ignore the 'reaction.substance'.
Control | 0..1
Terminology Binding | Substance Code (Example)
Type | CodeableConcept
Comments | Coding of the specific substance (or pharmaceutical product) with a terminology capable of triggering decision support should be used wherever possible. The 'code' element allows for the use of a specific substance or pharmaceutical product, or a group or class of substances. In the case of an allergy or intolerance to a class of substances, (for example, "penicillins"), the 'reaction.substance' element could be used to code the specific substance that was identifed as having caused the reaction (for example, "amoxycillin"). Duplication of the value in the 'code' and 'reaction.substance' elements is acceptable when a specific substance has been recorded in 'code'.
AllergyIntolerance.reaction.manifestation
Definition | Clinical symptoms and/or signs that are observed or associated with the adverse reaction event.
Control | 1..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Alternate Names | Symptoms; Signs
Comments | Manifestation can be expressed as a single word, phrase or brief description. For example: nausea, rash or no reaction. It is preferable that manifestation should be coded with a terminology, where possible. The values entered here may be used to display on an application screen as part of a list of adverse reactions, as recommended in the UK NHS CUI guidelines. Terminologies commonly used include, but are not limited to, SNOMED CT or ICD10.
AllergyIntolerance.reaction.description
Definition | Text description about the reaction as a whole, including details of the manifestation if required.
Control | 0..1
Type | string
Alternate Names | Narrative; Text
Comments | Use the description to provide any details of a particular event of the occurred reaction such as circumstances, reaction specifics, what happened before/after. Information, related to the event, but not describing a particular care should be captured in the comment field. For example: at the age of four, the patient was given penicillin for strep throat and subsequently developed severe hives.
AllergyIntolerance.reaction.onset
Definition | Record of the date and/or time of the onset of the Reaction.
Control | 0..1
Type | dateTime
AllergyIntolerance.reaction.severity
Definition | Clinical assessment of the severity of the reaction event as a whole, potentially considering multiple different manifestations.
Control | 0..1
Terminology Binding | AllergyIntoleranceSeverity (Required)
Type | code
Comments | It is acknowledged that this assessment is very subjective. There may be some some specific practice domains where objective scales have been applied. Objective scales can be included in this model as extensions.
AllergyIntolerance.reaction.exposureRoute
Definition | Identification of the route by which the subject was exposed to the substance.

Control | 0..1
Terminology Binding | SNOMED CT Route Codes (Example)
Type | CodeableConcept
Comments | Coding of the route of exposure with a terminology should be used wherever possible.
AllergyIntolerance.reaction.note
Definition | Additional text about the adverse reaction event not captured in other fields.
Control | 0..*
Type | Annotation
Comments | Use this field to record information indirectly related to a particular event and not captured in the description. For example: Clinical records are no longer available, recorded based on information provided to the patient by her mother and her mother is deceased.

9.1.7 Resource AllergyIntolerance - Mappings

URL: https://hl7.org/fhir/STU3/allergyintolerance-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.7 Resource AllergyIntolerance - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the allergyintolerance resource.

9.1.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

AllergyIntolerance | Observation[classCode=OBS, moodCode=EVN]
identifier | id
clinicalStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="clinicalStatus", moodCode=EVN].value
verificationStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="verificationStatus", moodCode=EVN].value
type | code
category | value < IntoleranceValue (Agent)
criticality | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=SEV, value <= SeverityObservation (Severity Level)]
code | substance/product: .participation[typeCode=CAGNT].role[classCode=ADMM].player[classCode=MAT, determinerCode=KIND, code <= ExposureAgentEntityType] negated/excluded substance/product: .participation[typeCode=CAGNT, negationInd=true].role[classCode=ADMM].player[classCode=MAT, determinerCode=KIND, code <= ExposureAgentEntityType] positive or negated/excluded condition/situation: Observation.code=ASSERTION; Observation.value
patient | .participation[typeCode=SBJ].role[classCode=PAT]
onset[x] | effectiveTime.low
assertedDate | .participation[typeCode=AUT].time
recorder | .participation[typeCode=AUT].role
asserter | .participation[typeCode=INF].role
lastOccurrence | outBoundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN, code <= CommonClinicalObservationType, value <= ObservationValue (Reaction Type)].effectiveTime
note | subjectOf.observationEvent[code="annotation"].value
reaction | outBoundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN, code <= CommonClinicalObservationType, value <= ObservationValue (Reaction Type)]
substance | outBoundRelationship[typeCode=SAS].target[classCode=SBADM, code <= ExposureCode].participation[typeCode=CSM].role[classCode=ADMM].player[classCode=MAT, determinerCode=KIND, code <= ExposureAgentEntityType]
manifestation | code
description | text
onset | effectiveTime.low
severity | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=SEV, value <= SeverityObservation (Severity Level)]
exposureRoute | outBoundRelationship[typeCode=SAS].target[classCode=SBADM, code <= ExposureCode].routeCode
note | subjectOf.observationEvent[code="annotation"].value

9.1.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

AllergyIntolerance
identifier | IAM-7
clinicalStatus
verificationStatus
type | IAM-9
category | AL1-2
criticality | AL1-4
code | AL1-3 / IAM-3
patient | (PID-3)
onset[x]

assertedDate | IAM-20
recorder
asserter | IAM-14 (if patient) / IAM-18 (if practitioner)
lastOccurrence
note
reaction
substance
manifestation | AL1-5
description
onset | AL1-6
severity
exposureRoute
note

9.1.8 Resource AllergyIntolerance - Design Notes

URL: https://hl7.org/fhir/STU3/allergyintolerance-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.8 Resource AllergyIntolerance - Design Notes

to do

9.1.9 Resource AllergyIntolerance - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/allergyintolerance-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.9 Resource AllergyIntolerance - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the AllergyIntolerance resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.1.9.1 Profiles

No Profiles defined for this resource

9.1.9.2 Extensions

allergyintolerance-certainty | certainty | for AllergyIntolerance HL7 Extensions
allergyintolerance-duration | duration | for AllergyIntolerance HL7 Extensions
allergyintolerance-reasonRefuted | reasonRefuted | for AllergyIntolerance HL7 Extensions
allergyintolerance-resolutionAge | resolutionAge | for AllergyIntolerance HL7 Extensions
allergyintolerance-substanceExposureRisk | substanceExposureRisk | for AllergyIntolerance HL7 Extensions
openEHR-administration | administration | for openEHR Archetype Profile
openEHR-careplan | careplan | for openEHR Archetype Profile
openEHR-exposureDate | exposureDate | for openEHR Archetype Profile
openEHR-exposureDescription | exposureDescription | for openEHR Archetype Profile
openEHR-exposureDuration | exposureDuration | for openEHR Archetype Profile
openEHR-location | location | for openEHR Archetype Profile
openEHR-management | management | for openEHR Archetype Profile
openEHR-test | test | for openEHR Archetype Profile

Extensions that reference this resource:

procedure-causedBy | causedBy | for Procedure HL7 Extensions

9.1.9.3 Search Extensions

No Search Extensions defined for this resource

9.1.10 openEHR Archetype Profile

URL: https://hl7.org/fhir/STU3/allergyintolerance-openehr.html

A set of extensions that define how the parts of the openEHR RiskOfAdverseReaction archetype that are not generaly encountered by implementers are represented

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.10 openEHR Archetype Profile

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.1.10.1 Content

Extensions:
openEHR-test | test : Observations that confirm or refute the risk and/or the substance.
openEHR-location | location : The anatomical location / body site(s) where the symptoms manifested.
openEHR-exposureDate | exposureDate : Record of the date and/or time of the first exposure to the Substance for this Reaction Event.
openEHR-exposureDuration | exposureDuration : The amount of time the individual was exposed to the Substance.
openEHR-exposureDescription | exposureDescription : Text description about exposure to the Substance.
openEHR-administration | administration : Link to an actual medication administration record with the full details of the administration, if a link is known.
openEHR-management | management : Text description about the clinical management provided.
openEHR-careplan | careplan : Additional details about the clinical management provided for this Reaction Event.

9.1.11 AllergyIntolerance HL7 Extensions

URL: https://hl7.org/fhir/STU3/allergyintolerance-extensions.html

Defines common extensions used with or related to the AllergyIntolerance resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.11 AllergyIntolerance HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.1.11.1 Content

Extensions:
allergyintolerance-resolutionAge | resolutionAge : The estimated patient age at which the allergy or intolerance resolved. Should be specified only if the status is resolved.
allergyintolerance-reasonRefuted | reasonRefuted : A code capturing the explanation of why the allergy or intolerance has been refuted. Should be specified only if the status is refuted.
allergyintolerance-duration | duration : The amount of time that the Adverse Reaction persisted.
allergyintolerance-substanceExposureRisk | substanceExposureRisk : A complex extension allowing structured capture of the exposure risk of the patient for an adverse reaction (allergy or intolerance) to the specified substance/product.
allergyintolerance-certainty | certainty : Statement about the degree of clinical certainty that the specific substance was the cause of the manifestation in this reaction event.

9.1.12 Resource AllergyIntolerance - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/allergyintolerance-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.1.12 Resource AllergyIntolerance - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for AllergyIntolerance.

Functional status for this map: 5 tests that all execute ok. 2 fail round-trip testing and 4 r3 resources are invalid (6 errors). (see documentation)

9.1.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/AllergyIntolerance2to3" = "R2 to R3 Conversions for AllergyIntolerance"

conceptmap "AllergyIntoleranceVerificationStatus" {
  prefix s = "http://hl7.org/fhir/allergy-intolerance-status"
  prefix t = "http://hl7.org/fhir/allergy-verification-status"

  s:unconfirmed = t:unconfirmed
  s:confirmed = t:confirmed
  s:inactive = t:unconfirmed
  s:refuted = t:refuted
  s:"entered-in-error" = t:"entered-in-error"
}

conceptmap "AllergyIntoleranceClinicalStatus" {
  prefix s = "http://hl7.org/fhir/allergy-intolerance-status"
  prefix t = "  http://hl7.org/fhir/allergy-clinical-status"

  s:active = t:active
  s:unconfirmed = t:active
  s:confirmed = t:active
  s:inactive = t:inactive
  s:resolved = t:resolved
}
conceptmap "AllergyIntoleranceCriticality" {
```

```
  prefix s = "http://hl7.org/fhir/allergy-intolerance-criticality"
  prefix t = "http://hl7.org/fhir/allergy-intolerance-criticality"

  s:CRITL = t:low
  s:CRITH = t:high
  s:CRITU = t:"unable-to-assess"
}

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/AllergyIntolerance" alias AllergyIntoleranceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/AllergyIntolerance" alias AllergyIntolerance as target

imports "http://hl7.org/fhir/StructureMap/*to3"



group for type+types  AllergyIntolerance extends DomainResource
  input src : AllergyIntoleranceR2 as source
  input tgt : AllergyIntolerance as target

  "AllergyIntolerance-identifier" : for src.identifier make tgt.identifier
  "AllergyIntolerance-onset" : for src.onset make tgt.onset
  "AllergyIntolerance-recordedDate" : for src.recordedDate make tgt.assertedDate
  "AllergyIntolerance-recorder" : for src.recorder make tgt.recorder
  "AllergyIntolerance-patient" : for src.patient make tgt.patient
  "AllergyIntolerance-reporter" : for src.reporter make tgt.asserter
  "AllergyIntolerance-substance" : for src.substance make tgt.code

  "AllergyIntolerance-status-clinical" :     for src.status as v where $this != 'refuted' and $this != 'entered-in-
    error' make tgt.clinicalStatus  = translate(v, "#AllergyIntoleranceClinicalStatus", "code")
  "AllergyIntolerance-Status-verification" : for src.status as v where $this != 'active' and $this != 'resolved' make
    tgt.verificationStatus = translate(v, "#AllergyIntoleranceVerificationStatus", "code")

  "AllergyIntolerance-criticality" : for src.criticality as v make tgt.criticality = translate(v,
    "#AllergyIntoleranceCriticality", "code")
  "AllergyIntolerance-type" : for src.type make tgt.type
  "AllergyIntolerance-category" : for src.category make tgt.category
  "AllergyIntolerance-lastOccurrence" : for src.lastOccurence make tgt.lastOccurrence
  "AllergyIntolerance-note" : for src.note make tgt.note
  "AllergyIntolerance-reaction" : for src.reaction as vs0 make tgt.reaction as vt0 then algReaction(vs0, vt0)

endgroup

group algReaction extends BackboneElement
  input src as source
  input tgt as target

  "AllergyIntolerance.reaction-substance" : for src.substance make tgt.substance
  "AllergyIntolerance.reaction-certainty" : for src.certainty as v make tgt.extension as t, t.url =
    'http://hl7.org/fhir/AllergyIntolerance-r2-certainty', t.value = v
  "AllergyIntolerance.reaction-manifestation" : for src.manifestation make tgt.manifestation
  "AllergyIntolerance.reaction-description" : for src.description make tgt.description
  "AllergyIntolerance.reaction-onset" : for src.onset make tgt.onset
  "AllergyIntolerance.reaction-severity" : for src.severity make tgt.severity
  "AllergyIntolerance.reaction-exposureRoute" : for src.exposureRoute make tgt.exposureRoute
  "AllergyIntolerance.reaction-note" : for src.note make tgt.note
endgroup
```

9.1.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/AllergyIntolerance3to2" = "R3 to R2 Conversion for AllergyIntolerance"

conceptmap "AllergyIntoleranceCriticality" {
  prefix s = "http://hl7.org/fhir/allergy-intolerance-criticality"
  prefix t = "http://hl7.org/fhir/allergy-intolerance-criticality"

  s:low = t:CRITL
  s:high = t:CRITH
  s:"unable-to-assess" = t:CRITU
}

uses "http://hl7.org/fhir/StructureDefinition/AllergyIntolerance" alias AllergyIntolerance as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/AllergyIntolerance" alias AllergyIntoleranceR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  AllergyIntolerance extends DomainResource
  input src : AllergyIntolerance as source
  input tgt : AllergyIntoleranceR2 as target

  "AllergyIntolerance-identifier" : for src.identifier make tgt.identifier
  "AllergyIntolerance-onset" : for src.onset make tgt.onset
  "AllergyIntolerance-recordedDate" : for src.assertedDate make tgt.recordedDate
  "AllergyIntolerance-recorder" : for src.recorder make tgt.recorder
  "AllergyIntolerance-patient" : for src.patient make tgt.patient
  "AllergyIntolerance-reporter" : for src.asserter make tgt.reporter
  "AllergyIntolerance-substance" : for src.code make tgt.substance

  "AllergyIntolerance-status-active" :           for src where src.clinicalStatus = 'active' and
    src.verificationStatus.empty() make tgt.status = 'active'
  "AllergyIntolerance-status-unconfirmed" :      for src where src.clinicalStatus = 'active' and src.verificationStatus
```

```
    = 'confirmed' make tgt.status = 'confirmed'
  "AllergyIntolerance-status-confirmed" :        for src where src.clinicalStatus = 'active' and src.verificationStatus
    = 'unconfirmed' make tgt.status = 'unconfirmed'
  "AllergyIntolerance-status-inactive" :         for src where src.clinicalStatus = 'inactive' make tgt.status =
    'inactive'
  "AllergyIntolerance-status-resolved" :         for src where src.clinicalStatus = 'resolved' make tgt.status =
    'resolved'
  "AllergyIntolerance-status-refuted" :          for src where src.verificationStatus = 'refuted' make tgt.status =
    'refuted'
  "AllergyIntolerance-status-entered-in-error" : for src where src.verificationStatus = 'entered-in-error' make
    tgt.status = 'entered-in-error'

  "AllergyIntolerance-criticality" : for src.criticality as v make tgt.criticality = translate(v,
    "#AllergyIntoleranceCriticality", "code")
  "AllergyIntolerance-type" : for src.type make tgt.type
  "AllergyIntolerance-category" : for src.category make tgt.category
  "AllergyIntolerance-lastOccurrence" : for src.lastOccurrence make tgt.lastOccurence
  "AllergyIntolerance-note" : for src.note make tgt.note
  "AllergyIntolerance-reaction" : for src.reaction as vs0 make tgt.reaction as vt0 then algReaction(vs0, vt0)

endgroup

group algReaction extends BackboneElement
  input src as source
  input tgt as target

  "AllergyIntolerance.reaction-substance" : for src.substance make tgt.substance
  "AllergyIntolerance.reaction-certainty" : for src.extension as e where e.url =
    'http://hl7.org/fhir/AllergyIntolerance-r2-certainty' make tgt.certainty = (e.value)
  "AllergyIntolerance.reaction-manifestation" : for src.manifestation make tgt.manifestation
  "AllergyIntolerance.reaction-description" : for src.description make tgt.description
  "AllergyIntolerance.reaction-onset" : for src.onset make tgt.onset
  "AllergyIntolerance.reaction-severity" : for src.severity make tgt.severity
  "AllergyIntolerance.reaction-exposureRoute" : for src.exposureRoute make tgt.exposureRoute
  "AllergyIntolerance.reaction-note" : for src.note make tgt.note
endgroup
```

9.1.12.3 R3 Validation Errors

allergyintolerance-example | ERROR @ AllergyIntolerance.reaction[1].extension: The extension http://hl7.org/fhir/AllergyIntolerance-r2-certainty is unknown, and not allowed here (src = InstanceValidator) ERROR @ AllergyIntolerance.reaction[2].extension: The extension http://hl7.org/fhir/AllergyIntolerance-r2-certainty is unknown, and not allowed here (src = InstanceValidator)

example | ERROR @ AllergyIntolerance.reaction[2].extension: The extension http://hl7.org/fhir/AllergyIntolerance-r2-certainty is unknown, and not allowed here (src = InstanceValidator)

allergyintolerance-example-refuted | ERROR @ AllergyIntolerance: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)

fishallergy | ERROR @ AllergyIntolerance: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ AllergyIntolerance: Profile http://hl7.org/fhir/StructureDefinition/AllergyIntolerance, Element 'AllergyIntolerance.verificationStatus': minimum required = 1, but only found 0 (src = InstanceValidator)

9.2 Resource Condition - Content

URL: https://hl7.org/fhir/STU3/condition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

A clinical condition, problem, diagnosis, or other event, situation, issue, or clinical concept that has risen to a level of concern.

9.2.1 Scope and Usage

Condition is one of the event resources in the FHIR workflow specification.

This resource is used to record detailed information about a condition, problem, diagnosis, or other event, situation, issue, or clinical concept that has risen to a level of concern. The condition could be a point in time diagnosis in context of an encounter, it could be an item on the practitioner's Problem List, or it could be a concern that doesn't exist on the practitioner's Problem List. Often times, a condition is about a clinician's assessment and assertion of a particular aspect of a patient's state of health. It can be used to record information about a disease/illness identified from application of clinical reasoning over the pathologic and pathophysiologic findings (diagnosis), or identification of health issues/situations that a practitioner considers harmful, potentially harmful and may be investigated and managed (problem), or other health issue/situation that may require ongoing monitoring and/or management (health issue/concern).

The condition resource may be used to record a certain health state of a patient which does not normally present a negative outcome, e.g. pregnancy. The condition resource may be used to record a condition following a procedure, such as the condition of Amputee-BKA following an amputation procedure.

While conditions are frequently a result of a clinician's assessment and assertion of a particular aspect of a patient's state of health, conditions can also be expressed by the patient, related person, or any care team member. A clinician may have a concern about a patient condition (e.g. anorexia) that the patient is not concerned about. Likewise, the patient may have a condition (e.g. hair loss) that does not rise to the level of importance such that it belongs on a practitioner's Problem List.

For example, each of the following conditions could rise to the level of importance such that it belongs on a problem or concern list due to its direct or indirect impact on the patient’s health. These examples may also be represented using other resources, such as FamilyMemberHistory, Observation, or Procedure.

* Unemployed
* Without transportation (or other barriers)
* Susceptibility to falls
* Exposure to communicable disease
* Family History of cardiovascular disease
* Fear of cancer
* Cardiac pacemaker
* Amputee-BKA
* Risk of Zika virus following travel to a country
* Former smoker
* Travel to a country planned (that warrants immunizations)
* Motor Vehicle Accident
* Patient has had coronary bypass graft

9.2.2 Boundaries and Relationships

The condition resource may be referenced by other resources as "reasons" for an action (e.g. MedicationRequest, Procedure, ProcedureRequest, etc.)

This resource is not typically used to record information about subjective and objective information that might lead to the recording of a Condition resource. Such signs and symptoms are typically captured using the Observation resource; although in some cases a persistent symptom, e.g. fever, headache may be captured as a condition before a definitive diagnosis can be discerned by a clinician.

Use the Observation resource when a symptom is resolved without long term management, tracking, or when a symptom contributes to the establishment of a condition.

Use Condition when a symptom requires long term management, tracking, or is used as a proxy for a diagnosis or problem that is not yet determined.

When the diagnosis is related to an allergy or intolerance, the Condition and AllergyIntolerance resources can both be used. However, using Condition alone is not sufficient as the allergy or intolerance needs to be represented as an AllergyIntolerance to be actionable for decision support

This resource is referenced by AdverseEvent, Appointment, CarePlan, CareTeam, Claim, ClinicalImpression, Communication, CommunicationRequest, Encounter, EpisodeOfCare, ExplanationOfBenefit, FamilyMemberHistory, Goal, MedicationAdministration, MedicationRequest, MedicationStatement, Procedure, ProcedureRequest, ReferralRequest, RiskAssessment and VisionPrescription

9.2.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Condition | I | | DomainResource | Detailed information about conditions, problems or diagnoses + If condition is abated, then clinicalStatus must be either inactive, resolved, or remission + Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this condition
clinicalStatus | ?!?I | 0..1 | code | active | recurrence | inactive | remission | resolved Condition Clinical Status Codes (Required)
verificationStatus | ?!?I | 0..1 | code | provisional | differential | confirmed | refuted | entered-in-error | unknown ConditionVerificationStatus (Required)
category | | 0..* | CodeableConcept | problem-list-item | encounter-diagnosis Condition Category Codes (Example)
severity | | 0..1 | CodeableConcept | Subjective severity of condition Condition/Diagnosis Severity (Preferred)
code | ? | 0..1 | CodeableConcept | Identification of the condition, problem or diagnosis Condition/Problem/Diagnosis Codes (Example)
bodySite | ? | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)
subject | ? | 1..1 | Reference(Patient | Group) | Who has the condition?
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode when condition first asserted
onset[x] | ? | 0..1 | | Estimated or actual date, date-time, or age
onsetDateTime | | | dateTime

onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
abatement[x] | I | 0..1 | | If/when in resolution/remission
abatementDateTime | | | dateTime
abatementAge | | | Age
abatementBoolean | | | boolean
abatementPeriod | | | Period
abatementRange | | | Range
abatementString | | | string
assertedDate | ? | 0..1 | dateTime | Date record was believed accurate
asserter | ? | 0..1 | Reference(Practitioner | Patient | RelatedPerson) | Person who asserts this condition
stage | I | 0..1 | BackboneElement | Stage/grade, usually assessed formally + Stage SHALL have summary or assessment
summary | I | 0..1 | CodeableConcept | Simple summary (disease specific) Condition Stage (Example)
assessment | I | 0..* | Reference(ClinicalImpression | DiagnosticReport | Observation) | Formal record of assessment
evidence | I | 0..* | BackboneElement | Supporting evidence + evidence SHALL have code or details
code | ?I | 0..* | CodeableConcept | Manifestation/symptom Manifestation and Symptom Codes (Example)
detail | ?I | 0..* | Reference(Any) | Supporting information found elsewhere
note | | 0..* | Annotation | Additional information about the Condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Condition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this condition --></identifier>
 <clinicalStatus value="[code]"/><!--  0..1 active | recurrence | inactive | remission | resolved -->
 <verificationStatus value="[code]"/><!--  0..1 provisional | differential | confirmed | refuted | entered-in-error |
    unknown -->
 <category><!-- 0..* CodeableConcept problem-list-item | encounter-diagnosis --></category>
 <severity><!-- 0..1 CodeableConcept Subjective severity of condition --></severity>
 <code><!-- 0..1 CodeableConcept Identification of the condition, problem or diagnosis --></code>
 <bodySite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodySite>
 <subject><!-- 1..1 Reference(Patient|Group) Who has the condition? --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode when condition first asserted --></context>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string Estimated or actual date,  date-time, or age --></onset[x]>
 <abatement[x]><!--  0..1 dateTime|Age|boolean|Period|Range|string If/when in resolution/remission --></abatement[x]>
 <assertedDate value="[dateTime]"/><!-- 0..1 Date record was believed accurate -->
 <asserter><!-- 0..1 Reference(Practitioner|Patient|RelatedPerson) Person who asserts this condition --></asserter>
 <stage>  <!-- 0..1 Stage/grade, usually assessed formally -->
  <summary><!--  0..1 CodeableConcept Simple summary (disease specific) --></summary>
  <assessment><!--  0..* Reference(ClinicalImpression|DiagnosticReport|Observation) Formal record of assessment
    --></assessment>
 </stage>
 <evidence>  <!-- 0..* Supporting evidence -->
  <code><!--  0..* CodeableConcept Manifestation/symptom --></code>
  <detail><!--  0..* Reference(Any) Supporting information found elsewhere --></detail>
 </evidence>
 <note><!-- 0..* Annotation Additional information about the Condition --></note>
</Condition>
```

JSON Template

```
{
  "resourceType" : "Condition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this condition
  "clinicalStatus" : "<code>", // C? active | recurrence | inactive | remission | resolved
  "verificationStatus" : "<code>", // C? provisional | differential | confirmed | refuted | entered-in-error | unknown
  "category" : [{ CodeableConcept }], // problem-list-item | encounter-diagnosis
  "severity" : { CodeableConcept }, // Subjective severity of condition
  "code" : { CodeableConcept }, // Identification of the condition, problem or diagnosis
  "bodySite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "subject" : { Reference(Patient|Group) }, // R!  Who has the condition?
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode when condition first asserted
  // onset[x]: Estimated or actual date,  date-time, or age. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  // abatement[x]: If/when in resolution/remission. One of these 6:
  "abatementDateTime" : "<dateTime>",
  "abatementAge" : { Age },
  "abatementBoolean" : <boolean>,
  "abatementPeriod" : { Period },
  "abatementRange" : { Range },
  "abatementString" : "<string>",
  "assertedDate" : "<dateTime>", // Date record was believed accurate
  "asserter" : { Reference(Practitioner|Patient|RelatedPerson) }, // Person who asserts this condition
  "stage" : { // Stage/grade, usually assessed formally
```

```
    "summary" : { CodeableConcept }, // C? Simple summary (disease specific)
    "assessment" : [{ Reference(ClinicalImpression|DiagnosticReport|Observation) }] // C? Formal record of assessment
  },
  "evidence" : [{ // Supporting evidence
    "code" : [{ CodeableConcept }], // C? Manifestation/symptom
    "detail" : [{ Reference(Any) }] // C? Supporting information found elsewhere
  }],
  "note" : [{ Annotation }] // Additional information about the Condition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Condition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Condition.identifier [ Identifier ], ... ; # 0..* External Ids for this condition
  fhir:Condition.clinicalStatus [ code ]; # 0..1 active | recurrence | inactive | remission | resolved
  fhir:Condition.verificationStatus [ code ]; # 0..1 provisional | differential | confirmed | refuted | entered-in-error
     | unknown
  fhir:Condition.category [ CodeableConcept ], ... ; # 0..* problem-list-item | encounter-diagnosis
  fhir:Condition.severity [ CodeableConcept ]; # 0..1 Subjective severity of condition
  fhir:Condition.code [ CodeableConcept ]; # 0..1 Identification of the condition, problem or diagnosis
  fhir:Condition.bodySite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:Condition.subject [ Reference(Patient|Group) ]; # 1..1 Who has the condition?
  fhir:Condition.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode when condition first
    asserted
  # Condition.onset[x] : 0..1 Estimated or actual date,  date-time, or age. One of these 5
    fhir:Condition.onsetDateTime [ dateTime ]
    fhir:Condition.onsetAge [ Age ]
    fhir:Condition.onsetPeriod [ Period ]
    fhir:Condition.onsetRange [ Range ]
    fhir:Condition.onsetString [ string ]
  # Condition.abatement[x] : 0..1 If/when in resolution/remission. One of these 6
    fhir:Condition.abatementDateTime [ dateTime ]
    fhir:Condition.abatementAge [ Age ]
    fhir:Condition.abatementBoolean [ boolean ]
    fhir:Condition.abatementPeriod [ Period ]
    fhir:Condition.abatementRange [ Range ]
    fhir:Condition.abatementString [ string ]
  fhir:Condition.assertedDate [ dateTime ]; # 0..1 Date record was believed accurate
  fhir:Condition.asserter [ Reference(Practitioner|Patient|RelatedPerson) ]; # 0..1 Person who asserts this condition
  fhir:Condition.stage [ # 0..1 Stage/grade, usually assessed formally
    fhir:Condition.stage.summary [ CodeableConcept ]; # 0..1 Simple summary (disease specific)
    fhir:Condition.stage.assessment [ Reference(ClinicalImpression|DiagnosticReport|Observation) ], ... ; # 0..* Formal
    record of assessment
  ];
  fhir:Condition.evidence [ # 0..* Supporting evidence
    fhir:Condition.evidence.code [ CodeableConcept ], ... ; # 0..* Manifestation/symptom
    fhir:Condition.evidence.detail [ Reference(Any) ], ... ; # 0..* Supporting information found elsewhere
  ], ...;
  fhir:Condition.note [ Annotation ], ... ; # 0..* Additional information about the Condition
]
```

Changes since DSTU2

Condition

Condition.clinicalStatus | Add Binding http://hl7.org/fhir/ValueSet/condition-clinical (required)

Condition.verificationStatus | Min Cardinality changed from 1 to 0 Default Value "unknown" added

Condition.category | Max Cardinality changed from 1 to *

Condition.code | Min Cardinality changed from 1 to 0

Condition.subject | Renamed from patient to subject Add Reference(Group)

Condition.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

Condition.onset[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age

Condition.abatement[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age

Condition.assertedDate | Renamed from dateRecorded to assertedDate Type changed from date to dateTime

Condition.asserter | Add Reference(RelatedPerson)

Condition.evidence.code | Max Cardinality changed from 1 to *

Condition.note | Renamed from notes to note Max Cardinality changed from 1 to * Type changed from string to Annotation

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 14 tests that all execute ok. 11 fail round-trip testing and 14 r3 resources are invalid (14 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

Condition | I | | DomainResource | Detailed information about conditions, problems or diagnoses + If condition is abated, then clinicalStatus must be either inactive, resolved, or remission + Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | External Ids for this condition

clinicalStatus | ?!?I | 0..1 | code | active | recurrence | inactive | remission | resolved Condition Clinical Status Codes (Required)
verificationStatus | ?!?I | 0..1 | code | provisional | differential | confirmed | refuted | entered-in-error | unknown ConditionVerificationStatus (Required)
category | | 0..* | CodeableConcept | problem-list-item | encounter-diagnosis Condition Category Codes (Example)
severity | | 0..1 | CodeableConcept | Subjective severity of condition Condition/Diagnosis Severity (Preferred)
code | ? | 0..1 | CodeableConcept | Identification of the condition, problem or diagnosis Condition/Problem/Diagnosis Codes (Example)
bodySite | ? | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)
subject | ? | 1..1 | Reference(Patient | Group) | Who has the condition?
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode when condition first asserted
onset[x] | ? | 0..1 | | Estimated or actual date, date-time, or age
onsetDateTime | | | dateTime
onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
abatement[x] | I | 0..1 | | If/when in resolution/remission
abatementDateTime | | | dateTime
abatementAge | | | Age
abatementBoolean | | | boolean
abatementPeriod | | | Period
abatementRange | | | Range
abatementString | | | string
assertedDate | ? | 0..1 | dateTime | Date record was believed accurate
asserter | ? | 0..1 | Reference(Practitioner | Patient | RelatedPerson) | Person who asserts this condition
stage | I | 0..1 | BackboneElement | Stage/grade, usually assessed formally + Stage SHALL have summary or assessment
summary | I | 0..1 | CodeableConcept | Simple summary (disease specific) Condition Stage (Example)
assessment | I | 0..* | Reference(ClinicalImpression | DiagnosticReport | Observation) | Formal record of assessment
evidence | I | 0..* | BackboneElement | Supporting evidence + evidence SHALL have code or details
code | ?I | 0..* | CodeableConcept | Manifestation/symptom Manifestation and Symptom Codes (Example)
detail | ?I | 0..* | Reference(Any) | Supporting information found elsewhere
note | | 0..* | Annotation | Additional information about the Condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Condition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this condition --></identifier>
 <clinicalStatus value="[code]"/><!--  0..1 active | recurrence | inactive | remission | resolved -->
 <verificationStatus value="[code]"/><!--  0..1 provisional | differential | confirmed | refuted | entered-in-error |
    unknown -->
 <category><!-- 0..* CodeableConcept problem-list-item | encounter-diagnosis --></category>
 <severity><!-- 0..1 CodeableConcept Subjective severity of condition --></severity>
 <code><!-- 0..1 CodeableConcept Identification of the condition, problem or diagnosis --></code>
 <bodySite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodySite>
 <subject><!-- 1..1 Reference(Patient|Group) Who has the condition? --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode when condition first asserted --></context>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string Estimated or actual date,  date-time, or age --></onset[x]>
 <abatement[x]><!--  0..1 dateTime|Age|boolean|Period|Range|string If/when in resolution/remission --></abatement[x]>
 <assertedDate value="[dateTime]"/><!-- 0..1 Date record was believed accurate -->
 <asserter><!-- 0..1 Reference(Practitioner|Patient|RelatedPerson) Person who asserts this condition --></asserter>
 <stage>  <!-- 0..1 Stage/grade, usually assessed formally -->
  <summary><!--  0..1 CodeableConcept Simple summary (disease specific) --></summary>
  <assessment><!--  0..* Reference(ClinicalImpression|DiagnosticReport|Observation) Formal record of assessment
     --></assessment>
 </stage>
 <evidence>  <!-- 0..* Supporting evidence -->
  <code><!--  0..* CodeableConcept Manifestation/symptom --></code>
  <detail><!--  0..* Reference(Any) Supporting information found elsewhere --></detail>
 </evidence>
 <note><!-- 0..* Annotation Additional information about the Condition --></note>
</Condition>
```

JSON Template

```
{
  "resourceType" : "Condition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this condition
  "clinicalStatus" : "<code>", // C? active | recurrence | inactive | remission | resolved
  "verificationStatus" : "<code>", // C? provisional | differential | confirmed | refuted | entered-in-error | unknown
  "category" : [{ CodeableConcept }], // problem-list-item | encounter-diagnosis
  "severity" : { CodeableConcept }, // Subjective severity of condition
  "code" : { CodeableConcept }, // Identification of the condition, problem or diagnosis
  "bodySite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "subject" : { Reference(Patient|Group) }, // R!  Who has the condition?
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode when condition first asserted
  // onset[x]: Estimated or actual date,  date-time, or age. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
```

```
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  // abatement[x]: If/when in resolution/remission. One of these 6:
  "abatementDateTime" : "<dateTime>",
  "abatementAge" : { Age },
  "abatementBoolean" : <boolean>,
  "abatementPeriod" : { Period },
  "abatementRange" : { Range },
  "abatementString" : "<string>",
  "assertedDate" : "<dateTime>", // Date record was believed accurate
  "asserter" : { Reference(Practitioner|Patient|RelatedPerson) }, // Person who asserts this condition
  "stage" : { // Stage/grade, usually assessed formally
    "summary" : { CodeableConcept }, // C? Simple summary (disease specific)
    "assessment" : [{ Reference(ClinicalImpression|DiagnosticReport|Observation) }] // C? Formal record of assessment
  },
  "evidence" : [{ // Supporting evidence
    "code" : [{ CodeableConcept }], // C? Manifestation/symptom
    "detail" : [{ Reference(Any) }] // C? Supporting information found elsewhere
  }],
  "note" : [{ Annotation }] // Additional information about the Condition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Condition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Condition.identifier [ Identifier ], ... ; # 0..* External Ids for this condition
  fhir:Condition.clinicalStatus [ code ]; # 0..1 active | recurrence | inactive | remission | resolved
  fhir:Condition.verificationStatus [ code ]; # 0..1 provisional | differential | confirmed | refuted | entered-in-error
     | unknown
  fhir:Condition.category [ CodeableConcept ], ... ; # 0..* problem-list-item | encounter-diagnosis
  fhir:Condition.severity [ CodeableConcept ]; # 0..1 Subjective severity of condition
  fhir:Condition.code [ CodeableConcept ]; # 0..1 Identification of the condition, problem or diagnosis
  fhir:Condition.bodySite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:Condition.subject [ Reference(Patient|Group) ]; # 1..1 Who has the condition?
  fhir:Condition.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode when condition first
    asserted
  # Condition.onset[x] : 0..1 Estimated or actual date,  date-time, or age. One of these 5
    fhir:Condition.onsetDateTime [ dateTime ]
    fhir:Condition.onsetAge [ Age ]
    fhir:Condition.onsetPeriod [ Period ]
    fhir:Condition.onsetRange [ Range ]
    fhir:Condition.onsetString [ string ]
  # Condition.abatement[x] : 0..1 If/when in resolution/remission. One of these 6
    fhir:Condition.abatementDateTime [ dateTime ]
    fhir:Condition.abatementAge [ Age ]
    fhir:Condition.abatementBoolean [ boolean ]
    fhir:Condition.abatementPeriod [ Period ]
    fhir:Condition.abatementRange [ Range ]
    fhir:Condition.abatementString [ string ]
  fhir:Condition.assertedDate [ dateTime ]; # 0..1 Date record was believed accurate
  fhir:Condition.asserter [ Reference(Practitioner|Patient|RelatedPerson) ]; # 0..1 Person who asserts this condition
  fhir:Condition.stage [ # 0..1 Stage/grade, usually assessed formally
    fhir:Condition.stage.summary [ CodeableConcept ]; # 0..1 Simple summary (disease specific)
    fhir:Condition.stage.assessment [ Reference(ClinicalImpression|DiagnosticReport|Observation) ], ... ; # 0..* Formal
    record of assessment
  ];
  fhir:Condition.evidence [ # 0..* Supporting evidence
    fhir:Condition.evidence.code [ CodeableConcept ], ... ; # 0..* Manifestation/symptom
    fhir:Condition.evidence.detail [ Reference(Any) ], ... ; # 0..* Supporting information found elsewhere
  ], ...;
  fhir:Condition.note [ Annotation ], ... ; # 0..* Additional information about the Condition
]
```

Changes since DSTU2

Condition

Condition.clinicalStatus | Add Binding http://hl7.org/fhir/ValueSet/condition-clinical (required)

Condition.verificationStatus | Min Cardinality changed from 1 to 0 Default Value "unknown" added

Condition.category | Max Cardinality changed from 1 to *

Condition.code | Min Cardinality changed from 1 to 0

Condition.subject | Renamed from patient to subject Add Reference(Group)

Condition.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

Condition.onset[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age

Condition.abatement[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age

Condition.assertedDate | Renamed from dateRecorded to assertedDate Type changed from date to dateTime

Condition.asserter | Add Reference(RelatedPerson)

Condition.evidence.code | Max Cardinality changed from 1 to *

Condition.note | Renamed from notes to note Max Cardinality changed from 1 to * Type changed from string to Annotation

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 14 tests that all execute ok. 11 fail round-trip testing and 14 r3 resources are invalid (14 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.2.3.1 Terminology Bindings

Path	Definition	Type	Reference
Condition.clinicalStatus	The clinical status of the condition or diagnosis.	Required	Condition Clinical Status Codes
Condition.verificationStatus	The verification status to support or decline the clinical status of the condition or diagnosis.	Required	ConditionVerificationStatus
Condition.category	A category assigned to the condition.	Example	Condition Category Codes
Condition.severity	A subjective assessment of the severity of the condition as evaluated by the clinician.	Preferred	Condition/Diagnosis Severity
Condition.code	Identification of the condition or diagnosis.	Example	Condition/Problem/Diagnosis Codes
Condition.bodySite	Codes describing anatomical locations. May include laterality.	Example	SNOMED CT Body Structures
Condition.stage.summary	Codes describing condition stages (e.g. Cancer stages).	Example	Condition Stage
Condition.evidence.code	Codes that describe the manifestation or symptoms of a condition.	Example	Manifestation and Symptom Codes

9.2.3.2 Constraints

* con-1: On Condition.stage: Stage SHALL have summary or assessment (expression on Condition.stage: summary.exists() or assessment.exists())

* con-2: On Condition.evidence: evidence SHALL have code or details (expression on Condition.evidence: code.exists() or detail.exists())

* con-3: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error (expression : verificationStatus='entered-in-error' or clinicalStatus.exists())

* con-4: If condition is abated, then clinicalStatus must be either inactive, resolved, or remission (expression : abatement.empty() or (abatement as boolean).not() or clinicalStatus='resolved' or clinicalStatus='remission' or clinicalStatus='inactive')

9.2.3.3 Use of Condition.code

Many of the code systems used for coding conditions will provide codes that define not only the condition itself, but may also specify a particular stage, location, or causality as part of the code. This is particularly true if SNOMED CT is used for the condition, and especially if expressions are allowed.

The Condition.code may also include such concepts as "history of X" and "good health", where it is useful or appropriate to make such assertions. It can also be used to capture "risk of" and "fear of", in addition to physical conditions, as well as "no known problems" or "negated" conditions (e.g., "no X" or "no history of X" - see the following section for "No Known Problems" and Negated Conditions).

When the Condition.code specifies additional properties of the condition, the other properties are not given a value - instead, the value must be understood from the Condition.code.

9.2.3.4 "No Known Problems" and Negated Conditions

Conditions/Problems Not Reviewed, Not Asked

When a sending system does not have any information about conditions/problems being reviewed or the statement is about conditions/problems not yet being asked, then the List resource should be used to indicate the List.emptyReason.code="notasked".

Conditions/Problems Reviewed, None Identified

Systems may use the List.emptyReason when a statement is about the full scope of the list (i.e. the patient has no conditions/problems of any type). However, it may be preferred to use a code for "no known problems" (e.g., SNOMED CT: 160245001 |No current problems or disability (situation)|), so that all condition/problem data will be available and queryable from Condition resource instances.

Also note that care should be used when adding new Condition resources to a list to ensure that any negation statements that are voided by the addition of a new record are removed from the list. E.g. If the list contains a "no known problems" record and you add a "diabetes" condition record, then be sure that you remove the "no known problems" record.

STU Note: There are two primary ways of reporting "no known problems" in the current specification: using the CodeableConcept, as described above, or using the List resource with emptyReason. During the STU period, feedback is sought regarding the preferred approach.

Provide feedback here .

Patient Denies Condition

When the patient denies a condition, that can be annotated in the Condition.note element.

9.2.3.5 Assertions of Condition Absence

Generally, electronic records do not contain assertions of conditions that a patient does not have. There are however two exceptions:

* It is appropriate to capture a "refuted" Condition record if the patient or anyone else had reason to believe that a patient did have a condition for a period of time and subsequent evidence has demonstrated that belief was mistaken. In this case, a concrete statement acknowledging the belief as well as the refutation of it is useful.

* It is common as part of checklists prior to admission, surgery, enrollment in trials, etc. to ask questions such as "are you pregnant", "do you have a history of hypertension", etc. This information should NOT be captured using the Condition resource but should instead be captured using QuestionnaireResponse or Observation. In this case, the combination of the question and answer would convey that a particular condition was not present.

9.2.3.6 Use of Condition.evidence

The Condition.evidence provides the basis for whatever is present in Condition.code.

9.2.3.7 Use of Condition.abatementRange

A range is used to communicate age period of subject at time of abatement.

9.2.3.8 Use of Condition.asserter

If the data enterer is different from the asserter and needs to be known, this could be captured using a Provenance instance pointing to the Condition. For example, it is possible that a nurse records the condition on behalf of a physician. The physician is taking responsibility, despite the nurse entering it into the medical record.

9.2.3.9 Use of Condition.clinicalStatus

The Condition.stage and Condition.clinicalStatus may have interdependencies. For example, some "stages" of cancer, etc. will be different for a remission than for the initial occurrence.

9.2.3.10 Known Issue

A known issue exists with circular references between Condition and ClinicalImpression, which is due to the low maturity level of ClinicalImpression. The Patient Care work group intends to address this issue when ClinicalImpression is considered substantially complete and ready for implementation.

9.2.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
abatement-age | quantity | Abatement as age or age range | Condition.abatement.as(Age) | Condition.abatement.as(Range) | Condition.abatement.as(Age)
abatement-boolean | token | Abatement boolean (boolean is true or non-boolean values are present) | Condition.abatement.as(boolean) | Condition.abatement.is(dateTime) | Condition.abatement.is(Age) | Condition.abatement.is(Period) | Condition.abatement.is(Range) | Condition.abatement.is(string)
abatement-date | date | Date-related abatements (dateTime and period) | Condition.abatement.as(dateTime) | Condition.abatement.as(Period)
abatement-string | string | Abatement as a string | Condition.abatement.as(string)
asserted-date | date | Date record was believed accurate | Condition.assertedDate
asserter | reference | Person who asserts this condition | Condition.asserter (Practitioner, Patient, RelatedPerson)
body-site | token | Anatomical location, if relevant | Condition.bodySite
category | token | The category of the condition | Condition.category
clinical-status | token | The clinical status of the condition | Condition.clinicalStatus
code | token | Code for the condition | Condition.code | 8 Resources
context | reference | Encounter or episode when condition first asserted | Condition.context (EpisodeOfCare, Encounter)
encounter | reference | Encounter when condition first asserted | Condition.context (Encounter)
evidence | token | Manifestation/symptom | Condition.evidence.code
evidence-detail | reference | Supporting information found elsewhere | Condition.evidence.detail (Any)
identifier | token | A unique identifier of the condition record | Condition.identifier | 26 Resources
onset-age | quantity | Onsets as age or age range | Condition.onset.as(Age) | Condition.onset.as(Range)
onset-date | date | Date related onsets (dateTime and Period) | Condition.onset.as(dateTime) | Condition.onset.as(Period)
onset-info | string | Onsets as a string | Condition.onset.as(string)
patient | reference | Who has the condition? | Condition.subject (Patient) | 31 Resources
severity | token | The severity of the condition | Condition.severity
stage | token | Simple summary (disease specific) | Condition.stage.summary
subject | reference | Who has the condition? | Condition.subject (Group, Patient)
verification-status | token | provisional | differential | confirmed | refuted | entered-in-error | unknown | Condition.verificationStatus

9.2.5 Resource Condition - Examples

URL: https://hl7.org/fhir/STU3/condition-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.5 Resource Condition - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example List:

General Condition Example | example | XML | JSON | Turtle
2nd Example | example2 | XML | JSON | Turtle
Real-word condition example (fever) | f201 | XML | JSON | Turtle
Real-word condition example (malignancy) | f202 | XML | JSON | Turtle
Real-word condition example (sepsis) | f203 | XML | JSON | Turtle
Real-word condition example (renal insufficiency) | f204 | XML | JSON | Turtle
Real-word condition example (Bacterial Infection) | f205 | XML | JSON | Turtle
Real-word condition example (heart) | f001 | XML | JSON | Turtle
Real-word condition example (lung) | f002 | XML | JSON | Turtle
Real-word condition example (abscess) | f003 | XML | JSON | Turtle
Stroke example (for prognosis) | stroke | XML | JSON | Turtle
Family history concern | family-history | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.2.6 Resource Condition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/condition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.6 Resource Condition - Detailed Descriptions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Condition resource.

Condition
Definition | A clinical condition, problem, diagnosis, or other event, situation, issue, or clinical concept that has risen to a level of concern.
Control | 1..1
Invariants | Defined on this element con-3: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error (expression : verificationStatus='entered-in-error' or clinicalStatus.exists(), xpath: f:verificationStatus/@value='entered-in-error' or exists(f:clinicalStatus)) con-4: If condition is abated, then clinicalStatus must be either inactive, resolved, or remission (expression : abatement.empty() or (abatement as boolean).not() or clinicalStatus='resolved' or clinicalStatus='remission' or clinicalStatus='inactive', xpath: not(exists(*[starts-with(local-name(.), 'abatement')])) or f:clinicalStatus/@value=('resolved', 'remission', 'inactive'))
Condition.identifier
Definition | This records identifiers associated with this condition that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Need to allow connection to a wider workflow.
Summary | true
Condition.clinicalStatus
Definition | The clinical status of the condition.
Control | 0..1
Terminology Binding | Condition Clinical Status Codes (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the condition as not currently valid or of concern.
Invariants | Affect this element con-3: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error (expression : verificationStatus='entered-in-error' or clinicalStatus.exists(), xpath: f:verificationStatus/@value='entered-in-error' or exists(f:clinicalStatus)) con-4: If condition is abated, then clinicalStatus must be either inactive, resolved, or remission (expression : abatement.empty() or (abatement as boolean).not() or clinicalStatus='resolved' or clinicalStatus='remission' or clinicalStatus='inactive', xpath: not(exists(*[starts-with(local-name(.), 'abatement')])) or f:clinicalStatus/@value=('resolved', 'remission', 'inactive'))
Condition.verificationStatus
Definition | The verification status to support the clinical status of the condition.
Control | 0..1
Terminology Binding | ConditionVerificationStatus (Required)
Type | code
Is Modifier | true
Default Value | unknown
Summary | true
Comments | verificationStatus is not required. For example, when a patient has abdominal pain in the ED, there is not likely going to be a verification status. This element is labeled as a modifier because the status contains the code refuted and entered-in-error that mark the Condition as not currently valid.
Invariants | Affect this element con-3: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error (expression : verificationStatus='entered-in-error' or clinicalStatus.exists(), xpath: f:verificationStatus/@value='entered-in-error' or exists(f:clinicalStatus))
Condition.category
Definition | A category assigned to the condition.
Control | 0..*
Terminology Binding | Condition Category Codes (Example)
Type | CodeableConcept

Comments | The categorization is often highly contextual and may appear poorly differentiated or not very useful in other contexts.
Condition.severity
Definition | A subjective assessment of the severity of the condition as evaluated by the clinician.
Control | 0..1
Terminology Binding | Condition/Diagnosis Severity (Preferred)
Type | CodeableConcept
Comments | Coding of the severity with a terminology is preferred, where possible.
Condition.code
Definition | Identification of the condition, problem or diagnosis.
Control | 0..1
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Requirements | 0..1 to account for primarily narrative only resources.
Alternate Names | type
Summary | true
Condition.bodySite
Definition | The anatomical location where this condition manifests itself.
Control | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
Comments | Only used if not implicit in code found in Condition.code. If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension body-site-instance. May be a summary code, or a reference to a very precise definition of the location, or both.
Condition.subject
Definition | Indicates the patient or group who the condition record is associated with.
Control | 1..1
Type | Reference(Patient | Group)
Requirements | Group is typically used for veterinary or public health use cases.
Alternate Names | patient
Summary | true
Condition.context
Definition | Encounter during which the condition was first asserted.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
Comments | This record indicates the encounter this particular record is associated with. In the case of a "new" diagnosis reflecting ongoing/revised information about the condition, this might be distinct from the first encounter in which the underlying condition was first "known".
Condition.onset[x]
Definition | Estimated or actual date or date-time the condition began, in the opinion of the clinician.
Control | 0..1
Type | dateTime|Age|Period|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Age is generally used when the patient reports an age at which the Condition began to occur.
Condition.abatement[x]
Definition | The date or estimated date that the condition resolved or went into remission. This is called "abatement" because of the many overloaded connotations associated with "remission" or "resolution" - Conditions are never really resolved, but they can abate.
Control | 0..1
Type | dateTime|Age|boolean|Period|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | There is no explicit distinction between resolution and remission because in many cases the distinction is not clear. Age is generally used when the patient reports an age at which the Condition abated. If there is no abatement element, it is unknown whether the condition has resolved or entered remission; applications and users should generally assume that the condition is still valid. When abatementString exists, it implies the condition is abated.
Invariants | Affect this element con-4: If condition is abated, then clinicalStatus must be either inactive, resolved, or remission (expression : abatement.empty() or (abatement as boolean).not() or clinicalStatus='resolved' or clinicalStatus='remission' or clinicalStatus='inactive', xpath: not(exists(*[starts-with(local-name(.), 'abatement')])) or f:clinicalStatus/@value=('resolved', 'remission', 'inactive'))
Condition.assertedDate
Definition | The date on which the existance of the Condition was first asserted or acknowledged.
Control | 0..1
Type | dateTime
Summary | true
Comments | The assertedDate represents the date when this particular Condition record was created in the EHR, not the date of the most recent update in terms of when severity, abatement, etc. were specified. The date of the last record modification can be retrieved from the resource metadata.
Condition.asserter
Definition | Individual who is making the condition statement.
Control | 0..1
Type | Reference(Practitioner | Patient | RelatedPerson)
Summary | true
Condition.stage

Definition | Clinical stage or grade of a condition. May include formal severity assessments.
Control | 0..1
Invariants | Defined on this element con-1: Stage SHALL have summary or assessment (expression : summary.exists() or assessment.exists(), xpath: exists(f:summary) or exists(f:assessment))
Condition.stage.summary
Definition | A simple summary of the stage such as "Stage 3". The determination of the stage is disease-specific.
Control | 0..1
Terminology Binding | Condition Stage (Example)
Type | CodeableConcept
Invariants | Affect this element con-1: Stage SHALL have summary or assessment (expression : summary.exists() or assessment.exists(), xpath: exists(f:summary) or exists(f:assessment))
Condition.stage.assessment
Definition | Reference to a formal record of the evidence on which the staging assessment is based.
Control | 0..*
Type | Reference(ClinicalImpression | DiagnosticReport | Observation)
Invariants | Affect this element con-1: Stage SHALL have summary or assessment (expression : summary.exists() or assessment.exists(), xpath: exists(f:summary) or exists(f:assessment))
Condition.evidence
Definition | Supporting Evidence / manifestations that are the basis on which this condition is suspected or confirmed.
Control | 0..*
Comments | The evidence may be a simple list of coded symptoms/manifestations, or references to observations or formal assessments, or both.
Invariants | Defined on this element con-2: evidence SHALL have code or details (expression : code.exists() or detail.exists(), xpath: exists(f:code) or exists(f:detail))
Condition.evidence.code
Definition | A manifestation or symptom that led to the recording of this condition.
Control | 0..*
Terminology Binding | Manifestation and Symptom Codes (Example)
Type | CodeableConcept
Summary | true
Invariants | Affect this element con-2: evidence SHALL have code or details (expression : code.exists() or detail.exists(), xpath: exists(f:code) or exists(f:detail))
Condition.evidence.detail
Definition | Links to other relevant information, including pathology reports.
Control | 0..*
Type | Reference(Any)
Summary | true
Invariants | Affect this element con-2: evidence SHALL have code or details (expression : code.exists() or detail.exists(), xpath: exists(f:code) or exists(f:detail))
Condition.note
Definition | Additional information about the Condition. This is a general notes/comments entry for description of the Condition, its diagnosis and prognosis.
Control | 0..*
Type | Annotation

9.2.7 Resource Condition - Mappings

URL: https://hl7.org/fhir/STU3/condition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.7 Resource Condition - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

SNOMED CT Attribute Binding

SNOMED CT Concept Domain Binding

HL7 v2 Mapping

Mappings for the condition resource.

9.2.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Condition | Observation[classCode=OBS, moodCode=EVN, code=ASSERTION, value<Diagnosis]
identifier | .id
clinicalStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="clinicalStatus", moodCode=EVN].value
verificationStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="verificationStatus", moodCode=EVN].value
category | .code
severity | Can be pre/post-coordinated into value. Or ./inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="severity"].value
code | .value
bodySite | .targetBodySiteCode

subject | .participation[typeCode=SBJ].role[classCode=PAT]
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
onset[x] | .effectiveTime.low or .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="age at onset"].value
abatement[x] | .effectiveTime.high or .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="age at remission"].value or .inboundRelationship[typeCode=SUBJ]source[classCode=CONC, moodCode=EVN].status=completed
assertedDate | .participation[typeCode=AUT].time
asserter | .participation[typeCode=AUT].role
stage | ./inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="stage/grade"]
summary | .value
assessment | .self
evidence | .outboundRelationship[typeCode=SPRT].target[classCode=OBS, moodCode=EVN]
code | [code="diagnosis"].value
detail | .self
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value

9.2.7.2 Mappings for SNOMED CT Attribute Binding (http://snomed.info/sct)

Attribute bindings link coded data elements in FHIR resources to a corresponding attribute in the SNOMED CT concept model. These bindings help to support:

* clarifying the intended meaning of the data element

* Quality checking the alignment between FHIR resource design and any coresponding SNOMED CT concept model

* Composition and decomposition of data instances by indicating the SNOMED CT concept model attribute whose value may be used to decompose a precoordinated concept into this data element

Condition
identifier
clinicalStatus
verificationStatus | 408729009
category
severity | 246112005
code | 246090004
bodySite | 363698007
subject
context
onset[x]
abatement[x]
assertedDate
asserter
stage
summary
assessment
evidence
code
detail
note

9.2.7.3 Mappings for SNOMED CT Concept Domain Binding (http://snomed.info/conceptdomain)

Concept domain bindings link a resource or an element to a set of SNOMED CT concepts that represent the intended semantics of the instances (whether or not SNOMED CT is used to encode that data element). This set of concepts is represented using a SNOMED CT expression constraint. Note that the 'Concept domain binding' may be a superset of the 'value set binding'. These bindings help to support:

* Quality checking FHIR resources by ensuring that (a) the intended semantics of the instances matches the valid range of the corresponding SNOMED CT attribute, and (b) the intended value set is appropriate for the intended semantics of the instances

* Semantic checking of data instances by helping to detect potential inconsistencies caused by overlap between the semantics incorporated in two concept domains

Condition | < 243796009 |Situation with explicit context|: 246090004 |Associated finding| = ((< 404684003 |Clinical finding| MINUS << 420134006 |Propensity to adverse reactions| MINUS << 473010000 |Hypersensitivity condition| MINUS << 79899007 |Drug interaction| MINUS << 69449002 |Drug action| MINUS << 441742003 |Evaluation finding| MINUS << 307824009 |Administrative status| MINUS << 385356007 |Tumor stage finding|) OR < 272379006 |Event|)
identifier
clinicalStatus | < 303105007 |Disease phases|
verificationStatus | < 410514004 |Finding context value|
category | < 404684003 |Clinical finding|
severity | < 272141005 |Severities|
code | code 246090004 |Associated finding| (< 404684003 |Clinical finding| MINUS << 420134006 |Propensity to adverse reactions| MINUS << 473010000 |Hypersensitivity condition| MINUS << 79899007 |Drug interaction| MINUS << 69449002 |Drug action| MINUS << 441742003 |Evaluation finding| MINUS << 307824009 |Administrative status| MINUS << 385356007 |Tumor stage finding|) OR < 413350009 |Finding with explicit context| OR < 272379006 |Event|
bodySite | < 442083009 |Anatomical or acquired body structure|
subject

context
onset[x]
abatement[x]
assertedDate
asserter
stage
summary | < 254291000 |Staging and scales|
assessment
evidence
code | < 404684003 |Clinical finding|
detail
note

9.2.7.4 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Condition | PPR message
identifier
clinicalStatus | PRB-14 / DG1-6
verificationStatus | PRB-13
category | 'problem' if from PRB-3. 'diagnosis' if from DG1 segment in PV1 message
severity | PRB-26 / ABS-3
code | PRB-3
bodySite
subject | PID-3
context | PV1-19 (+PV1-54)
onset[x] | PRB-16
abatement[x]
assertedDate | REL-11
asserter | REL-7.1 identifier + REL-7.12 type code
stage
summary | PRB-14
assessment
evidence
code
detail
note | NTE child of PRB

9.2.8 Resource Condition - Design Notes

URL: https://hl7.org/fhir/STU3/condition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.8 Resource Condition - Design Notes

to do

9.2.9 Resource Condition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/condition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.9 Resource Condition - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Condition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.2.9.1 Profiles

No Profiles defined for this resource

9.2.9.2 Extensions

condition-basedOn | basedOn | for Condition HL7 Extensions
condition-criticality | criticality | for Condition HL7 Extensions
condition-definition | definition | for Condition HL7 Extensions
condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
condition-outcome | outcome | for Condition HL7 Extensions
condition-partOf | partOf | for Condition HL7 Extensions
condition-ruledOut | ruledOut | for Condition HL7 Extensions

condition-targetBodySite | targetBodySite | for Condition HL7 Extensions

Extensions that reference this resource:

DiagnosticReport-geneticsAssessedCondition | AssessedCondition | for DiagnosticReport-Genetics
condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
condition-partOf | partOf | for Condition HL7 Extensions
condition-ruledOut | ruledOut | for Condition HL7 Extensions
event-reasonReference | reasonReference | for Event Pattern HL7 Extensions

9.2.9.3 Search Extensions

No Search Extensions defined for this resource

9.2.10 Condition HL7 Extensions

URL: https://hl7.org/fhir/STU3/condition-extensions.html

Defines common extensions used with or related to the Condition resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.10 Condition HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.2.10.1 Content

Extensions:

condition-criticality | criticality : Characterizes potential impact of the condition on life, durable physiological function, or quality of life. May include concepts such as self-resolving, life-threatening, requires hospitalization, or potential loss of function or capacity. Different from severity in that a moderate subarachnoid hemorrhage is likely to be highly critical, whereas a moderate headache is not as important.
condition-dueTo | dueTo : Further conditions, problems, diagnoses, procedures or events or the substance that caused/triggered this Condition.
condition-occurredFollowing | occurredFollowing : Further conditions, problems, diagnoses, procedures or events or the substance that preceded this Condition.
condition-outcome | outcome : A result of the condition. The "Cause of death" for a patient is typically captured as an Observation. The "outcome" doesn't imply causality. Some outcomes may not be assessable until the condition.clinicalStatus is no longer active.
condition-targetBodySite | targetBodySite :
condition-basedOn | basedOn : The order, plan or recommendation being satisfied in whole or in part by this condition record.
condition-partOf | partOf : The procedure, condition, or observation this condition was created as part of (such as elevated blood pressure is part of diabetes).
condition-ruledOut | ruledOut : Identifies what potential diagnoses have been ruled out for this condition.
condition-definition | definition : A protocol, guideline or other definition that was adhered to in whole or in part by this event.

9.2.10.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
based-on | reference | Matches on the basedon extension value | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-basedOn'] | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-basedOn'] | XML / JSON
definition | reference | Matches on the definition extension value | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-definition'] | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-definition'] | XML / JSON
part-of | reference | Matches on the partOf extension value | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-partOf'] | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-partOf'] | XML / JSON

9.2.10.3 Condition-extensions-Condition-based-on

URL: https://hl7.org/fhir/STU3/condition-extensions-Condition-based-on.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.10.3 Condition-extensions-Condition-based-on

Search Parameter based-on (type: reference) defined as part of the condition-extensions Profile.

Matches on the basedon extension value

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/condition-extensions-Condition-based-on

* Paths : f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-basedOn'] | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-basedOn']

* Targets :

9.2.10.4 Condition-extensions-Condition-part-of

URL: https://hl7.org/fhir/STU3/condition-extensions-Condition-part-of.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.10.4 Condition-extensions-Condition-part-of

Search Parameter part-of (type: reference) defined as part of the condition-extensions Profile.

Matches on the partOf extension value

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/condition-extensions-Condition-part-of

* Paths : f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-partOf'] | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-partOf']

* Targets :

9.2.10.5 Condition-extensions-Condition-definition

URL: https://hl7.org/fhir/STU3/condition-extensions-Condition-definition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.10.5 Condition-extensions-Condition-definition

Search Parameter definition (type: reference) defined as part of the condition-extensions Profile.

Matches on the definition extension value

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/condition-extensions-Condition-definition

* Paths : f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-definition'] | f:Condition/f:extension[@url='http://hl7.org/fhir/StructureDefinition/condition-definition']

* Targets :

9.2.11 Resource Condition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/condition-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.2.11 Resource Condition - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Condition.

Functional status for this map: 14 tests that all execute ok. 11 fail round-trip testing and 14 r3 resources are invalid (14 errors). (see documentation)

9.2.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Condition2to3" = "R2 to R3 Conversions for Condition"

conceptmap "ConditionClinicalStatus" {
  prefix s = "http://hl7.org/fhir/devicestatus"
  prefix t = "http://hl7.org/fhir/devicestatus"

  s:available  = t:active
  s:inactive = t:"not-available"
  s:"entered-in-error" = t:"entered-in-error"
}


uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Condition" alias ConditionR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Condition" alias Condition as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Condition extends DomainResource
  input src : ConditionR2 as source
  input tgt : Condition as target

  "Condition-identifier" : for src.identifier make tgt.identifier
  "Condition-patient" : for src.patient make tgt.subject
  "Condition-encounter" : for src.encounter make tgt.context
  "Condition-asserter" : for src.asserter make tgt.asserter
  "Condition-dateRecorded" : for src.dateRecorded make tgt.assertedDate
  "Condition-code" : for src.code make tgt.code
  "Condition-category" : for src.category make tgt.category
```

```
  "Condition-clinicalStatus" : for src.clinicalStatus as v make tgt.clinicalStatus = translate(v,
    "#ConditionClinicalStatus", "code")
  "Condition-verificationStatus" : for src.verificationStatus make tgt.verificationStatus
  "Condition-severity" : for src.severity make tgt.severity
  "Condition-onset" : for src.onset make tgt.onset
  "Condition-abatement" : for src.abatement make tgt.abatement
  "Condition-stage" : for src.stage as vs0 make tgt.stage as vt0 then cndStage(vs0, vt0)
  "Condition-evidence" : for src.evidence as vs0 make tgt.evidence as vt0 then cndEvidence(vs0, vt0)
  "Condition-bodySite" : for src.bodySite make tgt.bodySite
  "Condition-notes" : for src.notes as v make tgt.note as n, n.text = v
endgroup

group cndStage extends BackboneElement
  input src : ConditionR2 as source
  input tgt : Condition as target

  "Condition.stage-summary" : for src.summary make tgt.summary
  "Condition.stage-assessment" : for src.assessment make tgt.assessment
endgroup

group cndEvidence extends BackboneElement
  input src : ConditionR2 as source
  input tgt : Condition as target

  "Condition.evidence-code" : for src.code make tgt.code
  "Condition.evidence-detail" : for src.detail make tgt.detail
endgroup
```

9.2.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Condition3to2" = "R3 to R2 Conversion for Condition"

conceptmap "ConditionClinicalStatus" {
  prefix s = "http://hl7.org/fhir/devicestatus"
  prefix t = "http://hl7.org/fhir/devicestatus"

  s:available  = t:active
  s:"not-available" = t:inactive
  s:"entered-in-error" = t:"entered-in-error"
}


uses "http://hl7.org/fhir/StructureDefinition/Condition" alias Condition as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Condition" alias ConditionR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Condition extends DomainResource
  input src : Condition as source
  input tgt : ConditionR2 as target

  "Condition-identifier" : for src.identifier make tgt.identifier
  "Condition-patient" : for src.subject make tgt.patient
  "Condition-encounter" : for src.context make tgt.encounter
  "Condition-asserter" : for src.asserter make tgt.asserter
  "Condition-dateRecorded" : for src.assertedDate make tgt.dateRecorded
  "Condition-code" : for src.code make tgt.code
  "Condition-category" : for src.category make tgt.category
  "Condition-clinicalStatus" : for src.clinicalStatus as v check $this != 'inactive' make tgt.clinicalStatus =
    translate(v, "#ConditionClinicalStatus", "code")
  "Condition-verificationStatus" : for src.verificationStatus make tgt.verificationStatus
  "Condition-severity" : for src.severity make tgt.severity
  "Condition-onset" : for src.onset make tgt.onset
  "Condition-abatement" : for src.abatement make tgt.abatement
  "Condition-stage" : for src.stage as vs0 make tgt.stage as vt0 then cndStage(vs0, vt0)
  "Condition-evidence" : for src.evidence as vs0 make tgt.evidence as vt0 then cndEvidence(vs0, vt0)
  "Condition-bodySite" : for src.bodySite make tgt.bodySite
  "Condition-note" : for src.note as n then {
    "Condition.notes-text" : for n.text make tgt.notes
  }

endgroup

group cndStage extends BackboneElement
  input src : ConditionR2 as source
  input tgt : Condition as target

  "Condition.stage-summary" : for src.summary make tgt.summary
  "Condition.stage-assessment" : for src.assessment make tgt.assessment
endgroup

group cndEvidence extends BackboneElement
  input src : ConditionR2 as source
  input tgt : Condition as target

  "Condition.evidence-code" : for src.code make tgt.code
  "Condition.evidence-detail" : for src.detail make tgt.detail
endgroup
```

9.2.11.3 R3 Validation Errors

cond-uslab-example1 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
cond-uslab-example2 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
condition-example | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
example | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f001 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f002 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f003 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f201 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f202 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f203 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f204 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
f205 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
stroke | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)
example2 | ERROR @ Condition: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)

9.3 Resource Procedure - Content

URL: https://hl7.org/fhir/STU3/procedure.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

An action that is or was performed on a patient. This can be a physical intervention like an operation, or less invasive like counseling or hypnotherapy.

9.3.1 Scope and Usage

Procedure is one of the event resources in the FHIR workflow specification.

This resource is used to record the details of procedures performed on a patient. A procedure is an activity that is performed with or on a patient as part of the provision of care. Examples include surgical procedures, diagnostic procedures, endoscopic procedures, biopsies, counseling, physiotherapy, exercise, etc. Procedures may be performed by a healthcare professional, a friend or relative or in some cases by the patient themselves.

This resource provides summary information about the occurrence of the procedure and is not intended to provide real-time snapshots of a procedure as it unfolds, though for long-running procedures such as psychotherapy, it could represent summary level information about overall progress. The creation of a resource to support detailed real-time procedure information awaits the identification of a specific implementation use-case to share such information.

9.3.2 Boundaries and Relationships

The Procedure resource should not be used to capture an event if a more specific resource already exists - i.e. immunizations, drug administrations and communications. The boundary between determining whether an action is a Procedure (training or counseling) as opposed to a Communication is based on whether there's a specific intent to change the mind-set of the patient. Mere disclosure of information would be considered a Communication. A process that involves verification of the patient's comprehension or to change the patient's mental state would be a Procedure.

Note that many diagnostic processes are procedures that generate Observations and DiagnosticReports. In many cases, such an observation does not require an explicit representation of the procedure used to create the observation, but where there are details of interest about how the diagnostic procedure was performed, the procedure resource is used to describe the activity.

Some diagnostic procedures may not have a Procedure record. The Procedure record is only necessary when there is a need to capture information about the physical intervention that was performed to capture the diagnostic information (e.g. anesthetic, incision, scope size, etc.)

This resource is referenced by AdverseEvent, Appointment, ChargeItem, Claim, ClinicalImpression, Encounter, ExplanationOfBenefit, Flag, ImagingStudy, MedicationAdministration, MedicationDispense, MedicationStatement and QuestionnaireResponse

9.3.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Procedure | I | | DomainResource | An action that is being or was performed on a patient + Reason not done is only permitted if notDone indicator is true Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Identifiers for this procedure
definition | ? | 0..* | Reference(PlanDefinition | ActivityDefinition | HealthcareService) | Instantiates protocol or definition
basedOn | ? | 0..* | Reference(CarePlan | ProcedureRequest | ReferralRequest) | A request for this procedure
partOf | ? | 0..* | Reference(Procedure | Observation | MedicationAdministration) | Part of referenced event
status | ?!? | 1..1 | code | preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown EventStatus (Required)
notDone | ?!? | 0..1 | boolean | True if procedure was not performed as scheduled
notDoneReason | ?I | 0..1 | CodeableConcept | Reason procedure was not performed Procedure Not Performed Reason (SNOMED-CT) (Example)
category | ? | 0..1 | CodeableConcept | Classification of the procedure Procedure Category Codes (SNOMED CT) (Example)
code | ? | 0..1 | CodeableConcept | Identification of the procedure Procedure Codes (SNOMED CT) (Example)
subject | ? | 1..1 | Reference(Patient | Group) | Who the procedure was performed on
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode associated with the procedure
performed[x] | ? | 0..1 | | Date/Period the procedure was performed
performedDateTime | | | dateTime
performedPeriod | | | Period
performer | ? | 0..* | BackboneElement | The people who performed the procedure
role | ? | 0..1 | CodeableConcept | The role the actor was in Procedure Performer Role Codes (Example)
actor | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | The reference to the practitioner
onBehalfOf | | 0..1 | Reference(Organization) | Organization the device or practitioner was acting for
location | ? | 0..1 | Reference(Location) | Where the procedure happened
reasonCode | ? | 0..* | CodeableConcept | Coded reason procedure performed Procedure Reason Codes (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Condition that is the reason the procedure performed
bodySite | ? | 0..* | CodeableConcept | Target body sites SNOMED CT Body Structures (Example)
outcome | ? | 0..1 | CodeableConcept | The result of procedure Procedure Outcome Codes (SNOMED CT) (Example)
report | | 0..* | Reference(DiagnosticReport) | Any report resulting from the procedure
complication | | 0..* | CodeableConcept | Complication following the procedure Condition/Problem/Diagnosis Codes (Example)
complicationDetail | | 0..* | Reference(Condition) | A condition that is a result of the procedure
followUp | | 0..* | CodeableConcept | Instructions for follow up Procedure Follow up Codes (SNOMED CT) (Example)
note | | 0..* | Annotation | Additional information about the procedure
focalDevice | | 0..* | BackboneElement | Device changed in procedure
action | | 0..1 | CodeableConcept | Kind of change to device Procedure Device Action Codes (Preferred)
manipulated | | 1..1 | Reference(Device) | Device that was changed
usedReference | | 0..* | Reference(Device | Medication | Substance) | Items used during procedure
usedCode | | 0..* | CodeableConcept | Coded items used during the procedure FHIR Device Types (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Procedure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifiers for this procedure --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|ActivityDefinition|
   HealthcareService) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(CarePlan|ProcedureRequest|ReferralRequest) A request for this procedure --></basedOn>
 <partOf><!-- 0..* Reference(Procedure|Observation|MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-error |
    unknown -->
 <notDone value="[boolean]"/><!-- 0..1 True if procedure was not performed as scheduled -->
 <notDoneReason><!--  0..1 CodeableConcept Reason procedure was not performed --></notDoneReason>
 <category><!-- 0..1 CodeableConcept Classification of the procedure --></category>
 <code><!-- 0..1 CodeableConcept Identification of the procedure --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Who the procedure was performed on --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode associated with the procedure --></context>
 <performed[x]><!-- 0..1 dateTime|Period Date/Period the procedure was performed --></performed[x]>
 <performer>  <!-- 0..* The people who performed the procedure -->
  <role><!-- 0..1 CodeableConcept The role the actor was in --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) The reference to the practitioner --></actor>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization the device or practitioner was acting for --></onBehalfOf>
 </performer>
 <location><!-- 0..1 Reference(Location) Where the procedure happened --></location>
```

```
 <reasonCode><!-- 0..* CodeableConcept Coded reason procedure performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition that is the reason the procedure performed
    --></reasonReference>
 <bodySite><!-- 0..* CodeableConcept Target body sites --></bodySite>
 <outcome><!-- 0..1 CodeableConcept The result of procedure --></outcome>
 <report><!-- 0..* Reference(DiagnosticReport) Any report resulting from the procedure --></report>
 <complication><!-- 0..* CodeableConcept Complication following the procedure --></complication>
 <complicationDetail><!-- 0..* Reference(Condition) A condition that is a result of the procedure
    --></complicationDetail>
 <followUp><!-- 0..* CodeableConcept Instructions for follow up --></followUp>
 <note><!-- 0..* Annotation Additional information about the procedure --></note>
 <focalDevice>  <!-- 0..* Device changed in procedure -->
  <action><!-- 0..1 CodeableConcept Kind of change to device --></action>
  <manipulated><!-- 1..1 Reference(Device) Device that was changed --></manipulated>
 </focalDevice>
 <usedReference><!-- 0..* Reference(Device|Medication|Substance) Items used during procedure --></usedReference>
 <usedCode><!-- 0..* CodeableConcept Coded items used during the procedure --></usedCode>
</Procedure>
```

JSON Template

```
{
  "resourceType" : "Procedure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifiers for this procedure
  "definition" : [{ Reference(PlanDefinition|ActivityDefinition|
   HealthcareService) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(CarePlan|ProcedureRequest|ReferralRequest) }], // A request for this procedure
  "partOf" : [{ Reference(Procedure|Observation|MedicationAdministration) }], // Part of referenced event
  "status" : "<code>", // R!  preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown
  "notDone" : <boolean>, // True if procedure was not performed as scheduled
  "notDoneReason" : { CodeableConcept }, // C? Reason procedure was not performed
  "category" : { CodeableConcept }, // Classification of the procedure
  "code" : { CodeableConcept }, // Identification of the procedure
  "subject" : { Reference(Patient|Group) }, // R!  Who the procedure was performed on
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode associated with the procedure
  // performed[x]: Date/Period the procedure was performed. One of these 2:
  "performedDateTime" : "<dateTime>",
  "performedPeriod" : { Period },
  "performer" : [{ // The people who performed the procedure
    "role" : { CodeableConcept }, // The role the actor was in
    "actor" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  The reference to the practitioner
    "onBehalfOf" : { Reference(Organization) } // Organization the device or practitioner was acting for
  }],
  "location" : { Reference(Location) }, // Where the procedure happened
  "reasonCode" : [{ CodeableConcept }], // Coded reason procedure performed
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition that is the reason the procedure performed
  "bodySite" : [{ CodeableConcept }], // Target body sites
  "outcome" : { CodeableConcept }, // The result of procedure
  "report" : [{ Reference(DiagnosticReport) }], // Any report resulting from the procedure
  "complication" : [{ CodeableConcept }], // Complication following the procedure
  "complicationDetail" : [{ Reference(Condition) }], // A condition that is a result of the procedure
  "followUp" : [{ CodeableConcept }], // Instructions for follow up
  "note" : [{ Annotation }], // Additional information about the procedure
  "focalDevice" : [{ // Device changed in procedure
    "action" : { CodeableConcept }, // Kind of change to device
    "manipulated" : { Reference(Device) } // R!  Device that was changed
  }],
  "usedReference" : [{ Reference(Device|Medication|Substance) }], // Items used during procedure
  "usedCode" : [{ CodeableConcept }] // Coded items used during the procedure
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Procedure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Procedure.identifier [ Identifier ], ... ; # 0..* External Identifiers for this procedure
  fhir:Procedure.definition [ Reference(PlanDefinition|ActivityDefinition|HealthcareService) ], ... ; # 0..*
    Instantiates protocol or definition
  fhir:Procedure.basedOn [ Reference(CarePlan|ProcedureRequest|ReferralRequest) ], ... ; # 0..* A request for this
    procedure
  fhir:Procedure.partOf [ Reference(Procedure|Observation|MedicationAdministration) ], ... ; # 0..* Part of referenced
    event
  fhir:Procedure.status [ code ]; # 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-error
    | unknown
  fhir:Procedure.notDone [ boolean ]; # 0..1 True if procedure was not performed as scheduled
  fhir:Procedure.notDoneReason [ CodeableConcept ]; # 0..1 Reason procedure was not performed
  fhir:Procedure.category [ CodeableConcept ]; # 0..1 Classification of the procedure
  fhir:Procedure.code [ CodeableConcept ]; # 0..1 Identification of the procedure
  fhir:Procedure.subject [ Reference(Patient|Group) ]; # 1..1 Who the procedure was performed on
  fhir:Procedure.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode associated with the
    procedure
  # Procedure.performed[x] : 0..1 Date/Period the procedure was performed. One of these 2
```

```
    fhir:Procedure.performedDateTime [ dateTime ]
    fhir:Procedure.performedPeriod [ Period ]
  fhir:Procedure.performer [ # 0..* The people who performed the procedure
    fhir:Procedure.performer.role [ CodeableConcept ]; # 0..1 The role the actor was in
    fhir:Procedure.performer.actor [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1 The
    reference to the practitioner
    fhir:Procedure.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization the device or practitioner was
    acting for
  ], ...;
  fhir:Procedure.location [ Reference(Location) ]; # 0..1 Where the procedure happened
  fhir:Procedure.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason procedure performed
  fhir:Procedure.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition that is the reason the
    procedure performed
  fhir:Procedure.bodySite [ CodeableConcept ], ... ; # 0..* Target body sites
  fhir:Procedure.outcome [ CodeableConcept ]; # 0..1 The result of procedure
  fhir:Procedure.report [ Reference(DiagnosticReport) ], ... ; # 0..* Any report resulting from the procedure
  fhir:Procedure.complication [ CodeableConcept ], ... ; # 0..* Complication following the procedure
  fhir:Procedure.complicationDetail [ Reference(Condition) ], ... ; # 0..* A condition that is a result of the procedure
  fhir:Procedure.followUp [ CodeableConcept ], ... ; # 0..* Instructions for follow up
  fhir:Procedure.note [ Annotation ], ... ; # 0..* Additional information about the procedure
  fhir:Procedure.focalDevice [ # 0..* Device changed in procedure
    fhir:Procedure.focalDevice.action [ CodeableConcept ]; # 0..1 Kind of change to device
    fhir:Procedure.focalDevice.manipulated [ Reference(Device) ]; # 1..1 Device that was changed
  ], ...;
  fhir:Procedure.usedReference [ Reference(Device|Medication|Substance) ], ... ; # 0..* Items used during procedure
  fhir:Procedure.usedCode [ CodeableConcept ], ... ; # 0..* Coded items used during the procedure
]
```

Changes since DSTU2

Procedure
Procedure.definition | Added Element
Procedure.basedOn | Renamed from request to basedOn Max Cardinality changed from 1 to * Remove Reference(DiagnosticOrder)
Procedure.partOf | Added Element
Procedure.status | Change value set from http://hl7.org/fhir/ValueSet/procedure-status to http://hl7.org/fhir/ValueSet/event-status
Procedure.notDone | Renamed from notPerformed to notDone
Procedure.notDoneReason | Renamed from reasonNotPerformed to notDoneReason Max Cardinality changed from * to 1
Procedure.code | Min Cardinality changed from 1 to 0
Procedure.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Procedure.performer.actor | Min Cardinality changed from 0 to 1 Add Reference(Device)
Procedure.performer.onBehalfOf | Added Element
Procedure.reasonCode | Added Element
Procedure.reasonReference | Added Element
Procedure.complicationDetail | Added Element
Procedure.note | Renamed from notes to note
Procedure.focalDevice.action | Remove Binding http://hl7.org/fhir/ValueSet/device-action (required)
Procedure.usedReference | Renamed from used to usedReference
Procedure.usedCode | Added Element
Procedure.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 9 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
Procedure | I | | DomainResource | An action that is being or was performed on a patient + Reason not done is only permitted if notDone indicator is true Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Identifiers for this procedure
definition | ? | 0..* | Reference(PlanDefinition | ActivityDefinition | HealthcareService) | Instantiates protocol or definition
basedOn | ? | 0..* | Reference(CarePlan | ProcedureRequest | ReferralRequest) | A request for this procedure
partOf | ? | 0..* | Reference(Procedure | Observation | MedicationAdministration) | Part of referenced event
status | ?!? | 1..1 | code | preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown EventStatus (Required)
notDone | ?!? | 0..1 | boolean | True if procedure was not performed as scheduled
notDoneReason | ?I | 0..1 | CodeableConcept | Reason procedure was not performed Procedure Not Performed Reason (SNOMED-CT) (Example)
category | ? | 0..1 | CodeableConcept | Classification of the procedure Procedure Category Codes (SNOMED CT) (Example)
code | ? | 0..1 | CodeableConcept | Identification of the procedure Procedure Codes (SNOMED CT) (Example)
subject | ? | 1..1 | Reference(Patient | Group) | Who the procedure was performed on
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode associated with the procedure
performed[x] | ? | 0..1 | | Date/Period the procedure was performed
performedDateTime | | | dateTime
performedPeriod | | | Period
performer | ? | 0..* | BackboneElement | The people who performed the procedure
role | ? | 0..1 | CodeableConcept | The role the actor was in Procedure Performer Role Codes (Example)
actor | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | The reference to the practitioner
onBehalfOf | | 0..1 | Reference(Organization) | Organization the device or practitioner was acting for
location | ? | 0..1 | Reference(Location) | Where the procedure happened
reasonCode | ? | 0..* | CodeableConcept | Coded reason procedure performed Procedure Reason Codes (Example)

reasonReference | ? | 0..* | Reference(Condition | Observation) | Condition that is the reason the procedure performed
bodySite | ? | 0..* | CodeableConcept | Target body sites SNOMED CT Body Structures (Example)
outcome | ? | 0..1 | CodeableConcept | The result of procedure Procedure Outcome Codes (SNOMED CT) (Example)
report | | 0..* | Reference(DiagnosticReport) | Any report resulting from the procedure
complication | | 0..* | CodeableConcept | Complication following the procedure Condition/Problem/Diagnosis Codes (Example)
complicationDetail | | 0..* | Reference(Condition) | A condition that is a result of the procedure
followUp | | 0..* | CodeableConcept | Instructions for follow up Procedure Follow up Codes (SNOMED CT) (Example)
note | | 0..* | Annotation | Additional information about the procedure
focalDevice | | 0..* | BackboneElement | Device changed in procedure
action | | 0..1 | CodeableConcept | Kind of change to device Procedure Device Action Codes (Preferred)
manipulated | | 1..1 | Reference(Device) | Device that was changed
usedReference | | 0..* | Reference(Device | Medication | Substance) | Items used during procedure
usedCode | | 0..* | CodeableConcept | Coded items used during the procedure FHIR Device Types (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Procedure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifiers for this procedure --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|ActivityDefinition|
   HealthcareService) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(CarePlan|ProcedureRequest|ReferralRequest) A request for this procedure --></basedOn>
 <partOf><!-- 0..* Reference(Procedure|Observation|MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-error |
    unknown -->
 <notDone value="[boolean]"/><!-- 0..1 True if procedure was not performed as scheduled -->
 <notDoneReason><!--  0..1 CodeableConcept Reason procedure was not performed --></notDoneReason>
 <category><!-- 0..1 CodeableConcept Classification of the procedure --></category>
 <code><!-- 0..1 CodeableConcept Identification of the procedure --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Who the procedure was performed on --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode associated with the procedure --></context>
 <performed[x]><!-- 0..1 dateTime|Period Date/Period the procedure was performed --></performed[x]>
 <performer>  <!-- 0..* The people who performed the procedure -->
  <role><!-- 0..1 CodeableConcept The role the actor was in --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) The reference to the practitioner --></actor>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization the device or practitioner was acting for --></onBehalfOf>
 </performer>
 <location><!-- 0..1 Reference(Location) Where the procedure happened --></location>
 <reasonCode><!-- 0..* CodeableConcept Coded reason procedure performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition that is the reason the procedure performed
    --></reasonReference>
 <bodySite><!-- 0..* CodeableConcept Target body sites --></bodySite>
 <outcome><!-- 0..1 CodeableConcept The result of procedure --></outcome>
 <report><!-- 0..* Reference(DiagnosticReport) Any report resulting from the procedure --></report>
 <complication><!-- 0..* CodeableConcept Complication following the procedure --></complication>
 <complicationDetail><!-- 0..* Reference(Condition) A condition that is a result of the procedure
    --></complicationDetail>
 <followUp><!-- 0..* CodeableConcept Instructions for follow up --></followUp>
 <note><!-- 0..* Annotation Additional information about the procedure --></note>
 <focalDevice>  <!-- 0..* Device changed in procedure -->
  <action><!-- 0..1 CodeableConcept Kind of change to device --></action>
  <manipulated><!-- 1..1 Reference(Device) Device that was changed --></manipulated>
 </focalDevice>
 <usedReference><!-- 0..* Reference(Device|Medication|Substance) Items used during procedure --></usedReference>
 <usedCode><!-- 0..* CodeableConcept Coded items used during the procedure --></usedCode>
</Procedure>
```

JSON Template

```
{
  "resourceType" : "Procedure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifiers for this procedure
  "definition" : [{ Reference(PlanDefinition|ActivityDefinition|
   HealthcareService) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(CarePlan|ProcedureRequest|ReferralRequest) }], // A request for this procedure
  "partOf" : [{ Reference(Procedure|Observation|MedicationAdministration) }], // Part of referenced event
  "status" : "<code>", // R!  preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown
  "notDone" : <boolean>, // True if procedure was not performed as scheduled
  "notDoneReason" : { CodeableConcept }, // C? Reason procedure was not performed
  "category" : { CodeableConcept }, // Classification of the procedure
  "code" : { CodeableConcept }, // Identification of the procedure
  "subject" : { Reference(Patient|Group) }, // R!  Who the procedure was performed on
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode associated with the procedure
  // performed[x]: Date/Period the procedure was performed. One of these 2:
  "performedDateTime" : "<dateTime>",
  "performedPeriod" : { Period },
  "performer" : [{ // The people who performed the procedure
    "role" : { CodeableConcept }, // The role the actor was in
    "actor" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  The reference to the practitioner
    "onBehalfOf" : { Reference(Organization) } // Organization the device or practitioner was acting for
```

```
  }],
  "location" : { Reference(Location) }, // Where the procedure happened
  "reasonCode" : [{ CodeableConcept }], // Coded reason procedure performed
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition that is the reason the procedure performed
  "bodySite" : [{ CodeableConcept }], // Target body sites
  "outcome" : { CodeableConcept }, // The result of procedure
  "report" : [{ Reference(DiagnosticReport) }], // Any report resulting from the procedure
  "complication" : [{ CodeableConcept }], // Complication following the procedure
  "complicationDetail" : [{ Reference(Condition) }], // A condition that is a result of the procedure
  "followUp" : [{ CodeableConcept }], // Instructions for follow up
  "note" : [{ Annotation }], // Additional information about the procedure
  "focalDevice" : [{ // Device changed in procedure
    "action" : { CodeableConcept }, // Kind of change to device
    "manipulated" : { Reference(Device) } // R!  Device that was changed
  }],
  "usedReference" : [{ Reference(Device|Medication|Substance) }], // Items used during procedure
  "usedCode" : [{ CodeableConcept }] // Coded items used during the procedure
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Procedure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Procedure.identifier [ Identifier ], ... ; # 0..* External Identifiers for this procedure
  fhir:Procedure.definition [ Reference(PlanDefinition|ActivityDefinition|HealthcareService) ], ... ; # 0..*
    Instantiates protocol or definition
  fhir:Procedure.basedOn [ Reference(CarePlan|ProcedureRequest|ReferralRequest) ], ... ; # 0..* A request for this
    procedure
  fhir:Procedure.partOf [ Reference(Procedure|Observation|MedicationAdministration) ], ... ; # 0..* Part of referenced
    event
  fhir:Procedure.status [ code ]; # 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-error
    | unknown
  fhir:Procedure.notDone [ boolean ]; # 0..1 True if procedure was not performed as scheduled
  fhir:Procedure.notDoneReason [ CodeableConcept ]; # 0..1 Reason procedure was not performed
  fhir:Procedure.category [ CodeableConcept ]; # 0..1 Classification of the procedure
  fhir:Procedure.code [ CodeableConcept ]; # 0..1 Identification of the procedure
  fhir:Procedure.subject [ Reference(Patient|Group) ]; # 1..1 Who the procedure was performed on
  fhir:Procedure.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode associated with the
    procedure
  # Procedure.performed[x] : 0..1 Date/Period the procedure was performed. One of these 2
    fhir:Procedure.performedDateTime [ dateTime ]
    fhir:Procedure.performedPeriod [ Period ]
  fhir:Procedure.performer [ # 0..* The people who performed the procedure
    fhir:Procedure.performer.role [ CodeableConcept ]; # 0..1 The role the actor was in
    fhir:Procedure.performer.actor [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1 The
    reference to the practitioner
    fhir:Procedure.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization the device or practitioner was
    acting for
  ], ...;
  fhir:Procedure.location [ Reference(Location) ]; # 0..1 Where the procedure happened
  fhir:Procedure.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason procedure performed
  fhir:Procedure.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition that is the reason the
    procedure performed
  fhir:Procedure.bodySite [ CodeableConcept ], ... ; # 0..* Target body sites
  fhir:Procedure.outcome [ CodeableConcept ]; # 0..1 The result of procedure
  fhir:Procedure.report [ Reference(DiagnosticReport) ], ... ; # 0..* Any report resulting from the procedure
  fhir:Procedure.complication [ CodeableConcept ], ... ; # 0..* Complication following the procedure
  fhir:Procedure.complicationDetail [ Reference(Condition) ], ... ; # 0..* A condition that is a result of the procedure
  fhir:Procedure.followUp [ CodeableConcept ], ... ; # 0..* Instructions for follow up
  fhir:Procedure.note [ Annotation ], ... ; # 0..* Additional information about the procedure
  fhir:Procedure.focalDevice [ # 0..* Device changed in procedure
    fhir:Procedure.focalDevice.action [ CodeableConcept ]; # 0..1 Kind of change to device
    fhir:Procedure.focalDevice.manipulated [ Reference(Device) ]; # 1..1 Device that was changed
  ], ...;
  fhir:Procedure.usedReference [ Reference(Device|Medication|Substance) ], ... ; # 0..* Items used during procedure
  fhir:Procedure.usedCode [ CodeableConcept ], ... ; # 0..* Coded items used during the procedure
]
```

Changes since DSTU2

Procedure

Procedure.definition | Added Element

Procedure.basedOn | Renamed from request to basedOn Max Cardinality changed from 1 to * Remove Reference(DiagnosticOrder)

Procedure.partOf | Added Element

Procedure.status | Change value set from http://hl7.org/fhir/ValueSet/procedure-status to http://hl7.org/fhir/ValueSet/event-status

Procedure.notDone | Renamed from notPerformed to notDone

Procedure.notDoneReason | Renamed from reasonNotPerformed to notDoneReason Max Cardinality changed from * to 1

Procedure.code | Min Cardinality changed from 1 to 0

Procedure.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

Procedure.performer.actor | Min Cardinality changed from 0 to 1 Add Reference(Device)

Procedure.performer.onBehalfOf | Added Element

Procedure.reasonCode | Added Element

Procedure.reasonReference | Added Element
Procedure.complicationDetail | Added Element
Procedure.note | Renamed from notes to note
Procedure.focalDevice.action | Remove Binding http://hl7.org/fhir/ValueSet/device-action (required)
Procedure.usedReference | Renamed from used to usedReference
Procedure.usedCode | Added Element
Procedure.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 9 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.3.3.1 Terminology Bindings

Path | Definition | Type | Reference
Procedure.status | A code specifying the state of the procedure. | Required | EventStatus
Procedure.notDoneReason | A code that identifies the reason a procedure was not performed. | Example | Procedure Not Performed Reason (SNOMED-CT)
Procedure.category | A code that classifies a procedure for searching, sorting and display purposes. | Example | Procedure Category Codes (SNOMED CT)
Procedure.code | A code to identify a specific procedure . | Example | Procedure Codes (SNOMED CT)
Procedure.performer.role | A code that identifies the role of a performer of the procedure. | Example | Procedure Performer Role Codes
Procedure.reasonCode | A code that identifies the reason a procedure is required. | Example | Procedure Reason Codes
Procedure.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMED CT Body Structures
Procedure.outcome | An outcome of a procedure - whether it was resolved or otherwise. | Example | Procedure Outcome Codes (SNOMED CT)
Procedure.complication | Codes describing complications that resulted from a procedure. | Example | Condition/Problem/Diagnosis Codes
Procedure.followUp | Specific follow up required for a procedure e.g. removal of sutures. | Example | Procedure Follow up Codes (SNOMED CT)
Procedure.focalDevice.action | A kind of change that happened to the device during the procedure. | Preferred | Procedure Device Action Codes
Procedure.usedCode | Codes describing items used during a procedure | Example | FHIR Device Types

9.3.3.2 Constraints

* pro-1: Reason not done is only permitted if notDone indicator is true (expression : notDoneReason.empty() or notDone = true)

9.3.3.3 Use of Procedure properties

Many of the elements of Procedure have inherent relationships and may be conveyed by the Procedure.code or in the text element of the Procedure.code property. I.e. You may be able to infer category, bodySite and even indication. Whether these other properties will be populated may vary by implementation.

Care should be taken to avoid nonsensical combinations/statements; e.g. "name=amputation, bodySite=heart"

9.3.3.4 Use of Procedure.used

For devices, these are devices that are incidental to / or used to perform the procedure - scalpels, gauze, endoscopes, etc. Devices that are the focus of the procedure should appear in Procedure.device instead.

9.3.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | A request for this procedure | Procedure.basedOn (ReferralRequest, CarePlan, ProcedureRequest)
category | token | Classification of the procedure | Procedure.category
code | token | A code to identify a procedure | Procedure.code | 8 Resources
context | reference | Encounter or episode associated with the procedure | Procedure.context (EpisodeOfCare, Encounter)
date | date | Date/Period the procedure was performed | Procedure.performed | 18 Resources
definition | reference | Instantiates protocol or definition | Procedure.definition (PlanDefinition, HealthcareService, ActivityDefinition)
encounter | reference | Search by encounter | Procedure.context (Encounter) | 12 Resources
identifier | token | A unique identifier for a procedure | Procedure.identifier | 26 Resources
location | reference | Where the procedure happened | Procedure.location (Location)
part-of | reference | Part of referenced event | Procedure.partOf (Observation, Procedure, MedicationAdministration)
patient | reference | Search by subject - a patient | Procedure.subject (Patient) | 31 Resources
performer | reference | The reference to the practitioner | Procedure.performer.actor (Practitioner, Organization, Device, Patient, RelatedPerson)
status | token | preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown | Procedure.status
subject | reference | Search by subject | Procedure.subject (Group, Patient)

9.3.5 Resource Procedure - Examples

URL: https://hl7.org/fhir/STU3/procedure-examples.html

9.3.5 Resource Procedure - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example List:

General Procedure Example | example | XML | JSON | Turtle
Example of a Biopsy | biopsy | XML | JSON | Turtle
Real-world procedure example | f201 | XML | JSON | Turtle
Real-world procedure example | f001 | XML | JSON | Turtle
Real-world procedure example | f002 | XML | JSON | Turtle
Real-world procedure example | f003 | XML | JSON | Turtle
Real-world procedure example | f004 | XML | JSON | Turtle
Example of a device manipulation | example-implant | XML | JSON | Turtle
Example of ambulation procedure | ambulation | XML | JSON | Turtle
Example of biopsy procedure that was part of a colonoscopy | colon-biopsy | XML | JSON | Turtle
Example of colonoscopy procedure with complication | colonoscopy | XML | JSON | Turtle
Example of an appendectomy procedure that is primarily narrative | appendectomy-narrative | XML | JSON | Turtle
Example of an obstetric procedure | ob | XML | JSON | Turtle
Example of a physical therapy evaluation procedure | physical-therapy | XML | JSON | Turtle
Example of patient education | education | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.3.6 Resource Procedure - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/procedure-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.3.6 Resource Procedure - Detailed Descriptions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Procedure resource.

Procedure
Definition | An action that is or was performed on a patient. This can be a physical intervention like an operation, or less invasive like counseling or hypnotherapy.
Control | 1..1
Invariants | Defined on this element pro-1: Reason not done is only permitted if notDone indicator is true (expression : notDoneReason.empty() or notDone = true, xpath: not(exists(f:notDoneReason)) or f:notDone/@value=true())
Procedure.identifier
Definition | This records identifiers associated with this procedure that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Need to allow connection to a wider workflow.
Summary | true
Procedure.definition
Definition | A protocol, guideline, orderset or other definition that was adhered to in whole or in part by this procedure.
Control | 0..*
Type | Reference(PlanDefinition | ActivityDefinition | HealthcareService)
Summary | true
Procedure.basedOn
Definition | A reference to a resource that contains details of the request for this procedure.
Control | 0..*
Type | Reference(CarePlan | ProcedureRequest | ReferralRequest)
Alternate Names | fulfills
Summary | true
Procedure.partOf
Definition | A larger event of which this particular procedure is a component or step.
Control | 0..*
Type | Reference(Procedure | Observation | MedicationAdministration)
Alternate Names | container
Summary | true
Comments | The MedicationAdministration has a partOf reference to Procedure, but this is not a circular reference. For a surgical procedure, the anesthesia related medicationAdministration is part of the procedure. For an IV medication administration, the procedure to insert the IV port is part of the medication administration.
Procedure.status
Definition | A code specifying the state of the procedure. Generally this will be in-progress or completed state.
Control | 1..1
Terminology Binding | EventStatus (Required)
Type | code

Is Modifier | true
Summary | true
Comments | The unknown code is not to be used to convey other statuses. The unknown code should be used when one of the statuses applies, but the authoring system doesn't know the current state of the procedure. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Procedure.notDone
Definition | Set this to true if the record is saying that the procedure was NOT performed.
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | false
Summary | true
Comments | If true, it means the procedure did not occur as described. Typically it would be accompanied by attributes describing the type of activity. It might also be accompanied by body site information or time information (i.e. no procedure was done to the left arm or no procedure was done in this 2-year period). Specifying additional information such as performer, outcome, etc. is generally inappropriate. For example, it's not that useful to say "There was no appendectomy done at 12:03pm June 6th by Dr. Smith with a successful outcome" as it implies that there could have been an appendectomy done at any other time, by any other clinician or with any other outcome. This element is labeled as a modifier because it indicates that a procedure didn't happen.
Procedure.notDoneReason
Definition | A code indicating why the procedure was not performed.
Control | 0..1
Terminology Binding | Procedure Not Performed Reason (SNOMED-CT) (Example)
Type | CodeableConcept
Summary | true
Invariants | Affect this element pro-1: Reason not done is only permitted if notDone indicator is true (expression : notDoneReason.empty() or notDone = true, xpath: not(exists(f:notDoneReason)) or f:notDone/@value=true())
Procedure.category
Definition | A code that classifies the procedure for searching, sorting and display purposes (e.g. "Surgical Procedure").
Control | 0..1
Terminology Binding | Procedure Category Codes (SNOMED CT) (Example)
Type | CodeableConcept
Summary | true
Procedure.code
Definition | The specific procedure that is performed. Use text if the exact nature of the procedure cannot be coded (e.g. "Laparoscopic Appendectomy").
Control | 0..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
Requirements | 0..1 to account for primarily narrative only resources.
Alternate Names | type
Summary | true
Procedure.subject
Definition | The person, animal or group on which the procedure was performed.
Control | 1..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true
Procedure.context
Definition | The encounter during which the procedure was performed.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
Procedure.performed[x]
Definition | The date(time)/period over which the procedure was performed. Allows a period to support complex procedures that span more than one date, and also allows for the length of the procedure to be captured.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Procedure.performer
Definition | Limited to 'real' people rather than equipment.
Control | 0..*
Summary | true
Procedure.performer.role
Definition | For example: surgeon, anaethetist, endoscopist.
Control | 0..1
Terminology Binding | Procedure Performer Role Codes (Example)
Type | CodeableConcept
Summary | true
Procedure.performer.actor
Definition | The practitioner who was involved in the procedure.
Control | 1..1

Type | Reference(Practitioner | Organization | Patient | RelatedPerson | Device)
Requirements | A reference to Device supports use cases, such as pacemakers.
Summary | true
Procedure.performer.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when performing the action.
Procedure.location
Definition | The location where the procedure actually happened. E.g. a newborn at home, a tracheostomy at a restaurant.
Control | 0..1
Type | Reference(Location)
Requirements | Ties a procedure to where the records are likely kept.
Summary | true
Procedure.reasonCode
Definition | The coded reason why the procedure was performed. This may be coded entity of some type, or may simply be present as text.
Control | 0..*
Terminology Binding | Procedure Reason Codes (Example)
Type | CodeableConcept
Summary | true
Procedure.reasonReference
Definition | The condition that is the reason why the procedure was performed.
Control | 0..*
Type | Reference(Condition | Observation)
Summary | true
Comments | e.g. endoscopy for dilatation and biopsy, combination diagnosis and therapeutic.
Procedure.bodySite
Definition | Detailed and structured anatomical location information. Multiple locations are allowed - e.g. multiple punch biopsies of a lesion.
Control | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
Procedure.outcome
Definition | The outcome of the procedure - did it resolve reasons for the procedure being performed?
Control | 0..1
Terminology Binding | Procedure Outcome Codes (SNOMED CT) (Example)
Type | CodeableConcept
Summary | true
Comments | If outcome contains narrative text only, it can be captured using the CodeableConcept.text.
Procedure.report
Definition | This could be a histology result, pathology report, surgical report, etc..
Control | 0..*
Type | Reference(DiagnosticReport)
Comments | There could potentially be multiple reports - e.g. if this was a procedure which took multiple biopsies resulting in a number of anatomical pathology reports.
Procedure.complication
Definition | Any complications that occurred during the procedure, or in the immediate post-performance period. These are generally tracked separately from the notes, which will typically describe the procedure itself rather than any 'post procedure' issues.
Control | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Comments | If complications are only expressed by the narrative text, they can be captured using the CodeableConcept.text.
Procedure.complicationDetail
Definition | Any complications that occurred during the procedure, or in the immediate post-performance period.
Control | 0..*
Type | Reference(Condition)
Requirements | This is used to document a condition that is a result of the procedure, not the condition that was the reason for the procedure.
Procedure.followUp
Definition | If the procedure required specific follow up - e.g. removal of sutures. The followup may be represented as a simple note, or could potentially be more complex in which case the CarePlan resource can be used.
Control | 0..*
Terminology Binding | Procedure Follow up Codes (SNOMED CT) (Example)
Type | CodeableConcept
Procedure.note
Definition | Any other notes about the procedure. E.g. the operative notes.
Control | 0..*
Type | Annotation
Procedure.focalDevice
Definition | A device that is implanted, removed or otherwise manipulated (calibration, battery replacement, fitting a prosthesis, attaching a wound-vac, etc.) as a focal portion of the Procedure.
Control | 0..*
Procedure.focalDevice.action

Definition | The kind of change that happened to the device during the procedure.
Control | 0..1
Terminology Binding | Procedure Device Action Codes (Preferred)
Type | CodeableConcept
Procedure.focalDevice.manipulated
Definition | The device that was manipulated (changed) during the procedure.
Control | 1..1
Type | Reference(Device)
Procedure.usedReference
Definition | Identifies medications, devices and any other substance used as part of the procedure.
Control | 0..*
Type | Reference(Device | Medication | Substance)
Requirements | Used for tracking contamination, etc.
Comments | For devices actually implanted or removed, use Procedure.device.
Procedure.usedCode
Definition | Identifies coded items that were used as part of the procedure.
Control | 0..*
Terminology Binding | FHIR Device Types (Example)
Type | CodeableConcept
Comments | For devices actually implanted or removed, use Procedure.device.

9.3.7 Resource Procedure - Mappings

URL: https://hl7.org/fhir/STU3/procedure-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.3.7 Resource Procedure - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the procedure resource.

9.3.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Procedure | Procedure[moodCode=EVN]
identifier | .id
definition | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=(various e.g. PROC, OBS, PCPR, ACT, moodCode=RQO].code
partOf | .inboundRelationship[typeCode=COMP].source[classCode=SBADM or PROC or OBS, moodCode=EVN]
status | statusCode
notDone | .actionNegationInd
notDoneReason | .reason.Observation.value
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="EVN"].code
code | .code
subject | .participation[typeCode=SBJ].role
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
performed[x] | .effectiveTime
performer | .participation[typeCode=PRF]
role | .functionCode
actor | .role
onBehalfOf | .scoper
location | .participation[typeCode=LOC].role[classCode=SDLOC]
reasonCode | .reasonCode
reasonReference | .reasonCode
bodySite | .targetSiteCode
outcome | .outboundRelationship[typeCode=OUT].target.text
report | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN]
complication | .outboundRelationship[typeCode=OUTC].target[classCode=OBS, code="complication", moodCode=EVN].value
complicationDetail | .outboundRelationship[typeCode=OUTC].target[classCode=OBS, code="complication", moodCode=EVN].value
followUp | .outboundRelationship[typeCode=COMP].target[classCode=ACT, moodCode=INT].code
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value
focalDevice | .participation[typeCode=DEV].role[classCode=MANU]
action | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="procedure device action"].value=:procedure device action codes
manipulated | .participation[typeCode=DEV].role[classCode=SDLOC]
usedReference | .participation[typeCode=DEV].role[classCode=MANU] or .participation[typeCode=CSM].role[classCode=ADMM] (for Medication or Substance)
usedCode | participation[typeCode=Dev].role[classCode=MANU]

9.3.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Procedure
identifier | Some combination of ORC-2 / ORC-3 / OBR-2 / OBR-3 / IPC-1 / IPC-2 / IPC-3 / IPC-4
definition
basedOn
partOf
status
notDone
notDoneReason
category
code | OBR-44/OBR-45
subject | PID-3
context | PV1-19
performed[x] | OBR-7
performer
role | Some combination of STF-18 / PRA-3 / PRT-4 / ROL-3 / ORC-12 / OBR-16 / PV1-7 / PV1-8 / PV1-9 / PV1-17 / OBX-25
actor | ORC-19/PRT-5
onBehalfOf
location
reasonCode
reasonReference
bodySite | OBX-20
outcome
report
complication
complicationDetail
followUp
note | NTE
focalDevice
action
manipulated
usedReference
usedCode

9.3.8 Resource Procedure - Design Notes

URL: https://hl7.org/fhir/STU3/procedure-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.3.8 Resource Procedure - Design Notes

to do

9.3.9 Resource Procedure - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/procedure-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.3.9 Resource Procedure - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Procedure resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.3.9.1 Profiles

No Profiles defined for this resource

9.3.9.2 Extensions

procedure-approachBodySite | approachBodySite | for Procedure HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions
procedure-incisionDateTime | incisionDateTime | for Procedure HL7 Extensions
procedure-method | method | for Procedure HL7 Extensions
procedure-progressStatus | progressStatus | for Procedure HL7 Extensions
procedure-schedule | schedule | for Procedure HL7 Extensions
procedure-targetBodySite | targetBodySite | for Procedure HL7 Extensions

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions

condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
condition-partOf | partOf | for Condition HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions

9.3.9.3 Search Extensions

No Search Extensions defined for this resource

9.3.10 Procedure HL7 Extensions

URL: https://hl7.org/fhir/STU3/procedure-extensions.html

Defines common extensions used with or related to the Procedure resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.3.10 Procedure HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.3.10.1 Content

Extensions:
procedure-targetBodySite | targetBodySite : The target body site used for this procedure. Multiple locations are allowed.
procedure-approachBodySite | approachBodySite : The approach body site used for this procedure. Multiple locations are allowed.
procedure-method | method : The method used to perform this procedure.
procedure-progressStatus | progressStatus : A code to track a detailed progress of a procedure (e.g. In Recovery, Prepared for Surgery).
procedure-incisionDateTime | incisionDateTime : The time of the first incision.
procedure-causedBy | causedBy : This procedure is because of the related item.
procedure-schedule | schedule : The schedule that was followed.

9.3.11 Resource Procedure - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/procedure-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.3.11 Resource Procedure - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Procedure.

Functional status for this map: 9 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

9.3.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Procedure2to3" = "R2 to R3 Conversions for Procedure"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Procedure" alias ProcedureR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Procedure" alias Procedure as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Procedure extends DomainResource
  input src : ProcedureR2 as source
  input tgt : Procedure as target

  "Procedure-identifier" : for src.identifier make tgt.identifier
  "Procedure-subject" : for src.subject make tgt.subject
  "Procedure-status" : for src.status make tgt.status
  "Procedure-category" : for src.category make tgt.category
  "Procedure-code" : for src.code make tgt.code
  "Procedure-notPerformed" : for src.notPerformed make tgt.notDone
  "Procedure-reasonNotPerformed" : for src.reasonNotPerformed make tgt.notDoneReason
  "Procedure-bodySite" : for src.bodySite make tgt.bodySite
  "Procedure-reasonCode" : for src.reason where $this is CodeableConcept make tgt.reasonCode
  "Procedure-reasonReference" : for src.reason where $this is Reference make tgt.reasonReference
  "Procedure-performed" : for src.performed make tgt.performed
  "Procedure-performer" : for src.performer as vs0 make tgt.performer as vt0 then prcPerformer(vs0, vt0)
  "Procedure-encounter" : for src.encounter make tgt.context
  "Procedure-location" : for src.location make tgt.location
  "Procedure-outcome" : for src.outcome make tgt.outcome
  "Procedure-report" : for src.report make tgt.report
  "Procedure-complication" : for src.complication make tgt.complication
  "Procedure-followUp" : for src.followUp make tgt.followUp
  "Procedure-request" : for src.request make tgt.basedOn
  "Procedure-notes" : for src.notes make tgt.note
  "Procedure-focalDevice" : for src.focalDevice as vs0 make tgt.focalDevice as vt0 then prcFocalDevice(vs0, vt0)
  "Procedure-used" : for src.used make tgt.usedReference
endgroup
```

```
group prcPerformer extends BackboneElement
  input src : Procedure as source
  input tgt : ProcedureR2 as target

  "Procedure.performer-actor" : for src.actor make tgt.actor
  "Procedure.performer-role" : for src.role make tgt.role
endgroup

group prcFocalDevice extends BackboneElement
  input src : Procedure as source
  input tgt : ProcedureR2 as target

  "Procedure.focalDevice-action" : for src.action make tgt.action
  "Procedure.focalDevice-manipulated" : for src.manipulated make tgt.manipulated
endgroup
```

9.3.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Procedure3to2" = "R3 to R2 Conversion for Procedure"

uses "http://hl7.org/fhir/StructureDefinition/Procedure" alias  Procedure as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Procedure" alias  ProcedureR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Procedure extends DomainResource
  input src : Procedure as source
  input tgt : ProcedureR2 as target

  "Procedure-identifier" : for src.identifier make tgt.identifier
  "Procedure-subject" : for src.subject make tgt.subject
  "Procedure-status" : for src.status make tgt.status
  "Procedure-category" : for src.category make tgt.category
  "Procedure-code" : for src.code make tgt.code
  "Procedure-notPerformed" : for src.notDone make tgt.notPerformed
  "Procedure-reasonNotPerformed" : for src.notDoneReason make tgt.reasonNotPerformed
  "Procedure-bodySite" : for src.bodySite make tgt.bodySite
  "Procedure-reasonCode" : for src.reasonCode make tgt.reason
  "Procedure-reasonReference" : for src.reasonReference make tgt.reason
  "Procedure-performed" : for src.performed make tgt.performed
  "Procedure-performer" : for src.performer as vs0 make tgt.performer as vt0 then prcPerformer(vs0, vt0)
  "Procedure-encounter" : for src.context make tgt.encounter
  "Procedure-location" : for src.location make tgt.location
  "Procedure-outcome" : for src.outcome make tgt.outcome
  "Procedure-report" : for src.report make tgt.report
  "Procedure-complication" : for src.complication make tgt.complication
  "Procedure-followUp" : for src.followUp make tgt.followUp
  "Procedure-request" : for src.basedOn make tgt.request
  "Procedure-notes" : for src.note make tgt.notes
  "Procedure-focalDevice" : for src.focalDevice as vs0 make tgt.focalDevice as vt0 then prcFocalDevice(vs0, vt0)
  "Procedure-used" : for src.usedReference make tgt.used
endgroup

group prcPerformer extends BackboneElement
  input src : Procedure as source
  input tgt : ProcedureR2 as target

  "Procedure.performer-actor" : for src.actor make tgt.actor
  "Procedure.performer-role" : for src.role make tgt.role
endgroup

group prcFocalDevice extends BackboneElement
  input src : Procedure as source
  input tgt : ProcedureR2 as target

  "Procedure.focalDevice-action" : for src.action make tgt.action
  "Procedure.focalDevice-manipulated" : for src.manipulated make tgt.manipulated
endgroup
```

9.3.11.3 R3 Validation Errors

9.4 Resource FamilyMemberHistory - Content

URL: https://hl7.org/fhir/STU3/familymemberhistory.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Significant health events and conditions for a person related to the patient relevant in the context of care for the patient.

9.4.1 Scope and Usage

FamilyMemberHistory is one of the event resources in the FHIR workflow specification.

This resource records significant health events and conditions for a particular individual related to the subject. This information can be known to different levels of accuracy. Sometimes the exact condition ('asthma') is known, and sometimes

it is less precise ('some sort of cancer'). Equally, sometimes the person can be identified ('my aunt Agatha') and sometimes all that is known is that the person was an uncle.

This resource represents a simple structure used to capture an 'elementary' family history for a particular family member. However, it can also be the basis for capturing a more rigorous history useful for genetic and other analysis - refer to the Genetic Pedigree profile for an example.

The entire family history for an individual can be represented by combining references to FamilyMemberHistory instances into a List resource instance.

This resource is referenced by AdverseEvent and ClinicalImpression

9.4.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | I | | DomainResource | Information about patient's relatives, relevant for patient + Can only have estimatedAge if age[x] is present + Not Done Reason can only be specified if NotDone is "true" + Can have age[x] or born[x], but not both Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Id(s) for this record
definition | ? | 0..* | Reference(PlanDefinition | Questionnaire) | Instantiates protocol or definition
status | ?!? | 1..1 | code | partial | completed | entered-in-error | health-unknown FamilyHistoryStatus (Required)
notDone | ?!? | 0..1 | boolean | The taking of a family member's history did not occur
notDoneReason | ?I | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred FamilyHistoryNotDoneReason (Example)
patient | ? | 1..1 | Reference(Patient) | Patient history is about
date | ? | 0..1 | dateTime | When history was captured/updated
name | ? | 0..1 | string | The family member described
relationship | ? | 1..1 | CodeableConcept | Relationship to the subject FamilyMember (Example)
gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
born[x] | I | 0..1 | | (approximate) date of birth
bornPeriod | | | Period
bornDate | | | date
bornString | | | string
age[x] | ?I | 0..1 | | (approximate) age
ageAge | | | Age
ageRange | | | Range
ageString | | | string
estimatedAge | ?!?I | 0..1 | boolean | Age is estimated?
deceased[x] | ? | 0..1 | | Dead? How old/when?
deceasedBoolean | | | boolean
deceasedAge | | | Age
deceasedRange | | | Range
deceasedDate | | | date
deceasedString | | | string
reasonCode | ? | 0..* | CodeableConcept | Why was family member history performed? SNOMED CT Clinical Findings (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse) | Why was family member history performed?
note | | 0..* | Annotation | General note about related person
condition | | 0..* | BackboneElement | Condition that the related person had
code | | 1..1 | CodeableConcept | Condition suffered by relation Condition/Problem/Diagnosis Codes (Example)
outcome | | 0..1 | CodeableConcept | deceased | permanent disability | etc. Condition Outcome Codes (Example)
onset[x] | | 0..1 | | When condition first manifested
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | | 0..* | Annotation | Extra information about condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<FamilyMemberHistory xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Id(s) for this record --></identifier>
```

```
 <definition><!-- 0..* Reference(PlanDefinition|Questionnaire) Instantiates protocol or definition --></definition>
 <status value="[code]"/><!-- 1..1 partial | completed | entered-in-error | health-unknown -->
 <notDone value="[boolean]"/><!-- 0..1 The taking of a family member's history did not occur -->
 <notDoneReason><!--  0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred --></notDoneReason>
 <patient><!-- 1..1 Reference(Patient) Patient history is about --></patient>
 <date value="[dateTime]"/><!-- 0..1 When history was captured/updated -->
 <name value="[string]"/><!-- 0..1 The family member described -->
 <relationship><!-- 1..1 CodeableConcept Relationship to the subject --></relationship>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <born[x]><!--  0..1 Period|date|string (approximate) date of birth --></born[x]>
 <age[x]><!--  0..1 Age|Range|string (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!--  0..1 Age is estimated? -->
 <deceased[x]><!-- 0..1 boolean|Age|Range|date|string Dead? How old/when? --></deceased[x]>
 <reasonCode><!-- 0..* CodeableConcept Why was family member history performed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse) Why was family member history performed? --></reasonReference>
 <note><!-- 0..* Annotation General note about related person --></note>
 <condition>  <!-- 0..* Condition that the related person had -->
  <code><!-- 1..1 CodeableConcept Condition suffered by relation --></code>
  <outcome><!-- 0..1 CodeableConcept deceased | permanent disability | etc. --></outcome>
  <onset[x]><!-- 0..1 Age|Range|Period|string When condition first manifested --></onset[x]>
  <note><!-- 0..* Annotation Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

JSON Template

```
{
  "resourceType" : "FamilyMemberHistory",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Id(s) for this record
  "definition" : [{ Reference(PlanDefinition|Questionnaire) }], // Instantiates protocol or definition
  "status" : "<code>", // R!  partial | completed | entered-in-error | health-unknown
  "notDone" : <boolean>, // The taking of a family member's history did not occur
  "notDoneReason" : { CodeableConcept }, // C? subject-unknown | withheld | unable-to-obtain | deferred
  "patient" : { Reference(Patient) }, // R!  Patient history is about
  "date" : "<dateTime>", // When history was captured/updated
  "name" : "<string>", // The family member described
  "relationship" : { CodeableConcept }, // R!  Relationship to the subject
  "gender" : "<code>", // male | female | other | unknown
  // born[x]: (approximate) date of birth. One of these 3:
  "bornPeriod" : { Period },
  "bornDate" : "<date>",
  "bornString" : "<string>",
  // age[x]: (approximate) age. One of these 3:
  "ageAge" : { Age },
  "ageRange" : { Range },
  "ageString" : "<string>",
  "estimatedAge" : <boolean>, // C? Age is estimated?
  // deceased[x]: Dead? How old/when?. One of these 5:
  "deceasedBoolean" : <boolean>,
  "deceasedAge" : { Age },
  "deceasedRange" : { Range },
  "deceasedDate" : "<date>",
  "deceasedString" : "<string>",
  "reasonCode" : [{ CodeableConcept }], // Why was family member history performed?
  "reasonReference" : [{ Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse) }], // Why was family member history performed?
  "note" : [{ Annotation }], // General note about related person
  "condition" : [{ // Condition that the related person had
    "code" : { CodeableConcept }, // R!  Condition suffered by relation
    "outcome" : { CodeableConcept }, // deceased | permanent disability | etc.
    // onset[x]: When condition first manifested. One of these 4:
    "onsetAge" : { Age },
    "onsetRange" : { Range },
    "onsetPeriod" : { Period },
    "onsetString" : "<string>",
    "note" : [{ Annotation }] // Extra information about condition
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:FamilyMemberHistory;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:FamilyMemberHistory.identifier [ Identifier ], ... ; # 0..* External Id(s) for this record
  fhir:FamilyMemberHistory.definition [ Reference(PlanDefinition|Questionnaire) ], ... ; # 0..* Instantiates protocol or
     definition
  fhir:FamilyMemberHistory.status [ code ]; # 1..1 partial | completed | entered-in-error | health-unknown
  fhir:FamilyMemberHistory.notDone [ boolean ]; # 0..1 The taking of a family member's history did not occur
  fhir:FamilyMemberHistory.notDoneReason [ CodeableConcept ]; # 0..1 subject-unknown | withheld | unable-to-obtain |
    deferred
  fhir:FamilyMemberHistory.patient [ Reference(Patient) ]; # 1..1 Patient history is about
  fhir:FamilyMemberHistory.date [ dateTime ]; # 0..1 When history was captured/updated
```

```
  fhir:FamilyMemberHistory.name [ string ]; # 0..1 The family member described
  fhir:FamilyMemberHistory.relationship [ CodeableConcept ]; # 1..1 Relationship to the subject
  fhir:FamilyMemberHistory.gender [ code ]; # 0..1 male | female | other | unknown
  # FamilyMemberHistory.born[x] : 0..1 (approximate) date of birth. One of these 3
    fhir:FamilyMemberHistory.bornPeriod [ Period ]
    fhir:FamilyMemberHistory.bornDate [ date ]
    fhir:FamilyMemberHistory.bornString [ string ]
  # FamilyMemberHistory.age[x] : 0..1 (approximate) age. One of these 3
    fhir:FamilyMemberHistory.ageAge [ Age ]
    fhir:FamilyMemberHistory.ageRange [ Range ]
    fhir:FamilyMemberHistory.ageString [ string ]
  fhir:FamilyMemberHistory.estimatedAge [ boolean ]; # 0..1 Age is estimated?
  # FamilyMemberHistory.deceased[x] : 0..1 Dead? How old/when?. One of these 5
    fhir:FamilyMemberHistory.deceasedBoolean [ boolean ]
    fhir:FamilyMemberHistory.deceasedAge [ Age ]
    fhir:FamilyMemberHistory.deceasedRange [ Range ]
    fhir:FamilyMemberHistory.deceasedDate [ date ]
    fhir:FamilyMemberHistory.deceasedString [ string ]
  fhir:FamilyMemberHistory.reasonCode [ CodeableConcept ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.reasonReference [ Reference(Condition|Observation|AllergyIntolerance|QuestionnaireResponse)
    ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.note [ Annotation ], ... ; # 0..* General note about related person
  fhir:FamilyMemberHistory.condition [ # 0..* Condition that the related person had
    fhir:FamilyMemberHistory.condition.code [ CodeableConcept ]; # 1..1 Condition suffered by relation
    fhir:FamilyMemberHistory.condition.outcome [ CodeableConcept ]; # 0..1 deceased | permanent disability | etc.
    # FamilyMemberHistory.condition.onset[x] : 0..1 When condition first manifested. One of these 4
      fhir:FamilyMemberHistory.condition.onsetAge [ Age ]
      fhir:FamilyMemberHistory.condition.onsetRange [ Range ]
      fhir:FamilyMemberHistory.condition.onsetPeriod [ Period ]
      fhir:FamilyMemberHistory.condition.onsetString [ string ]
    fhir:FamilyMemberHistory.condition.note [ Annotation ], ... ; # 0..* Extra information about condition
  ], ...;
]
```

Changes since DSTU2

FamilyMemberHistory
FamilyMemberHistory.definition | Added Element
FamilyMemberHistory.notDone | Added Element
FamilyMemberHistory.notDoneReason | Added Element
FamilyMemberHistory.age[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.estimatedAge | Added Element
FamilyMemberHistory.deceased[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.reasonCode | Added Element
FamilyMemberHistory.reasonReference | Added Element
FamilyMemberHistory.note | Max Cardinality changed from 1 to *
FamilyMemberHistory.condition.onset[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.condition.note | Max Cardinality changed from 1 to *

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | I | | DomainResource | Information about patient's relatives, relevant for patient + Can only have estimatedAge if age[x] is present + Not Done Reason can only be specified if NotDone is "true" + Can have age[x] or born[x], but not both Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Id(s) for this record
definition | ? | 0..* | Reference(PlanDefinition | Questionnaire) | Instantiates protocol or definition
status | ?!? | 1..1 | code | partial | completed | entered-in-error | health-unknown FamilyHistoryStatus (Required)
notDone | ?!? | 0..1 | boolean | The taking of a family member's history did not occur
notDoneReason | ?I | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred FamilyHistoryNotDoneReason (Example)
patient | ? | 1..1 | Reference(Patient) | Patient history is about
date | ? | 0..1 | dateTime | When history was captured/updated
name | ? | 0..1 | string | The family member described
relationship | ? | 1..1 | CodeableConcept | Relationship to the subject FamilyMember (Example)
gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
born[x] | I | 0..1 | | (approximate) date of birth
bornPeriod | | | Period
bornDate | | | date
bornString | | | string
age[x] | ?I | 0..1 | | (approximate) age
ageAge | | | Age
ageRange | | | Range
ageString | | | string
estimatedAge | ?!?I | 0..1 | boolean | Age is estimated?
deceased[x] | ? | 0..1 | | Dead? How old/when?
deceasedBoolean | | | boolean

deceasedAge | | | Age
deceasedRange | | | Range
deceasedDate | | | date
deceasedString | | | string
reasonCode | ? | 0..* | CodeableConcept | Why was family member history performed? SNOMED CT Clinical Findings (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse) | Why was family member history performed?
note | | 0..* | Annotation | General note about related person
condition | | 0..* | BackboneElement | Condition that the related person had
code | | 1..1 | CodeableConcept | Condition suffered by relation Condition/Problem/Diagnosis Codes (Example)
outcome | | 0..1 | CodeableConcept | deceased | permanent disability | etc. Condition Outcome Codes (Example)
onset[x] | | 0..1 | | When condition first manifested
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | | 0..* | Annotation | Extra information about condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<FamilyMemberHistory xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Id(s) for this record --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|Questionnaire) Instantiates protocol or definition --></definition>
 <status value="[code]"/><!-- 1..1 partial | completed | entered-in-error | health-unknown -->
 <notDone value="[boolean]"/><!-- 0..1 The taking of a family member's history did not occur -->
 <notDoneReason><!--  0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred --></notDoneReason>
 <patient><!-- 1..1 Reference(Patient) Patient history is about --></patient>
 <date value="[dateTime]"/><!-- 0..1 When history was captured/updated -->
 <name value="[string]"/><!-- 0..1 The family member described -->
 <relationship><!-- 1..1 CodeableConcept Relationship to the subject --></relationship>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <born[x]><!--  0..1 Period|date|string (approximate) date of birth --></born[x]>
 <age[x]><!--  0..1 Age|Range|string (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!--  0..1 Age is estimated? -->
 <deceased[x]><!-- 0..1 boolean|Age|Range|date|string Dead? How old/when? --></deceased[x]>
 <reasonCode><!-- 0..* CodeableConcept Why was family member history performed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse) Why was family member history performed? --></reasonReference>
 <note><!-- 0..* Annotation General note about related person --></note>
 <condition>  <!-- 0..* Condition that the related person had -->
  <code><!-- 1..1 CodeableConcept Condition suffered by relation --></code>
  <outcome><!-- 0..1 CodeableConcept deceased | permanent disability | etc. --></outcome>
  <onset[x]><!-- 0..1 Age|Range|Period|string When condition first manifested --></onset[x]>
  <note><!-- 0..* Annotation Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

JSON Template

```
{
  "resourceType" : "FamilyMemberHistory",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Id(s) for this record
  "definition" : [{ Reference(PlanDefinition|Questionnaire) }], // Instantiates protocol or definition
  "status" : "<code>", // R!  partial | completed | entered-in-error | health-unknown
  "notDone" : <boolean>, // The taking of a family member's history did not occur
  "notDoneReason" : { CodeableConcept }, // C? subject-unknown | withheld | unable-to-obtain | deferred
  "patient" : { Reference(Patient) }, // R!  Patient history is about
  "date" : "<dateTime>", // When history was captured/updated
  "name" : "<string>", // The family member described
  "relationship" : { CodeableConcept }, // R!  Relationship to the subject
  "gender" : "<code>", // male | female | other | unknown
  // born[x]: (approximate) date of birth. One of these 3:
  "bornPeriod" : { Period },
  "bornDate" : "<date>",
  "bornString" : "<string>",
  // age[x]: (approximate) age. One of these 3:
  "ageAge" : { Age },
  "ageRange" : { Range },
  "ageString" : "<string>",
  "estimatedAge" : <boolean>, // C? Age is estimated?
  // deceased[x]: Dead? How old/when?. One of these 5:
  "deceasedBoolean" : <boolean>,
  "deceasedAge" : { Age },
  "deceasedRange" : { Range },
  "deceasedDate" : "<date>",
  "deceasedString" : "<string>",
  "reasonCode" : [{ CodeableConcept }], // Why was family member history performed?
  "reasonReference" : [{ Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse) }], // Why was family member history performed?
  "note" : [{ Annotation }], // General note about related person
```

```
  "condition" : [{ // Condition that the related person had
    "code" : { CodeableConcept }, // R!  Condition suffered by relation
    "outcome" : { CodeableConcept }, // deceased | permanent disability | etc.
    // onset[x]: When condition first manifested. One of these 4:
    "onsetAge" : { Age },
    "onsetRange" : { Range },
    "onsetPeriod" : { Period },
    "onsetString" : "<string>",
    "note" : [{ Annotation }] // Extra information about condition
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:FamilyMemberHistory;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:FamilyMemberHistory.identifier [ Identifier ], ... ; # 0..* External Id(s) for this record
  fhir:FamilyMemberHistory.definition [ Reference(PlanDefinition|Questionnaire) ], ... ; # 0..* Instantiates protocol or
     definition
  fhir:FamilyMemberHistory.status [ code ]; # 1..1 partial | completed | entered-in-error | health-unknown
  fhir:FamilyMemberHistory.notDone [ boolean ]; # 0..1 The taking of a family member's history did not occur
  fhir:FamilyMemberHistory.notDoneReason [ CodeableConcept ]; # 0..1 subject-unknown | withheld | unable-to-obtain |
    deferred
  fhir:FamilyMemberHistory.patient [ Reference(Patient) ]; # 1..1 Patient history is about
  fhir:FamilyMemberHistory.date [ dateTime ]; # 0..1 When history was captured/updated
  fhir:FamilyMemberHistory.name [ string ]; # 0..1 The family member described
  fhir:FamilyMemberHistory.relationship [ CodeableConcept ]; # 1..1 Relationship to the subject
  fhir:FamilyMemberHistory.gender [ code ]; # 0..1 male | female | other | unknown
  # FamilyMemberHistory.born[x] : 0..1 (approximate) date of birth. One of these 3
    fhir:FamilyMemberHistory.bornPeriod [ Period ]
    fhir:FamilyMemberHistory.bornDate [ date ]
    fhir:FamilyMemberHistory.bornString [ string ]
  # FamilyMemberHistory.age[x] : 0..1 (approximate) age. One of these 3
    fhir:FamilyMemberHistory.ageAge [ Age ]
    fhir:FamilyMemberHistory.ageRange [ Range ]
    fhir:FamilyMemberHistory.ageString [ string ]
  fhir:FamilyMemberHistory.estimatedAge [ boolean ]; # 0..1 Age is estimated?
  # FamilyMemberHistory.deceased[x] : 0..1 Dead? How old/when?. One of these 5
    fhir:FamilyMemberHistory.deceasedBoolean [ boolean ]
    fhir:FamilyMemberHistory.deceasedAge [ Age ]
    fhir:FamilyMemberHistory.deceasedRange [ Range ]
    fhir:FamilyMemberHistory.deceasedDate [ date ]
    fhir:FamilyMemberHistory.deceasedString [ string ]
  fhir:FamilyMemberHistory.reasonCode [ CodeableConcept ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.reasonReference [ Reference(Condition|Observation|AllergyIntolerance|QuestionnaireResponse)
    ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.note [ Annotation ], ... ; # 0..* General note about related person
  fhir:FamilyMemberHistory.condition [ # 0..* Condition that the related person had
    fhir:FamilyMemberHistory.condition.code [ CodeableConcept ]; # 1..1 Condition suffered by relation
    fhir:FamilyMemberHistory.condition.outcome [ CodeableConcept ]; # 0..1 deceased | permanent disability | etc.
    # FamilyMemberHistory.condition.onset[x] : 0..1 When condition first manifested. One of these 4
      fhir:FamilyMemberHistory.condition.onsetAge [ Age ]
      fhir:FamilyMemberHistory.condition.onsetRange [ Range ]
      fhir:FamilyMemberHistory.condition.onsetPeriod [ Period ]
      fhir:FamilyMemberHistory.condition.onsetString [ string ]
    fhir:FamilyMemberHistory.condition.note [ Annotation ], ... ; # 0..* Extra information about condition
  ], ...;
]
```

Changes since DSTU2

FamilyMemberHistory
FamilyMemberHistory.definition | Added Element
FamilyMemberHistory.notDone | Added Element
FamilyMemberHistory.notDoneReason | Added Element
FamilyMemberHistory.age[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.estimatedAge | Added Element
FamilyMemberHistory.deceased[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.reasonCode | Added Element
FamilyMemberHistory.reasonReference | Added Element
FamilyMemberHistory.note | Max Cardinality changed from 1 to *
FamilyMemberHistory.condition.onset[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.condition.note | Max Cardinality changed from 1 to *

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.4.2.1 Terminology Bindings

Path | Definition | Type | Reference
FamilyMemberHistory.status | A code that identifies the status of the family history record. | Required | FamilyHistoryStatus
FamilyMemberHistory.notDoneReason | Codes describing the reason why a family member history was not done. | Example | FamilyHistoryNotDoneReason
FamilyMemberHistory.relationship | The nature of the relationship between the patient and the related person being described in the family member history. | Example | FamilyMember
FamilyMemberHistory.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender
FamilyMemberHistory.reasonCode | Codes indicating why the family member history was done. | Example | SNOMED CT Clinical Findings
FamilyMemberHistory.condition.code | Identification of the Condition or diagnosis. | Example | Condition/Problem/Diagnosis Codes
FamilyMemberHistory.condition.outcome | The result of the condition for the patient; e.g. death, permanent disability, temporary disability, etc. | Example | Condition Outcome Codes

9.4.2.2 Constraints

* fhs-1: Can have age[x] or born[x], but not both (expression : age.empty() or born.empty())

* fhs-2: Can only have estimatedAge if age[x] is present (expression : age.exists() or estimatedAge.empty())

* fhs-3: Not Done Reason can only be specified if NotDone is "true" (expression : notDone or notDoneReason.exists().not())

9.4.2.3 Processing information about the Family Member History

The Family Member History List may contain other than FamilyMemberHistory resources. For example, a full Family History could be a List that might include a mixture of FamilyMemberHistory records as well as Observation records of things like "maternal family history of breast cancer", "number of siblings", "number of female family members with breast cancer" etc.

The List representing a patient's "family history" can include Condition and Observation records that capture "family-history" relevant assertions about the patient themselves that would typically be captured as part of a family history.

9.4.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | A search by a condition code | FamilyMemberHistory.condition.code | 8 Resources
date | date | When history was captured/updated | FamilyMemberHistory.date | 18 Resources
definition | reference | Instantiates protocol or definition | FamilyMemberHistory.definition (Questionnaire, PlanDefinition)
gender | token | A search by a gender code of a family member | FamilyMemberHistory.gender
identifier | token | A search by a record identifier | FamilyMemberHistory.identifier | 26 Resources
patient | reference | The identity of a subject to list family member history items for | FamilyMemberHistory.patient (Patient) | 31 Resources
relationship | token | A search by a relationship type | FamilyMemberHistory.relationship
status | token | partial | completed | entered-in-error | health-unknown | FamilyMemberHistory.status

9.4.4 Resource FamilyMemberHistory - Examples

URL: https://hl7.org/fhir/STU3/familymemberhistory-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.4 Resource FamilyMemberHistory - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Examples of the Family History, using the List resource:

* Real-world patient example

* Simple genetic family member history

* Example for risk assessment (The source picture can be checked as a binary file in the source xml or json file)

* Example for double cousin relationship

Example List:

Basic Example. Describes the fathers death at age 74 from a heart attack | father | XML | JSON | Turtle
Mother died from a stroke aged 56. Brother with diabetes. | mother | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.4.5 Resource FamilyMemberHistory - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/familymemberhistory-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.5 Resource FamilyMemberHistory - Detailed Descriptions

Detailed Descriptions for the elements in the FamilyMemberHistory resource.

FamilyMemberHistory

Definition | Significant health events and conditions for a person related to the patient relevant in the context of care for the patient.

Control | 1..1

Invariants | Defined on this element fhs-1: Can have age[x] or born[x], but not both (expression : age.empty() or born.empty(), xpath: not (*[starts-with(local-name(.), 'age')] and *[starts-with(local-name(.), 'birth')])) fhs-2: Can only have estimatedAge if age[x] is present (expression : age.exists() or estimatedAge.empty(), xpath: exists(*[starts-with(local-name(.), 'age')]) or not(exists(f:estimatedAge))) fhs-3: Not Done Reason can only be specified if NotDone is "true" (expression : notDone or notDoneReason.exists().not(), xpath: f:notDone/@value=true() or not(exists(f:notDoneReason)))

FamilyMemberHistory.identifier

Definition | This records identifiers associated with this family member history record that are defined by business processes and/ or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).

Note | This is a business identifer, not a resource identifier (see discussion)

Control | 0..*

Type | Identifier

Requirements | Need to allow connection to a wider workflow.

Summary | true

FamilyMemberHistory.definition

Definition | A protocol or questionnaire that was adhered to in whole or in part by this event.

Control | 0..*

Type | Reference(PlanDefinition | Questionnaire)

Summary | true

FamilyMemberHistory.status

Definition | A code specifying the status of the record of the family history of a specific family member.

Control | 1..1

Terminology Binding | FamilyHistoryStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.

FamilyMemberHistory.notDone

Definition | If true, indicates the taking of an individual family member's history did not occur. The notDone element should not be used to document negated conditions, such as a family member that did not have a condition.

Control | 0..1

Type | boolean

Is Modifier | true

Default Value | false

Requirements | Creating a FamilyMemberHistory where notDone is true is intended for situations where there's a need for a specific statement in the record about something not being done. If the need is merely to indicate that a request wasn't fulfilled, that should be handled using Task.

Summary | true

Comments | This element is labeled as a modifier because it marks the family member history as a family member history that did not occur. The more attributes are populated, the more constrained the negated statement is. This notDone element is being evaluated and will likely be removed in a subsequent release.

FamilyMemberHistory.notDoneReason

Definition | Describes why the family member's history is absent.

Control | 0..1

Terminology Binding | FamilyHistoryNotDoneReason (Example)

Type | CodeableConcept

Requirements | This is a separate element to allow it to have a distinct binding from reasonCode.

Summary | true

Comments | This notDoneReason element is being evaluated and will likely be replaced in a subsequent release (e.g. dataAbsentReason).

Invariants | Affect this element fhs-3: Not Done Reason can only be specified if NotDone is "true" (expression : notDone or notDoneReason.exists().not(), xpath: f:notDone/@value=true() or not(exists(f:notDoneReason)))

FamilyMemberHistory.patient

Definition | The person who this history concerns.

Control | 1..1

Type | Reference(Patient)

Alternate Names | Proband

Summary | true

FamilyMemberHistory.date

Definition | The date (and possibly time) when the family member history was taken.

Control | 0..1

Type | dateTime

Requirements | Allows determination of how current the summary is.

Summary | true

Comments | This should be captured even if the same as the date on the List aggregating the full family history.

FamilyMemberHistory.name

Definition | This will either be a name or a description; e.g. "Aunt Susan", "my cousin with the red hair".

Control | 0..1

Type | string

Requirements | Allows greater ease in ensuring the same person is being talked about.

Summary | true
FamilyMemberHistory.relationship
Definition | The type of relationship this person has to the patient (father, mother, brother etc.).
Control | 1..1
Terminology Binding | FamilyMember (Example)
Type | CodeableConcept
Summary | true
FamilyMemberHistory.gender
Definition | Administrative Gender - the gender that the relative is considered to have for administration and record keeping purposes.
Control | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Not all relationship codes imply gender and the relative's gender can be relevant for risk assessments.
Summary | true
FamilyMemberHistory.born[x]
Definition | The actual or approximate date of birth of the relative.
Control | 0..1
Type | Period|date|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Allows calculation of the relative's age.
Invariants | Affect this element fhs-1: Can have age[x] or born[x], but not both (expression : age.empty() or born.empty(), xpath: not (*[starts-with(local-name(.), 'age')] and *[starts-with(local-name(.), 'birth')]))
FamilyMemberHistory.age[x]
Definition | The age of the relative at the time the family member history is recorded.
Control | 0..1
Type | Age|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | While age can be calculated from date of birth, sometimes recording age directly is more natureal for clinicians.
Summary | true
Comments | use estimatedAge to indicate whether the age is actual or not.
Invariants | Affect this element fhs-1: Can have age[x] or born[x], but not both (expression : age.empty() or born.empty(), xpath: not (*[starts-with(local-name(.), 'age')] and *[starts-with(local-name(.), 'birth')]))
FamilyMemberHistory.estimatedAge
Definition | If true, indicates that the age value specified is an estimated value.
Control | 0..1
Type | boolean
Is Modifier | true
Meaning if Missing | It is unknown whether the age is an estimate or not
Requirements | Clinicians often prefer to specify an estimaged age rather than an age range.
Summary | true
Comments | This element is labeled as a modifier because the fact tha age is estimated can/should change the results of any algorithm that calculates based on the specified age.
Invariants | Affect this element fhs-2: Can only have estimatedAge if age[x] is present (expression : age.exists() or estimatedAge.empty(), xpath: exists(*[starts-with(local-name(.), 'age')]) or not(exists(f:estimatedAge)))
FamilyMemberHistory.deceased[x]
Definition | Deceased flag or the actual or approximate age of the relative at the time of death for the family member history record.
Control | 0..1
Type | boolean|Age|Range|date|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
FamilyMemberHistory.reasonCode
Definition | Describes why the family member history occurred in coded or textual form.
Control | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Summary | true
Comments | Textual reasons can be caprued using reasonCode.text.
FamilyMemberHistory.reasonReference
Definition | Indicates a Condition, Observation, AllergyIntolerance, or QuestionnaireResponse that justifies this family member history event.
Control | 0..*
Type | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse)
Summary | true
FamilyMemberHistory.note
Definition | This property allows a non condition-specific note to the made about the related person. Ideally, the note would be in the condition property, but this is not always possible.
Control | 0..*
Type | Annotation
FamilyMemberHistory.condition
Definition | The significant Conditions (or condition) that the family member had. This is a repeating section to allow a system to represent more than one condition per resource, though there is nothing stopping multiple resources - one per condition.
Control | 0..*
Comments | If none of the conditions listed have an outcome of "death" specified, that indicates that none of the specified conditions are known to have been the primary cause of death.

FamilyMemberHistory.condition.code
Definition | The actual condition specified. Could be a coded condition (like MI or Diabetes) or a less specific string like 'cancer' depending on how much is known about the condition and the capabilities of the creating system.
Control | 1..1
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
FamilyMemberHistory.condition.outcome
Definition | Indicates what happened as a result of this condition. If the condition resulted in death, deceased date is captured on the relation.
Control | 0..1
Terminology Binding | Condition Outcome Codes (Example)
Type | CodeableConcept
FamilyMemberHistory.condition.onset[x]
Definition | Either the age of onset, range of approximate age or descriptive string can be recorded. For conditions with multiple occurrences, this describes the first known occurrence.
Control | 0..1
Type | Age|Range|Period|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Age of onset of a condition in relatives is predictive of risk for the patient.
FamilyMemberHistory.condition.note
Definition | An area where general notes can be placed about this specific condition.
Control | 0..*
Type | Annotation

9.4.6 Resource FamilyMemberHistory - Mappings

URL: https://hl7.org/fhir/STU3/familymemberhistory-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.6 Resource FamilyMemberHistory - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the familymemberhistory resource.

9.4.6.1 Mappings for RIM Mapping (http://hl7.org/v3)

FamilyMemberHistory | Observation[classCode=OBS, moodCode=EVN]
identifier | id
definition | .outboundRelationship[typeCode=DEFN].target
status
notDone | .actionNegationInd
notDoneReason | .reasonCode
patient | participation[typeCode=SBJ].role
date | participation[typeCode=AUT].time
name | name
relationship | code
gender
born[x] | player[classCode=LIV, determinerCode=INSTANCE]. birthDate (could be URG)
age[x] | participation[typeCode=SBJ].act[classCode=OBS,moodCode=EVN, code="age"].value
estimatedAge
deceased[x] | player[classCode=LIV, determinerCode=INSTANCE].deceasedInd, deceasedDate (could be URG) For age, you'd hang an observation off the role
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
note | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=ActCode#ANNGEN].value
condition | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=ActCode#ASSERTION, value<Diagnosis]
code | .value
outcome | outboundRelationship[typeCode=OUTC)].target[classCode=OBS, moodCode=EVN, code=ActCode#ASSERTION].value
onset[x] | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="Subject Age at measurement", value<Diagnosis].value[@xsi:typeCode='TS' or 'IVL_TS'] Use originalText for string
note | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=ActCode#ANNGEN].value

9.4.6.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

FamilyMemberHistory | Not in scope for v2
identifier
definition
status
notDone
notDoneReason

patient
date
name
relationship
gender
born[x]
age[x]
estimatedAge
deceased[x]
reasonCode
reasonReference
note
condition
code
outcome
onset[x]
note

9.4.7 Resource FamilyMemberHistory - Design Notes

URL: https://hl7.org/fhir/STU3/familymemberhistory-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.7 Resource FamilyMemberHistory - Design Notes

to do

9.4.8 Resource FamilyMemberHistory - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/familymemberhistory-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.8 Resource FamilyMemberHistory - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the FamilyMemberHistory resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.4.8.1 Profiles

FamilyMemberHistory-Genetic | Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients | for Family member history for genetics analysis

9.4.8.2 Extensions

family-member-history-genetics-observation | observation | for Family member history for genetics analysis
family-member-history-genetics-parent | parent | for Family member history for genetics analysis
family-member-history-genetics-sibling | sibling | for Family member history for genetics analysis
familymemberhistory-abatement | abatement | for Family Member History HL7 Extensions
familymemberhistory-patient-record | patient-record | for Family Member History HL7 Extensions
familymemberhistory-severity | severity | for Family Member History HL7 Extensions
familymemberhistory-type | type | for Family Member History HL7 Extensions

Extensions that reference this resource:

DiagnosticReport-geneticsFamilyMemberHistory | FamilyMemberHistory | for DiagnosticReport-Genetics
family-member-history-genetics-parent | parent | for Family member history for genetics analysis
family-member-history-genetics-sibling | sibling | for Family member history for genetics analysis
procedure-causedBy | causedBy | for Procedure HL7 Extensions

9.4.8.3 Search Extensions

No Search Extensions defined for this resource

9.4.9 Family member history for genetics analysis

URL: https://hl7.org/fhir/STU3/family-member-history-genetics.html

Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.9 Family member history for genetics analysis

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.4.9.1 Content

Profiles:

FamilyMemberHistory-Genetic | Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients

Extensions:

family-member-history-genetics-parent | parent : Identifies a parent of the relative.

family-member-history-genetics-sibling | sibling : Identifies a sibling of the relative.

family-member-history-genetics-observation | observation : Allows capturing risk-relevant observations about the relative that aren't themselves a specific health condition; e.g. Certain ethnic ancestries that are disease-relevant, presence of particular genetic markers, etc.

9.4.9.2 StructureDefinition: FamilyMemberHistory-Genetic

URL: https://hl7.org/fhir/STU3/familymemberhistory-genetic.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.9.2 StructureDefinition: FamilyMemberHistory-Genetic

(No assigned work group) | Maturity Level: 1 | Informative | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/familymemberhistory-genetic`

Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients

This profile was published on Wed, Oct 2, 2013 00:00+1000 as a draft by Health Level Seven International (Clinical Genomics).

9.4.9.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

* All

This structure is derived from FamilyMemberHistory.

Summary

Mandatory: 1 element (+1 nested mandatory element) Must-Support: 12 elements

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent

* http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling

* http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation

This structure is derived from FamilyMemberHistory.

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | | 0..*
relationship | S | 1..1 | CodeableConcept
gender | S | 0..1 | code | Binding: AdministrativeGender (required)
born[x] | S | 0..1 | Period, date, string
age[x] | S | 0..1 | Age, Range, string
deceased[x] | S | 0..1 | boolean, Age, Range, date, string
condition | S | 0..* | BackboneElement
code | S | 1..1 | CodeableConcept
outcome | S | 0..1 | CodeableConcept
onset[x] | S | 0..1 | Age, Range, Period, string
note | | 0..1 | Annotation
family-member-history-genetics-parent | S | 0..* | (Complex) | Mother(s) & Father(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling | S | 0..* | (Complex) | natural brother(s) & natural sister(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation | S | 0..* | Reference(Observation) | Genetic markers, ethnicity, etc. URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | I | 0..* | | Information about patient's relatives, relevant for patient
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
family-member-history-genetics-parent | SI | 0..* | (Complex) | Extension URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling | SI | 0..* | (Complex) | Extension URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation | SI | 0..* | Reference(Observation) | Extension URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | External Id(s) for this record
definition | ? | 0..* | Reference(PlanDefinition), Reference(Questionnaire) | Instantiates protocol or definition
status | ?!? | 1..1 | code | partial | completed | entered-in-error | health-unknown Binding: FamilyHistoryStatus (required)
notDone | ?!? | 0..1 | boolean | The taking of a family member's history did not occur
notDoneReason | ?! | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred Binding: FamilyHistoryNotDoneReason (example)
patient | ? | 1..1 | Reference(Patient) | Patient history is about
date | ? | 0..1 | dateTime | When history was captured/updated
name | ? | 0..1 | string | The family member described
relationship | S? | 1..1 | CodeableConcept | Relationship to the subject Binding: FamilyMember (example)
gender | S? | 0..1 | code | male | female | other | unknown Binding: AdministrativeGender (required)
born[x] | SI | 0..1 | Period, date, string | (approximate) date of birth
age[x] | S?I | 0..1 | Age, Range, string | (approximate) age
estimatedAge | ?!?I | 0..1 | boolean | Age is estimated?
deceased[x] | S? | 0..1 | boolean, Age, Range, date, string | Dead? How old/when?
reasonCode | ? | 0..* | CodeableConcept | Why was family member history performed? Binding: SNOMED CT Clinical Findings (example)
reasonReference | ? | 0..* | Reference(Condition), Reference(Observation), Reference(AllergyIntolerance), Reference(QuestionnaireResponse) | Why was family member history performed?
note | | 0..* | Annotation | General note about related person
condition | SI | 0..* | BackboneElement | Condition that the related person had
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S | 1..1 | CodeableConcept | Condition suffered by relation Binding: Condition/Problem/Diagnosis Codes (example)
outcome | S | 0..1 | CodeableConcept | deceased | permanent disability | etc. Binding: Condition Outcome Codes (example)
onset[x] | S | 0..1 | Age, Range, Period, string | When condition first manifested
note | | 0..1 | Annotation | Extra information about condition

Documentation for this format

```
<!-- Family member history for genetics analysis -->

<FamilyMemberHistory xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
     External Id(s) for this record --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition)|Reference(Questionnaire)
     Instantiates protocol or definition --></definition>
 <status value="[code]"/><!-- 1..1 partial | completed | entered-in-error | health-unknown  -->
 <notDone value="[boolean]"/><!-- 0..1 The taking of a family member's history did not occur -->
 <notDoneReason><!-- 0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred  --></notDoneReason>
 <patient><!-- 1..1 Reference(Patient)
     Patient history is about --></patient>
 <date value="[dateTime]"/><!-- 0..1 When history was captured/updated -->
 <name value="[string]"/><!-- 0..1 The family member described -->
 <relationship><!-- 1..1 CodeableConcept Relationship to the subject  --></relationship>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown  -->
 <born[x]><!-- 0..1 Period|date|string
     (approximate) date of birth --></born[x]>
 <age[x]><!-- 0..1 Age|Range|string
     (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!-- 0..1 Age is estimated? -->
 <deceased[x]><!-- 0..1 boolean|Age|Range|date|string
     Dead? How old/when? --></deceased[x]>
 <reasonCode><!-- 0..* CodeableConcept Why was family member history performed?  --></reasonCode>
```

```
 <reasonReference><!-- 0..* Reference(Condition)|Reference(Observation)|Reference(AllergyIntolerance)|
   Reference(QuestionnaireResponse) Why was family member history performed? --></reasonReference>
 <note><!-- 0..* Annotation
     General note about related person --></note>
 <condition>  0..* BackboneElement  <!--  0..* Condition that the related person had -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Condition suffered by relation  --></code>
  <outcome><!-- 0..1 CodeableConcept deceased | permanent disability | etc.  --></outcome>
  <onset[x]><!-- 0..1 Age|Range|Period|string
      When condition first manifested --></onset[x]>
  <note><!-- 0..1 Annotation
      Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

```
{ // Family member history for genetics analysis
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // Extension // C?
        // from Element: extension
      },
      { // Extension // C?
        // from Element: extension
      },
      { // Extension // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //External Id(s) for this record
// value[x]: Instantiates protocol or definition. One of these 2:
    "definition" : [{ Reference(PlanDefinition) }], //Instantiates protocol or definition
    "definition" : [{ Reference(Questionnaire) }], //Instantiates protocol or definition
    "status" : "<code>", // R! partial | completed | entered-in-error | health-unknown
    "notDone" : <boolean>, //The taking of a family member's history did not occur
    "notDoneReason" : { CodeableConcept }, //subject-unknown | withheld | unable-to-obtain | deferred
    "patient" : { Reference(Patient) }, // R! Patient history is about
    "date" : "<dateTime>", //When history was captured/updated
    "name" : "<string>", //The family member described
    "relationship" : { CodeableConcept }, // R! Relationship to the subject
    "gender" : "<code>", //male | female | other | unknown
// value[x]: (approximate) date of birth. One of these 3:
    "bornPeriod" : { Period }, //(approximate) date of birth
    "bornDate" : "<date>", //(approximate) date of birth
    "bornString" : "<string>", //(approximate) date of birth
// value[x]: (approximate) age. One of these 3:
    "ageAge" : { Age }, //(approximate) age
    "ageRange" : { Range }, //(approximate) age
    "ageString" : "<string>", //(approximate) age
    "estimatedAge" : <boolean>, //Age is estimated?
// value[x]: Dead? How old/when?. One of these 5:
    "deceasedBoolean" : <boolean>, //Dead? How old/when?
    "deceasedAge" : { Age }, //Dead? How old/when?
    "deceasedRange" : { Range } //Dead? How old/when?
    "deceasedDate" : "<date>", //Dead? How old/when?
    "deceasedString" : "<string>", //Dead? How old/when?
    "reasonCode" : [{ CodeableConcept }], //Why was family member history performed?
// value[x]: Why was family member history performed?. One of these 4:
    "reasonReference" : [{ Reference(Condition) }], //Why was family member history performed?
    "reasonReference" : [{ Reference(Observation) }], //Why was family member history performed?
    "reasonReference" : [{ Reference(AllergyIntolerance) }], //Why was family member history performed?
    "reasonReference" : [{ Reference(QuestionnaireResponse) }], //Why was family member history performed?
    "note" : [{ Annotation }], //General note about related person
    "condition" : [{ BackboneElement }], // C?Condition that the related person had
  }
```

This structure is derived from FamilyMemberHistory.

Summary

Mandatory: 1 element (+1 nested mandatory element) Must-Support: 12 elements

Extensions

This structure refers to these extensions:

* http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent

* http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling

* http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation

Differential View

This structure is derived from FamilyMemberHistory.

Name	Flags	Card.	Type	Description & Constraints
FamilyMemberHistory		0..*		

relationship | S | 1..1 | CodeableConcept
gender | S | 0..1 | code | Binding: AdministrativeGender (required)
born[x] | S | 0..1 | Period, date, string
age[x] | S | 0..1 | Age, Range, string
deceased[x] | S | 0..1 | boolean, Age, Range, date, string
condition | S | 0..* | BackboneElement
code | S | 1..1 | CodeableConcept
outcome | S | 0..1 | CodeableConcept
onset[x] | S | 0..1 | Age, Range, Period, string
note | | 0..1 | Annotation
family-member-history-genetics-parent | S | 0..* | (Complex) | Mother(s) & Father(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling | S | 0..* | (Complex) | natural brother(s) & natural sister(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation | S | 0..* | Reference(Observation) | Genetic markers, ethnicity, etc. URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | I | 0..* | | Information about patient's relatives, relevant for patient
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Binding: Common Languages (extensible)
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
family-member-history-genetics-parent | SI | 0..* | (Complex) | Extension URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling | SI | 0..* | (Complex) | Extension URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation | SI | 0..* | Reference(Observation) | Extension URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
identifier | ? | 0..* | Identifier | External Id(s) for this record
definition | ? | 0..* | Reference(PlanDefinition), Reference(Questionnaire) | Instantiates protocol or definition
status | ?!? | 1..1 | code | partial | completed | entered-in-error | health-unknown Binding: FamilyHistoryStatus (required)
notDone | ?!? | 0..1 | boolean | The taking of a family member's history did not occur
notDoneReason | ?I | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred Binding: FamilyHistoryNotDoneReason (example)
patient | ? | 1..1 | Reference(Patient) | Patient history is about
date | ? | 0..1 | dateTime | When history was captured/updated
name | ? | 0..1 | string | The family member described
relationship | S? | 1..1 | CodeableConcept | Relationship to the subject Binding: FamilyMember (example)
gender | S? | 0..1 | code | male | female | other | unknown Binding: AdministrativeGender (required)
born[x] | SI | 0..1 | Period, date, string | (approximate) date of birth
age[x] | S?I | 0..1 | Age, Range, string | (approximate) age
estimatedAge | ?!?I | 0..1 | boolean | Age is estimated?
deceased[x] | S? | 0..1 | boolean, Age, Range, date, string | Dead? How old/when?
reasonCode | ? | 0..* | CodeableConcept | Why was family member history performed? Binding: SNOMED CT Clinical Findings (example)
reasonReference | ? | 0..* | Reference(Condition), Reference(Observation), Reference(AllergyIntolerance), Reference(QuestionnaireResponse) | Why was family member history performed?
note | | 0..* | Annotation | General note about related person
condition | SI | 0..* | BackboneElement | Condition that the related person had
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored
code | S | 1..1 | CodeableConcept | Condition suffered by relation Binding: Condition/Problem/Diagnosis Codes (example)
outcome | S | 0..1 | CodeableConcept | deceased | permanent disability | etc. Binding: Condition Outcome Codes (example)
onset[x] | S | 0..1 | Age, Range, Period, string | When condition first manifested
note | | 0..1 | Annotation | Extra information about condition
Documentation for this format

XML Template

```
<!-- Family member history for genetics analysis -->

<FamilyMemberHistory xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content  -->
 <text><!-- 0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
```

```
    Contained, inline Resources --></contained>
 <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!-- 0..* Identifier
    External Id(s) for this record --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition)|Reference(Questionnaire)
    Instantiates protocol or definition --></definition>
 <status value="[code]"/><!-- 1..1 partial | completed | entered-in-error | health-unknown  -->
 <notDone value="[boolean]"/><!-- 0..1 The taking of a family member's history did not occur -->
 <notDoneReason><!-- 0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred  --></notDoneReason>
 <patient><!-- 1..1 Reference(Patient)
    Patient history is about --></patient>
 <date value="[dateTime]"/><!-- 0..1 When history was captured/updated -->
 <name value="[string]"/><!-- 0..1 The family member described -->
 <relationship><!-- 1..1 CodeableConcept Relationship to the subject  --></relationship>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown  -->
 <born[x]><!-- 0..1 Period|date|string
    (approximate) date of birth --></born[x]>
 <age[x]><!-- 0..1 Age|Range|string
    (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!-- 0..1 Age is estimated? -->
 <deceased[x]><!-- 0..1 boolean|Age|Range|date|string
    Dead? How old/when? --></deceased[x]>
 <reasonCode><!-- 0..* CodeableConcept Why was family member history performed?  --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition)|Reference(Observation)|Reference(AllergyIntolerance)|
   Reference(QuestionnaireResponse) Why was family member history performed? --></reasonReference>
 <note><!-- 0..* Annotation
    General note about related person --></note>
 <condition>  0..* BackboneElement  <!--  0..* Condition that the related person had -->
  <modifierExtension><!-- 0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
  <code><!-- 1..1 CodeableConcept Condition suffered by relation  --></code>
  <outcome><!-- 0..1 CodeableConcept deceased | permanent disability | etc.  --></outcome>
  <onset[x]><!-- 0..1 Age|Range|Period|string
     When condition first manifested --></onset[x]>
  <note><!-- 0..1 Annotation
     Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

JSON Template

```
{ // Family member history for genetics analysis
  // from Element: extension
    "meta" : { Meta }, //Metadata about the resource
    "implicitRules" : "<uri>", //A set of rules under which this content was created
    "language" : "<code>", //Language of the resource content
    "text" : { Narrative }, //Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified orderOpen
      { // Extension // C?
        // from Element: extension
      },
      { // Extension // C?
        // from Element: extension
      },
      { // Extension // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], //Extensions that cannot be ignored
    "identifier" : [{ Identifier }], //External Id(s) for this record
// value[x]: Instantiates protocol or definition. One of these 2:
    "definition" : [{ Reference(PlanDefinition) }], //Instantiates protocol or definition
    "definition" : [{ Reference(Questionnaire) }], //Instantiates protocol or definition
    "status" : "<code>", // R! partial | completed | entered-in-error | health-unknown
    "notDone" : <boolean>, //The taking of a family member's history did not occur
    "notDoneReason" : { CodeableConcept }, //subject-unknown | withheld | unable-to-obtain | deferred
    "patient" : { Reference(Patient) }, // R! Patient history is about
    "date" : "<dateTime>", //When history was captured/updated
    "name" : "<string>", //The family member described
    "relationship" : { CodeableConcept }, // R! Relationship to the subject
    "gender" : "<code>", //male | female | other | unknown
// value[x]: (approximate) date of birth. One of these 3:
    "bornPeriod" : { Period }, //(approximate) date of birth
    "bornDate" : "<date>", //(approximate) date of birth
    "bornString" : "<string>", //(approximate) date of birth
// value[x]: (approximate) age. One of these 3:
    "ageAge" : { Age }, //(approximate) age
    "ageRange" : { Range }, //(approximate) age
    "ageString" : "<string>", //(approximate) age
    "estimatedAge" : <boolean>, //Age is estimated?
// value[x]: Dead? How old/when?. One of these 5:
    "deceasedBoolean" : <boolean>, //Dead? How old/when?
    "deceasedAge" : { Age }, //Dead? How old/when?
    "deceasedRange" : { Range } //Dead? How old/when?
    "deceasedDate" : "<date>", //Dead? How old/when?
    "deceasedString" : "<string>", //Dead? How old/when?
    "reasonCode" : [{ CodeableConcept }], //Why was family member history performed?
// value[x]: Why was family member history performed?. One of these 4:
    "reasonReference" : [{ Reference(Condition) }], //Why was family member history performed?
    "reasonReference" : [{ Reference(Observation) }], //Why was family member history performed?
```

```
    "reasonReference" : [{ Reference(AllergyIntolerance) }], //Why was family member history performed?
    "reasonReference" : [{ Reference(QuestionnaireResponse) }], //Why was family member history performed?
    "note" : [{ Annotation }], //General note about related person
    "condition" : [{ BackboneElement }], // C?Condition that the related person had
  }
```

Other representations of profile: Schematron

9.4.9.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet

FamilyMemberHistory.language | Common Languages | extensible | Common Languages

FamilyMemberHistory.status | FamilyHistoryStatus | required | FamilyHistoryStatus

FamilyMemberHistory.notDoneReason | FamilyHistoryNotDoneReason | example | FamilyHistoryNotDoneReason

FamilyMemberHistory.relationship | FamilyMember | example | FamilyMember

FamilyMemberHistory.gender | AdministrativeGender | required | AdministrativeGender

FamilyMemberHistory.reasonCode | SNOMED CT Clinical Findings | example | SNOMED CT Clinical Findings

FamilyMemberHistory.condition.code | Condition/Problem/Diagnosis Codes | example | Condition/Problem/Diagnosis Codes

FamilyMemberHistory.condition.outcome | Condition Outcome Codes | example | Condition Outcome Codes

9.4.10 Family Member History HL7 Extensions

URL: https://hl7.org/fhir/STU3/familymemberhistory-extensions.html

Defines common extensions used with or related to the FamilyMemberHistory resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.10 Family Member History HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.4.10.1 Content

Extensions:

familymemberhistory-abatement | abatement : The approximate date, age, or flag indicating that the condition of the family member resolved. The abatement should only be specified if the condition is stated in the positive sense, i.e., the didNotHave flag is false.

familymemberhistory-severity | severity : A qualification of the seriousness or impact on health of the family member condition.

familymemberhistory-patient-record | patient-record : A link to one to more patient records for the relation.

familymemberhistory-type | type : Purpose of the family member history or why it was created, such as when family member history is targeted for cardiovascular health, mental health, or genetic counseling.

9.4.11 Resource FamilyMemberHistory - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/familymemberhistory-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.4.11 Resource FamilyMemberHistory - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

R2 : R3 Converson maps for FamilyMemberHistory.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

9.4.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/FamilyMemberHistory2to3" = "R2 to R3 Conversions for FamilyMemberHistory"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistoryR2 as source
uses "http://hl7.org/fhir/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistory as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  FamilyMemberHistory extends DomainResource
  input src : FamilyMemberHistoryR2 as source
  input tgt : FamilyMemberHistory as target

  "FamilyMemberHistory-identifier" : for src.identifier make tgt.identifier
  "FamilyMemberHistory-patient" : for src.patient make tgt.patient
  "FamilyMemberHistory-date" : for src.date make tgt.date
  "FamilyMemberHistory-status" : for src.status make tgt.status
  "FamilyMemberHistory-name" : for src.name make tgt.name
  "FamilyMemberHistory-relationship" : for src.relationship make tgt.relationship
  "FamilyMemberHistory-gender" : for src.gender make tgt.gender
  "FamilyMemberHistory-born" : for src.born make tgt.born
  "FamilyMemberHistory-age" : for src.age make tgt.age
  "FamilyMemberHistory-deceased" : for src.deceased make tgt.deceased
  "FamilyMemberHistory-note" : for src.note make tgt.note
  "FamilyMemberHistory-condition" : for src.condition as vs0 make tgt.condition as vt0 then fmhCondition(vs0, vt0)
```

```
endgroup

group fmhCondition extends BackboneElement
  input src as source
  input tgt as target

  "FamilyMemberHistory.condition-code" : for src.code make tgt.code
  "FamilyMemberHistory.condition-outcome" : for src.outcome make tgt.outcome
  "FamilyMemberHistory.condition-onset" : for src.onset make tgt.onset
  "FamilyMemberHistory.condition-note" : for src.note make tgt.note
endgroup
```

9.4.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/FamilyMemberHistory3to2" = "R3 to R2 Conversion for FamilyMemberHistory"

uses "http://hl7.org/fhir/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistory as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistoryR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  FamilyMemberHistory extends DomainResource
  input src : FamilyMemberHistory as source
  input tgt : FamilyMemberHistoryR2 as target

  "FamilyMemberHistory-identifier" : for src.identifier make tgt.identifier
  "FamilyMemberHistory-patient" : for src.patient make tgt.patient
  "FamilyMemberHistory-date" : for src.date make tgt.date
  "FamilyMemberHistory-status" : for src.status make tgt.status
  "FamilyMemberHistory-name" : for src.name make tgt.name
  "FamilyMemberHistory-relationship" : for src.relationship make tgt.relationship
  "FamilyMemberHistory-gender" : for src.gender make tgt.gender
  "FamilyMemberHistory-born" : for src.born make tgt.born
  "FamilyMemberHistory-age" : for src.age make tgt.age
  "FamilyMemberHistory-deceased" : for src.deceased make tgt.deceased
  "FamilyMemberHistory-note" : for src.note make tgt.note
  "FamilyMemberHistory-condition" : for src.condition as vs0 make tgt.condition as vt0 then fmhCondition(vs0, vt0)
endgroup

group fmhCondition extends BackboneElement
  input src as source
  input tgt as target

  "FamilyMemberHistory.condition-code" : for src.code make tgt.code
  "FamilyMemberHistory.condition-outcome" : for src.outcome make tgt.outcome
  "FamilyMemberHistory.condition-onset" : for src.onset make tgt.onset
  "FamilyMemberHistory.condition-note" : for src.note make tgt.note
endgroup
```

9.4.11.3 R3 Validation Errors

9.5 Resource CarePlan - Content

URL: https://hl7.org/fhir/STU3/careplan.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Describes the intention of how one or more practitioners intend to deliver care for a particular patient, group or community for a period of time, possibly limited to care for a specific condition or set of conditions.

9.5.1 Scope and Usage

CarePlan is one of the request resources in the FHIR workflow specification.

Care Plans are used in many areas of healthcare with a variety of scopes. They can be as simple as a general practitioner keeping track of when their patient is next due for a tetanus immunization through to a detailed plan for an oncology patient covering diet, chemotherapy, radiation, lab work and counseling with detailed timing relationships, pre-conditions and goals. They may be used in veterinary care or clinical research to describe the care of a herd or other collection of animals. In public health, they may describe education or immunization campaigns.

This resource takes an intermediate approach to complexity. It captures basic details about who is involved and what actions are intended without dealing in discrete data about dependencies and timing relationships. These can be supported where necessary using the extension mechanism.

The scope of care plans may vary widely. Examples include:

* Multi-disciplinary cross-organizational care plans; e.g. An oncology plan including the oncologist, home nursing staff, pharmacy and others

* Plans to manage specific disease/condition(s) (e.g. nutritional plan for a patient post bowel resection, neurological plan post head injury, pre-natal plan, post-partum plan, grief management plan, etc.)

* Decision support-generated plans following specific practice guidelines (e.g. stroke care plan, diabetes plan, falls prevention, etc.)

* Self-maintained patient or care-giver authored plans identifying their goals and an integrated understanding of actions to be taken

This resource can be used to represent both proposed plans (for example, recommendations from a decision support engine or returned as part of a consult report) as well as active plans. The nature of the plan is communicated by the status. Some systems may need to filter CarePlans to ensure that only appropriate plans are exposed via a given user interface.

9.5.2 Boundaries and Relationships

For simplicity sake, CarePlan allows the in-line definition of activities as part of a plan using the activity.detail element. However, activities can also be defined using references to the various "request" resources. These references could be to resources with a status of "planned" or to an active order. It is possible for planned activities to exist (e.g. appointments) without needing a CarePlan at all. CarePlans are used when there's a need to group activities, goals and/or participants together to provide some degree of context.

CarePlans can be tied to specific Conditions however they can also be condition-independent and instead focused on a particular type of care (e.g. psychological, nutritional) or the care delivered by a particular practitioner or group of practitioners.

An ImmunizationRecommendation can be interpreted as a narrow type of Care Plan dealing only with immunization events. Where such information could appear in either resource, the immunization-specific resource is preferred.

CarePlans represent a specific plan instance for a particular patient or group. It is not intended to be used to define generic plans or protocols that are independent of a specific individual or group. CarePlan represents a specific intent, not a general definition. Protocols and order sets are supported through PlanDefinition.

This resource is referenced by DiagnosticReport, GuidanceResponse, ImagingStudy, MedicationRequest, MedicationStatement, Observation, Procedure, QuestionnaireResponse and ReferralRequest

9.5.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
CarePlan | | | DomainResource | Healthcare plan for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this plan
definition | ? | 0..* | Reference(PlanDefinition | Questionnaire) | Protocol or definition
basedOn | ? | 0..* | Reference(CarePlan) | Fulfills care plan
replaces | ? | 0..* | Reference(CarePlan) | CarePlan replaced by this CarePlan
partOf | ? | 0..* | Reference(CarePlan) | Part of referenced CarePlan
status | ?!? | 1..1 | code | draft | active | suspended | completed | entered-in-error | cancelled | unknown CarePlanStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order | option CarePlanIntent (Required)
category | ? | 0..* | CodeableConcept | Type of plan Care Plan Category (Example)
title | ? | 0..1 | string | Human-friendly name for the CarePlan
description | ? | 0..1 | string | Summary of nature of plan
subject | ? | 1..1 | Reference(Patient | Group) | Who care plan is for
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Created in context of
period | ? | 0..1 | Period | Time period plan covers
author | ? | 0..* | Reference(Patient | Practitioner | RelatedPerson | Organization | CareTeam) | Who is responsible for contents of the plan
careTeam | | 0..* | Reference(CareTeam) | Who's involved in plan?
addresses | ? | 0..* | Reference(Condition) | Health issues this plan addresses
supportingInfo | | 0..* | Reference(Any) | Information considered as part of plan
goal | | 0..* | Reference(Goal) | Desired outcome of plan
activity | I | 0..* | BackboneElement | Action to occur as part of plan + Provide a reference or detail, not both
outcomeCodeableConcept | | 0..* | CodeableConcept | Results of the activity Care Plan Activity Outcome (Example)
outcomeReference | | 0..* | Reference(Any) | Appointment, Encounter, Procedure, etc.
progress | | 0..* | Annotation | Comments about the activity status/progress
reference | I | 0..1 | Reference(Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ProcedureRequest | ReferralRequest | VisionPrescription | RequestGroup) | Activity details defined in specific resource
detail | I | 0..1 | BackboneElement | In-line definition of activity
category | | 0..1 | CodeableConcept | diet | drug | encounter | observation | procedure | supply | other CarePlanActivityCategory (Example)
definition | | 0..1 | Reference(PlanDefinition | ActivityDefinition | Questionnaire) | Protocol or definition
code | | 0..1 | CodeableConcept | Detail type of activity Care Plan Activity (Example)
reasonCode | | 0..* | CodeableConcept | Why activity should be done or why activity was prohibited Activity Reason (Example)
reasonReference | | 0..* | Reference(Condition) | Condition triggering need for activity

goal | | 0..* | Reference(Goal) | Goals this activity relates to
status | ?! | 1..1 | code | not-started | scheduled | in-progress | on-hold | completed | cancelled | unknown CarePlanActivityStatus (Required)
statusReason | | 0..1 | string | Reason for current status
prohibited | ?! | 0..1 | boolean | Do NOT do
scheduled[x] | | 0..1 | | When activity is to occur
scheduledTiming | | | Timing
scheduledPeriod | | | Period
scheduledString | | | string
location | | 0..1 | Reference(Location) | Where it should happen
performer | | 0..* | Reference(Practitioner | Organization | RelatedPerson | Patient | CareTeam) | Who will be responsible?
product[x] | | 0..1 | | What is to be administered/supplied SNOMED CT Medication Codes (Example)
productCodeableConcept | | | CodeableConcept
productReference | | | Reference(Medication | Substance)
dailyAmount | | 0..1 | SimpleQuantity | How to consume/day?
quantity | | 0..1 | SimpleQuantity | How much to administer/supply/consume
description | | 0..1 | string | Extra info describing activity to perform
note | | 0..* | Annotation | Comments about the plan
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CarePlan xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this plan --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|Questionnaire) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(CarePlan) Fulfills care plan --></basedOn>
 <replaces><!-- 0..* Reference(CarePlan) CarePlan replaced by this CarePlan --></replaces>
 <partOf><!-- 0..* Reference(CarePlan) Part of referenced CarePlan --></partOf>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | completed | entered-in-error | cancelled | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | option -->
 <category><!-- 0..* CodeableConcept Type of plan --></category>
 <title value="[string]"/><!-- 0..1 Human-friendly name for the CarePlan -->
 <description value="[string]"/><!-- 0..1 Summary of nature of plan -->
 <subject><!-- 1..1 Reference(Patient|Group) Who care plan is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Created in context of --></context>
 <period><!-- 0..1 Period Time period plan covers --></period>
 <author><!-- 0..* Reference(Patient|Practitioner|RelatedPerson|Organization|
   CareTeam) Who is responsible for contents of the plan --></author>
 <careTeam><!-- 0..* Reference(CareTeam) Who's involved in plan? --></careTeam>
 <addresses><!-- 0..* Reference(Condition) Health issues this plan addresses --></addresses>
 <supportingInfo><!-- 0..* Reference(Any) Information considered as part of plan --></supportingInfo>
 <goal><!-- 0..* Reference(Goal) Desired outcome of plan --></goal>
 <activity>  <!-- 0..* Action to occur as part of plan -->
  <outcomeCodeableConcept><!-- 0..* CodeableConcept Results of the activity --></outcomeCodeableConcept>
  <outcomeReference><!-- 0..* Reference(Any) Appointment, Encounter, Procedure, etc. --></outcomeReference>
  <progress><!-- 0..* Annotation Comments about the activity status/progress --></progress>
  <reference><!--  0..1 Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ProcedureRequest|ReferralRequest|
    VisionPrescription|RequestGroup) Activity details defined in specific resource --></reference>
  <detail>  <!--  0..1 In-line definition of activity -->
   <category><!-- 0..1 CodeableConcept diet | drug | encounter | observation | procedure | supply | other --></category>
   <definition><!-- 0..1 Reference(PlanDefinition|ActivityDefinition|
     Questionnaire) Protocol or definition --></definition>
   <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
   <reasonCode><!-- 0..* CodeableConcept Why activity should be done or why activity was prohibited --></reasonCode>
   <reasonReference><!-- 0..* Reference(Condition) Condition triggering need for activity --></reasonReference>
   <goal><!-- 0..* Reference(Goal) Goals this activity relates to --></goal>
   <status value="[code]"/><!-- 1..1 not-started | scheduled | in-progress | on-hold | completed | cancelled | unknown
    -->
   <statusReason value="[string]"/><!-- 0..1 Reason for current status -->
   <prohibited value="[boolean]"/><!-- 0..1 Do NOT do -->
   <scheduled[x]><!-- 0..1 Timing|Period|string When activity is to occur --></scheduled[x]>
   <location><!-- 0..1 Reference(Location) Where it should happen --></location>
   <performer><!-- 0..* Reference(Practitioner|Organization|RelatedPerson|Patient|
     CareTeam) Who will be responsible? --></performer>
   <product[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance) What is to be administered/supplied
    --></product[x]>
   <dailyAmount><!-- 0..1 Quantity(SimpleQuantity) How to consume/day? --></dailyAmount>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) How much to administer/supply/consume --></quantity>
   <description value="[string]"/><!-- 0..1 Extra info describing activity to perform -->
  </detail>
 </activity>
 <note><!-- 0..* Annotation Comments about the plan --></note>
</CarePlan>
```

JSON Template

```
{
  "resourceType" : "CarePlan",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this plan
  "definition" : [{ Reference(PlanDefinition|Questionnaire) }], // Protocol or definition
  "basedOn" : [{ Reference(CarePlan) }], // Fulfills care plan
```

```
  "replaces" : [{ Reference(CarePlan) }], // CarePlan replaced by this CarePlan
  "partOf" : [{ Reference(CarePlan) }], // Part of referenced CarePlan
  "status" : "<code>", // R!  draft | active | suspended | completed | entered-in-error | cancelled | unknown
  "intent" : "<code>", // R!  proposal | plan | order | option
  "category" : [{ CodeableConcept }], // Type of plan
  "title" : "<string>", // Human-friendly name for the CarePlan
  "description" : "<string>", // Summary of nature of plan
  "subject" : { Reference(Patient|Group) }, // R!  Who care plan is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Created in context of
  "period" : { Period }, // Time period plan covers
  "author" : [{ Reference(Patient|Practitioner|RelatedPerson|Organization|
   CareTeam) }], // Who is responsible for contents of the plan
  "careTeam" : [{ Reference(CareTeam) }], // Who's involved in plan?
  "addresses" : [{ Reference(Condition) }], // Health issues this plan addresses
  "supportingInfo" : [{ Reference(Any) }], // Information considered as part of plan
  "goal" : [{ Reference(Goal) }], // Desired outcome of plan
  "activity" : [{ // Action to occur as part of plan
    "outcomeCodeableConcept" : [{ CodeableConcept }], // Results of the activity
    "outcomeReference" : [{ Reference(Any) }], // Appointment, Encounter, Procedure, etc.
    "progress" : [{ Annotation }], // Comments about the activity status/progress
    "reference" : { Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ProcedureRequest|ReferralRequest|
    VisionPrescription|RequestGroup) }, // C? Activity details defined in specific resource
    "detail" : { // C? In-line definition of activity
      "category" : { CodeableConcept }, // diet | drug | encounter | observation | procedure | supply | other
      "definition" : { Reference(PlanDefinition|ActivityDefinition|
     Questionnaire) }, // Protocol or definition
      "code" : { CodeableConcept }, // Detail type of activity
      "reasonCode" : [{ CodeableConcept }], // Why activity should be done or why activity was prohibited
      "reasonReference" : [{ Reference(Condition) }], // Condition triggering need for activity
      "goal" : [{ Reference(Goal) }], // Goals this activity relates to
      "status" : "<code>", // R!  not-started | scheduled | in-progress | on-hold | completed | cancelled | unknown
      "statusReason" : "<string>", // Reason for current status
      "prohibited" : <boolean>, // Do NOT do
      // scheduled[x]: When activity is to occur. One of these 3:
      "scheduledTiming" : { Timing },
      "scheduledPeriod" : { Period },
      "scheduledString" : "<string>",
      "location" : { Reference(Location) }, // Where it should happen
      "performer" : [{ Reference(Practitioner|Organization|RelatedPerson|Patient|
     CareTeam) }], // Who will be responsible?
      // product[x]: What is to be administered/supplied. One of these 2:
      "productCodeableConcept" : { CodeableConcept },
      "productReference" : { Reference(Medication|Substance) },
      "dailyAmount" : { Quantity(SimpleQuantity) }, // How to consume/day?
      "quantity" : { Quantity(SimpleQuantity) }, // How much to administer/supply/consume
      "description" : "<string>" // Extra info describing activity to perform
    }
  }],
  "note" : [{ Annotation }] // Comments about the plan
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CarePlan;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CarePlan.identifier [ Identifier ], ... ; # 0..* External Ids for this plan
  fhir:CarePlan.definition [ Reference(PlanDefinition|Questionnaire) ], ... ; # 0..* Protocol or definition
  fhir:CarePlan.basedOn [ Reference(CarePlan) ], ... ; # 0..* Fulfills care plan
  fhir:CarePlan.replaces [ Reference(CarePlan) ], ... ; # 0..* CarePlan replaced by this CarePlan
  fhir:CarePlan.partOf [ Reference(CarePlan) ], ... ; # 0..* Part of referenced CarePlan
  fhir:CarePlan.status [ code ]; # 1..1 draft | active | suspended | completed | entered-in-error | cancelled | unknown
  fhir:CarePlan.intent [ code ]; # 1..1 proposal | plan | order | option
  fhir:CarePlan.category [ CodeableConcept ], ... ; # 0..* Type of plan
  fhir:CarePlan.title [ string ]; # 0..1 Human-friendly name for the CarePlan
  fhir:CarePlan.description [ string ]; # 0..1 Summary of nature of plan
  fhir:CarePlan.subject [ Reference(Patient|Group) ]; # 1..1 Who care plan is for
  fhir:CarePlan.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Created in context of
  fhir:CarePlan.period [ Period ]; # 0..1 Time period plan covers
  fhir:CarePlan.author [ Reference(Patient|Practitioner|RelatedPerson|Organization|CareTeam) ], ... ; # 0..* Who is
    responsible for contents of the plan
  fhir:CarePlan.careTeam [ Reference(CareTeam) ], ... ; # 0..* Who's involved in plan?
  fhir:CarePlan.addresses [ Reference(Condition) ], ... ; # 0..* Health issues this plan addresses
  fhir:CarePlan.supportingInfo [ Reference(Any) ], ... ; # 0..* Information considered as part of plan
  fhir:CarePlan.goal [ Reference(Goal) ], ... ; # 0..* Desired outcome of plan
  fhir:CarePlan.activity [ # 0..* Action to occur as part of plan
    fhir:CarePlan.activity.outcomeCodeableConcept [ CodeableConcept ], ... ; # 0..* Results of the activity
    fhir:CarePlan.activity.outcomeReference [ Reference(Any) ], ... ; # 0..* Appointment, Encounter, Procedure, etc.
    fhir:CarePlan.activity.progress [ Annotation ], ... ; # 0..* Comments about the activity status/progress
    fhir:CarePlan.activity.reference [
    Reference(Appointment|CommunicationRequest|DeviceRequest|MedicationRequest|NutritionOrder|
  Task|ProcedureRequest|ReferralRequest|VisionPrescription|RequestGroup) ]; # 0..1 Activity details defined in specific
    resource
    fhir:CarePlan.activity.detail [ # 0..1 In-line definition of activity
```

```
    fhir:CarePlan.activity.detail.category [ CodeableConcept ]; # 0..1 diet | drug | encounter | observation |
  procedure | supply | other
    fhir:CarePlan.activity.detail.definition [ Reference(PlanDefinition|ActivityDefinition|Questionnaire) ]; # 0..1
  Protocol or definition
    fhir:CarePlan.activity.detail.code [ CodeableConcept ]; # 0..1 Detail type of activity
    fhir:CarePlan.activity.detail.reasonCode [ CodeableConcept ], ... ; # 0..* Why activity should be done or why
  activity was prohibited
    fhir:CarePlan.activity.detail.reasonReference [ Reference(Condition) ], ... ; # 0..* Condition triggering need for
   activity
    fhir:CarePlan.activity.detail.goal [ Reference(Goal) ], ... ; # 0..* Goals this activity relates to
    fhir:CarePlan.activity.detail.status [ code ]; # 1..1 not-started | scheduled | in-progress | on-hold | completed
  | cancelled | unknown
    fhir:CarePlan.activity.detail.statusReason [ string ]; # 0..1 Reason for current status
    fhir:CarePlan.activity.detail.prohibited [ boolean ]; # 0..1 Do NOT do
    # CarePlan.activity.detail.scheduled[x] : 0..1 When activity is to occur. One of these 3
      fhir:CarePlan.activity.detail.scheduledTiming [ Timing ]
      fhir:CarePlan.activity.detail.scheduledPeriod [ Period ]
      fhir:CarePlan.activity.detail.scheduledString [ string ]
    fhir:CarePlan.activity.detail.location [ Reference(Location) ]; # 0..1 Where it should happen
    fhir:CarePlan.activity.detail.performer [ Reference(Practitioner|Organization|RelatedPerson|Patient|CareTeam) ],
  ... ; # 0..* Who will be responsible?
    # CarePlan.activity.detail.product[x] : 0..1 What is to be administered/supplied. One of these 2
      fhir:CarePlan.activity.detail.productCodeableConcept [ CodeableConcept ]
      fhir:CarePlan.activity.detail.productReference [ Reference(Medication|Substance) ]
    fhir:CarePlan.activity.detail.dailyAmount [ Quantity(SimpleQuantity) ]; # 0..1 How to consume/day?
    fhir:CarePlan.activity.detail.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much to administer/supply/consume
    fhir:CarePlan.activity.detail.description [ string ]; # 0..1 Extra info describing activity to perform
  ];
 ], ...;
 fhir:CarePlan.note [ Annotation ], ... ; # 0..* Comments about the plan
]
```

Changes since DSTU2

CarePlan
CarePlan.definition | Added Element
CarePlan.basedOn | Added Element
CarePlan.replaces | Added Element
CarePlan.partOf | Added Element
CarePlan.intent | Added Element
CarePlan.title | Added Element
CarePlan.subject | Min Cardinality changed from 0 to 1
CarePlan.author | Add Reference(CareTeam)
CarePlan.careTeam | Added Element
CarePlan.supportingInfo | Renamed from support to supportingInfo
CarePlan.activity.outcomeCodeableConcept | Added Element
CarePlan.activity.outcomeReference | Renamed from actionResulting to outcomeReference
CarePlan.activity.reference | Remove Reference(DiagnosticOrder), Remove Reference(MedicationOrder), Remove Reference(Order), Remove Reference(ProcessRequest), Remove Reference(SupplyRequest), Add Reference(MedicationRequest), Add Reference(Task), Add Reference(RequestGroup)
CarePlan.activity.detail.definition | Added Element
CarePlan.activity.detail.status | Min Cardinality changed from 0 to 1
CarePlan.activity.detail.statusReason | Type changed from CodeableConcept to string
CarePlan.activity.detail.prohibited | Min Cardinality changed from 1 to 0 Default Value "false" added
CarePlan.activity.detail.performer | Add Reference(CareTeam)
CarePlan.note | Max Cardinality changed from 1 to *
CarePlan.modified | deleted
CarePlan.relatedPlan | deleted
CarePlan.participant | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 10 tests that all execute ok. 8 fail round-trip testing and 10 r3 resources are invalid (33 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
CarePlan | | | DomainResource | Healthcare plan for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this plan
definition | ? | 0..* | Reference(PlanDefinition | Questionnaire) | Protocol or definition
basedOn | ? | 0..* | Reference(CarePlan) | Fulfills care plan
replaces | ? | 0..* | Reference(CarePlan) | CarePlan replaced by this CarePlan
partOf | ? | 0..* | Reference(CarePlan) | Part of referenced CarePlan
status | ?!? | 1..1 | code | draft | active | suspended | completed | entered-in-error | cancelled | unknown CarePlanStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order | option CarePlanIntent (Required)
category | ? | 0..* | CodeableConcept | Type of plan Care Plan Category (Example)
title | ? | 0..1 | string | Human-friendly name for the CarePlan
description | ? | 0..1 | string | Summary of nature of plan
subject | ? | 1..1 | Reference(Patient | Group) | Who care plan is for

context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Created in context of
period | ? | 0..1 | Period | Time period plan covers
author | ? | 0..* | Reference(Patient | Practitioner | RelatedPerson | Organization | CareTeam) | Who is responsible for contents of the plan
careTeam | | 0..* | Reference(CareTeam) | Who's involved in plan?
addresses | ? | 0..* | Reference(Condition) | Health issues this plan addresses
supportingInfo | | 0..* | Reference(Any) | Information considered as part of plan
goal | | 0..* | Reference(Goal) | Desired outcome of plan
activity | I | 0..* | BackboneElement | Action to occur as part of plan + Provide a reference or detail, not both
outcomeCodeableConcept | | 0..* | CodeableConcept | Results of the activity Care Plan Activity Outcome (Example)
outcomeReference | | 0..* | Reference(Any) | Appointment, Encounter, Procedure, etc.
progress | | 0..* | Annotation | Comments about the activity status/progress
reference | I | 0..1 | Reference(Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ProcedureRequest | ReferralRequest | VisionPrescription | RequestGroup) | Activity details defined in specific resource
detail | I | 0..1 | BackboneElement | In-line definition of activity
category | | 0..1 | CodeableConcept | diet | drug | encounter | observation | procedure | supply | other CarePlanActivityCategory (Example)
definition | | 0..1 | Reference(PlanDefinition | ActivityDefinition | Questionnaire) | Protocol or definition
code | | 0..1 | CodeableConcept | Detail type of activity Care Plan Activity (Example)
reasonCode | | 0..* | CodeableConcept | Why activity should be done or why activity was prohibited Activity Reason (Example)
reasonReference | | 0..* | Reference(Condition) | Condition triggering need for activity
goal | | 0..* | Reference(Goal) | Goals this activity relates to
status | ?! | 1..1 | code | not-started | scheduled | in-progress | on-hold | completed | cancelled | unknown CarePlanActivityStatus (Required)
statusReason | | 0..1 | string | Reason for current status
prohibited | ?! | 0..1 | boolean | Do NOT do
scheduled[x] | | 0..1 | | When activity is to occur
scheduledTiming | | | Timing
scheduledPeriod | | | Period
scheduledString | | | string
location | | 0..1 | Reference(Location) | Where it should happen
performer | | 0..* | Reference(Practitioner | Organization | RelatedPerson | Patient | CareTeam) | Who will be responsible?
product[x] | | 0..1 | | What is to be administered/supplied SNOMED CT Medication Codes (Example)
productCodeableConcept | | | CodeableConcept
productReference | | | Reference(Medication | Substance)
dailyAmount | | 0..1 | SimpleQuantity | How to consume/day?
quantity | | 0..1 | SimpleQuantity | How much to administer/supply/consume
description | | 0..1 | string | Extra info describing activity to perform
note | | 0..* | Annotation | Comments about the plan
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CarePlan xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this plan --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|Questionnaire) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(CarePlan) Fulfills care plan --></basedOn>
 <replaces><!-- 0..* Reference(CarePlan) CarePlan replaced by this CarePlan --></replaces>
 <partOf><!-- 0..* Reference(CarePlan) Part of referenced CarePlan --></partOf>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | completed | entered-in-error | cancelled | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | option -->
 <category><!-- 0..* CodeableConcept Type of plan --></category>
 <title value="[string]"/><!-- 0..1 Human-friendly name for the CarePlan -->
 <description value="[string]"/><!-- 0..1 Summary of nature of plan -->
 <subject><!-- 1..1 Reference(Patient|Group) Who care plan is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Created in context of --></context>
 <period><!-- 0..1 Period Time period plan covers --></period>
 <author><!-- 0..* Reference(Patient|Practitioner|RelatedPerson|Organization|
   CareTeam) Who is responsible for contents of the plan --></author>
 <careTeam><!-- 0..* Reference(CareTeam) Who's involved in plan? --></careTeam>
 <addresses><!-- 0..* Reference(Condition) Health issues this plan addresses --></addresses>
 <supportingInfo><!-- 0..* Reference(Any) Information considered as part of plan --></supportingInfo>
 <goal><!-- 0..* Reference(Goal) Desired outcome of plan --></goal>
 <activity>  <!-- 0..* Action to occur as part of plan -->
  <outcomeCodeableConcept><!-- 0..* CodeableConcept Results of the activity --></outcomeCodeableConcept>
  <outcomeReference><!-- 0..* Reference(Any) Appointment, Encounter, Procedure, etc. --></outcomeReference>
  <progress><!-- 0..* Annotation Comments about the activity status/progress --></progress>
  <reference><!--  0..1 Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ProcedureRequest|ReferralRequest|
    VisionPrescription|RequestGroup) Activity details defined in specific resource --></reference>
  <detail>  <!--  0..1 In-line definition of activity -->
   <category><!-- 0..1 CodeableConcept diet | drug | encounter | observation | procedure | supply | other --></category>
   <definition><!-- 0..1 Reference(PlanDefinition|ActivityDefinition|
     Questionnaire) Protocol or definition --></definition>
   <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
   <reasonCode><!-- 0..* CodeableConcept Why activity should be done or why activity was prohibited --></reasonCode>
   <reasonReference><!-- 0..* Reference(Condition) Condition triggering need for activity --></reasonReference>
   <goal><!-- 0..* Reference(Goal) Goals this activity relates to --></goal>
   <status value="[code]"/><!-- 1..1 not-started | scheduled | in-progress | on-hold | completed | cancelled | unknown
    -->
   <statusReason value="[string]"/><!-- 0..1 Reason for current status -->
```

```
   <prohibited value="[boolean]"/><!-- 0..1 Do NOT do -->
   <scheduled[x]><!-- 0..1 Timing|Period|string When activity is to occur --></scheduled[x]>
   <location><!-- 0..1 Reference(Location) Where it should happen --></location>
   <performer><!-- 0..* Reference(Practitioner|Organization|RelatedPerson|Patient|
     CareTeam) Who will be responsible? --></performer>
   <product[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance) What is to be administered/supplied
    --></product[x]>
   <dailyAmount><!-- 0..1 Quantity(SimpleQuantity) How to consume/day? --></dailyAmount>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) How much to administer/supply/consume --></quantity>
   <description value="[string]"/><!-- 0..1 Extra info describing activity to perform -->
  </detail>
 </activity>
 <note><!-- 0..* Annotation Comments about the plan --></note>
</CarePlan>
```

JSON Template

```
{
  "resourceType" : "CarePlan",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this plan
  "definition" : [{ Reference(PlanDefinition|Questionnaire) }], // Protocol or definition
  "basedOn" : [{ Reference(CarePlan) }], // Fulfills care plan
  "replaces" : [{ Reference(CarePlan) }], // CarePlan replaced by this CarePlan
  "partOf" : [{ Reference(CarePlan) }], // Part of referenced CarePlan
  "status" : "<code>", // R!  draft | active | suspended | completed | entered-in-error | cancelled | unknown
  "intent" : "<code>", // R!  proposal | plan | order | option
  "category" : [{ CodeableConcept }], // Type of plan
  "title" : "<string>", // Human-friendly name for the CarePlan
  "description" : "<string>", // Summary of nature of plan
  "subject" : { Reference(Patient|Group) }, // R!  Who care plan is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Created in context of
  "period" : { Period }, // Time period plan covers
  "author" : [{ Reference(Patient|Practitioner|RelatedPerson|Organization|
   CareTeam) }], // Who is responsible for contents of the plan
  "careTeam" : [{ Reference(CareTeam) }], // Who's involved in plan?
  "addresses" : [{ Reference(Condition) }], // Health issues this plan addresses
  "supportingInfo" : [{ Reference(Any) }], // Information considered as part of plan
  "goal" : [{ Reference(Goal) }], // Desired outcome of plan
  "activity" : [{ // Action to occur as part of plan
    "outcomeCodeableConcept" : [{ CodeableConcept }], // Results of the activity
    "outcomeReference" : [{ Reference(Any) }], // Appointment, Encounter, Procedure, etc.
    "progress" : [{ Annotation }], // Comments about the activity status/progress
    "reference" : { Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ProcedureRequest|ReferralRequest|
    VisionPrescription|RequestGroup) }, // C? Activity details defined in specific resource
    "detail" : { // C? In-line definition of activity
      "category" : { CodeableConcept }, // diet | drug | encounter | observation | procedure | supply | other
      "definition" : { Reference(PlanDefinition|ActivityDefinition|
     Questionnaire) }, // Protocol or definition
      "code" : { CodeableConcept }, // Detail type of activity
      "reasonCode" : [{ CodeableConcept }], // Why activity should be done or why activity was prohibited
      "reasonReference" : [{ Reference(Condition) }], // Condition triggering need for activity
      "goal" : [{ Reference(Goal) }], // Goals this activity relates to
      "status" : "<code>", // R!  not-started | scheduled | in-progress | on-hold | completed | cancelled | unknown
      "statusReason" : "<string>", // Reason for current status
      "prohibited" : <boolean>, // Do NOT do
      // scheduled[x]: When activity is to occur. One of these 3:
      "scheduledTiming" : { Timing },
      "scheduledPeriod" : { Period },
      "scheduledString" : "<string>",
      "location" : { Reference(Location) }, // Where it should happen
      "performer" : [{ Reference(Practitioner|Organization|RelatedPerson|Patient|
     CareTeam) }], // Who will be responsible?
      // product[x]: What is to be administered/supplied. One of these 2:
      "productCodeableConcept" : { CodeableConcept },
      "productReference" : { Reference(Medication|Substance) },
      "dailyAmount" : { Quantity(SimpleQuantity) }, // How to consume/day?
      "quantity" : { Quantity(SimpleQuantity) }, // How much to administer/supply/consume
      "description" : "<string>" // Extra info describing activity to perform
    }
  }],
  "note" : [{ Annotation }] // Comments about the plan
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CarePlan;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CarePlan.identifier [ Identifier ], ... ; # 0..* External Ids for this plan
  fhir:CarePlan.definition [ Reference(PlanDefinition|Questionnaire) ], ... ; # 0..* Protocol or definition
  fhir:CarePlan.basedOn [ Reference(CarePlan) ], ... ; # 0..* Fulfills care plan
  fhir:CarePlan.replaces [ Reference(CarePlan) ], ... ; # 0..* CarePlan replaced by this CarePlan
  fhir:CarePlan.partOf [ Reference(CarePlan) ], ... ; # 0..* Part of referenced CarePlan
```

```
  fhir:CarePlan.status [ code ]; # 1..1 draft | active | suspended | completed | entered-in-error | cancelled | unknown
  fhir:CarePlan.intent [ code ]; # 1..1 proposal | plan | order | option
  fhir:CarePlan.category [ CodeableConcept ], ... ; # 0..* Type of plan
  fhir:CarePlan.title [ string ]; # 0..1 Human-friendly name for the CarePlan
  fhir:CarePlan.description [ string ]; # 0..1 Summary of nature of plan
  fhir:CarePlan.subject [ Reference(Patient|Group) ]; # 1..1 Who care plan is for
  fhir:CarePlan.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Created in context of
  fhir:CarePlan.period [ Period ]; # 0..1 Time period plan covers
  fhir:CarePlan.author [ Reference(Patient|Practitioner|RelatedPerson|Organization|CareTeam) ], ... ; # 0..* Who is
    responsible for contents of the plan
  fhir:CarePlan.careTeam [ Reference(CareTeam) ], ... ; # 0..* Who's involved in plan?
  fhir:CarePlan.addresses [ Reference(Condition) ], ... ; # 0..* Health issues this plan addresses
  fhir:CarePlan.supportingInfo [ Reference(Any) ], ... ; # 0..* Information considered as part of plan
  fhir:CarePlan.goal [ Reference(Goal) ], ... ; # 0..* Desired outcome of plan
  fhir:CarePlan.activity [ # 0..* Action to occur as part of plan
    fhir:CarePlan.activity.outcomeCodeableConcept [ CodeableConcept ], ... ; # 0..* Results of the activity
    fhir:CarePlan.activity.outcomeReference [ Reference(Any) ], ... ; # 0..* Appointment, Encounter, Procedure, etc.
    fhir:CarePlan.activity.progress [ Annotation ], ... ; # 0..* Comments about the activity status/progress
    fhir:CarePlan.activity.reference [
    Reference(Appointment|CommunicationRequest|DeviceRequest|MedicationRequest|NutritionOrder|
  Task|ProcedureRequest|ReferralRequest|VisionPrescription|RequestGroup) ]; # 0..1 Activity details defined in specific
    resource
    fhir:CarePlan.activity.detail [ # 0..1 In-line definition of activity
      fhir:CarePlan.activity.detail.category [ CodeableConcept ]; # 0..1 diet | drug | encounter | observation |
    procedure | supply | other
      fhir:CarePlan.activity.detail.definition [ Reference(PlanDefinition|ActivityDefinition|Questionnaire) ]; # 0..1
    Protocol or definition
      fhir:CarePlan.activity.detail.code [ CodeableConcept ]; # 0..1 Detail type of activity
      fhir:CarePlan.activity.detail.reasonCode [ CodeableConcept ], ... ; # 0..* Why activity should be done or why
    activity was prohibited
      fhir:CarePlan.activity.detail.reasonReference [ Reference(Condition) ], ... ; # 0..* Condition triggering need for
     activity
      fhir:CarePlan.activity.detail.goal [ Reference(Goal) ], ... ; # 0..* Goals this activity relates to
      fhir:CarePlan.activity.detail.status [ code ]; # 1..1 not-started | scheduled | in-progress | on-hold | completed
    | cancelled | unknown
      fhir:CarePlan.activity.detail.statusReason [ string ]; # 0..1 Reason for current status
      fhir:CarePlan.activity.detail.prohibited [ boolean ]; # 0..1 Do NOT do
      # CarePlan.activity.detail.scheduled[x] : 0..1 When activity is to occur. One of these 3
        fhir:CarePlan.activity.detail.scheduledTiming [ Timing ]
        fhir:CarePlan.activity.detail.scheduledPeriod [ Period ]
        fhir:CarePlan.activity.detail.scheduledString [ string ]
      fhir:CarePlan.activity.detail.location [ Reference(Location) ]; # 0..1 Where it should happen
      fhir:CarePlan.activity.detail.performer [ Reference(Practitioner|Organization|RelatedPerson|Patient|CareTeam) ],
    ... ; # 0..* Who will be responsible?
      # CarePlan.activity.detail.product[x] : 0..1 What is to be administered/supplied. One of these 2
        fhir:CarePlan.activity.detail.productCodeableConcept [ CodeableConcept ]
        fhir:CarePlan.activity.detail.productReference [ Reference(Medication|Substance) ]
      fhir:CarePlan.activity.detail.dailyAmount [ Quantity(SimpleQuantity) ]; # 0..1 How to consume/day?
      fhir:CarePlan.activity.detail.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much to administer/supply/consume
      fhir:CarePlan.activity.detail.description [ string ]; # 0..1 Extra info describing activity to perform
    ];
  ], ...;
  fhir:CarePlan.note [ Annotation ], ... ; # 0..* Comments about the plan
]
```

Changes since DSTU2

CarePlan

CarePlan.definition | Added Element

CarePlan.basedOn | Added Element

CarePlan.replaces | Added Element

CarePlan.partOf | Added Element

CarePlan.intent | Added Element

CarePlan.title | Added Element

CarePlan.subject | Min Cardinality changed from 0 to 1

CarePlan.author | Add Reference(CareTeam)

CarePlan.careTeam | Added Element

CarePlan.supportingInfo | Renamed from support to supportingInfo

CarePlan.activity.outcomeCodeableConcept | Added Element

CarePlan.activity.outcomeReference | Renamed from actionResulting to outcomeReference

CarePlan.activity.reference | Remove Reference(DiagnosticOrder), Remove Reference(MedicationOrder), Remove Reference(Order), Remove Reference(ProcessRequest), Remove Reference(SupplyRequest), Add Reference(MedicationRequest), Add Reference(Task), Add Reference(RequestGroup)

CarePlan.activity.detail.definition | Added Element

CarePlan.activity.detail.status | Min Cardinality changed from 0 to 1

CarePlan.activity.detail.statusReason | Type changed from CodeableConcept to string

CarePlan.activity.detail.prohibited | Min Cardinality changed from 1 to 0 Default Value "false" added

CarePlan.activity.detail.performer | Add Reference(CareTeam)

CarePlan.note | Max Cardinality changed from 1 to *

CarePlan.modified | deleted

CarePlan.relatedPlan | deleted

CarePlan.participant | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 10 tests that all execute ok. 8 fail round-trip testing and 10 r3 resources are invalid (33 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.5.3.1 Terminology Bindings

Path | Definition | Type | Reference
CarePlan.status | Indicates whether the plan is currently being acted upon, represents future intentions or is now a historical record. | Required | CarePlanStatus
CarePlan.intent | Codes indicating the degree of authority/intentionality associated with a care plan | Required | CarePlanIntent
CarePlan.category | Identifies what "kind" of plan this is to support differentiation between multiple co-existing plans; e.g. "Home health", "psychiatric", "asthma", "disease management", etc. | Example | Care Plan Category
CarePlan.activity.outcomeCodeableConcept | Identifies the results of the activity | Example | Care Plan Activity Outcome
CarePlan.activity.detail.category | High-level categorization of the type of activity in a care plan. | Example | CarePlanActivityCategory
CarePlan.activity.detail.code | Detailed description of the type of activity; e.g. What lab test, what procedure, what kind of encounter. | Example | Care Plan Activity
CarePlan.activity.detail.reasonCode | Identifies why a care plan activity is needed. Can include any health condition codes as well as such concepts as "general wellness", prophylaxis, surgical preparation, etc. | Example | Activity Reason
CarePlan.activity.detail.status | Indicates where the activity is at in its overall life cycle. | Required | CarePlanActivityStatus
CarePlan.activity.detail.product[x] | A product supplied or administered as part of a care plan activity. | Example | SNOMED CT Medication Codes

9.5.3.2 Constraints

* cpl-3: On CarePlan.activity: Provide a reference or detail, not both (expression on CarePlan.activity: detail.empty() or reference.empty())

9.5.3.3 Notes

The Provenance resource can be used for detailed review information, such as when the care plan was last reviewed and by whom.

9.5.4 Open Issues

9.5.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
activity-code | token | Detail type of activity | CarePlan.activity.detail.code
activity-date | date | Specified date occurs within period specified by CarePlan.activity.timingSchedule | CarePlan.activity.detail.scheduled
activity-reference | reference | Activity details defined in specific resource | CarePlan.activity.reference (Appointment, ReferralRequest, MedicationRequest, Task, NutritionOrder, RequestGroup, VisionPrescription, ProcedureRequest, DeviceRequest, CommunicationRequest)
based-on | reference | Fulfills care plan | CarePlan.basedOn (CarePlan)
care-team | reference | Who's involved in plan? | CarePlan.careTeam (CareTeam)
category | token | Type of plan | CarePlan.category
condition | reference | Health issues this plan addresses | CarePlan.addresses (Condition)
context | reference | Created in context of | CarePlan.context (EpisodeOfCare, Encounter)
date | date | Time period plan covers | CarePlan.period | 18 Resources
definition | reference | Protocol or definition | CarePlan.definition (Questionnaire, PlanDefinition)
encounter | reference | Created in context of | CarePlan.context (Encounter)
goal | reference | Desired outcome of plan | CarePlan.goal (Goal)
identifier | token | External Ids for this plan | CarePlan.identifier | 26 Resources
intent | token | proposal | plan | order | option | CarePlan.intent
part-of | reference | Part of referenced CarePlan | CarePlan.partOf (CarePlan)
patient | reference | Who care plan is for | CarePlan.subject (Patient) | 31 Resources
performer | reference | Matches if the practitioner is listed as a performer in any of the "simple" activities. (For performers of the detailed activities, chain through the activitydetail search parameter.) | CarePlan.activity.detail.performer (Practitioner, Organization, CareTeam, Patient, RelatedPerson)
replaces | reference | CarePlan replaced by this CarePlan | CarePlan.replaces (CarePlan)
status | token | draft | active | suspended | completed | entered-in-error | cancelled | unknown | CarePlan.status
subject | reference | Who care plan is for | CarePlan.subject (Group, Patient)

9.5.6 Resource CarePlan - Examples

URL: https://hl7.org/fhir/STU3/careplan-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.5.6 Resource CarePlan - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Example List:

General Person Example | example | XML | JSON | Turtle
Patient 1 for linking | preg | XML | JSON | Turtle
Simple integrated care plan with multiple goals & conditions (from Mayo) | integrate | XML | JSON | Turtle
Encounter to GP (Primary care Practitioner) | gpvisit | XML | JSON | Turtle
Real-world patient example | f001 | XML | JSON | Turtle
Real-world patient example | f002 | XML | JSON | Turtle
Real-world patient example | f003 | XML | JSON | Turtle
Real-world patient example | f201 | XML | JSON | Turtle
Real-world patient example | f202 | XML | JSON | Turtle
Real-world patient example | f203 | XML | JSON | Turtle
Care plan to address obesity that is primarily narrative | obesity-narrative | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.5.7 Resource CarePlan - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/careplan-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.5.7 Resource CarePlan - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the CarePlan resource.

CarePlan
Definition | Describes the intention of how one or more practitioners intend to deliver care for a particular patient, group or community for a period of time, possibly limited to care for a specific condition or set of conditions.
Control | 1..1
Alternate Names | Care Team
CarePlan.identifier
Definition | This records identifiers associated with this care plan that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Need to allow connection to a wider workflow.
Summary | true
CarePlan.definition
Definition | Identifies the protocol, questionnaire, guideline or other specification the care plan should be conducted in accordance with.
Control | 0..*
Type | Reference(PlanDefinition | Questionnaire)
Summary | true
CarePlan.basedOn
Definition | A care plan that is fulfilled in whole or in part by this care plan.
Control | 0..*
Type | Reference(CarePlan)
Requirements | Allows tracing of the care plan and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
Summary | true
CarePlan.replaces
Definition | Completed or terminated care plan whose function is taken by this new care plan.
Control | 0..*
Type | Reference(CarePlan)
Requirements | Allows tracing the continuation of a therapy or administrative process instantiated through multiple care plans.
Alternate Names | supersedes
Summary | true
Comments | The replacement could be because the initial care plan was immediately rejected (due to an issue) or because the previous care plan was completed, but the need for the action described by the care plan remains ongoing.
CarePlan.partOf
Definition | A larger care plan of which this particular care plan is a component or step.
Control | 0..*
Type | Reference(CarePlan)
Summary | true
Comments | Each care plan is an independent request, such that having a care plan be part of another care plan can cause issues with cascading statuses. As such, this element is still being discussed.
CarePlan.status
Definition | Indicates whether the plan is currently being acted upon, represents future intentions or is now a historical record.
Control | 1..1
Terminology Binding | CarePlanStatus (Required)
Type | code
Is Modifier | true
Requirements | Allows clinicians to determine whether the plan is actionable or not.

Summary | true
Comments | The unknown code is not to be used to convey other statuses. The unknown code should be used when one of the statuses applies, but the authoring system doesn't know the current state of the care plan. This element is labeled as a modifier because the status contains the code entered-in-error] that marks the plan as not currently valid.
CarePlan.intent
Definition | Indicates the level of authority/intentionality associated with the care plan and where the care plan fits into the workflow chain.
Control | 1..1
Terminology Binding | CarePlanIntent (Required)
Type | code
Is Modifier | true
Requirements | Proposals/recommendations, plans and orders all use the same structure and can exist in the same fulfillment chain.
Summary | true
Comments | This element is labeled as a modifier because the intent alters when and how the resource is actually applicable.
CarePlan.category
Definition | Identifies what "kind" of plan this is to support differentiation between multiple co-existing plans; e.g. "Home health", "psychiatric", "asthma", "disease management", "wellness plan", etc.
Control | 0..*
Terminology Binding | Care Plan Category (Example)
Type | CodeableConcept
Requirements | Used for filtering what plan(s) are retrieved and displayed to different types of users.
Summary | true
Comments | There may be multiple axis of categorization and one plan may serve multiple purposes. In some cases, this may be redundant with references to CarePlan.concern.
CarePlan.title
Definition | Human-friendly name for the CarePlan.
Control | 0..1
Type | string
Summary | true
CarePlan.description
Definition | A description of the scope and nature of the plan.
Control | 0..1
Type | string
Requirements | Provides more detail than conveyed by category.
Summary | true
CarePlan.subject
Definition | Identifies the patient or group whose intended care is described by the plan.
Control | 1..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true
CarePlan.context
Definition | Identifies the original context in which this particular CarePlan was created.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
Comments | Activities conducted as a result of the care plan may well occur as part of other encounters/episodes.
CarePlan.period
Definition | Indicates when the plan did (or is intended to) come into effect and end.
Control | 0..1
Type | Period
Requirements | Allows tracking what plan(s) are in effect at a particular time.
Alternate Names | timing
Summary | true
Comments | Any activities scheduled as part of the plan should be constrained to the specified period regardless of whether the activities are planned within a single encounter/episode or across multiple encounters/episodes (e.g. the longitudinal management of a chronic condition).
CarePlan.author
Definition | Identifies the individual(s) or ogranization who is responsible for the content of the care plan.
Control | 0..*
Type | Reference(Patient | Practitioner | RelatedPerson | Organization | CareTeam)
Summary | true
Comments | Collaborative care plans may have multiple authors.
CarePlan.careTeam
Definition | Identifies all people and organizations who are expected to be involved in the care envisioned by this plan.
Control | 0..*
Type | Reference(CareTeam)
Requirements | Allows representation of care teams, helps scope care plan. In some cases may be a determiner of access permissions.
CarePlan.addresses
Definition | Identifies the conditions/problems/concerns/diagnoses/etc. whose management and/or mitigation are handled by this plan.
Control | 0..*
Type | Reference(Condition)
Requirements | Links plan to the conditions it manages. The element can identify risks addressed by the plan as well as active conditions. (The Condition resource can include things like "at risk for hypertension" or "fall risk".) Also scopes plans - multiple plans may exist addressing different

concerns.
Summary | true
CarePlan.supportingInfo
Definition | Identifies portions of the patient's record that specifically influenced the formation of the plan. These might include co-morbidities, recent procedures, limitations, recent assessments, etc.
Control | 0..*
Type | Reference(Any)
Requirements | Identifies barriers and other considerations associated with the care plan.
Comments | Use "concern" to identify specific conditions addressed by the care plan.
CarePlan.goal
Definition | Describes the intended objective(s) of carrying out the care plan.
Control | 0..*
Type | Reference(Goal)
Requirements | Provides context for plan. Allows plan effectiveness to be evaluated by clinicians.
Comments | Goal can be achieving a particular change or merely maintaining a current state or even slowing a decline.
CarePlan.activity
Definition | Identifies a planned action to occur as part of the plan. For example, a medication to be used, lab tests to perform, self-monitoring, education, etc.
Control | 0..*
Requirements | Allows systems to prompt for performance of planned activities, and validate plans against best practice.
Invariants | Defined on this element cpl-3: Provide a reference or detail, not both (expression : detail.empty() or reference.empty(), xpath: not(exists(f:detail)) or not(exists(f:reference)))
CarePlan.activity.outcomeCodeableConcept
Definition | Identifies the outcome at the point when the status of the activity is assessed. For example, the outcome of an education activity could be patient understands (or not).
Control | 0..*
Terminology Binding | Care Plan Activity Outcome (Example)
Type | CodeableConcept
Comments | Note that this should not duplicate the activity status (e.g. completed or in progress).
CarePlan.activity.outcomeReference
Definition | Details of the outcome or action resulting from the activity. The reference to an "event" resource, such as Procedure or Encounter or Observation, is the result/outcome of the activity itself. The activity can be conveyed using CarePlan.activity.detail OR using the CarePlan.activity.reference (a reference to a “request” resource).
Control | 0..*
Type | Reference(Any)
Requirements | Links plan to resulting actions.
Comments | The activity outcome is independent of the outcome of the related goal(s). For example, if the goal is to achieve a target body weight of 150 lb and an activity is defined to diet, then the activity outcome could be calories consumed whereas the goal outcome is an observation for the actual body weight measured.
CarePlan.activity.progress
Definition | Notes about the adherence/status/progress of the activity.
Control | 0..*
Type | Annotation
Requirements | Can be used to capture information about adherence, progress, concerns, etc.
Comments | This element should NOT be used to describe the activity to be performed - that occurs either within the resource pointed to by activity.detail.reference or in activity.detail.description.
CarePlan.activity.reference
Definition | The details of the proposed activity represented in a specific resource.
Control | 0..1
Type | Reference(Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ProcedureRequest | ReferralRequest | VisionPrescription | RequestGroup)
Requirements | Details in a form consistent with other applications and contexts of use.
Comments | Standard extension exists (goal-pertainstogoal) that allows goals to be referenced from any of the referenced resources in CarePlan.activity.reference. The goal should be visible when the resource referenced by CarePlan.activity.reference is viewed indepedently from the CarePlan. Requests that are pointed to by a CarePlan using this element should not point to this CarePlan using the "basedOn" element. i.e. Requests that are part of a CarePlan are not "based on" the CarePlan.
Invariants | Affect this element cpl-3: Provide a reference or detail, not both (expression : detail.empty() or reference.empty(), xpath: not(exists(f:detail)) or not(exists(f:reference)))
CarePlan.activity.detail
Definition | A simple summary of a planned activity suitable for a general care plan system (e.g. form driven) that doesn't know about specific resources such as procedure etc.
Control | 0..1
Requirements | Details in a simple form for generic care plan systems.
Invariants | Affect this element cpl-3: Provide a reference or detail, not both (expression : detail.empty() or reference.empty(), xpath: not(exists(f:detail)) or not(exists(f:reference)))
CarePlan.activity.detail.category
Definition | High-level categorization of the type of activity in a care plan.
Control | 0..1
Terminology Binding | CarePlanActivityCategory (Example)
Type | CodeableConcept
Requirements | May determine what types of extensions are permitted.
CarePlan.activity.detail.definition
Definition | Identifies the protocol, questionnaire, guideline or other specification the planned activity should be conducted in accordance with.
Control | 0..1

Type | Reference(PlanDefinition | ActivityDefinition | Questionnaire)
Requirements | Allows Questionnaires that the patient (or practitioner) should fill in to fulfill the care plan activity.
CarePlan.activity.detail.code
Definition | Detailed description of the type of planned activity; e.g. What lab test, what procedure, what kind of encounter.
Control | 0..1
Terminology Binding | Care Plan Activity (Example)
Type | CodeableConcept
Requirements | Allows matching performed to planned as well as validation against protocols.
Comments | Tends to be less relevant for activities involving particular products. Codes should not convey negation - use "prohibited" instead.
CarePlan.activity.detail.reasonCode
Definition | Provides the rationale that drove the inclusion of this particular activity as part of the plan or the reason why the activity was prohibited.
Control | 0..*
Terminology Binding | Activity Reason (Example)
Type | CodeableConcept
Comments | This could be a diagnosis code. If a full condition record exists or additional detail is needed, use reasonCondition instead.
CarePlan.activity.detail.reasonReference
Definition | Provides the health condition(s) that drove the inclusion of this particular activity as part of the plan.
Control | 0..*
Type | Reference(Condition)
Comments | Conditions can be identified at the activity level that are not identified as reasons for the overall plan.
CarePlan.activity.detail.goal
Definition | Internal reference that identifies the goals that this activity is intended to contribute towards meeting.
Control | 0..*
Type | Reference(Goal)
Requirements | So that participants know the link explicitly.
CarePlan.activity.detail.status
Definition | Identifies what progress is being made for the specific activity.
Control | 1..1
Terminology Binding | CarePlanActivityStatus (Required)
Type | code
Is Modifier | true
Requirements | Indicates progress against the plan, whether the activity is still relevant for the plan.
Comments | Some aspects of status can be inferred based on the resources linked in actionTaken. Note that "status" is only as current as the plan was most recently updated. The unknown code is not to be used to convey other statuses. The unknown code should be used when one of the statuses applies, but the authoring system doesn't know the current state of the activity.
CarePlan.activity.detail.statusReason
Definition | Provides reason why the activity isn't yet started, is on hold, was cancelled, etc.
Control | 0..1
Type | string
Comments | Will generally not be present if status is "complete". Be sure to prompt to update this (or at least remove the existing value) if the status is changed.
CarePlan.activity.detail.prohibited
Definition | If true, indicates that the described activity is one that must NOT be engaged in when following the plan. If false, indicates that the described activity is one that should be engaged in when following the plan.
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | false
Requirements | Captures intention to not do something that may have been previously typical.
Comments | This element is labeled as a modifier because it marks an activity as an activity that is not to be performed.
CarePlan.activity.detail.scheduled[x]
Definition | The period, timing or frequency upon which the described activity is to occur.
Control | 0..1
Type | Timing|Period|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Allows prompting for activities and detection of missed planned activities.
CarePlan.activity.detail.location
Definition | Identifies the facility where the activity will occur; e.g. home, hospital, specific clinic, etc.
Control | 0..1
Type | Reference(Location)
Requirements | Helps in planning of activity.
Comments | May reference a specific clinical location or may identify a type of location.
CarePlan.activity.detail.performer
Definition | Identifies who's expected to be involved in the activity.
Control | 0..*
Type | Reference(Practitioner | Organization | RelatedPerson | Patient | CareTeam)
Requirements | Helps in planning of activity.
Comments | A performer MAY also be a participant in the care plan.
CarePlan.activity.detail.product[x]
Definition | Identifies the food, drug or other product to be consumed or supplied in the activity.
Control | 0..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication | Substance)

[x] Note | See Choice of Data Types for further information about how to use [x]
CarePlan.activity.detail.dailyAmount
Definition | Identifies the quantity expected to be consumed in a given day.
Control | 0..1
Type | SimpleQuantity
Requirements | Allows rough dose checking.
Alternate Names | daily dose
CarePlan.activity.detail.quantity
Definition | Identifies the quantity expected to be supplied, administered or consumed by the subject.
Control | 0..1
Type | SimpleQuantity
CarePlan.activity.detail.description
Definition | This provides a textual description of constraints on the intended activity occurrence, including relation to other activities. It may also include objectives, pre-conditions and end-conditions. Finally, it may convey specifics about the activity such as body site, method, route, etc.
Control | 0..1
Type | string
CarePlan.note
Definition | General notes about the care plan not covered elsewhere.
Control | 0..*
Type | Annotation
Requirements | Used to capture information that applies to the plan as a whole that doesn't fit into discrete elements.

9.5.8 Resource CarePlan - Mappings

URL: https://hl7.org/fhir/STU3/careplan-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.5.8 Resource CarePlan - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the careplan resource.

9.5.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

CarePlan | Act[classCode=PCPR, moodCode=INT]
identifier | .id
definition
basedOn
replaces
partOf
status | .statusCode planned = new active = active completed = completed
intent
category
title
description
subject | .participation[typeCode=PAT].role[classCode=PAT]
context | .
period | .effectiveTime
author
careTeam
addresses | .actRelationship[typeCode=SUBJ].target[classCode=CONC, moodCode=EVN]
supportingInfo
goal | .outboundRelationship[typeCode<=OBJ].
activity | .outboundRelationship[typeCode=COMP].target
outcomeCodeableConcept
outcomeReference | .inboundRelationship[typeCode=FLFS].source
progress | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value
reference | .outboundRelationship[typeCode=COMP].target
detail | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target
category | .inboundRelationship[typeCode=COMP].source[classCode=LIST].code
definition
code | .code
reasonCode
reasonReference
goal | .outboundRelationship[typeCode<=OBJ].

status | .statusCode not-started = new scheduled = not-started (and fulfillment relationship to appointent) in-progress = active on-hold = suspended completed = completed cancelled = aborted
statusReason
prohibited | actionNegationInd
scheduled[x] | .effectiveTime
location | .participation[typeCode=LOC].role
performer | .participation[typeCode=PFM]
product[x] | .participation[typeCode=PRD].role
dailyAmount | .outboundRelationship[typeCode=COMP][classCode=SBADM].doseQuantity
quantity | .outboundRelationship[typeCode=COMP][classCode=SPLY].quantity
description | .text
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value

9.5.8.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

CarePlan
identifier | PTH-3
definition
basedOn
replaces
partOf
status | PTH-5
intent
category
title
description
subject | PID-3
context | Associated PV1
period | GOL-7 / GOL-8
author
careTeam
addresses | PRB-4
supportingInfo
goal | GOL.1
activity
outcomeCodeableConcept
outcomeReference
progress | NTE?
reference
detail
category
definition
code | OBR-4 / RXE-2 / RXO-1 / RXD-2
reasonCode
reasonReference
goal
status | ORC-5?
statusReason
prohibited
scheduled[x] | TQ1
location | OBR-24(???!!)
performer | PRT-5 : (PRV-4 e {provider participations}) / PRT-5 : (PRV-4 e {non-provider person participations}) / PRT-5 : (PRV-4 = (patient non-subject of care)) / PRT-8
product[x] | RXE-2 / RXO-1 / RXD-2
dailyAmount | RXO-23 / RXE-19 / RXD-12
quantity | RXO-11 / RXE-10 / RXD-4 / RXG-5 / RXA-6 / TQ1-2.1 *and* RXO-12 / RXE-11 / RXD-5 / RXG-7 / RXA-7 / TQ1-2.2
description | NTE?
note | NTE?

9.5.8.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

CarePlan | Request
identifier | Request.identifier
definition | Request.definition
basedOn | Request.basedOn
replaces | Request.replaces
partOf
status | Request.status (uses different ValueSet)
intent | Request.intent
category | Request.supportingInfo
title
description
subject | Request.subject
context | Request.context

period | Request.occurrencePeriod
author | Request.requester.agent (if only one author)
careTeam | Request.performer (similar but does not entail CareTeam)
addresses | Request.reasonReference
supportingInfo | Request.supportingInfo
goal
activity | no mapping NOTE: This is a list of contained Request-Event tuples!
outcomeCodeableConcept
outcomeReference | Event that is outcome of Request in activity.reference
progress | Event.status (Annotation instead of CodableConcept)
reference | Request that resulted in Event in activity.actionResulting
detail
category | Request.supportingInformation
definition | Request.definition
code | Request.code
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
goal | Request.goal
status | Request.status (uses different ValueSet)
statusReason
prohibited
scheduled[x] | Request.occurrence
location
performer | Request.performer
product[x]
dailyAmount
quantity
description | Request.note
note | Request.note

9.5.9 Resource CarePlan - Design Notes

URL: https://hl7.org/fhir/STU3/careplan-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.5.9 Resource CarePlan - Design Notes

to do

9.5.10 Resource CarePlan - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/careplan-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.5.10 Resource CarePlan - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the CarePlan resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.5.10.1 Profiles

No Profiles defined for this resource

9.5.10.2 Extensions

careplan-activity-title | activity-title | for Common Care Plan Extensions

Extensions that reference this resource:

condition-basedOn | basedOn | for Condition HL7 Extensions
openEHR-careplan | careplan | for openEHR Archetype Profile

9.5.10.3 Search Extensions

No Search Extensions defined for this resource

9.5.11 Common Care Plan Extensions

URL: https://hl7.org/fhir/STU3/careplan-extensions.html

A set of commonly used extensions that don't make the usage requirements for inclusion in the resource itself

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.5.11 Common Care Plan Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.5.11.1 Content

Extensions:
careplan-activity-title | activity-title : Human-friendly name for the activity.

9.5.12 Resource CarePlan - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/careplan-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.5.12 Resource CarePlan - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for CarePlan.

Functional status for this map: 10 tests that all execute ok. 8 fail round-trip testing and 10 r3 resources are invalid (33 errors). (see documentation)

9.5.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/CarePlan2to3" = "R2 to R3 Conversions for CarePlan"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/CarePlan" alias CarePlanR2 as source
uses "http://hl7.org/fhir/StructureDefinition/CarePlan" alias CarePlan as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  CarePlan extends DomainResource
  input src : CarePlanR2 as source
  input tgt : CarePlan as target

  "CarePlan-identifier" : for src.identifier make tgt.identifier
  "CarePlan-subject" : for src.subject make tgt.subject
  "CarePlan-status" : for src.status make tgt.status
  "CarePlan-context" : for src.context make tgt.context
  "CarePlan-period" : for src.period make tgt.period
  "CarePlan-author" : for src.author make tgt.author
  "CarePlan-modified" : for src.modified as v make tgt.extension as t, t.url =
    'http://hl7.org/fhir/CarePlan-r2-modified', t.value = v
  "CarePlan-category" : for src.category make tgt.category
  "CarePlan-description" : for src.description make tgt.description
  "CarePlan-addresses" : for src.addresses make tgt.addresses
  "CarePlan-support" : for src.support make tgt.supportingInfo
  // "CarePlan-relatedPlan1" : for src.relatedPlan as vs0 where vs0.code = 'includes' make tgt.basedOn = (vs0.plan) this
     relationship is inverted to partOf, and that's beyond the scope of the transforms to do
  "CarePlan-relatedPlan2" : for src.relatedPlan as vs0 where vs0.code = 'replaces' make tgt.basedOn = (vs0.plan)
  "CarePlan-relatedPlan3" : for src.relatedPlan as vs0 where vs0.code = 'fulfills' make tgt.replaces = (vs0.plan)
  "CarePlan-participant" : for src where src.participant.exists() make create("CareTeam") as vt0 then
    plnParticipant(src, vt0, tgt)
  "CarePlan-goal" : for src.goal make tgt.goal
  "CarePlan-activity" : for src.activity as vs0 make tgt.activity as vt0 then plnActivity(vs0, vt0)
  "CarePlan-note" : for src.note make tgt.note
endgroup

group plnRelated extends BackboneElement
  input src as source
  input tgt as target

  "CarePlan.relatedPlan-code" : for src.code make tgt.code
  "CarePlan.relatedPlan-plan" : for src.plan make tgt.plan
endgroup

group plnActivity extends BackboneElement
  input src as source
  input tgt as target

  "CarePlan.activity-actionResulting" : for src.actionResulting make tgt.outcomeReference
  "CarePlan.activity-progress" : for src.progress make tgt.progress
  "CarePlan.activity-reference" : for src.reference make tgt.reference
  "CarePlan.activity-detail" : for src.detail as vs0 make tgt.detail as vt0 then plnDetail(vs0, vt0)
endgroup

group plnDetail extends BackboneElement
  input src as source
  input tgt as target

  "CarePlan.activity.detail-category" : for src.category make tgt.category
```

```
  "CarePlan.activity.detail-code" : for src.code make tgt.code
  "CarePlan.activity.detail-reasonCode" : for src.reasonCode make tgt.reasonCode
  "CarePlan.activity.detail-reasonReference" : for src.reasonReference make tgt.reasonReference
  "CarePlan.activity.detail-goal" : for src.goal make tgt.goal
  "CarePlan.activity.detail-status" : for src.status make tgt.status
  "CarePlan.activity.detail-statusReason" : for src.statusReason make tgt.statusReason
  "CarePlan.activity.detail-prohibited" : for src.prohibited make tgt.prohibited
  "CarePlan.activity.detail-scheduled" : for src.scheduled make tgt.scheduled
  "CarePlan.activity.detail-location" : for src.location make tgt.location
  "CarePlan.activity.detail-performer" : for src.performer make tgt.performer
  "CarePlan.activity.detail-product" : for src.product make tgt.product
  "CarePlan.activity.detail-dailyAmount" : for src.dailyAmount make tgt.dailyAmount
  "CarePlan.activity.detail-quantity" : for src.quantity make tgt.quantity
  "CarePlan.activity.detail-description" : for src.description make tgt.description
endgroup

group plnParticipant
  input src as source
  input tgt as target
  input tgtE as target

  "CarePlan.participant-link" : for src make tgtE.careTeam = reference(tgt)
  "CarePlan.participant-member" : for src.participant as vs make tgt.participant as vt then plnParticipantMember(vs, vt)
endgroup

group plnParticipantMember extends BackboneElement
  input src as source
  input tgt as target

  "CarePlan.participant-role" : for src.role make tgt.role
  "CarePlan.participant-member" : for src.member make tgt.member
endgroup
```

9.5.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/CarePlan3to2" = "R3 to R2 Conversion for CarePlan"

uses "http://hl7.org/fhir/StructureDefinition/CarePlan" alias CarePlan as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/CarePlan" alias CarePlanR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  CarePlan extends DomainResource
  input src : CarePlan as source
  input tgt : CarePlanR2 as target

  "CarePlan-identifier" : for src.identifier make tgt.identifier
  "CarePlan-subject" : for src.subject make tgt.subject
  "CarePlan-status" : for src.status make tgt.status
  "CarePlan-context" : for src.context make tgt.context
  "CarePlan-period" : for src.period make tgt.period
  "CarePlan-author" : for src.author make tgt.author
  "CarePlan-modified" : for src.extension as e where e.url = 'http://hl7.org/fhir/CarePlan-r2-modified' make
    tgt.modified = (e.value)
  "CarePlan-category" : for src.category make tgt.category
  "CarePlan-description" : for src.description make tgt.description
  "CarePlan-addresses" : for src.addresses make tgt.addresses
  "CarePlan-support" : for src.supportingInfo make tgt.support
  "CarePlan-relatedPlan2" : for src.basedOn as vs0 make tgt.relatedPlan as rp, rp.code = 'fulfills', rp.plan = v
  "CarePlan-relatedPlan3" : for src.replaces as vs0 make tgt.relatedPlan as rp, rp.code = 'replaces', rp.plan = v
  "CarePlan-participant" : for src.careTeam as vs where reference.resolve().exists() make
    (src.careTeam.reference.resolve()) as ct then plnParticipant(ct, tgt)
  "CarePlan-goal" : for src.goal make tgt.goal
  "CarePlan-activity" : for src.activity as vs0 make tgt.activity as vt0 then plnActivity(vs0, vt0)
  "CarePlan-note" : for src.note make tgt.note
endgroup

group plnActivity extends BackboneElement
  input src as source
  input tgt as target

  "CarePlan.activity-actionResulting" : for src.outcomeReference make tgt.actionResulting
  "CarePlan.activity-progress" : for src.progress make tgt.progress
  "CarePlan.activity-reference" : for src.reference make tgt.reference
  "CarePlan.activity-detail" : for src.detail as vs0 make tgt.detail as vt0 then plnDetail(vs0, vt0)
endgroup

group plnDetail extends BackboneElement
  input src as source
  input tgt as target

  "CarePlan.activity.detail-category" : for src.category make tgt.category
  "CarePlan.activity.detail-code" : for src.code make tgt.code
  "CarePlan.activity.detail-reasonCode" : for src.reasonCode make tgt.reasonCode
  "CarePlan.activity.detail-reasonReference" : for src.reasonReference make tgt.reasonReference
  "CarePlan.activity.detail-goal" : for src.goal make tgt.goal
  "CarePlan.activity.detail-status" : for src.status make tgt.status
  "CarePlan.activity.detail-statusReason" : for src.statusReason make tgt.statusReason
  "CarePlan.activity.detail-prohibited" : for src.prohibited make tgt.prohibited
  "CarePlan.activity.detail-scheduled" : for src.scheduled make tgt.scheduled
```

```
  "CarePlan.activity.detail-location" : for src.location make tgt.location
  "CarePlan.activity.detail-performer" : for src.performer make tgt.performer
  "CarePlan.activity.detail-product" : for src.product make tgt.product
  "CarePlan.activity.detail-dailyAmount" : for src.dailyAmount make tgt.dailyAmount
  "CarePlan.activity.detail-quantity" : for src.quantity make tgt.quantity
  "CarePlan.activity.detail-description" : for src.description make tgt.description
endgroup


group plnParticipant
  input src as source
  input tgt as target

  "CarePlan.participant-unlink" : for src.participant as vs make tgt.participant as vt then plnParticipantMember(vs, vt)
endgroup

group plnParticipantMember extends BackboneElement
  input src as source
  input tgt as target

  "CarePlan.participant-role" : for src.role make tgt.role
  "CarePlan.participant-member" : for src.member make tgt.member
endgroup
```

9.5.12.3 R3 Validation Errors

example | ERROR @ CarePlan: If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource [contained.where(('#'+id in %resource.descendants().reference).not()).empty()] (src = InstanceValidator) ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.contained[1]: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.contained[3].extension: The extension http://hl7.org/fhir/StructureDefinition/goal-target is unknown, and not allowed here (src = InstanceValidator) ERROR @ CarePlan.addresses.contained: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.goal.contained.extension: The extension http://hl7.org/fhir/StructureDefinition/goal-target is unknown, and not allowed here (src = InstanceValidator) ERROR @ CarePlan.activity.detail: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.activity.detail.status': minimum required = 1, but only found 0 (src = InstanceValidator)

gpvisit | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.contained[1]: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.addresses.contained: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)

f001 | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.extension: The extension http://hl7.org/fhir/CarePlan-r2-modified is unknown, and not allowed here (src = InstanceValidator)

f002 | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.extension: The extension http://hl7.org/fhir/CarePlan-r2-modified is unknown, and not allowed here (src = InstanceValidator)

f003 | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.extension: The extension http://hl7.org/fhir/CarePlan-r2-modified is unknown, and not allowed here (src = InstanceValidator)

f201 | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.extension: The extension http://hl7.org/fhir/CarePlan-r2-modified is unknown, and not allowed here (src = InstanceValidator)

f202 | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator)

f203 | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.extension: The extension http://hl7.org/fhir/CarePlan-r2-modified is unknown, and not allowed here (src = InstanceValidator)

integrate | ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.contained[1]: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.contained[2]: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.contained[3]: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.extension: The extension http://hl7.org/fhir/CarePlan-r2-modified is unknown, and not allowed here (src = InstanceValidator) ERROR @ CarePlan.addresses[1].contained: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.addresses[2].contained: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.addresses[3].contained: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)

preg | ERROR @ CarePlan: If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource [contained.where(('#'+id in %resource.descendants().reference).not()).empty()] (src = InstanceValidator) ERROR @ CarePlan: Profile http://hl7.org/fhir/StructureDefinition/CarePlan, Element 'CarePlan.intent': minimum required = 1, but only found 0 (src = InstanceValidator) ERROR @ CarePlan.contained[1]: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator) ERROR @ CarePlan.addresses.contained: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error [verificationStatus='entered-in-error' or clinicalStatus.exists()] (src = InstanceValidator)

9.6 Resource Goal - Content

URL: https://hl7.org/fhir/STU3/goal.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Describes the intended objective(s) for a patient, group or organization care, for example, weight loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process improvement objective, etc.

9.6.1 Scope and Usage

A Goal in health care services delivery is generally an expressed desired health state to be achieved by a subject of care (or family/group) over a period or at a specific point of time. This desired target health state may be achieved as a result of health care intervention(s) or resulting from natural recovery over time. For example:

* A goal of a plan for a condition such as a diabetes might specify desired outcome(s) (e.g. HgbA1c level =<5.6% in 3 months) as a result of interventions such as medication therapy, nutritional management and/or increase physical activity.

* A goal of a procedure might be to meet the intended objective of the procedure (e.g. wet-dry-dressing changes twice a day; goal: wound healed completely in 2 weeks) or to prevent an unintended complication (e.g. repositioning a patient every two hours: goal to maintain skin integrity)

Goals may address the prevention of illness, cure or mitigation of a condition, prolongation of life, or mitigation of pain and discomfort.

When dealing with groups, goals may also reflect health state, such as a reduction of addiction behaviors. However, they may also reflect population health objectives such as education, screening, etc.

Organizational goals are typically not health state specific but may instead identify measurement targets such as infection control, cost management, patient satisfaction, etc.

9.6.2 Boundaries and Relationships

Goals are typically established in the context of a CarePlan. However, goals may also be directly referenced by request-type resources (e.g. MedicationRequest or ReferralRequest) by using an extension.

A goal represents a specific goal instance for a particular patient, group, etc. It is not intended to be used to define types of potential goals as part of an order set or protocol definition. Protocol definitions and order sets are supported through PlanDefinition. The Goal resource is intended to be used once an order set is instantiated or assigned to a patient, which is when the potential goals become the actual goals, if not changed or deleted.

Goals are often evaluated using Observations.

This resource is referenced by careplan

9.6.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Goal | | | DomainResource | Describes the intended objective(s) for a patient, group or organization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External Ids for this goal
status | ?!? | 1..1 | code | proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected GoalStatus (Required)
category | ? | 0..* | CodeableConcept | E.g. Treatment, dietary, behavioral, etc. GoalCategory (Example)
priority | ? | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority GoalPriority (Preferred)
description | ? | 1..1 | CodeableConcept | Code or text describing goal SNOMED CT Clinical Findings (Example)
subject | ? | 0..1 | Reference(Patient | Group | Organization) | Who this goal is intended for
start[x] | ? | 0..1 | | When goal pursuit begins GoalStartEvent (Example)
startDate | | | date
startCodeableConcept | | | CodeableConcept
target | I | 0..1 | BackboneElement | Target outcome for the goal + Goal.target.measure is required if Goal.target.detail is populated
measure | ?I | 0..1 | CodeableConcept | The parameter whose value is being tracked LOINC Codes (Example)
detail[x] | ?I | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due[x] | ? | 0..1 | | Reach goal on or before
dueDate | | | date
dueDuration | | | Duration

statusDate | ? | 0..1 | date | When goal status took effect
statusReason | | 0..1 | string | Reason for current status
expressedBy | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson) | Who's responsible for creating Goal?
addresses | | 0..* | Reference(Condition | Observation | MedicationStatement | NutritionOrder | ProcedureRequest | RiskAssessment) | Issues addressed by this goal
note | | 0..* | Annotation | Comments about the goal
outcomeCode | | 0..* | CodeableConcept | What result was achieved regarding the goal? SNOMED CT Clinical Findings (Example)
outcomeReference | | 0..* | Reference(Observation) | Observation that resulted from goal
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Goal xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this goal --></identifier>
 <status value="[code]"/><!-- 1..1 proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-
    target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected -->
 <category><!-- 0..* CodeableConcept E.g. Treatment, dietary, behavioral, etc. --></category>
 <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
 <description><!-- 1..1 CodeableConcept Code or text describing goal --></description>
 <subject><!-- 0..1 Reference(Patient|Group|Organization) Who this goal is intended for --></subject>
 <start[x]><!-- 0..1 date|CodeableConcept When goal pursuit begins --></start[x]>
 <target>  <!--  0..1 Target outcome for the goal -->
  <measure><!--  0..1 CodeableConcept The parameter whose value is being tracked --></measure>
  <detail[x]><!--  0..1 Quantity|Range|CodeableConcept The target value to be achieved --></detail[x]>
  <due[x]><!-- 0..1 date|Duration Reach goal on or before --></due[x]>
 </target>
 <statusDate value="[date]"/><!-- 0..1 When goal status took effect -->
 <statusReason value="[string]"/><!-- 0..1 Reason for current status -->
 <expressedBy><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson) Who's responsible for creating Goal?
    --></expressedBy>
 <addresses><!-- 0..* Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ProcedureRequest|RiskAssessment) Issues addressed by this goal --></addresses>
 <note><!-- 0..* Annotation Comments about the goal --></note>
 <outcomeCode><!-- 0..* CodeableConcept What result was achieved regarding the goal? --></outcomeCode>
 <outcomeReference><!-- 0..* Reference(Observation) Observation that resulted from goal --></outcomeReference>
</Goal>
```

JSON Template

```
{
  "resourceType" : "Goal",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this goal
  "status" : "<code>", // R!  proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-target
    | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected
  "category" : [{ CodeableConcept }], // E.g. Treatment, dietary, behavioral, etc.
  "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
  "description" : { CodeableConcept }, // R!  Code or text describing goal
  "subject" : { Reference(Patient|Group|Organization) }, // Who this goal is intended for
  // start[x]: When goal pursuit begins. One of these 2:
  "startDate" : "<date>",
  "startCodeableConcept" : { CodeableConcept },
  "target" : { // C? Target outcome for the goal
    "measure" : { CodeableConcept }, // C? The parameter whose value is being tracked
    // detail[x]: The target value to be achieved. One of these 3:
    "detailQuantity" : { Quantity },
    "detailRange" : { Range },
    "detailCodeableConcept" : { CodeableConcept },
    // due[x]: Reach goal on or before. One of these 2:
    "dueDate" : "<date>"
    "dueDuration" : { Duration }
  },
  "statusDate" : "<date>", // When goal status took effect
  "statusReason" : "<string>", // Reason for current status
  "expressedBy" : { Reference(Patient|Practitioner|RelatedPerson) }, // Who's responsible for creating Goal?
  "addresses" : [{ Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ProcedureRequest|RiskAssessment) }], // Issues addressed by this goal
  "note" : [{ Annotation }], // Comments about the goal
  "outcomeCode" : [{ CodeableConcept }], // What result was achieved regarding the goal?
  "outcomeReference" : [{ Reference(Observation) }] // Observation that resulted from goal
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Goal;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Goal.identifier [ Identifier ], ... ; # 0..* External Ids for this goal
  fhir:Goal.status [ code ]; # 1..1 proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-
    target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected
```

```
  fhir:Goal.category [ CodeableConcept ], ... ; # 0..* E.g. Treatment, dietary, behavioral, etc.
  fhir:Goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
  fhir:Goal.description [ CodeableConcept ]; # 1..1 Code or text describing goal
  fhir:Goal.subject [ Reference(Patient|Group|Organization) ]; # 0..1 Who this goal is intended for
  # Goal.start[x] : 0..1 When goal pursuit begins. One of these 2
    fhir:Goal.startDate [ date ]
    fhir:Goal.startCodeableConcept [ CodeableConcept ]
  fhir:Goal.target [ # 0..1 Target outcome for the goal
    fhir:Goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is being tracked
    # Goal.target.detail[x] : 0..1 The target value to be achieved. One of these 3
      fhir:Goal.target.detailQuantity [ Quantity ]
      fhir:Goal.target.detailRange [ Range ]
      fhir:Goal.target.detailCodeableConcept [ CodeableConcept ]
    # Goal.target.due[x] : 0..1 Reach goal on or before. One of these 2
      fhir:Goal.target.dueDate [ date ]
      fhir:Goal.target.dueDuration [ Duration ]
  ];
  fhir:Goal.statusDate [ date ]; # 0..1 When goal status took effect
  fhir:Goal.statusReason [ string ]; # 0..1 Reason for current status
  fhir:Goal.expressedBy [ Reference(Patient|Practitioner|RelatedPerson) ]; # 0..1 Who's responsible for creating Goal?
  fhir:Goal.addresses [ Reference(Condition|Observation|MedicationStatement|NutritionOrder|ProcedureRequest|
  RiskAssessment) ], ... ; # 0..* Issues addressed by this goal
  fhir:Goal.note [ Annotation ], ... ; # 0..* Comments about the goal
  fhir:Goal.outcomeCode [ CodeableConcept ], ... ; # 0..* What result was achieved regarding the goal?
  fhir:Goal.outcomeReference [ Reference(Observation) ], ... ; # 0..* Observation that resulted from goal
]
```

Changes since DSTU2

Goal

Goal.description | Type changed from string to CodeableConcept

Goal.target | Renamed from target[x] to target Remove date, Remove Quantity{http://hl7.org/fhir/StructureDefinition/Duration}

Goal.target.measure | Added Element

Goal.target.detail[x] | Added Element

Goal.target.due[x] | Added Element

Goal.statusReason | Type changed from CodeableConcept to string

Goal.expressedBy | Renamed from author to expressedBy

Goal.outcomeCode | Added Element

Goal.outcomeReference | Added Element

Goal.outcome | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (2 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

Goal | | | DomainResource | Describes the intended objective(s) for a patient, group or organization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | External Ids for this goal

status | ?!? | 1..1 | code | proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected GoalStatus (Required)

category | ? | 0..* | CodeableConcept | E.g. Treatment, dietary, behavioral, etc. GoalCategory (Example)

priority | ? | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority GoalPriority (Preferred)

description | ? | 1..1 | CodeableConcept | Code or text describing goal SNOMED CT Clinical Findings (Example)

subject | ? | 0..1 | Reference(Patient | Group | Organization) | Who this goal is intended for

start[x] | ? | 0..1 | | When goal pursuit begins GoalStartEvent (Example)

startDate | | | date

startCodeableConcept | | | CodeableConcept

target | I | 0..1 | BackboneElement | Target outcome for the goal + Goal.target.measure is required if Goal.target.detail is populated

measure | ?I | 0..1 | CodeableConcept | The parameter whose value is being tracked LOINC Codes (Example)

detail[x] | ?I | 0..1 | | The target value to be achieved

detailQuantity | | | Quantity

detailRange | | | Range

detailCodeableConcept | | | CodeableConcept

due[x] | ? | 0..1 | | Reach goal on or before

dueDate | | | date

dueDuration | | | Duration

statusDate | ? | 0..1 | date | When goal status took effect

statusReason | | 0..1 | string | Reason for current status

expressedBy | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson) | Who's responsible for creating Goal?

addresses | | 0..* | Reference(Condition | Observation | MedicationStatement | NutritionOrder | ProcedureRequest | RiskAssessment) | Issues addressed by this goal

note | | 0..* | Annotation | Comments about the goal

outcomeCode | | 0..* | CodeableConcept | What result was achieved regarding the goal? SNOMED CT Clinical Findings (Example)

outcomeReference | | 0..* | Reference(Observation) | Observation that resulted from goal

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Goal xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this goal --></identifier>
 <status value="[code]"/><!-- 1..1 proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-
    target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected -->
 <category><!-- 0..* CodeableConcept E.g. Treatment, dietary, behavioral, etc. --></category>
 <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
 <description><!-- 1..1 CodeableConcept Code or text describing goal --></description>
 <subject><!-- 0..1 Reference(Patient|Group|Organization) Who this goal is intended for --></subject>
 <start[x]><!-- 0..1 date|CodeableConcept When goal pursuit begins --></start[x]>
 <target>  <!--  0..1 Target outcome for the goal -->
  <measure><!--  0..1 CodeableConcept The parameter whose value is being tracked --></measure>
  <detail[x]><!--  0..1 Quantity|Range|CodeableConcept The target value to be achieved --></detail[x]>
  <due[x]><!-- 0..1 date|Duration Reach goal on or before --></due[x]>
 </target>
 <statusDate value="[date]"/><!-- 0..1 When goal status took effect -->
 <statusReason value="[string]"/><!-- 0..1 Reason for current status -->
 <expressedBy><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson) Who's responsible for creating Goal?
    --></expressedBy>
 <addresses><!-- 0..* Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ProcedureRequest|RiskAssessment) Issues addressed by this goal --></addresses>
 <note><!-- 0..* Annotation Comments about the goal --></note>
 <outcomeCode><!-- 0..* CodeableConcept What result was achieved regarding the goal? --></outcomeCode>
 <outcomeReference><!-- 0..* Reference(Observation) Observation that resulted from goal --></outcomeReference>
</Goal>
```

JSON Template

```
{
  "resourceType" : "Goal",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this goal
  "status" : "<code>", // R!  proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-target
    | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected
  "category" : [{ CodeableConcept }], // E.g. Treatment, dietary, behavioral, etc.
  "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
  "description" : { CodeableConcept }, // R!  Code or text describing goal
  "subject" : { Reference(Patient|Group|Organization) }, // Who this goal is intended for
  // start[x]: When goal pursuit begins. One of these 2:
  "startDate" : "<date>",
  "startCodeableConcept" : { CodeableConcept },
  "target" : { // C? Target outcome for the goal
    "measure" : { CodeableConcept }, // C? The parameter whose value is being tracked
    // detail[x]: The target value to be achieved. One of these 3:
    "detailQuantity" : { Quantity },
    "detailRange" : { Range },
    "detailCodeableConcept" : { CodeableConcept },
    // due[x]: Reach goal on or before. One of these 2:
    "dueDate" : "<date>"
    "dueDuration" : { Duration }
  },
  "statusDate" : "<date>", // When goal status took effect
  "statusReason" : "<string>", // Reason for current status
  "expressedBy" : { Reference(Patient|Practitioner|RelatedPerson) }, // Who's responsible for creating Goal?
  "addresses" : [{ Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ProcedureRequest|RiskAssessment) }], // Issues addressed by this goal
  "note" : [{ Annotation }], // Comments about the goal
  "outcomeCode" : [{ CodeableConcept }], // What result was achieved regarding the goal?
  "outcomeReference" : [{ Reference(Observation) }] // Observation that resulted from goal
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Goal;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Goal.identifier [ Identifier ], ... ; # 0..* External Ids for this goal
  fhir:Goal.status [ code ]; # 1..1 proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-
    target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected
  fhir:Goal.category [ CodeableConcept ], ... ; # 0..* E.g. Treatment, dietary, behavioral, etc.
  fhir:Goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
  fhir:Goal.description [ CodeableConcept ]; # 1..1 Code or text describing goal
  fhir:Goal.subject [ Reference(Patient|Group|Organization) ]; # 0..1 Who this goal is intended for
  # Goal.start[x] : 0..1 When goal pursuit begins. One of these 2
    fhir:Goal.startDate [ date ]
    fhir:Goal.startCodeableConcept [ CodeableConcept ]
  fhir:Goal.target [ # 0..1 Target outcome for the goal
    fhir:Goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is being tracked
    # Goal.target.detail[x] : 0..1 The target value to be achieved. One of these 3
      fhir:Goal.target.detailQuantity [ Quantity ]
```

```
      fhir:Goal.target.detailRange [ Range ]
      fhir:Goal.target.detailCodeableConcept [ CodeableConcept ]
    # Goal.target.due[x] : 0..1 Reach goal on or before. One of these 2
      fhir:Goal.target.dueDate [ date ]
      fhir:Goal.target.dueDuration [ Duration ]
  ];
  fhir:Goal.statusDate [ date ]; # 0..1 When goal status took effect
  fhir:Goal.statusReason [ string ]; # 0..1 Reason for current status
  fhir:Goal.expressedBy [ Reference(Patient|Practitioner|RelatedPerson) ]; # 0..1 Who's responsible for creating Goal?
  fhir:Goal.addresses [ Reference(Condition|Observation|MedicationStatement|NutritionOrder|ProcedureRequest|
  RiskAssessment) ], ... ; # 0..* Issues addressed by this goal
  fhir:Goal.note [ Annotation ], ... ; # 0..* Comments about the goal
  fhir:Goal.outcomeCode [ CodeableConcept ], ... ; # 0..* What result was achieved regarding the goal?
  fhir:Goal.outcomeReference [ Reference(Observation) ], ... ; # 0..* Observation that resulted from goal
]
```

Changes since DSTU2

Goal
Goal.description | Type changed from string to CodeableConcept
Goal.target | Renamed from target[x] to target Remove date, Remove Quantity{http://hl7.org/fhir/StructureDefinition/Duration}
Goal.target.measure | Added Element
Goal.target.detail[x] | Added Element
Goal.target.due[x] | Added Element
Goal.statusReason | Type changed from CodeableConcept to string
Goal.expressedBy | Renamed from author to expressedBy
Goal.outcomeCode | Added Element
Goal.outcomeReference | Added Element
Goal.outcome | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (2 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.6.3.1 Terminology Bindings

Path | Definition | Type | Reference
Goal.status | Indicates whether the goal has been met and is still being targeted | Required | GoalStatus
Goal.category | Codes for grouping and sorting goals | Example | GoalCategory
Goal.priority | The level of importance associated with a goal | Preferred | GoalPriority
Goal.description | Codes providing the details of a particular goal. This will generally be system or implementation guide-specific. In many systems, only the text element will be used. | Example | SNOMED CT Clinical Findings
Goal.start[x] | Codes describing events that can trigger the initiation of a goal | Example | GoalStartEvent
Goal.target.measure | Codes to identify the value being tracked, e.g. body weight, blood pressure, or hemoglobin A1c level. | Example | LOINC Codes
Goal.target.detail[x] | Codes to identify the target value of the focus to be achieved to signify the fulfillment of the goal. | Unknown | No details provided yet
Goal.outcomeCode | The result of the goal; e.g. "25% increase in shoulder mobility", "Anxiety reduced to moderate levels". "15 kg weight loss sustained over 6 months" | Example | SNOMED CT Clinical Findings

9.6.3.2 Constraints

* gol-1: On Goal.target: Goal.target.measure is required if Goal.target.detail is populated (expression on Goal.target: (detail.exists() and measure.exists()) or detail.exists().not())

9.6.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | E.g. Treatment, dietary, behavioral, etc. | Goal.category
identifier | token | External Ids for this goal | Goal.identifier | 26 Resources
patient | reference | Who this goal is intended for | Goal.subject (Patient) | 31 Resources
start-date | date | When goal pursuit begins | Goal.start.as(Date)
status | token | proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected | Goal.status
subject | reference | Who this goal is intended for | Goal.subject (Group, Organization, Patient)
target-date | date | Reach goal on or before | Goal.target.due.as(Date)

9.6.5 Resource Goal - Examples

URL: https://hl7.org/fhir/STU3/goal-examples.html

9.6.5 Resource Goal - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Example List:

Goal of Weight Loss | example | XML | JSON | Turtle
Goal to Stop Smoking | stop-smoking | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.6.6 Resource Goal - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/goal-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.6.6 Resource Goal - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Detailed Descriptions for the elements in the Goal resource.

Goal
Definition | Describes the intended objective(s) for a patient, group or organization care, for example, weight loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process improvement objective, etc.
Control | 1..1
Comments | Goal can be achieving a particular change or merely maintaining a current state or even slowing a decline.
Goal.identifier
Definition | This records identifiers associated with this care plan that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Need to allow connection to a wider workflow.
Goal.status
Definition | Indicates whether the goal has been reached and is still considered relevant.
Control | 1..1
Terminology Binding | GoalStatus (Required)
Type | code
Is Modifier | true
Requirements | Allows measuring outcome and whether goal needs to be further tracked.
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Goal.category
Definition | Indicates a category the goal falls within.
Control | 0..*
Terminology Binding | GoalCategory (Example)
Type | CodeableConcept
Requirements | Allows goals to be filtered and sorted.
Summary | true
Goal.priority
Definition | Identifies the mutually agreed level of importance associated with reaching/sustaining the goal.
Control | 0..1
Terminology Binding | GoalPriority (Preferred)
Type | CodeableConcept
Requirements | Used for sorting and presenting goals.
Summary | true
Comments | Extensions are available to track priorities as established by each participant (i.e. Priority from the patient's perspective, different practitioners' perspectives, family member's perspectives) The ordinal extension on Coding can be used to convey a numerically comparable ranking to priority. (Keep in mind that different coding systems may use a "low value=important".
Goal.description
Definition | Human-readable and/or coded description of a specific desired objective of care, such as "control blood pressure" or "negotiate an obstacle course" or "dance with child at wedding".
Control | 1..1
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Requirements | Without a description of what's trying to be achieved, element has no purpose.
Summary | true
Comments | If no code is available, use CodeableConcept.text.
Goal.subject
Definition | Identifies the patient, group or organization for whom the goal is being established.
Control | 0..1
Type | Reference(Patient | Group | Organization)
Requirements | Subject is optional to support annonymized reporting.

Summary | true
Goal.start[x]
Definition | The date or event after which the goal should begin being pursued.
Control | 0..1
Terminology Binding | GoalStartEvent (Example)
Type | date|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Goals can be established prior to there being an intention to start pursuing them; e.g. Goals for post-surgical recovery established prior to surgery.
Summary | true
Goal.target
Definition | Indicates what should be done by when.
Control | 0..1
Requirements | Allows the progress of the goal to be monitored against an observation or due date.
Invariants | Defined on this element gol-1: Goal.target.measure is required if Goal.target.detail is populated (expression : (detail.exists() and measure.exists()) or detail.exists().not(), xpath: (exists(f:*[starts-with(local-name(.), 'detail')]) and exists(f:measure)) or not(exists(f:*[starts-with(local-name(.), 'detail')]))) Affect this element
Goal.target.measure
Definition | The parameter whose value is being tracked, e.g. body weight, blood pressure, or hemoglobin A1c level.
Control | 0..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
Summary | true
Invariants | Affect this element gol-1: Goal.target.measure is required if Goal.target.detail is populated (expression : (detail.exists() and measure.exists()) or detail.exists().not(), xpath: (exists(f:*[starts-with(local-name(.), 'detail')]) and exists(f:measure)) or not(exists(f:*[starts-with(local-name(.), 'detail')])))
Goal.target.detail[x]
Definition | The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150 pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is missing, it indicates that the goal is achieved at any focus value at or above the low value.
Control | 0..1
Terminology Binding | GoalTargetDetail:
Type | Quantity|Range|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Invariants | Affect this element gol-1: Goal.target.measure is required if Goal.target.detail is populated (expression : (detail.exists() and measure.exists()) or detail.exists().not(), xpath: (exists(f:*[starts-with(local-name(.), 'detail')]) and exists(f:measure)) or not(exists(f:*[starts-with(local-name(.), 'detail')])))
Goal.target.due[x]
Definition | Indicates either the date or the duration after start by which the goal should be met.
Control | 0..1
Type | date|Duration
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Identifies when the goal should be evaluated.
Summary | true
Goal.statusDate
Definition | Identifies when the current status. I.e. When initially created, when achieved, when cancelled, etc.
Control | 0..1
Type | date
Summary | true
Comments | To see the date for past statuses, query history.
Goal.statusReason
Definition | Captures the reason for the current status.
Control | 0..1
Type | string
Comments | This will typically be captured for statuses such as rejected, on-hold or cancelled, but could be present for others.
Goal.expressedBy
Definition | Indicates whose goal this is - patient goal, practitioner goal, etc.
Control | 0..1
Type | Reference(Patient | Practitioner | RelatedPerson)
Summary | true
Comments | This is the individual reponsible for establishing the goal, not necessarily who recorded it. (For that, use the Provenance resource.).
Goal.addresses
Definition | The identified conditions and other health record elements that are intended to be addressed by the goal.
Control | 0..*
Type | Reference(Condition | Observation | MedicationStatement | NutritionOrder | ProcedureRequest | RiskAssessment)
Requirements | Allows specific goals to explicitly linked to the concerns they're dealing with - makes the goal more understandable.
Goal.note
Definition | Any comments related to the goal.
Control | 0..*
Type | Annotation
Requirements | There's a need to capture information about the goal that doesn't actually describe the goal.
Comments | May be used for progress notes, concerns or other related information that doesn't actually describe the goal itself.

Goal.outcomeCode
Definition | Identifies the change (or lack of change) at the point when the status of the goal is assessed.
Control | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Requirements | Outcome tracking is a key aspect of care planning.
Comments | Note that this should not duplicate the goal status.
Goal.outcomeReference
Definition | Details of what's changed (or not changed).
Control | 0..*
Type | Reference(Observation)
Requirements | Outcome tracking is a key aspect of care planning.
Comments | The goal outcome is independent of the outcome of the related activities. For example, if the Goal is to achieve a target body weight of 150 lb and a care plan activity is defined to diet, then the care plan's activity outcome could be calories consumed whereas goal outcome is an observation for the actual body weight measured.

9.6.7 Resource Goal - Mappings

URL: https://hl7.org/fhir/STU3/goal-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.6.7 Resource Goal - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the goal resource.

9.6.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Goal | .outboundRelationship[typeCode<=OBJ].
identifier | .id
status | .statusCode in-progress = active (classCode = OBJ) achieved = complete sustaining = active (classCode=OBJC) cancelled = aborted
category
priority | .priorityCode
description | .text
subject | .participation[typeCode=PAT].role[classCode=PAT]
start[x]
target
measure
detail[x]
due[x]
statusDate
statusReason
expressedBy
addresses | .outboundRelationship[typeCode=SUBJ].target[classCode=CONC]
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value
outcomeCode
outcomeReference

9.6.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Goal | GOL.1
identifier
status | GOL-18-goal life cycle status
category
priority
description | GOL-3.2-goal ID.text
subject | PID-3-patient ID list
start[x]
target
measure
detail[x]
due[x]
statusDate
statusReason
expressedBy
addresses
note | GOL-16-goal evaluation + NTE?
outcomeCode

outcomeReference

9.6.8 Resource Goal - Design Notes

URL: https://hl7.org/fhir/STU3/goal-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.6.8 Resource Goal - Design Notes

to do

9.6.9 Resource Goal - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/goal-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.6.9 Resource Goal - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Goal resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.6.9.1 Profiles

No Profiles defined for this resource

9.6.9.2 Extensions

goal-acceptance | acceptance | for FHIR Core Goal Profile
goal-reasonRejected | reasonRejected | for FHIR Core Goal Profile
goal-relationship | relationship | for FHIR Core Goal Profile

Extensions that reference this resource:

goal-pertainsToGoal | pertainsToGoal | for FHIR Core Goal Profile
goal-relationship | relationship | for FHIR Core Goal Profile

9.6.9.3 Search Extensions

No Search Extensions defined for this resource

9.6.10 FHIR Core Goal Profile

URL: https://hl7.org/fhir/STU3/goal-extensions.html

This profile holds core Goal extensions

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.6.10 FHIR Core Goal Profile

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

9.6.10.1 Content

Extensions:

goal-reasonRejected | reasonRejected : The reason the goal was not accepted. Applies only if the status of the goal is rejected.

goal-relationship | relationship : Establishes a relationship between this goal and other goals.

goal-pertainsToGoal | pertainsToGoal : Indicates that the resource is related to either the measurement, achievement or progress towards the referenced goal.

goal-acceptance | acceptance : Information about the acceptance and relative priority assigned to the goal by the patient, practitioners and other stake-holders.

9.6.11 Resource Goal - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/goal-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.6.11 Resource Goal - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

R2 : R3 Converson maps for Goal.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (2 errors). (see documentation)

9.6.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Goal2to3" = "R2 to R3 Conversions for Goal"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Goal" alias GoalR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Goal" alias Goal as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Goal extends DomainResource
  input src : GoalR2 as source
  input tgt : Goal as target

  "Goal-identifier" : for src.identifier make tgt.identifier
  "Goal-subject" : for src.subject make tgt.subject
  "Goal-start" : for src.start make tgt.start
  "Goal-target" : for src.target make tgt.target
  "Goal-category" : for src.category make tgt.category
  "Goal-description" : for src.description make tgt.description
  "Goal-status" : for src.status make tgt.status
  "Goal-statusDate" : for src.statusDate make tgt.statusDate
  "Goal-statusReason" : for src.statusReason make tgt.statusReason
  "Goal-author" : for src.author make tgt.expressedBy
  "Goal-priority" : for src.priority make tgt.priority
  "Goal-addresses" : for src.addresses make tgt.addresses
  "Goal-note" : for src.note make tgt.note
  "Goal-outcome" : for src.outcome as vs0 then goalOutcome(vs0, tgt)
endgroup

group goalOutcome extends BackboneElement
  input src as source
  input tgt as target

  "Goal.outcome-result" : for src.result : CodeableConcept 0..1 make tgt.outcomeCode
  "Goal.outcome-result" : for src.result : Reference 0..1 make tgt.outcomeReference
endgroup
```

9.6.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Goal3to2" = "R3 to R2 Conversion for Goal"

uses "http://hl7.org/fhir/StructureDefinition/Goal" alias Goal as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Goal" alias GoalR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Goal extends DomainResource
  input src : Goal as source
  input tgt : GoalR2 as target

  "Goal-identifier" : for src.identifier make tgt.identifier
  "Goal-subject" : for src.subject make tgt.subject
  "Goal-start" : for src.start make tgt.start
  "Goal-target" : for src.target make tgt.target
  "Goal-category" : for src.category make tgt.category
  "Goal-description" : for src.description make tgt.description
  "Goal-status" : for src.status make tgt.status
  "Goal-statusDate" : for src.statusDate make tgt.statusDate
  "Goal-statusReason" : for src.statusReason make tgt.statusReason
  "Goal-author" : for src.expressedBy make tgt.author
  "Goal-priority" : for src.priority make tgt.priority
  "Goal-addresses" : for src.addresses make tgt.addresses
  "Goal-note" : for src.note make tgt.note
  "Goal-outcome" : for src.outcomeCode as vs0 make tgt.outcome as vt0, vt0.result = vs0
  "Goal-outcome" : for src.outcomeReference as vs0 make tgt.outcome as vt0, vt0.result = vs0
endgroup
```

9.6.11.3 R3 Validation Errors

example | ERROR @ Goal.extension: The extension http://hl7.org/fhir/StructureDefinition/goal-target is unknown, and not allowed here (src = InstanceValidator)

goal-example | ERROR @ Goal.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/goal-target is unknown, and not allowed here (src = InstanceValidator)

9.7 Resource CareTeam - Content

URL: https://hl7.org/fhir/STU3/careteam.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

The Care Team includes all the people and organizations who plan to participate in the coordination and delivery of care for a patient.

9.7.1 Scope and Usage

The Care Team resource includes all the people and/or organizations who plan to participate in the coordination and delivery of care for a patient. This is not limited to practitioners, but may include other caregivers such as family members, guardians, the patient themself, or others. The Care Team, depending on where used, may include care team members specific to a particular care plan, an episode, an encounter, or may reflect all known team members across these perspectives.

9.7.2 Boundaries and Relationships

Care Team is distinct from Group. Group is patient-independent and identifies an undifferentiated set of individuals who are intended to be the target of one or more clinical activities (e.g. set of clinical trial participants, set of individuals impacted by or at risk of a public health event, a herd or flock, etc.) The CareTeam resource establishes a set of relationships and roles and is specific to a particular Patient. The actors are the individual members or organized group of individuals. CareTeam can be referenced by EpisodeOfCare, Encounter, or CarePlan to identify the set of individuals (and their respective roles) who are intended to be involved in providing the care defined by those resources.

This resource is referenced by CarePlan, CommunicationRequest, Consent and EpisodeOfCare

9.7.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
CareTeam | | | DomainResource | Planned participants in the coordination and delivery of care for a patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this team
status | ?!? | 0..1 | code | proposed | active | suspended | inactive | entered-in-error CareTeamStatus (Required)
category | ? | 0..* | CodeableConcept | Type of team CareTeamCategory (Example)
name | ? | 0..1 | string | Name of the team, such as crisis assessment team
subject | ? | 0..1 | Reference(Patient | Group) | Who care team is for
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode associated with CareTeam
period | ? | 0..1 | Period | Time period team covers
participant | I | 0..* | BackboneElement | Members of the team + CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner
role | ? | 0..1 | CodeableConcept | Type of involvement Participant Roles (Example)
member | ? | 0..1 | Reference(Practitioner | RelatedPerson | Patient | Organization | CareTeam) | Who is involved
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization of the practitioner
period | | 0..1 | Period | Time period of participant
reasonCode | | 0..* | CodeableConcept | Why the care team exists SNOMED CT Clinical Findings (Example)
reasonReference | | 0..* | Reference(Condition) | Why the care team exists
managingOrganization | ? | 0..* | Reference(Organization) | Organization responsible for the care team
note | | 0..* | Annotation | Comments made about the CareTeam
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CareTeam xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this team --></identifier>
 <status value="[code]"/><!-- 0..1 proposed | active | suspended | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Type of team --></category>
 <name value="[string]"/><!-- 0..1 Name of the team, such as crisis assessment team -->
 <subject><!-- 0..1 Reference(Patient|Group) Who care team is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode associated with CareTeam --></context>
 <period><!-- 0..1 Period Time period team covers --></period>
 <participant>  <!--  0..* Members of the team -->
  <role><!-- 0..1 CodeableConcept Type of involvement --></role>
  <member><!-- 0..1 Reference(Practitioner|RelatedPerson|Patient|Organization|
    CareTeam) Who is involved --></member>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization of the practitioner --></onBehalfOf>
  <period><!-- 0..1 Period Time period of participant --></period>
 </participant>
 <reasonCode><!-- 0..* CodeableConcept Why the care team exists --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition) Why the care team exists --></reasonReference>
 <managingOrganization><!-- 0..* Reference(Organization) Organization responsible for the care team
    --></managingOrganization>
 <note><!-- 0..* Annotation Comments made about the CareTeam --></note>
</CareTeam>
```

JSON Template

```
{
  "resourceType" : "CareTeam",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this team
  "status" : "<code>", // proposed | active | suspended | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Type of team
  "name" : "<string>", // Name of the team, such as crisis assessment team
  "subject" : { Reference(Patient|Group) }, // Who care team is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode associated with CareTeam
  "period" : { Period }, // Time period team covers
  "participant" : [{ // C? Members of the team
    "role" : { CodeableConcept }, // Type of involvement
    "member" : { Reference(Practitioner|RelatedPerson|Patient|Organization|
    CareTeam) }, // Who is involved
    "onBehalfOf" : { Reference(Organization) }, // Organization of the practitioner
    "period" : { Period } // Time period of participant
  }],
  "reasonCode" : [{ CodeableConcept }], // Why the care team exists
  "reasonReference" : [{ Reference(Condition) }], // Why the care team exists
  "managingOrganization" : [{ Reference(Organization) }], // Organization responsible for the care team
  "note" : [{ Annotation }] // Comments made about the CareTeam
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CareTeam;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CareTeam.identifier [ Identifier ], ... ; # 0..* External Ids for this team
  fhir:CareTeam.status [ code ]; # 0..1 proposed | active | suspended | inactive | entered-in-error
  fhir:CareTeam.category [ CodeableConcept ], ... ; # 0..* Type of team
  fhir:CareTeam.name [ string ]; # 0..1 Name of the team, such as crisis assessment team
  fhir:CareTeam.subject [ Reference(Patient|Group) ]; # 0..1 Who care team is for
  fhir:CareTeam.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode associated with CareTeam
  fhir:CareTeam.period [ Period ]; # 0..1 Time period team covers
  fhir:CareTeam.participant [ # 0..* Members of the team
    fhir:CareTeam.participant.role [ CodeableConcept ]; # 0..1 Type of involvement
    fhir:CareTeam.participant.member [ Reference(Practitioner|RelatedPerson|Patient|Organization|CareTeam) ]; # 0..1 Who
     is involved
    fhir:CareTeam.participant.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization of the practitioner
    fhir:CareTeam.participant.period [ Period ]; # 0..1 Time period of participant
  ], ...;
  fhir:CareTeam.reasonCode [ CodeableConcept ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.reasonReference [ Reference(Condition) ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.managingOrganization [ Reference(Organization) ], ... ; # 0..* Organization responsible for the care
    team
  fhir:CareTeam.note [ Annotation ], ... ; # 0..* Comments made about the CareTeam
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
CareTeam | | | DomainResource | Planned participants in the coordination and delivery of care for a patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External Ids for this team
status | ?!? | 0..1 | code | proposed | active | suspended | inactive | entered-in-error CareTeamStatus (Required)
category | ? | 0..* | CodeableConcept | Type of team CareTeamCategory (Example)
name | ? | 0..1 | string | Name of the team, such as crisis assessment team
subject | ? | 0..1 | Reference(Patient | Group) | Who care team is for
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode associated with CareTeam
period | ? | 0..1 | Period | Time period team covers
participant | I | 0..* | BackboneElement | Members of the team + CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner
role | ? | 0..1 | CodeableConcept | Type of involvement Participant Roles (Example)
member | ? | 0..1 | Reference(Practitioner | RelatedPerson | Patient | Organization | CareTeam) | Who is involved
onBehalfOf | ? | 0..1 | Reference(Organization) | Organization of the practitioner
period | | 0..1 | Period | Time period of participant
reasonCode | | 0..* | CodeableConcept | Why the care team exists SNOMED CT Clinical Findings (Example)
reasonReference | | 0..* | Reference(Condition) | Why the care team exists
managingOrganization | ? | 0..* | Reference(Organization) | Organization responsible for the care team
note | | 0..* | Annotation | Comments made about the CareTeam
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CareTeam xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this team --></identifier>
 <status value="[code]"/><!-- 0..1 proposed | active | suspended | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Type of team --></category>
 <name value="[string]"/><!-- 0..1 Name of the team, such as crisis assessment team -->
 <subject><!-- 0..1 Reference(Patient|Group) Who care team is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode associated with CareTeam --></context>
 <period><!-- 0..1 Period Time period team covers --></period>
 <participant>  <!--  0..* Members of the team -->
  <role><!-- 0..1 CodeableConcept Type of involvement --></role>
  <member><!-- 0..1 Reference(Practitioner|RelatedPerson|Patient|Organization|
    CareTeam) Who is involved --></member>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization of the practitioner --></onBehalfOf>
  <period><!-- 0..1 Period Time period of participant --></period>
 </participant>
 <reasonCode><!-- 0..* CodeableConcept Why the care team exists --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition) Why the care team exists --></reasonReference>
 <managingOrganization><!-- 0..* Reference(Organization) Organization responsible for the care team
    --></managingOrganization>
 <note><!-- 0..* Annotation Comments made about the CareTeam --></note>
</CareTeam>
```

JSON Template

```
{
  "resourceType" : "CareTeam",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this team
  "status" : "<code>", // proposed | active | suspended | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Type of team
  "name" : "<string>", // Name of the team, such as crisis assessment team
  "subject" : { Reference(Patient|Group) }, // Who care team is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode associated with CareTeam
  "period" : { Period }, // Time period team covers
  "participant" : [{ // C? Members of the team
    "role" : { CodeableConcept }, // Type of involvement
    "member" : { Reference(Practitioner|RelatedPerson|Patient|Organization|
    CareTeam) }, // Who is involved
    "onBehalfOf" : { Reference(Organization) }, // Organization of the practitioner
    "period" : { Period } // Time period of participant
  }],
  "reasonCode" : [{ CodeableConcept }], // Why the care team exists
  "reasonReference" : [{ Reference(Condition) }], // Why the care team exists
  "managingOrganization" : [{ Reference(Organization) }], // Organization responsible for the care team
  "note" : [{ Annotation }] // Comments made about the CareTeam
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CareTeam;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CareTeam.identifier [ Identifier ], ... ; # 0..* External Ids for this team
  fhir:CareTeam.status [ code ]; # 0..1 proposed | active | suspended | inactive | entered-in-error
  fhir:CareTeam.category [ CodeableConcept ], ... ; # 0..* Type of team
  fhir:CareTeam.name [ string ]; # 0..1 Name of the team, such as crisis assessment team
  fhir:CareTeam.subject [ Reference(Patient|Group) ]; # 0..1 Who care team is for
  fhir:CareTeam.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode associated with CareTeam
  fhir:CareTeam.period [ Period ]; # 0..1 Time period team covers
  fhir:CareTeam.participant [ # 0..* Members of the team
    fhir:CareTeam.participant.role [ CodeableConcept ]; # 0..1 Type of involvement
    fhir:CareTeam.participant.member [ Reference(Practitioner|RelatedPerson|Patient|Organization|CareTeam) ]; # 0..1 Who
     is involved
    fhir:CareTeam.participant.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization of the practitioner
    fhir:CareTeam.participant.period [ Period ]; # 0..1 Time period of participant
  ], ...;
  fhir:CareTeam.reasonCode [ CodeableConcept ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.reasonReference [ Reference(Condition) ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.managingOrganization [ Reference(Organization) ], ... ; # 0..* Organization responsible for the care
    team
  fhir:CareTeam.note [ Annotation ], ... ; # 0..* Comments made about the CareTeam
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.7.3.1 Terminology Bindings

Path | Definition | Type | Reference
CareTeam.status | Indicates the status of the care team. | Required | CareTeamStatus
CareTeam.category | Indicates the type of care team. | Example | CareTeamCategory
CareTeam.participant.role | Indicates specific responsibility of an individual within the care team, such as "Primary physician", "Team coordinator", "Caregiver", etc. | Example | Participant Roles
CareTeam.reasonCode | Indicates the reason for the care team. | Example | SNOMED CT Clinical Findings

9.7.3.2 Constraints

* ctm-1: On CareTeam.participant: CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner (expression on CareTeam.participant: onBehalfOf.exists() implies (member.resolve() is Practitioner))

9.7.3.3 Notes

The Provenance resource can be used for detailed review information, such as when the care team was last reviewed and by whom.

9.7.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | Type of team | CareTeam.category
context | reference | Encounter or episode associated with CareTeam | CareTeam.context (EpisodeOfCare, Encounter)
date | date | Time period team covers | CareTeam.period | 18 Resources
encounter | reference | Encounter or episode associated with CareTeam | CareTeam.context (Encounter)
identifier | token | External Ids for this team | CareTeam.identifier | 26 Resources
participant | reference | Who is involved | CareTeam.participant.member (Practitioner, Organization, CareTeam, Patient, RelatedPerson)
patient | reference | Who care team is for | CareTeam.subject (Patient) | 31 Resources
status | token | proposed | active | suspended | inactive | entered-in-error | CareTeam.status
subject | reference | Who care team is for | CareTeam.subject (Group, Patient)

9.7.5 Resource CareTeam - Examples

URL: https://hl7.org/fhir/STU3/careteam-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.7.5 Resource CareTeam - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Example List:

Care Team to address obesity | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.7.6 Resource CareTeam - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/careteam-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.7.6 Resource CareTeam - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the CareTeam resource.

CareTeam
Definition | The Care Team includes all the people and organizations who plan to participate in the coordination and delivery of care for a patient.
Control | 1..1
CareTeam.identifier
Definition | This records identifiers associated with this care team that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
CareTeam.status
Definition | Indicates the current state of the care team.
Control | 0..1
Terminology Binding | CareTeamStatus (Required)
Type | code

Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the code entered-in-error that marks the care team as not currently valid.
CareTeam.category
Definition | Identifies what kind of team. This is to support differentiation between multiple co-existing teams, such as care plan team, episode of care team, longitudinal care team.
Control | 0..*
Terminology Binding | CareTeamCategory (Example)
Type | CodeableConcept
Requirements | Used for filtering what teams(s) are retrieved and displayed to different types of users.
Summary | true
Comments | There may be multiple axis of categorization and one team may serve multiple purposes.
CareTeam.name
Definition | A label for human use intended to distinguish like teams. E.g. the "red" vs. "green" trauma teams.
Control | 0..1
Type | string
Summary | true
Comments | The meaning/purpose of the team is conveyed in CareTeam.category. This element may also convey semantics of the team (e.g. "Red trauma team"), but its primary purpose is to distinguish between identical teams in a human-friendly way. ("Team 18735" isn't as friendly .).
CareTeam.subject
Definition | Identifies the patient or group whose intended care is handled by the team.
Control | 0..1
Type | Reference(Patient | Group)
Requirements | Allows the team to care for a group (e.g. marriage) therapy.
Alternate Names | patient
Summary | true
CareTeam.context
Definition | The encounter or episode of care that establishes the context for this care team.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
CareTeam.period
Definition | Indicates when the team did (or is intended to) come into effect and end.
Control | 0..1
Type | Period
Requirements | Allows tracking what team(s) are in effect at a particular time.
Summary | true
CareTeam.participant
Definition | Identifies all people and organizations who are expected to be involved in the care team.
Control | 0..*
Invariants | Defined on this element ctm-1: CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner (expression : onBehalfOf.exists() implies (member.resolve() is Practitioner), xpath: contains(f:member/f:reference/@value, '/Practitioner/') or not(exists(f:onBehalfOf))) Affect this element
CareTeam.participant.role
Definition | Indicates specific responsibility of an individual within the care team, such as "Primary care physician", "Trained social worker counselor", "Caregiver", etc.
Control | 0..1
Terminology Binding | Participant Roles (Example)
Type | CodeableConcept
Summary | true
Comments | Roles may sometimes be inferred by type of Practitioner. These are relationships that hold only within the context of the care team. General relationships should be handled as properties of the Patient resource directly.
CareTeam.participant.member
Definition | The specific person or organization who is participating/expected to participate in the care team.
Control | 0..1
Type | Reference(Practitioner | RelatedPerson | Patient | Organization | CareTeam)
Summary | true
Comments | Patient only needs to be listed if they have a role other than "subject of care". Member is optional because some participants may be known only by their role, particularly in draft plans.
CareTeam.participant.onBehalfOf
Definition | The organization of the practitioner.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners can be associated with multiple organizations. This element indicates which organization they were acting on behalf of.
Summary | true
CareTeam.participant.period
Definition | Indicates when the specific member or organization did (or is intended to) come into effect and end.
Control | 0..1
Type | Period
CareTeam.reasonCode
Definition | Describes why the care team exists.

Control | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
CareTeam.reasonReference
Definition | Condition(s) that this care team addresses.
Control | 0..*
Type | Reference(Condition)
CareTeam.managingOrganization
Definition | The organization responsible for the care team.
Control | 0..*
Type | Reference(Organization)
Requirements | Allows for multiple organizations to collaboratively manage cross-organizational, longitudinal care plan.
Summary | true
CareTeam.note
Definition | Comments made about the CareTeam.
Control | 0..*
Type | Annotation

9.7.7 Resource CareTeam - Mappings

URL: https://hl7.org/fhir/STU3/careteam-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.7.7 Resource CareTeam - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the careteam resource.

9.7.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

CareTeam
identifier
status
category
name
subject
context
period
participant | .participation[typeCode=PRF]
role | .functionCode
member | .role
onBehalfOf
period
reasonCode
reasonReference
managingOrganization
note

9.7.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

CareTeam
identifier
status
category
name
subject
context
period
participant | REL (REL.4 is always the Patient) (or PRT?)
role | REL.2 (or PRT-4?)
member | REL.5 (or PRT-5 : (PRV-4 {provider participations}) / PRT-5 : (PRV-4 {non-provider person participations}) / PRT-5 : (PRV-4 = (patient non-subject of care)) / PRT-8?)
onBehalfOf
period
reasonCode
reasonReference
managingOrganization
note

9.7.8 Resource CareTeam - Design Notes

URL: https://hl7.org/fhir/STU3/careteam-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.7.8 Resource CareTeam - Design Notes

to do

9.7.9 Resource CareTeam - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/careteam-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.7.9 Resource CareTeam - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the CareTeam resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.7.9.1 Profiles

No Profiles defined for this resource

9.7.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

9.7.9.3 Search Extensions

No Search Extensions defined for this resource

9.8 Resource ClinicalImpression - Content

URL: https://hl7.org/fhir/STU3/clinicalimpression.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 0 | Draft | Compartments: Encounter, Patient, Practitioner

A record of a clinical assessment performed to determine what problem(s) may affect the patient and before planning the treatments or management strategies that are best to manage a patient's condition. Assessments are often 1:1 with a clinical consultation / encounter, but this varies greatly depending on the clinical workflow. This resource is called "ClinicalImpression" rather than "ClinicalAssessment" to avoid confusion with the recording of assessment tools such as Apgar score.

9.8.1 Scope and Usage

Performing a clinical assessment is a fundamental part of a clinician's workflow, performed repeatedly throughout the day. In spite of this - or perhaps, because of it - there is wide variance in how clinical impressions are recorded. Some clinical assessments simply result in an impression recorded as a single text note in the patient 'record' (e.g. "Progress satisfactory, continue with treatment"), while others are associated with careful, detailed record keeping of the evidence gathered and the reasoning leading to a differential diagnosis, and there is a continuum between these. This resource is intended to be used to cover all these use cases.

The assessment is intimately linked to the process of care. It may occur in the context of a care plan, and it very often results in a new (or revised) care plan. Normally. clinical assessments are part of an ongoing process of care, and the patient will be re-assessed repeatedly. For this reason, the clinical impression can explicit reference both care plans (preceeding and resulting) and reference a previous impression that this impression follows on from.

An impression is a clinical summation of information and/or an opinion formed, which is the outcome of the clinical assessment process. The ClinicalImpression may lead to a statement of a Condition about a patient.

In FHIR, an assessment is typically an instrument or tool used to collect information about a patient.

STU Note: Unlike many other resources, there is little prior art with regard to exchanging records of clinical assessments. For this reason, this resource should be regarded as particularly prone to ongoing revision. In terms of scope and usage, the Patient Care workgroup wishes to draw the attention of reviewers and implementers to the following issues:

* When is an existing clinical impression revised, rather than a new one created (that references the existing one)? How does that affect the status? what's the interplay between the status of the diagnosis and the status of the impression? (e.g. for a

'provisional' impression, which bit is provisional?)

* This structure doesn't differentiate between a working and a final diagnosis. Given an answer to the previous question, should it?

* Further clarify around the relationship between care plan and impression is needed. Both answers to the previous questions and ongoing discussions around revisions to the care plan will influence the design of clinical impression

* Should prognosis be represented, and if so, how much structure should it have?

* Should an impression reference other impressions that are related? (how related?)

* Investigations - the specification needs a good value set for the code for the group, and will be considering the name "investigations" further

Feedback is welcome here .

9.8.2 Boundaries and Relationships

There is another related clinical concept often called an "assessment": assessment Tools such as Apgar (also known as "Assessment Scales"). This is not what the ClinicalImpression resource is about; assessment tools such as Apgar are represented as Observations, and Questionnaires may be used to help generate these. Clinical Impressions may refer to these assessment tools as one of the investigations that was performed during the assessment process.

9.8.3 Background and Context

An important background to understanding this resource is the FHIR wiki page for clinical assessment . In particular, the storyboards there drove the design of the resource, and will be the basis for all examples created.

PLANNED CHANGE:

ClinicalImpression is one of the Event resources in the FHIR Workflow specification. As such, it is expected to be adjusted to align with the Event workflow pattern which will involve adding a number of additional data elements and potentially renaming a few elements. Any concerns about performing such alignment are welcome as ballot comments and/or tracker items.

This resource is referenced by condition

9.8.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ClinicalImpression | | | DomainResource | A clinical assessment performed when planning treatments and management strategies for a patient
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier
status | ?!? | 1..1 | code | draft | completed | entered-in-error ClinicalImpressionStatus (Required)
code | ? | 0..1 | CodeableConcept | Kind of assessment performed
description | ? | 0..1 | string | Why/how the assessment was performed
subject | ? | 1..1 | Reference(Patient | Group) | Patient or group assessed
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode created from
effective[x] | ? | 0..1 | | Time of assessment
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
date | ? | 0..1 | dateTime | When the assessment was documented
assessor | ? | 0..1 | Reference(Practitioner) | The clinician performing the assessment
previous | | 0..1 | Reference(ClinicalImpression) | Reference to last assessment
problem | ? | 0..* | Reference(Condition | AllergyIntolerance) | Relevant impressions of patient state
investigation | | 0..* | BackboneElement | One or more sets of investigations (signs, symptions, etc.)
code | | 1..1 | CodeableConcept | A name/code for the set Investigation Type (Example)
item | | 0..* | Reference(Observation | QuestionnaireResponse | FamilyMemberHistory | DiagnosticReport | RiskAssessment | ImagingStudy) |
Record of a specific investigation
protocol | | 0..* | uri | Clinical Protocol followed
summary | | 0..1 | string | Summary of the assessment
finding | | 0..* | BackboneElement | Possible or likely findings and diagnoses
item[x] | | 1..1 | | What was found Condition/Problem/Diagnosis Codes (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Condition | Observation)
basis | | 0..1 | string | Which investigations support finding
prognosisCodeableConcept | | 0..* | CodeableConcept | Estimate of likely outcome Clinical Impression Prognosis (Example)

prognosisReference | | 0..* | Reference(RiskAssessment) | RiskAssessment expressing likely outcome
action | | 0..* | Reference(ReferralRequest | ProcedureRequest | Procedure | MedicationRequest | Appointment) | Action taken as part of assessment procedure
note | | 0..* | Annotation | Comments made about the ClinicalImpression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ClinicalImpression xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 draft | completed | entered-in-error -->
 <code><!-- 0..1 CodeableConcept Kind of assessment performed --></code>
 <description value="[string]"/><!-- 0..1 Why/how the assessment was performed -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient or group assessed --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode created from --></context>
 <effective[x]><!-- 0..1 dateTime|Period Time of assessment --></effective[x]>
 <date value="[dateTime]"/><!-- 0..1 When the assessment was documented -->
 <assessor><!-- 0..1 Reference(Practitioner) The clinician performing the assessment --></assessor>
 <previous><!-- 0..1 Reference(ClinicalImpression) Reference to last assessment --></previous>
 <problem><!-- 0..* Reference(Condition|AllergyIntolerance) Relevant impressions of patient state --></problem>
 <investigation>  <!-- 0..* One or more sets of investigations (signs, symptions, etc.) -->
  <code><!-- 1..1 CodeableConcept A name/code for the set --></code>
  <item><!-- 0..* Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy) Record of a specific investigation --></item>
 </investigation>
 <protocol value="[uri]"/><!-- 0..* Clinical Protocol followed -->
 <summary value="[string]"/><!-- 0..1 Summary of the assessment -->
 <finding>  <!-- 0..* Possible or likely findings and diagnoses -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Condition|Observation) What was found --></item[x]>
  <basis value="[string]"/><!-- 0..1 Which investigations support finding -->
 </finding>
 <prognosisCodeableConcept><!-- 0..* CodeableConcept Estimate of likely outcome --></prognosisCodeableConcept>
 <prognosisReference><!-- 0..* Reference(RiskAssessment) RiskAssessment expressing likely outcome
    --></prognosisReference>
 <action><!-- 0..* Reference(ReferralRequest|ProcedureRequest|Procedure|
   MedicationRequest|Appointment) Action taken as part of assessment procedure --></action>
 <note><!-- 0..* Annotation Comments made about the ClinicalImpression --></note>
</ClinicalImpression>
```

JSON Template

```
{
  "resourceType" : "ClinicalImpression",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  draft | completed | entered-in-error
  "code" : { CodeableConcept }, // Kind of assessment performed
  "description" : "<string>", // Why/how the assessment was performed
  "subject" : { Reference(Patient|Group) }, // R!  Patient or group assessed
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode created from
  // effective[x]: Time of assessment. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "date" : "<dateTime>", // When the assessment was documented
  "assessor" : { Reference(Practitioner) }, // The clinician performing the assessment
  "previous" : { Reference(ClinicalImpression) }, // Reference to last assessment
  "problem" : [{ Reference(Condition|AllergyIntolerance) }], // Relevant impressions of patient state
  "investigation" : [{ // One or more sets of investigations (signs, symptions, etc.)
    "code" : { CodeableConcept }, // R!  A name/code for the set
    "item" : [{ Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy) }] // Record of a specific investigation
  }],
  "protocol" : ["<uri>"], // Clinical Protocol followed
  "summary" : "<string>", // Summary of the assessment
  "finding" : [{ // Possible or likely findings and diagnoses
    // item[x]: What was found. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Condition|Observation) },
    "basis" : "<string>" // Which investigations support finding
  }],
  "prognosisCodeableConcept" : [{ CodeableConcept }], // Estimate of likely outcome
  "prognosisReference" : [{ Reference(RiskAssessment) }], // RiskAssessment expressing likely outcome
  "action" : [{ Reference(ReferralRequest|ProcedureRequest|Procedure|
   MedicationRequest|Appointment) }], // Action taken as part of assessment procedure
  "note" : [{ Annotation }] // Comments made about the ClinicalImpression
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ClinicalImpression;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
```

```
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ClinicalImpression.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ClinicalImpression.status [ code ]; # 1..1 draft | completed | entered-in-error
  fhir:ClinicalImpression.code [ CodeableConcept ]; # 0..1 Kind of assessment performed
  fhir:ClinicalImpression.description [ string ]; # 0..1 Why/how the assessment was performed
  fhir:ClinicalImpression.subject [ Reference(Patient|Group) ]; # 1..1 Patient or group assessed
  fhir:ClinicalImpression.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode created from
  # ClinicalImpression.effective[x] : 0..1 Time of assessment. One of these 2
    fhir:ClinicalImpression.effectiveDateTime [ dateTime ]
    fhir:ClinicalImpression.effectivePeriod [ Period ]
  fhir:ClinicalImpression.date [ dateTime ]; # 0..1 When the assessment was documented
  fhir:ClinicalImpression.assessor [ Reference(Practitioner) ]; # 0..1 The clinician performing the assessment
  fhir:ClinicalImpression.previous [ Reference(ClinicalImpression) ]; # 0..1 Reference to last assessment
  fhir:ClinicalImpression.problem [ Reference(Condition|AllergyIntolerance) ], ... ; # 0..* Relevant impressions of
    patient state
  fhir:ClinicalImpression.investigation [ # 0..* One or more sets of investigations (signs, symptions, etc.)
    fhir:ClinicalImpression.investigation.code [ CodeableConcept ]; # 1..1 A name/code for the set
    fhir:ClinicalImpression.investigation.item [
    Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|DiagnosticReport|
  RiskAssessment|ImagingStudy) ], ... ; # 0..* Record of a specific investigation
  ], ...;
  fhir:ClinicalImpression.protocol [ uri ], ... ; # 0..* Clinical Protocol followed
  fhir:ClinicalImpression.summary [ string ]; # 0..1 Summary of the assessment
  fhir:ClinicalImpression.finding [ # 0..* Possible or likely findings and diagnoses
    # ClinicalImpression.finding.item[x] : 1..1 What was found. One of these 2
      fhir:ClinicalImpression.finding.itemCodeableConcept [ CodeableConcept ]
      fhir:ClinicalImpression.finding.itemReference [ Reference(Condition|Observation) ]
    fhir:ClinicalImpression.finding.basis [ string ]; # 0..1 Which investigations support finding
  ], ...;
  fhir:ClinicalImpression.prognosisCodeableConcept [ CodeableConcept ], ... ; # 0..* Estimate of likely outcome
  fhir:ClinicalImpression.prognosisReference [ Reference(RiskAssessment) ], ... ; # 0..* RiskAssessment expressing
    likely outcome
  fhir:ClinicalImpression.action [ Reference(ReferralRequest|ProcedureRequest|Procedure|MedicationRequest|Appointment)
    ], ... ; # 0..* Action taken as part of assessment procedure
  fhir:ClinicalImpression.note [ Annotation ], ... ; # 0..* Comments made about the ClinicalImpression
]
```

Changes since DSTU2

ClinicalImpression
ClinicalImpression.identifier | Added Element
ClinicalImpression.code | Added Element
ClinicalImpression.subject | Renamed from patient to subject Add Reference(Group)
ClinicalImpression.context | Added Element
ClinicalImpression.effective[x] | Added Element
ClinicalImpression.investigation | Added Element
ClinicalImpression.investigation.code | Added Element
ClinicalImpression.investigation.item | Added Element
ClinicalImpression.protocol | Max Cardinality changed from 1 to *
ClinicalImpression.finding.item[x] | Renamed from item to item[x] Add Reference(Condition), Add Reference(Observation)
ClinicalImpression.finding.basis | Added Element
ClinicalImpression.prognosisCodeableConcept | Added Element
ClinicalImpression.prognosisReference | Added Element
ClinicalImpression.action | Remove Reference(MedicationOrder), Remove Reference(DiagnosticOrder), Remove Reference(NutritionOrder), Remove Reference(SupplyRequest), Add Reference(MedicationRequest)
ClinicalImpression.note | Added Element
ClinicalImpression.trigger[x] | deleted
ClinicalImpression.investigations | deleted
ClinicalImpression.finding.cause | deleted
ClinicalImpression.resolved | deleted
ClinicalImpression.ruledOut | deleted
ClinicalImpression.prognosis | deleted
ClinicalImpression.plan | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
ClinicalImpression | | | DomainResource | A clinical assessment performed when planning treatments and management strategies for a patient
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier
status | ?!? | 1..1 | code | draft | completed | entered-in-error ClinicalImpressionStatus (Required)
code | ? | 0..1 | CodeableConcept | Kind of assessment performed
description | ? | 0..1 | string | Why/how the assessment was performed
subject | ? | 1..1 | Reference(Patient | Group) | Patient or group assessed
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode created from
effective[x] | ? | 0..1 | | Time of assessment
effectiveDateTime | | | dateTime
effectivePeriod | | | Period

date | ? | 0..1 | dateTime | When the assessment was documented
assessor | ? | 0..1 | Reference(Practitioner) | The clinician performing the assessment
previous | | 0..1 | Reference(ClinicalImpression) | Reference to last assessment
problem | ? | 0..* | Reference(Condition | AllergyIntolerance) | Relevant impressions of patient state
investigation | | 0..* | BackboneElement | One or more sets of investigations (signs, symptions, etc.)
code | | 1..1 | CodeableConcept | A name/code for the set Investigation Type (Example)
item | | 0..* | Reference(Observation | QuestionnaireResponse | FamilyMemberHistory | DiagnosticReport | RiskAssessment | ImagingStudy) | Record of a specific investigation
protocol | | 0..* | uri | Clinical Protocol followed
summary | | 0..1 | string | Summary of the assessment
finding | | 0..* | BackboneElement | Possible or likely findings and diagnoses
item[x] | | 1..1 | | What was found Condition/Problem/Diagnosis Codes (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Condition | Observation)
basis | | 0..1 | string | Which investigations support finding
prognosisCodeableConcept | | 0..* | CodeableConcept | Estimate of likely outcome Clinical Impression Prognosis (Example)
prognosisReference | | 0..* | Reference(RiskAssessment) | RiskAssessment expressing likely outcome
action | | 0..* | Reference(ReferralRequest | ProcedureRequest | Procedure | MedicationRequest | Appointment) | Action taken as part of assessment procedure
note | | 0..* | Annotation | Comments made about the ClinicalImpression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ClinicalImpression xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 draft | completed | entered-in-error -->
 <code><!-- 0..1 CodeableConcept Kind of assessment performed --></code>
 <description value="[string]"/><!-- 0..1 Why/how the assessment was performed -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient or group assessed --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode created from --></context>
 <effective[x]><!-- 0..1 dateTime|Period Time of assessment --></effective[x]>
 <date value="[dateTime]"/><!-- 0..1 When the assessment was documented -->
 <assessor><!-- 0..1 Reference(Practitioner) The clinician performing the assessment --></assessor>
 <previous><!-- 0..1 Reference(ClinicalImpression) Reference to last assessment --></previous>
 <problem><!-- 0..* Reference(Condition|AllergyIntolerance) Relevant impressions of patient state --></problem>
 <investigation>  <!-- 0..* One or more sets of investigations (signs, symptions, etc.) -->
  <code><!-- 1..1 CodeableConcept A name/code for the set --></code>
  <item><!-- 0..* Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy) Record of a specific investigation --></item>
 </investigation>
 <protocol value="[uri]"/><!-- 0..* Clinical Protocol followed -->
 <summary value="[string]"/><!-- 0..1 Summary of the assessment -->
 <finding>  <!-- 0..* Possible or likely findings and diagnoses -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Condition|Observation) What was found --></item[x]>
  <basis value="[string]"/><!-- 0..1 Which investigations support finding -->
 </finding>
 <prognosisCodeableConcept><!-- 0..* CodeableConcept Estimate of likely outcome --></prognosisCodeableConcept>
 <prognosisReference><!-- 0..* Reference(RiskAssessment) RiskAssessment expressing likely outcome
    --></prognosisReference>
 <action><!-- 0..* Reference(ReferralRequest|ProcedureRequest|Procedure|
   MedicationRequest|Appointment) Action taken as part of assessment procedure --></action>
 <note><!-- 0..* Annotation Comments made about the ClinicalImpression --></note>
</ClinicalImpression>
```

JSON Template

```
{
  "resourceType" : "ClinicalImpression",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  draft | completed | entered-in-error
  "code" : { CodeableConcept }, // Kind of assessment performed
  "description" : "<string>", // Why/how the assessment was performed
  "subject" : { Reference(Patient|Group) }, // R!  Patient or group assessed
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode created from
  // effective[x]: Time of assessment. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "date" : "<dateTime>", // When the assessment was documented
  "assessor" : { Reference(Practitioner) }, // The clinician performing the assessment
  "previous" : { Reference(ClinicalImpression) }, // Reference to last assessment
  "problem" : [{ Reference(Condition|AllergyIntolerance) }], // Relevant impressions of patient state
  "investigation" : [{ // One or more sets of investigations (signs, symptions, etc.)
    "code" : { CodeableConcept }, // R!  A name/code for the set
    "item" : [{ Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy) }] // Record of a specific investigation
  }],
  "protocol" : ["<uri>"], // Clinical Protocol followed
  "summary" : "<string>", // Summary of the assessment
  "finding" : [{ // Possible or likely findings and diagnoses
```

```
    // item[x]: What was found. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Condition|Observation) },
    "basis" : "<string>" // Which investigations support finding
  }],
  "prognosisCodeableConcept" : [{ CodeableConcept }], // Estimate of likely outcome
  "prognosisReference" : [{ Reference(RiskAssessment) }], // RiskAssessment expressing likely outcome
  "action" : [{ Reference(ReferralRequest|ProcedureRequest|Procedure|
   MedicationRequest|Appointment) }], // Action taken as part of assessment procedure
  "note" : [{ Annotation }] // Comments made about the ClinicalImpression
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ClinicalImpression;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ClinicalImpression.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ClinicalImpression.status [ code ]; # 1..1 draft | completed | entered-in-error
  fhir:ClinicalImpression.code [ CodeableConcept ]; # 0..1 Kind of assessment performed
  fhir:ClinicalImpression.description [ string ]; # 0..1 Why/how the assessment was performed
  fhir:ClinicalImpression.subject [ Reference(Patient|Group) ]; # 1..1 Patient or group assessed
  fhir:ClinicalImpression.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode created from
  # ClinicalImpression.effective[x] : 0..1 Time of assessment. One of these 2
    fhir:ClinicalImpression.effectiveDateTime [ dateTime ]
    fhir:ClinicalImpression.effectivePeriod [ Period ]
  fhir:ClinicalImpression.date [ dateTime ]; # 0..1 When the assessment was documented
  fhir:ClinicalImpression.assessor [ Reference(Practitioner) ]; # 0..1 The clinician performing the assessment
  fhir:ClinicalImpression.previous [ Reference(ClinicalImpression) ]; # 0..1 Reference to last assessment
  fhir:ClinicalImpression.problem [ Reference(Condition|AllergyIntolerance) ], ... ; # 0..* Relevant impressions of
    patient state
  fhir:ClinicalImpression.investigation [ # 0..* One or more sets of investigations (signs, symptions, etc.)
    fhir:ClinicalImpression.investigation.code [ CodeableConcept ]; # 1..1 A name/code for the set
    fhir:ClinicalImpression.investigation.item [
    Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|DiagnosticReport|
  RiskAssessment|ImagingStudy) ], ... ; # 0..* Record of a specific investigation
  ], ...;
  fhir:ClinicalImpression.protocol [ uri ], ... ; # 0..* Clinical Protocol followed
  fhir:ClinicalImpression.summary [ string ]; # 0..1 Summary of the assessment
  fhir:ClinicalImpression.finding [ # 0..* Possible or likely findings and diagnoses
    # ClinicalImpression.finding.item[x] : 1..1 What was found. One of these 2
      fhir:ClinicalImpression.finding.itemCodeableConcept [ CodeableConcept ]
      fhir:ClinicalImpression.finding.itemReference [ Reference(Condition|Observation) ]
    fhir:ClinicalImpression.finding.basis [ string ]; # 0..1 Which investigations support finding
  ], ...;
  fhir:ClinicalImpression.prognosisCodeableConcept [ CodeableConcept ], ... ; # 0..* Estimate of likely outcome
  fhir:ClinicalImpression.prognosisReference [ Reference(RiskAssessment) ], ... ; # 0..* RiskAssessment expressing
    likely outcome
  fhir:ClinicalImpression.action [ Reference(ReferralRequest|ProcedureRequest|Procedure|MedicationRequest|Appointment)
    ], ... ; # 0..* Action taken as part of assessment procedure
  fhir:ClinicalImpression.note [ Annotation ], ... ; # 0..* Comments made about the ClinicalImpression
]
```

Changes since DSTU2

ClinicalImpression
ClinicalImpression.identifier | Added Element
ClinicalImpression.code | Added Element
ClinicalImpression.subject | Renamed from patient to subject Add Reference(Group)
ClinicalImpression.context | Added Element
ClinicalImpression.effective[x] | Added Element
ClinicalImpression.investigation | Added Element
ClinicalImpression.investigation.code | Added Element
ClinicalImpression.investigation.item | Added Element
ClinicalImpression.protocol | Max Cardinality changed from 1 to *
ClinicalImpression.finding.item[x] | Renamed from item to item[x] Add Reference(Condition), Add Reference(Observation)
ClinicalImpression.finding.basis | Added Element
ClinicalImpression.prognosisCodeableConcept | Added Element
ClinicalImpression.prognosisReference | Added Element
ClinicalImpression.action | Remove Reference(MedicationOrder), Remove Reference(DiagnosticOrder), Remove Reference(NutritionOrder), Remove Reference(SupplyRequest), Add Reference(MedicationRequest)
ClinicalImpression.note | Added Element
ClinicalImpression.trigger[x] | deleted
ClinicalImpression.investigations | deleted
ClinicalImpression.finding.cause | deleted
ClinicalImpression.resolved | deleted
ClinicalImpression.ruledOut | deleted
ClinicalImpression.prognosis | deleted
ClinicalImpression.plan | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.8.4.1 Terminology Bindings

Path | Definition | Type | Reference
ClinicalImpression.status | The workflow state of a clinical impression. | Required | ClinicalImpressionStatus
ClinicalImpression.code | Identifies categories of clinical impressions. This is a place-holder only. It may be removed | Unknown | No details provided yet
ClinicalImpression.investigation.code | A name/code for a set of investigations. | Example | Investigation Type
ClinicalImpression.finding.item[x] | Identification of the Condition or diagnosis. | Example | Condition/Problem/Diagnosis Codes
ClinicalImpression.prognosisCodeableConcept | Prognosis or outlook findings | Example | Clinical Impression Prognosis

9.8.4.2 Known Issue

A known issue exists with circular references between Condition and ClinicalImpression, which is due to the low maturity level of ClinicalImpression. The Patient Care work group intends to address this issue when ClinicalImpression is considered substantially complete and ready for implementation.

9.8.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
action | reference | Action taken as part of assessment procedure | ClinicalImpression.action (Appointment, ReferralRequest, MedicationRequest, ProcedureRequest, Procedure)
assessor | reference | The clinician performing the assessment | ClinicalImpression.assessor (Practitioner)
context | reference | Encounter or Episode created from | ClinicalImpression.context (EpisodeOfCare, Encounter)
date | date | When the assessment was documented | ClinicalImpression.date | 18 Resources
finding-code | token | What was found | ClinicalImpression.finding.item.as(CodeableConcept)
finding-ref | reference | What was found | ClinicalImpression.finding.item.as(Reference) (Condition, Observation)
identifier | token | Business identifier | ClinicalImpression.identifier
investigation | reference | Record of a specific investigation | ClinicalImpression.investigation.item (RiskAssessment, FamilyMemberHistory, Observation, DiagnosticReport, ImagingStudy, QuestionnaireResponse)
patient | reference | Patient or group assessed | ClinicalImpression.subject (Patient) | 31 Resources
previous | reference | Reference to last assessment | ClinicalImpression.previous (ClinicalImpression)
problem | reference | Relevant impressions of patient state | ClinicalImpression.problem (Condition, AllergyIntolerance)
status | token | draft | completed | entered-in-error | ClinicalImpression.status
subject | reference | Patient or group assessed | ClinicalImpression.subject (Group, Patient)

9.8.6 Resource ClinicalImpression - Examples

URL: https://hl7.org/fhir/STU3/clinicalimpression-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.8.6 Resource ClinicalImpression - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

Example List:

Example of clinicalimpression | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.8.7 Resource ClinicalImpression - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/clinicalimpression-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.8.7 Resource ClinicalImpression - Detailed Descriptions

Patient Care Work Group | Maturity Level: 0 | Draft | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the ClinicalImpression resource.

ClinicalImpression
Definition | A record of a clinical assessment performed to determine what problem(s) may affect the patient and before planning the treatments or management strategies that are best to manage a patient's condition. Assessments are often 1:1 with a clinical consultation / encounter, but this varies greatly depending on the clinical workflow. This resource is called "ClinicalImpression" rather than "ClinicalAssessment" to avoid confusion with the recording of assessment tools such as Apgar score.
Control | 1..1
ClinicalImpression.identifier

Definition | A unique identifier assigned to the clinical impression that remains consistent regardless of what server the impression is stored on.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
ClinicalImpression.status
Definition | Identifies the workflow status of the assessment.
Control | 1..1
Terminology Binding | ClinicalImpressionStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the code entered-in-error that marks the clinical impression as not currently valid.
ClinicalImpression.code
Definition | Categorizes the type of clinical assessment performed.
Control | 0..1
Terminology Binding | ClinicalImpressionCode:
Type | CodeableConcept
Alternate Names | type
Summary | true
Comments | This is present as a place-holder only and may be removed based on feedback/work group opinion.
ClinicalImpression.description
Definition | A summary of the context and/or cause of the assessment - why / where was it performed, and what patient events/status prompted it.
Control | 0..1
Type | string
Summary | true
ClinicalImpression.subject
Definition | The patient or group of individuals assessed as part of this record.
Control | 1..1
Type | Reference(Patient | Group)
Requirements | Group is typically for veterinary and/or public health purposes.
Summary | true
ClinicalImpression.context
Definition | The encounter or episode of care this impression was created as part of.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
ClinicalImpression.effective[x]
Definition | The point in time or period over which the subject was assessed.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | This SHOULD be accurate to at least the minute, though some assessments only have a known date.
ClinicalImpression.date
Definition | Indicates when the documentation of the assessment was complete.
Control | 0..1
Type | dateTime
Summary | true
ClinicalImpression.assessor
Definition | The clinician performing the assessment.
Control | 0..1
Type | Reference(Practitioner)
Summary | true
ClinicalImpression.previous
Definition | A reference to the last assesment that was conducted bon this patient. Assessments are often/usually ongoing in nature; a care provider (practitioner or team) will make new assessments on an ongoing basis as new data arises or the patient's conditions changes.
Control | 0..1
Type | Reference(ClinicalImpression)
Comments | It is always likely that multiple previous assessments exist for a patient. The point of quoting a previous assessment is that this assessment is relative to it (see resolved).
ClinicalImpression.problem
Definition | This a list of the relevant problems/conditions for a patient.
Control | 0..*
Type | Reference(Condition | AllergyIntolerance)
Summary | true
Comments | e.g. The patient is a pregnant, and cardiac congestive failure, ?Adenocarcinoma, and is allergic to penicillin.
ClinicalImpression.investigation
Definition | One or more sets of investigations (signs, symptions, etc.). The actual grouping of investigations vary greatly depending on the type and context of the assessment. These investigations may include data generated during the assessment process, or data previously generated and recorded that is pertinent to the outcomes.

Control | 0..*
ClinicalImpression.investigation.code
Definition | A name/code for the group ("set") of investigations. Typically, this will be something like "signs", "symptoms", "clinical", "diagnostic", but the list is not constrained, and others such groups such as (exposure|family|travel|nutitirional) history may be used.
Control | 1..1
Terminology Binding | Investigation Type (Example)
Type | CodeableConcept
ClinicalImpression.investigation.item
Definition | A record of a specific investigation that was undertaken.
Control | 0..*
Type | Reference(Observation | QuestionnaireResponse | FamilyMemberHistory | DiagnosticReport | RiskAssessment | ImagingStudy)
Comments | Most investigations are observations of one kind of or another but some other specific types of data collection resources can also be used.
ClinicalImpression.protocol
Definition | Reference to a specific published clinical protocol that was followed during this assessment, and/or that provides evidence in support of the diagnosis.
Control | 0..*
Type | uri
ClinicalImpression.summary
Definition | A text summary of the investigations and the diagnosis.
Control | 0..1
Type | string
ClinicalImpression.finding
Definition | Specific findings or diagnoses that was considered likely or relevant to ongoing treatment.
Control | 0..*
ClinicalImpression.finding.item[x]
Definition | Specific text, code or reference for finding or diagnosis, which may include ruled-out or resolved conditions.
Control | 1..1
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept|Reference(Condition | Observation)
[x] Note | See Choice of Data Types for further information about how to use [x]
ClinicalImpression.finding.basis
Definition | Which investigations support finding or diagnosis.
Control | 0..1
Type | string
ClinicalImpression.prognosisCodeableConcept
Definition | Estimate of likely outcome.
Control | 0..*
Terminology Binding | Clinical Impression Prognosis (Example)
Type | CodeableConcept
ClinicalImpression.prognosisReference
Definition | RiskAssessment expressing likely outcome.
Control | 0..*
Type | Reference(RiskAssessment)
ClinicalImpression.action
Definition | Action taken as part of assessment procedure.
Control | 0..*
Type | Reference(ReferralRequest | ProcedureRequest | Procedure | MedicationRequest | Appointment)
Comments | Actions recommended as a result of the impression should not be captured using the 'action' element.
ClinicalImpression.note
Definition | Commentary about the impression, typically recorded after the impression itself was made, though supplemental notes by the original author could also appear.
Control | 0..*
Type | Annotation
Comments | Don't use this element for content that should more properly appear as one of the specific elements of the impression.

9.8.8 Resource ClinicalImpression - Mappings

URL: https://hl7.org/fhir/STU3/clinicalimpression-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.8.8 Resource ClinicalImpression - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

Mappings:

HL7 v2 Mapping

Mappings for the clinicalimpression resource.

9.8.8.1 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ClinicalImpression | Partial mapping for problem evaluation

identifier | PID-3
status | PRB-14
code
description
subject | PID-3
context
effective[x] | PRB-2
date
assessor | ROL-4
previous
problem | PRB-3 / IAM-7
investigation
code
item | OBX-21
protocol
summary | OBX
finding
item[x] | OBX
basis
prognosisCodeableConcept | PRB-22
prognosisReference
action
note

9.8.9 Resource ClinicalImpression - Design Notes

URL: https://hl7.org/fhir/STU3/clinicalimpression-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.8.9 Resource ClinicalImpression - Design Notes

to do

9.8.10 Resource ClinicalImpression - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/clinicalimpression-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.8.10 Resource ClinicalImpression - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ClinicalImpression resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.8.10.1 Profiles

No Profiles defined for this resource

9.8.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

9.8.10.3 Search Extensions

No Search Extensions defined for this resource

9.8.11 Resource ClinicalImpression - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/clinicalimpression-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.8.11 Resource ClinicalImpression - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

R2 : R3 Converson maps for ClinicalImpression.

Functional status for this map: 1 test that all execute ok. 1 fail round-trip testing and all r3 resources are valid. (see documentation)

9.8.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/ClinicalImpression2to3" = "R2 to R3 Conversions for ClinicalImpression"

conceptmap "ClinicalImpressionStatus" {
  prefix s = "http://hl7.org/fhir/clinical-impression-status"
  prefix t = "http://hl7.org/fhir/clinical-impression-status"

  s:"in-progress" = t:draft
  s:completed  = t:completed
  s:"entered-in-error" = t:"entered-in-error"
}

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ClinicalImpression" alias ClinicalImpressionR2 as source
uses "http://hl7.org/fhir/StructureDefinition/ClinicalImpression" alias ClinicalImpression as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  ClinicalImpression extends DomainResource
  input src : ClinicalImpressionR2 as source
  input tgt : ClinicalImpression as target

  "ClinicalImpression-subject" : for src.patient make tgt.subject
  "ClinicalImpression-assessor" : for src.assessor make tgt.assessor
  "ClinicalImpression-status" : for src.status as v make tgt.status = translate(v, "#ClinicalImpressionStatus", "code")
  "ClinicalImpression-date" : for src.date make tgt.date
  "ClinicalImpression-description" : for src.description make tgt.description
  "ClinicalImpression-previous" : for src.previous make tgt.previous
  "ClinicalImpression-problem" : for src.problem make tgt.problem
  "ClinicalImpression-investigations" : for src.investigations as vs0 make tgt.investigation as vt0 then
    cimInvestigation(vs0, vt0)
  "ClinicalImpression-protocol" : for src.protocol make tgt.protocol
  "ClinicalImpression-summary" : for src.summary make tgt.summary
  "ClinicalImpression-finding" : for src.finding as vs0 make tgt.finding as vt0 then cimFinding(vs0, vt0)
  "ClinicalImpression-action" : for src.action make tgt.action
endgroup

group cimFinding extends BackboneElement
  input src as source
  input tgt as target

  "ClinicalImpression.finding-item" : for src.item make tgt.item
  "ClinicalImpression.finding-reason" : for src.reason make tgt.basis
endgroup

group cimInvestigation extends BackboneElement
  input src as source
  input tgt as target

  "ClinicalImpression.finding-code" : for src.code make tgt.code
  "ClinicalImpression.finding-item" : for src.item make tgt.item
endgroup
```

9.8.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/ClinicalImpression3to2" = "R3 to R2 Conversion for ClinicalImpression"

conceptmap "ClinicalImpressionStatus" {
  prefix s = "http://hl7.org/fhir/clinical-impression-status"
  prefix t = "http://hl7.org/fhir/clinical-impression-status"

  s:draft = t:"in-progress"
  s:completed  = t:completed
  s:"entered-in-error" = t:"entered-in-error"
}


uses "http://hl7.org/fhir/StructureDefinition/ClinicalImpression" alias ClinicalImpression as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ClinicalImpression" alias ClinicalImpressionR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  ClinicalImpression extends DomainResource
  input src : ClinicalImpression as source
  input tgt : ClinicalImpressionR2 as target

  "ClinicalImpression-subject" : for src.subject make tgt.patient
  "ClinicalImpression-assessor" : for src.assessor make tgt.assessor
  "ClinicalImpression-status" : for src.status as v make tgt.status = translate(v, "#ClinicalImpressionStatus", "code")
  "ClinicalImpression-date" : for src.date make tgt.date
  "ClinicalImpression-description" : for src.description make tgt.description
  "ClinicalImpression-previous" : for src.previous make tgt.previous
  "ClinicalImpression-problem" : for src.problem make tgt.problem
  "ClinicalImpression-investigations" : for src.investigation as vs0 make tgt.investigations as vt0 then
    cimInvestigation(vs0, vt0)
  "ClinicalImpression-protocol" : for src.protocol make tgt.protocol
```

```
  "ClinicalImpression-summary" : for src.summary make tgt.summary
  "ClinicalImpression-finding" : for src.finding as vs0 make tgt.finding as vt0 then cimFinding(vs0, vt0)
  "ClinicalImpression-action" : for src.action make tgt.action
endgroup

group cimFinding extends BackboneElement
  input src as source
  input tgt as target

  "ClinicalImpression.finding-item" : for src.item make tgt.item
  "ClinicalImpression.finding-reason" : for src.basis make tgt.reason
endgroup

group cimInvestigation extends BackboneElement
  input src as source
  input tgt as target

  "ClinicalImpression.finding-code" : for src.code make tgt.code
  "ClinicalImpression.finding-item" : for src.item make tgt.item
endgroup
```

9.8.11.3 R3 Validation Errors

9.9 Resource AdverseEvent - Content

URL: https://hl7.org/fhir/STU3/adverseevent.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Regulated Clinical Research Information Management Work Group | Maturity Level: 0 | Draft | Compartments: Patient, Practitioner, RelatedPerson

Actual or potential/avoided event causing unintended physical injury resulting from or contributed to by medical care, a research study or other healthcare setting factors that requires additional monitoring, treatment, or hospitalization, or that results in death.

9.9.1 Scope and Usage

AdverseEvent is an event resource from a FHIR workflow perspective - see Workflow Event

This resource applies to events that occur during the course of medical care or medical research which may impact an individual as the recipient of care or the participant in a research study. There are also events that occur within a care setting that may or may not impact an individual but had the potential to cause an adverse event. Health care organizations monitor and report both adverse events as well as events that had the potential to cause patient harm. Data are often aggregated for reporting purposes.

An adverse event is the result of an intervention that caused unintentional harm to a specific subject or group of subjects. Examples of adverse events include the administration of an incorrect drug or an incorrect dose of a drug causing an adverse reaction, the use of an implanted device that causes an infection, or a biologic used during a research study that causes unanticipated renal failure. These events are characterized by the need to capture cause and effect (although they may not be known at the time of the event), severity, and outcome.

The context of an adverse event is also important. A subject may have condition(s) or current treatments (medications, diet, devices) that impact their response to a newly introduced medication, device or procedure. Knowledge of these variables is essential in establishing a cause and effect relationship for an adverse event.

A potential adverse event may also be called a near miss or an error. These are also events but because they were detected did not cause harm to a subject. Examples of potential adverse events include a product problem such as a faulty pacemaker that is detected prior implantation, a doctor working simultaneously on two electronic health records realizing the order for a drug was entered on the incorrect patient and then canceling the order, or a patient with a peanut allergy notices that his hospital dinner tray includes peanuts, and he does not eat the peanuts.

9.9.2 Boundaries and Relationships

The AdverseEvent resource is designed to represent events that have a harmful impact on a subject, or had the potential to cause harm to a subject but were avoided. In the course of medical care there are many actions that may impact how a subject responds to a particular treatment impacting patient safety. Therefore the AdverseEvent resource may reference multiple other resources to represent the context or details of an adverse event including but not limited to Observation, Condition, MedicationAdminsitration, Immunization, Procedure, or ResearchStudy.

A DetectedIssue reference is also related to the context of an AdverseEvent to the extent that a known risk for a potential issue such as a drug-drug interaction is documented. If in the context of a known issue, and adverse event occurs, citing this relationship is important for preventing such an occurrence in the future.

The AdverseEvent resource should not be used when a more specific resource exists.

* The AllergyIntolerance resource is a case specific means of capturing the condition of an allergy or intolerance and the criticality (or potential for future harm) based on the response of a particular individual.

* The Clinical Reasoning module provides resources and operations to enable the representation, distribution, and evaluation of clinical knowledge artifacts such as clinical decision support rules, quality measures, order sets, and protocols. The suite of resources within the clinical reasoning module should be used to capture clinical quality measures and clinical protocols that

help drive clinical best practices.

* The AdverseEvent resource is not intended to be used to capture potential subject risk in a prospective manner. A more appropriate resource for this purpose would be Risk Assessment which captures predicted outcomes for a patient or population on the basis of source information. Examples include a prognosis statement for a particular condition, risk of health outcome (heart attack, particular type of cancer) on the basis of lifestyle factors and/or family history or list of potential health risks based on a patient's genetic analysis.

9.9.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
AdverseEvent | ? | | DomainResource | Medical care, research study or other healthcare event causing physical injury Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Business identifier for the event
category | ? | 0..1 | code | AE | PAE An adverse event is an event that caused harm to a patient, an adverse reaction is a something that is a subject-specific event that is a result of an exposure to a medication, food, device or environmental substance, a potential adverse event is something that occurred and that could have caused harm to a patient but did not AdverseEventCategory (Required)
type | ? | 0..1 | CodeableConcept | actual | potential SNOMED CT Clinical Findings (Example)
subject | ? | 0..1 | Reference(Patient | ResearchSubject | Medication | Device) | Subject or group impacted by event
date | ? | 0..1 | dateTime | When the event occurred
reaction | ? | 0..* | Reference(Condition) | Adverse Reaction Events linked to exposure to substance
location | ? | 0..1 | Reference(Location) | Location where adverse event occurred
seriousness | ? | 0..1 | CodeableConcept | Mild | Moderate | Severe AdverseEventSeriousness (Example)
outcome | ? | 0..1 | CodeableConcept | resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown AdverseEventOutcome (Required)
recorder | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson) | Who recorded the adverse event
eventParticipant | ? | 0..1 | Reference(Practitioner | Device) | Who was involved in the adverse event or the potential adverse event
description | ? | 0..1 | string | Description of the adverse event
suspectEntity | ? | 0..* | BackboneElement | The suspected agent causing the adverse event
instance | ? | 1..1 | Reference(Substance | Medication | MedicationAdministration | MedicationStatement | Device) | Refers to the specific entity that caused the adverse event
causality | ? | 0..1 | code | causality1 | causality2 AdverseEventCausality (Required)
causalityAssessment | ? | 0..1 | CodeableConcept | assess1 | assess2 AdverseEventCausalityAssessment (Example)
causalityProductRelatedness | ? | 0..1 | string | AdverseEvent.suspectEntity.causalityProductRelatedness
causalityMethod | ? | 0..1 | CodeableConcept | method1 | method2 AdverseEventCausalityMethod (Example)
causalityAuthor | ? | 0..1 | Reference(Practitioner | PractitionerRole) | AdverseEvent.suspectEntity.causalityAuthor
causalityResult | ? | 0..1 | CodeableConcept | result1 | result2 AdverseEventCausalityResult (Example)
subjectMedicalHistory | ? | 0..* | Reference(Condition | Observation | AllergyIntolerance | FamilyMemberHistory | Immunization | Procedure) | AdverseEvent.subjectMedicalHistory
referenceDocument | ? | 0..* | Reference(DocumentReference) | AdverseEvent.referenceDocument
study | ? | 0..* | Reference(ResearchStudy) | AdverseEvent.study
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AdverseEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business identifier for the event --></identifier>
 <category value="[code]"/><!-- 0..1 AE | PAE  An adverse event is an event that caused harm to a patient,  an adverse
    reaction is a something that is a subject-specific event that is a result of an exposure to a medication, food,
    device or environmental substance, a potential adverse event is something that occurred and that could have caused
    harm to a patient but did not -->
 <type><!-- 0..1 CodeableConcept actual | potential --></type>
 <subject><!-- 0..1 Reference(Patient|ResearchSubject|Medication|Device) Subject or group impacted by event
    --></subject>
 <date value="[dateTime]"/><!-- 0..1 When the event occurred -->
 <reaction><!-- 0..* Reference(Condition) Adverse Reaction Events linked to exposure to substance --></reaction>
 <location><!-- 0..1 Reference(Location) Location where adverse event occurred --></location>
 <seriousness><!-- 0..1 CodeableConcept Mild | Moderate | Severe --></seriousness>
 <outcome><!-- 0..1 CodeableConcept resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
    --></outcome>
 <recorder><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson) Who recorded the adverse event --></recorder>
 <eventParticipant><!-- 0..1 Reference(Practitioner|Device) Who  was involved in the adverse event or the potential
    adverse event --></eventParticipant>
 <description value="[string]"/><!-- 0..1 Description of the adverse event -->
 <suspectEntity>  <!-- 0..* The suspected agent causing the adverse event -->
```

```
  <instance><!-- 1..1 Reference(Substance|Medication|MedicationAdministration|
    MedicationStatement|Device) Refers to the specific entity that caused the adverse event --></instance>
  <causality value="[code]"/><!-- 0..1 causality1 | causality2 -->
  <causalityAssessment><!-- 0..1 CodeableConcept assess1 | assess2 --></causalityAssessment>
  <causalityProductRelatedness value="[string]"/><!-- 0..1 AdverseEvent.suspectEntity.causalityProductRelatedness -->
  <causalityMethod><!-- 0..1 CodeableConcept method1 | method2 --></causalityMethod>
  <causalityAuthor><!-- 0..1 Reference(Practitioner|PractitionerRole) AdverseEvent.suspectEntity.causalityAuthor
    --></causalityAuthor>
  <causalityResult><!-- 0..1 CodeableConcept result1 | result2 --></causalityResult>
 </suspectEntity>
 <subjectMedicalHistory><!-- 0..* Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure) AdverseEvent.subjectMedicalHistory
    --></subjectMedicalHistory>
 <referenceDocument><!-- 0..* Reference(DocumentReference) AdverseEvent.referenceDocument --></referenceDocument>
 <study><!-- 0..* Reference(ResearchStudy) AdverseEvent.study --></study>
</AdverseEvent>
```

JSON Template

```
{
  "resourceType" : "AdverseEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business identifier for the event
  "category" : "<code>", // AE | PAE  An adverse event is an event that caused harm to a patient,  an adverse reaction
    is a something that is a subject-specific event that is a result of an exposure to a medication, food, device or
    environmental substance, a potential adverse event is something that occurred and that could have caused harm to a
    patient but did not
  "type" : { CodeableConcept }, // actual | potential
  "subject" : { Reference(Patient|ResearchSubject|Medication|Device) }, // Subject or group impacted by event
  "date" : "<dateTime>", // When the event occurred
  "reaction" : [{ Reference(Condition) }], // Adverse Reaction Events linked to exposure to substance
  "location" : { Reference(Location) }, // Location where adverse event occurred
  "seriousness" : { CodeableConcept }, // Mild | Moderate | Severe
  "outcome" : { CodeableConcept }, // resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
  "recorder" : { Reference(Patient|Practitioner|RelatedPerson) }, // Who recorded the adverse event
  "eventParticipant" : { Reference(Practitioner|Device) }, // Who  was involved in the adverse event or the potential
    adverse event
  "description" : "<string>", // Description of the adverse event
  "suspectEntity" : [{ // The suspected agent causing the adverse event
    "instance" : { Reference(Substance|Medication|MedicationAdministration|
    MedicationStatement|Device) }, // R!  Refers to the specific entity that caused the adverse event
    "causality" : "<code>", // causality1 | causality2
    "causalityAssessment" : { CodeableConcept }, // assess1 | assess2
    "causalityProductRelatedness" : "<string>", // AdverseEvent.suspectEntity.causalityProductRelatedness
    "causalityMethod" : { CodeableConcept }, // method1 | method2
    "causalityAuthor" : { Reference(Practitioner|PractitionerRole) }, // AdverseEvent.suspectEntity.causalityAuthor
    "causalityResult" : { CodeableConcept } // result1 | result2
  }],
  "subjectMedicalHistory" : [{ Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure) }], // AdverseEvent.subjectMedicalHistory
  "referenceDocument" : [{ Reference(DocumentReference) }], // AdverseEvent.referenceDocument
  "study" : [{ Reference(ResearchStudy) }] // AdverseEvent.study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AdverseEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AdverseEvent.identifier [ Identifier ]; # 0..1 Business identifier for the event
  fhir:AdverseEvent.category [ code ]; # 0..1 AE | PAE  An adverse event is an event that caused harm to a patient,  an
    adverse reaction is a something that is a subject-specific event that is a result of an exposure to a medication,
    food, device or environmental substance, a potential adverse event is something that occurred and that could have
    caused harm to a patient but did not
  fhir:AdverseEvent.type [ CodeableConcept ]; # 0..1 actual | potential
  fhir:AdverseEvent.subject [ Reference(Patient|ResearchSubject|Medication|Device) ]; # 0..1 Subject or group impacted
    by event
  fhir:AdverseEvent.date [ dateTime ]; # 0..1 When the event occurred
  fhir:AdverseEvent.reaction [ Reference(Condition) ], ... ; # 0..* Adverse Reaction Events linked to exposure to
    substance
  fhir:AdverseEvent.location [ Reference(Location) ]; # 0..1 Location where adverse event occurred
  fhir:AdverseEvent.seriousness [ CodeableConcept ]; # 0..1 Mild | Moderate | Severe
  fhir:AdverseEvent.outcome [ CodeableConcept ]; # 0..1 resolved | recovering | ongoing | resolvedWithSequelae | fatal |
     unknown
  fhir:AdverseEvent.recorder [ Reference(Patient|Practitioner|RelatedPerson) ]; # 0..1 Who recorded the adverse event
  fhir:AdverseEvent.eventParticipant [ Reference(Practitioner|Device) ]; # 0..1 Who  was involved in the adverse event
    or the potential adverse event
  fhir:AdverseEvent.description [ string ]; # 0..1 Description of the adverse event
  fhir:AdverseEvent.suspectEntity [ # 0..* The suspected agent causing the adverse event
    fhir:AdverseEvent.suspectEntity.instance [
    Reference(Substance|Medication|MedicationAdministration|MedicationStatement|Device) ]; # 1..1 Refers to the specific
     entity that caused the adverse event
    fhir:AdverseEvent.suspectEntity.causality [ code ]; # 0..1 causality1 | causality2
    fhir:AdverseEvent.suspectEntity.causalityAssessment [ CodeableConcept ]; # 0..1 assess1 | assess2
    fhir:AdverseEvent.suspectEntity.causalityProductRelatedness [ string ]; # 0..1
```

```
    AdverseEvent.suspectEntity.causalityProductRelatedness
    fhir:AdverseEvent.suspectEntity.causalityMethod [ CodeableConcept ]; # 0..1 method1 | method2
    fhir:AdverseEvent.suspectEntity.causalityAuthor [ Reference(Practitioner|PractitionerRole) ]; # 0..1
    AdverseEvent.suspectEntity.causalityAuthor
    fhir:AdverseEvent.suspectEntity.causalityResult [ CodeableConcept ]; # 0..1 result1 | result2
  ], ...;
  fhir:AdverseEvent.subjectMedicalHistory [
    Reference(Condition|Observation|AllergyIntolerance|FamilyMemberHistory|Immunization|
  Procedure) ], ... ; # 0..* AdverseEvent.subjectMedicalHistory
  fhir:AdverseEvent.referenceDocument [ Reference(DocumentReference) ], ... ; # 0..* AdverseEvent.referenceDocument
  fhir:AdverseEvent.study [ Reference(ResearchStudy) ], ... ; # 0..* AdverseEvent.study
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

AdverseEvent | ? | | DomainResource | Medical care, research study or other healthcare event causing physical injury Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..1 | Identifier | Business identifier for the event

category | ? | 0..1 | code | AE | PAE An adverse event is an event that caused harm to a patient, an adverse reaction is a something that is a subject-specific event that is a result of an exposure to a medication, food, device or environmental substance, a potential adverse event is something that occurred and that could have caused harm to a patient but did not AdverseEventCategory (Required)

type | ? | 0..1 | CodeableConcept | actual | potential SNOMED CT Clinical Findings (Example)

subject | ? | 0..1 | Reference(Patient | ResearchSubject | Medication | Device) | Subject or group impacted by event

date | ? | 0..1 | dateTime | When the event occurred

reaction | ? | 0..* | Reference(Condition) | Adverse Reaction Events linked to exposure to substance

location | ? | 0..1 | Reference(Location) | Location where adverse event occurred

seriousness | ? | 0..1 | CodeableConcept | Mild | Moderate | Severe AdverseEventSeriousness (Example)

outcome | ? | 0..1 | CodeableConcept | resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown AdverseEventOutcome (Required)

recorder | ? | 0..1 | Reference(Patient | Practitioner | RelatedPerson) | Who recorded the adverse event

eventParticipant | ? | 0..1 | Reference(Practitioner | Device) | Who was involved in the adverse event or the potential adverse event

description | ? | 0..1 | string | Description of the adverse event

suspectEntity | ? | 0..* | BackboneElement | The suspected agent causing the adverse event

instance | ? | 1..1 | Reference(Substance | Medication | MedicationAdministration | MedicationStatement | Device) | Refers to the specific entity that caused the adverse event

causality | ? | 0..1 | code | causality1 | causality2 AdverseEventCausality (Required)

causalityAssessment | ? | 0..1 | CodeableConcept | assess1 | assess2 AdverseEventCausalityAssessment (Example)

causalityProductRelatedness | ? | 0..1 | string | AdverseEvent.suspectEntity.causalityProductRelatedness

causalityMethod | ? | 0..1 | CodeableConcept | method1 | method2 AdverseEventCausalityMethod (Example)

causalityAuthor | ? | 0..1 | Reference(Practitioner | PractitionerRole) | AdverseEvent.suspectEntity.causalityAuthor

causalityResult | ? | 0..1 | CodeableConcept | result1 | result2 AdverseEventCausalityResult (Example)

subjectMedicalHistory | ? | 0..* | Reference(Condition | Observation | AllergyIntolerance | FamilyMemberHistory | Immunization | Procedure) | AdverseEvent.subjectMedicalHistory

referenceDocument | ? | 0..* | Reference(DocumentReference) | AdverseEvent.referenceDocument

study | ? | 0..* | Reference(ResearchStudy) | AdverseEvent.study

Documentation for this format

UML Diagram (Legend)

XML Template

```
<AdverseEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business identifier for the event --></identifier>
 <category value="[code]"/><!-- 0..1 AE | PAE  An adverse event is an event that caused harm to a patient,  an adverse
    reaction is a something that is a subject-specific event that is a result of an exposure to a medication, food,
    device or environmental substance, a potential adverse event is something that occurred and that could have caused
    harm to a patient but did not -->
 <type><!-- 0..1 CodeableConcept actual | potential --></type>
 <subject><!-- 0..1 Reference(Patient|ResearchSubject|Medication|Device) Subject or group impacted by event
    --></subject>
 <date value="[dateTime]"/><!-- 0..1 When the event occurred -->
 <reaction><!-- 0..* Reference(Condition) Adverse Reaction Events linked to exposure to substance --></reaction>
 <location><!-- 0..1 Reference(Location) Location where adverse event occurred --></location>
 <seriousness><!-- 0..1 CodeableConcept Mild | Moderate | Severe --></seriousness>
 <outcome><!-- 0..1 CodeableConcept resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
    --></outcome>
 <recorder><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson) Who recorded the adverse event --></recorder>
 <eventParticipant><!-- 0..1 Reference(Practitioner|Device) Who  was involved in the adverse event or the potential
    adverse event --></eventParticipant>
 <description value="[string]"/><!-- 0..1 Description of the adverse event -->
 <suspectEntity>  <!-- 0..* The suspected agent causing the adverse event -->
  <instance><!-- 1..1 Reference(Substance|Medication|MedicationAdministration|
    MedicationStatement|Device) Refers to the specific entity that caused the adverse event --></instance>
  <causality value="[code]"/><!-- 0..1 causality1 | causality2 -->
  <causalityAssessment><!-- 0..1 CodeableConcept assess1 | assess2 --></causalityAssessment>
  <causalityProductRelatedness value="[string]"/><!-- 0..1 AdverseEvent.suspectEntity.causalityProductRelatedness -->
  <causalityMethod><!-- 0..1 CodeableConcept method1 | method2 --></causalityMethod>
```

```
  <causalityAuthor><!-- 0..1 Reference(Practitioner|PractitionerRole) AdverseEvent.suspectEntity.causalityAuthor
    --></causalityAuthor>
  <causalityResult><!-- 0..1 CodeableConcept result1 | result2 --></causalityResult>
 </suspectEntity>
 <subjectMedicalHistory><!-- 0..* Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure) AdverseEvent.subjectMedicalHistory
    --></subjectMedicalHistory>
 <referenceDocument><!-- 0..* Reference(DocumentReference) AdverseEvent.referenceDocument --></referenceDocument>
 <study><!-- 0..* Reference(ResearchStudy) AdverseEvent.study --></study>
</AdverseEvent>
```

JSON Template

```
{
  "resourceType" : "AdverseEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business identifier for the event
  "category" : "<code>", // AE | PAE  An adverse event is an event that caused harm to a patient,  an adverse reaction
    is a something that is a subject-specific event that is a result of an exposure to a medication, food, device or
    environmental substance, a potential adverse event is something that occurred and that could have caused harm to a
    patient but did not
  "type" : { CodeableConcept }, // actual | potential
  "subject" : { Reference(Patient|ResearchSubject|Medication|Device) }, // Subject or group impacted by event
  "date" : "<dateTime>", // When the event occurred
  "reaction" : [{ Reference(Condition) }], // Adverse Reaction Events linked to exposure to substance
  "location" : { Reference(Location) }, // Location where adverse event occurred
  "seriousness" : { CodeableConcept }, // Mild | Moderate | Severe
  "outcome" : { CodeableConcept }, // resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
  "recorder" : { Reference(Patient|Practitioner|RelatedPerson) }, // Who recorded the adverse event
  "eventParticipant" : { Reference(Practitioner|Device) }, // Who  was involved in the adverse event or the potential
    adverse event
  "description" : "<string>", // Description of the adverse event
  "suspectEntity" : [{ // The suspected agent causing the adverse event
    "instance" : { Reference(Substance|Medication|MedicationAdministration|
    MedicationStatement|Device) }, // R!  Refers to the specific entity that caused the adverse event
    "causality" : "<code>", // causality1 | causality2
    "causalityAssessment" : { CodeableConcept }, // assess1 | assess2
    "causalityProductRelatedness" : "<string>", // AdverseEvent.suspectEntity.causalityProductRelatedness
    "causalityMethod" : { CodeableConcept }, // method1 | method2
    "causalityAuthor" : { Reference(Practitioner|PractitionerRole) }, // AdverseEvent.suspectEntity.causalityAuthor
    "causalityResult" : { CodeableConcept } // result1 | result2
  }],
  "subjectMedicalHistory" : [{ Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure) }], // AdverseEvent.subjectMedicalHistory
  "referenceDocument" : [{ Reference(DocumentReference) }], // AdverseEvent.referenceDocument
  "study" : [{ Reference(ResearchStudy) }] // AdverseEvent.study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AdverseEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AdverseEvent.identifier [ Identifier ]; # 0..1 Business identifier for the event
  fhir:AdverseEvent.category [ code ]; # 0..1 AE | PAE  An adverse event is an event that caused harm to a patient,  an
    adverse reaction is a something that is a subject-specific event that is a result of an exposure to a medication,
    food, device or environmental substance, a potential adverse event is something that occurred and that could have
    caused harm to a patient but did not
  fhir:AdverseEvent.type [ CodeableConcept ]; # 0..1 actual | potential
  fhir:AdverseEvent.subject [ Reference(Patient|ResearchSubject|Medication|Device) ]; # 0..1 Subject or group impacted
    by event
  fhir:AdverseEvent.date [ dateTime ]; # 0..1 When the event occurred
  fhir:AdverseEvent.reaction [ Reference(Condition) ], ... ; # 0..* Adverse Reaction Events linked to exposure to
    substance
  fhir:AdverseEvent.location [ Reference(Location) ]; # 0..1 Location where adverse event occurred
  fhir:AdverseEvent.seriousness [ CodeableConcept ]; # 0..1 Mild | Moderate | Severe
  fhir:AdverseEvent.outcome [ CodeableConcept ]; # 0..1 resolved | recovering | ongoing | resolvedWithSequelae | fatal |
     unknown
  fhir:AdverseEvent.recorder [ Reference(Patient|Practitioner|RelatedPerson) ]; # 0..1 Who recorded the adverse event
  fhir:AdverseEvent.eventParticipant [ Reference(Practitioner|Device) ]; # 0..1 Who  was involved in the adverse event
    or the potential adverse event
  fhir:AdverseEvent.description [ string ]; # 0..1 Description of the adverse event
  fhir:AdverseEvent.suspectEntity [ # 0..* The suspected agent causing the adverse event
    fhir:AdverseEvent.suspectEntity.instance [
    Reference(Substance|Medication|MedicationAdministration|MedicationStatement|Device) ]; # 1..1 Refers to the specific
     entity that caused the adverse event
    fhir:AdverseEvent.suspectEntity.causality [ code ]; # 0..1 causality1 | causality2
    fhir:AdverseEvent.suspectEntity.causalityAssessment [ CodeableConcept ]; # 0..1 assess1 | assess2
    fhir:AdverseEvent.suspectEntity.causalityProductRelatedness [ string ]; # 0..1
    AdverseEvent.suspectEntity.causalityProductRelatedness
    fhir:AdverseEvent.suspectEntity.causalityMethod [ CodeableConcept ]; # 0..1 method1 | method2
    fhir:AdverseEvent.suspectEntity.causalityAuthor [ Reference(Practitioner|PractitionerRole) ]; # 0..1
    AdverseEvent.suspectEntity.causalityAuthor
    fhir:AdverseEvent.suspectEntity.causalityResult [ CodeableConcept ]; # 0..1 result1 | result2
  ], ...;
```

```
  fhir:AdverseEvent.subjectMedicalHistory [
    Reference(Condition|Observation|AllergyIntolerance|FamilyMemberHistory|Immunization|
  Procedure) ], ... ; # 0..* AdverseEvent.subjectMedicalHistory
  fhir:AdverseEvent.referenceDocument [ Reference(DocumentReference) ], ... ; # 0..* AdverseEvent.referenceDocument
  fhir:AdverseEvent.study [ Reference(ResearchStudy) ], ... ; # 0..* AdverseEvent.study
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.9.3.1 Terminology Bindings

Path | Definition | Type | Reference
AdverseEvent.category | Overall categorization of the event, e.g. real or potential | Required | AdverseEventCategory
AdverseEvent.type | Detailed type of event | Example | SNOMED CT Clinical Findings
AdverseEvent.seriousness | Overall seriousness of this event for the patient | Example | AdverseEventSeriousness
AdverseEvent.outcome | TODO (and should this be required?) | Required | AdverseEventOutcome
AdverseEvent.suspectEntity.causality | TODO | Required | AdverseEventCausality
AdverseEvent.suspectEntity.causalityAssessment | TODO | Example | AdverseEventCausalityAssessment
AdverseEvent.suspectEntity.causalityMethod | TODO | Example | AdverseEventCausalityMethod
AdverseEvent.suspectEntity.causalityResult | TODO | Example | AdverseEventCausalityResult

9.9.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | AE | PAE An adverse event is an event that caused harm to a patient, an adverse reaction is a something that is a subject-specific event that is a result of an exposure to a medication, food, device or environmental substance, a potential adverse event is something that occurred and that could have caused harm to a patient but did not | AdverseEvent.category
date | date | When the event occurred | AdverseEvent.date
location | reference | Location where adverse event occurred | AdverseEvent.location (Location)
reaction | reference | Adverse Reaction Events linked to exposure to substance | AdverseEvent.reaction (Condition)
recorder | reference | Who recorded the adverse event | AdverseEvent.recorder (Practitioner, Patient, RelatedPerson)
seriousness | token | Mild | Moderate | Severe | AdverseEvent.seriousness
study | reference | AdverseEvent.study | AdverseEvent.study (ResearchStudy)
subject | reference | Subject or group impacted by event | AdverseEvent.subject (Device, Medication, Patient, ResearchSubject)
substance | reference | Refers to the specific entity that caused the adverse event | AdverseEvent.suspectEntity.instance (Device, Medication, Substance, MedicationAdministration, MedicationStatement)
type | token | actual | potential | AdverseEvent.type

9.9.5 Resource AdverseEvent - Examples

URL: https://hl7.org/fhir/STU3/adverseevent-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.9.5 Resource AdverseEvent - Examples

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Example List:

Example of adverseevent | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.9.6 Resource AdverseEvent - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/adverseevent-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.9.6 Resource AdverseEvent - Detailed Descriptions

Regulated Clinical Research Information Management Work Group | Maturity Level: 0 | Draft | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the AdverseEvent resource.

AdverseEvent
Definition | Actual or potential/avoided event causing unintended physical injury resulting from or contributed to by medical care, a research study or other healthcare setting factors that requires additional monitoring, treatment, or hospitalization, or that results in death.
Control | 1..1

Alternate Names | AE
Summary | true
AdverseEvent.identifier
Definition | The identifier(s) of this adverse event that are assigned by business processes and/or used to refer to it when a direct URL reference to the resource itsefl is not appropriate.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
AdverseEvent.category
Definition | The type of event which is important to characterize what occurred and caused harm to the subject, or had the potential to cause harm to the subject.
Control | 0..1
Terminology Binding | AdverseEventCategory (Required)
Type | code
Summary | true
AdverseEvent.type
Definition | This element defines the specific type of event that occurred or that was prevented from occurring.
Control | 0..1
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Summary | true
AdverseEvent.subject
Definition | This subject or group impacted by the event. With a prospective adverse event, there will be no subject as the adverse event was prevented.
Control | 0..1
Type | Reference(Patient | ResearchSubject | Medication | Device)
Alternate Names | patient
Summary | true
AdverseEvent.date
Definition | The date (and perhaps time) when the adverse event occurred.
Control | 0..1
Type | dateTime
Summary | true
AdverseEvent.reaction
Definition | Includes information about the reaction that occurred as a result of exposure to a substance (for example, a drug or a chemical).
Control | 0..*
Type | Reference(Condition)
Summary | true
AdverseEvent.location
Definition | The information about where the adverse event occurred.
Control | 0..1
Type | Reference(Location)
Summary | true
AdverseEvent.seriousness
Definition | Describes the seriousness or severity of the adverse event.
Control | 0..1
Terminology Binding | AdverseEventSeriousness (Example)
Type | CodeableConcept
Summary | true
AdverseEvent.outcome
Definition | Describes the type of outcome from the adverse event.
Control | 0..1
Terminology Binding | AdverseEventOutcome (Required)
Type | CodeableConcept
Summary | true
AdverseEvent.recorder
Definition | Information on who recorded the adverse event. May be the patient or a practitioner.
Control | 0..1
Type | Reference(Patient | Practitioner | RelatedPerson)
Summary | true
AdverseEvent.eventParticipant
Definition | Parties that may or should contribute or have contributed information to the Act. Such information includes information leading to the decision to perform the Act and how to perform the Act (e.g. consultant), information that the Act itself seeks to reveal (e.g. informant of clinical history), or information about what Act was performed (e.g. informant witness).
Control | 0..1
Type | Reference(Practitioner | Device)
Summary | true
AdverseEvent.description
Definition | Describes the adverse event in text.
Control | 0..1
Type | string
Summary | true

AdverseEvent.suspectEntity
Definition | Describes the entity that is suspected to have caused the adverse event.
Control | 0..*
Summary | true
AdverseEvent.suspectEntity.instance
Definition | Identifies the actual instance of what caused the adverse event. May be a substance, medication, medication administration, medication statement or a device.
Control | 1..1
Type | Reference(Substance | Medication | MedicationAdministration | MedicationStatement | Device)
Alternate Names | Substance or Medication or Device
Summary | true
AdverseEvent.suspectEntity.causality
Definition | causality1 | causality2.
Control | 0..1
Terminology Binding | AdverseEventCausality (Required)
Type | code
Summary | true
To Do | is this meant to be a backbone element?
AdverseEvent.suspectEntity.causalityAssessment
Definition | assess1 | assess2.
Control | 0..1
Terminology Binding | AdverseEventCausalityAssessment (Example)
Type | CodeableConcept
Summary | true
AdverseEvent.suspectEntity.causalityProductRelatedness
Definition | AdverseEvent.suspectEntity.causalityProductRelatedness.
Control | 0..1
Type | string
Summary | true
AdverseEvent.suspectEntity.causalityMethod
Definition | method1 | method2.
Control | 0..1
Terminology Binding | AdverseEventCausalityMethod (Example)
Type | CodeableConcept
Summary | true
AdverseEvent.suspectEntity.causalityAuthor
Definition | AdverseEvent.suspectEntity.causalityAuthor.
Control | 0..1
Type | Reference(Practitioner | PractitionerRole)
Summary | true
AdverseEvent.suspectEntity.causalityResult
Definition | result1 | result2.
Control | 0..1
Terminology Binding | AdverseEventCausalityResult (Example)
Type | CodeableConcept
Summary | true
To Do | what is difference between assessment and result?
AdverseEvent.subjectMedicalHistory
Definition | AdverseEvent.subjectMedicalHistory.
Control | 0..*
Type | Reference(Condition | Observation | AllergyIntolerance | FamilyMemberHistory | Immunization | Procedure)
Summary | true
AdverseEvent.referenceDocument
Definition | AdverseEvent.referenceDocument.
Control | 0..*
Type | Reference(DocumentReference)
Summary | true
AdverseEvent.study
Definition | AdverseEvent.study.
Control | 0..*
Type | Reference(ResearchStudy)
Summary | true

9.9.7 Resource AdverseEvent - Mappings

URL: https://hl7.org/fhir/STU3/adverseevent-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.9.7 Resource AdverseEvent - Mappings

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Mappings for the adverseevent resource.

9.9.8 Resource AdverseEvent - Design Notes

URL: https://hl7.org/fhir/STU3/adverseevent-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.9.8 Resource AdverseEvent - Design Notes

to do

9.9.9 Resource AdverseEvent - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/adverseevent-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.9.9 Resource AdverseEvent - Extensions & Profiles

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the AdverseEvent resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.9.9.1 Profiles

No Profiles defined for this resource

9.9.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

9.9.9.3 Search Extensions

No Search Extensions defined for this resource

9.10 Resource DetectedIssue - Content

URL: https://hl7.org/fhir/STU3/detectedissue.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, Ineffective treatment frequency, Procedure-condition conflict, etc.

9.10.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow, specifically Event.

This resource applies to various circumstances where there is a concern about an existing or proposed set of clinical activity. The issue could relate to single, proposed, or multiple actions. It does not apply to technical issues (e.g. lack of user permissions) but could relate to violation of patient consent limitations. Examples include:

* Drug-drug interactions

* Inappropriate therapy (wrong dose, frequency, body site)

* Duplicate therapy

This resource represents a specific instance of a potential issue for a particular patient. It is not intended to represent general patient-independent knowledge. This resource is also not intended to be used in defining general prohibitions on actions such as "No NSAIDs", "No solid oral dose forms" or "No MRIs - metalic tatoos". These guidelines can be captured using the AllergyIntolerance, and/or Flag resources. Similarly this resource is not to be used to capture clinical facts that may imply contraindications such as pregnancy, breast feeding, patient preferences, past procedures, etc. These would be represented using Condition, Procedure or other resources.

9.10.2 Boundaries and Relationships

This resource only applies to documenting a risk associated with a specific planned or ongoing action, not a general propensity to risk. The latter would be handled using AllergyIntolerance for substance-specific issues or Flag for other types

of issues.

This resource is limited to clinical issues associated with a proposed or ongoing action. It does not cover technical issues such as lack of permission, duplicate identifiers and other business rule violations. Technical issues are conveyed using the OperationOutcome resource. It is possible to have both OperationOutcome and DetectedIssue together, where the OperationOutcome might indicate that a requested action was rejected due to a clinical issue and the DetectedIssue provides the details of the issue.

9.10.3 Background and Context

Detected issues are typically identified by decision support systems. However, they may also be captured directly by clinicians. The latter typically happens for one of two reasons:

* A clinician wishes to communicate an issue to another clinician whose responsibility would be to resolve it (e.g. a pharmacist identifying an issue with a prescription prior to putting it on hold)

* A clinician wishes to pre-emptively identify that an issue is known and is being managed (to avoid red flags being raised as part of downstream workflow); e.g. Submitting a new order and including a link to a "duplicate therapy" issue with mitigation indicating that the therapy is not considered to be duplicate.

Decision-support generated issues can result from calling a decision-support engine directly (e.g. via a custom OperationDefinition) or as part of an attempt to perform some other function (creating an order, submitting an insurance claim, capturing a medication list). When the issues are generated as a by-product of performing some other sort of action, they may be included in the "response" to the requested action in the same manner as an OperationOutcome. In fact, both may be present - the OperationOutcome indicating that there was a warning or error associated with the request and a DetectedIssue providing the clinical details. (The OperationOutcome could point to the DetectedIssue via an extension.)

In those circumstances where requested operations are rejected as a result of a detected issue, the workflow may support allowing the operation to be re-tried, provided that the identified issue is included as part of the submission (possibly also including a mitigation). In doing so, the sender acknowledges the issue and takes responsibility for it, thus allowing the requested operation to proceed. See Linking to Detected Issues for guidance on how a DetectedIssue instance might be included as part of another operation.

Systems that require such workflows should document expected behavior as part of their CapabilityStatement declarations.

This resource is referenced by MedicationDispense and MedicationRequest

9.10.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
DetectedIssue | | | DomainResource | Clinical issue with action Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Unique id for the detected issue
status | ?!? | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
category | ? | 0..1 | CodeableConcept | Issue Category, e.g. drug-drug, duplicate therapy, etc. Detected Issue Category (Preferred)
severity | ? | 0..1 | code | high | moderate | low DetectedIssueSeverity (Required)
patient | ? | 0..1 | Reference(Patient) | Associated patient
date | ? | 0..1 | dateTime | When identified
author | ? | 0..1 | Reference(Practitioner | Device) | The provider or device that identified the issue
implicated | ? | 0..* | Reference(Any) | Problem resource
detail | | 0..1 | string | Description and context
reference | | 0..1 | uri | Authority for issue
mitigation | | 0..* | BackboneElement | Step taken to address
action | | 1..1 | CodeableConcept | What mitigation? Detected Issue Mitigation Action (Preferred)
date | | 0..1 | dateTime | Date committed
author | | 0..1 | Reference(Practitioner) | Who is committing?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DetectedIssue xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique id for the detected issue --></identifier>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <category><!-- 0..1 CodeableConcept Issue Category, e.g. drug-drug, duplicate therapy, etc. --></category>
```

```
 <severity value="[code]"/><!-- 0..1 high | moderate | low -->
 <patient><!-- 0..1 Reference(Patient) Associated patient --></patient>
 <date value="[dateTime]"/><!-- 0..1 When identified -->
 <author><!-- 0..1 Reference(Practitioner|Device) The provider or device that identified the issue --></author>
 <implicated><!-- 0..* Reference(Any) Problem resource --></implicated>
 <detail value="[string]"/><!-- 0..1 Description and context -->
 <reference value="[uri]"/><!-- 0..1 Authority for issue -->
 <mitigation>  <!-- 0..* Step taken to address -->
  <action><!-- 1..1 CodeableConcept What mitigation? --></action>
  <date value="[dateTime]"/><!-- 0..1 Date committed -->
  <author><!-- 0..1 Reference(Practitioner) Who is committing? --></author>
 </mitigation>
</DetectedIssue>
```

JSON Template

```
{
  "resourceType" : "DetectedIssue",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique id for the detected issue
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "category" : { CodeableConcept }, // Issue Category, e.g. drug-drug, duplicate therapy, etc.
  "severity" : "<code>", // high | moderate | low
  "patient" : { Reference(Patient) }, // Associated patient
  "date" : "<dateTime>", // When identified
  "author" : { Reference(Practitioner|Device) }, // The provider or device that identified the issue
  "implicated" : [{ Reference(Any) }], // Problem resource
  "detail" : "<string>", // Description and context
  "reference" : "<uri>", // Authority for issue
  "mitigation" : [{ // Step taken to address
    "action" : { CodeableConcept }, // R!  What mitigation?
    "date" : "<dateTime>", // Date committed
    "author" : { Reference(Practitioner) } // Who is committing?
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DetectedIssue;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DetectedIssue.identifier [ Identifier ]; # 0..1 Unique id for the detected issue
  fhir:DetectedIssue.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:DetectedIssue.category [ CodeableConcept ]; # 0..1 Issue Category, e.g. drug-drug, duplicate therapy, etc.
  fhir:DetectedIssue.severity [ code ]; # 0..1 high | moderate | low
  fhir:DetectedIssue.patient [ Reference(Patient) ]; # 0..1 Associated patient
  fhir:DetectedIssue.date [ dateTime ]; # 0..1 When identified
  fhir:DetectedIssue.author [ Reference(Practitioner|Device) ]; # 0..1 The provider or device that identified the issue
  fhir:DetectedIssue.implicated [ Reference(Any) ], ... ; # 0..* Problem resource
  fhir:DetectedIssue.detail [ string ]; # 0..1 Description and context
  fhir:DetectedIssue.reference [ uri ]; # 0..1 Authority for issue
  fhir:DetectedIssue.mitigation [ # 0..* Step taken to address
    fhir:DetectedIssue.mitigation.action [ CodeableConcept ]; # 1..1 What mitigation?
    fhir:DetectedIssue.mitigation.date [ dateTime ]; # 0..1 Date committed
    fhir:DetectedIssue.mitigation.author [ Reference(Practitioner) ]; # 0..1 Who is committing?
  ], ...;
]
```

Changes since DSTU2

DetectedIssue

DetectedIssue.status | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (4 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

DetectedIssue | | | DomainResource | Clinical issue with action Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..1 | Identifier | Unique id for the detected issue

status | ?!? | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)

category | ? | 0..1 | CodeableConcept | Issue Category, e.g. drug-drug, duplicate therapy, etc. Detected Issue Category (Preferred)

severity | ? | 0..1 | code | high | moderate | low DetectedIssueSeverity (Required)

patient | ? | 0..1 | Reference(Patient) | Associated patient

date | ? | 0..1 | dateTime | When identified

author | ? | 0..1 | Reference(Practitioner | Device) | The provider or device that identified the issue

implicated | ? | 0..* | Reference(Any) | Problem resource

detail | | 0..1 | string | Description and context

reference | | 0..1 | uri | Authority for issue
mitigation | | 0..* | BackboneElement | Step taken to address
action | | 1..1 | CodeableConcept | What mitigation? Detected Issue Mitigation Action (Preferred)
date | | 0..1 | dateTime | Date committed
author | | 0..1 | Reference(Practitioner) | Who is committing?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DetectedIssue xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique id for the detected issue --></identifier>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <category><!-- 0..1 CodeableConcept Issue Category, e.g. drug-drug, duplicate therapy, etc. --></category>
 <severity value="[code]"/><!-- 0..1 high | moderate | low -->
 <patient><!-- 0..1 Reference(Patient) Associated patient --></patient>
 <date value="[dateTime]"/><!-- 0..1 When identified -->
 <author><!-- 0..1 Reference(Practitioner|Device) The provider or device that identified the issue --></author>
 <implicated><!-- 0..* Reference(Any) Problem resource --></implicated>
 <detail value="[string]"/><!-- 0..1 Description and context -->
 <reference value="[uri]"/><!-- 0..1 Authority for issue -->
 <mitigation>  <!-- 0..* Step taken to address -->
  <action><!-- 1..1 CodeableConcept What mitigation? --></action>
  <date value="[dateTime]"/><!-- 0..1 Date committed -->
  <author><!-- 0..1 Reference(Practitioner) Who is committing? --></author>
 </mitigation>
</DetectedIssue>
```

JSON Template

```
{
  "resourceType" : "DetectedIssue",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique id for the detected issue
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "category" : { CodeableConcept }, // Issue Category, e.g. drug-drug, duplicate therapy, etc.
  "severity" : "<code>", // high | moderate | low
  "patient" : { Reference(Patient) }, // Associated patient
  "date" : "<dateTime>", // When identified
  "author" : { Reference(Practitioner|Device) }, // The provider or device that identified the issue
  "implicated" : [{ Reference(Any) }], // Problem resource
  "detail" : "<string>", // Description and context
  "reference" : "<uri>", // Authority for issue
  "mitigation" : [{ // Step taken to address
    "action" : { CodeableConcept }, // R!  What mitigation?
    "date" : "<dateTime>", // Date committed
    "author" : { Reference(Practitioner) } // Who is committing?
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DetectedIssue;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DetectedIssue.identifier [ Identifier ]; # 0..1 Unique id for the detected issue
  fhir:DetectedIssue.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:DetectedIssue.category [ CodeableConcept ]; # 0..1 Issue Category, e.g. drug-drug, duplicate therapy, etc.
  fhir:DetectedIssue.severity [ code ]; # 0..1 high | moderate | low
  fhir:DetectedIssue.patient [ Reference(Patient) ]; # 0..1 Associated patient
  fhir:DetectedIssue.date [ dateTime ]; # 0..1 When identified
  fhir:DetectedIssue.author [ Reference(Practitioner|Device) ]; # 0..1 The provider or device that identified the issue
  fhir:DetectedIssue.implicated [ Reference(Any) ], ... ; # 0..* Problem resource
  fhir:DetectedIssue.detail [ string ]; # 0..1 Description and context
  fhir:DetectedIssue.reference [ uri ]; # 0..1 Authority for issue
  fhir:DetectedIssue.mitigation [ # 0..* Step taken to address
    fhir:DetectedIssue.mitigation.action [ CodeableConcept ]; # 1..1 What mitigation?
    fhir:DetectedIssue.mitigation.date [ dateTime ]; # 0..1 Date committed
    fhir:DetectedIssue.mitigation.author [ Reference(Practitioner) ]; # 0..1 Who is committing?
  ], ...;
]
```

Changes since DSTU2

DetectedIssue
DetectedIssue.status | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (4 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.10.4.1 Terminology Bindings

Path | Definition | Type | Reference
DetectedIssue.status | Indicates the status of the identified issue | Required | ObservationStatus
DetectedIssue.category | Codes identifying the general type of detected issue; e.g. Drug-drug interaction, Timing issue, Duplicate therapy, etc. | Preferred | Detected Issue Category
DetectedIssue.severity | Indicates the potential degree of impact of the identified issue on the patient. | Required | DetectedIssueSeverity
DetectedIssue.mitigation.action | Codes describing steps taken to resolve the issue or other circumstances that mitigate the risk associated with the issue; e.g. 'added concurrent therapy', 'prior therapy documented', etc. | Preferred | Detected Issue Mitigation Action

9.10.5 Linking to Detected Issues

DetectedIssue follows the pattern of linking from the resource created "second". As DetectedIssue originates in response to one or more other existing records, it points to those records rather than being pointed to from them.

In some cases, a detected issue might be associated with a single record. When this occurs, it may be stored as a contained resource within the implicated resource provided that there is no expected need to search for the detected issue directly. However, with detected issues that implicate multiple records, containment is more problematic. In some workflows, a detected issue might be deemed to be "owned" by the record whose creation triggers the contraindication being created - i.e. the "second" or "last" record. However, where multiple actions are proposed as part of a single submission, there can be no single owner and containment will not be feasible.

If there is a strong need to point from an implicated resource to DetectedIssue and containment is not appropriate, an extension can be used.

9.10.6 Workflow Challenges

DetectedIssue is a resource that is frequently associated with workflow challenges where frequent alerts that are not clinically relevant result in clinicians tuning out (or turning off) the content and thus missing relevant alerts. Give consideration to this issue before making heavy use of this resource.

9.10.7 Open Issues

* Are author, reference and/or mitigation (and its various parts) all part of the 80%?

9.10.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | The provider or device that identified the issue | DetectedIssue.author (Practitioner, Device)
category | token | Issue Category, e.g. drug-drug, duplicate therapy, etc. | DetectedIssue.category
date | date | When identified | DetectedIssue.date | 18 Resources
identifier | token | Unique id for the detected issue | DetectedIssue.identifier | 26 Resources
implicated | reference | Problem resource | DetectedIssue.implicated (Any)
patient | reference | Associated patient | DetectedIssue.patient (Patient) | 31 Resources

9.10.9 Resource DetectedIssue - Examples

URL: https://hl7.org/fhir/STU3/detectedissue-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.10.9 Resource DetectedIssue - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Example List:

Drug-drug interaction between prescription and medication statement | ddi | XML | JSON | Turtle
Diagnostic order is placed when a recent image of the same body site already exists | duplicate | XML | JSON | Turtle
Patient is allergic to an inactive ingredient in an ordered medication | allergy | XML | JSON | Turtle
Lab order is contraindicated based on a patient's existing medication | lab | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.10.10 Resource DetectedIssue - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/detectedissue-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.10.10 Resource DetectedIssue - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the DetectedIssue resource.

DetectedIssue
Definition | Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, Ineffective treatment frequency, Procedure-condition conflict, etc.
Control | 1..1
Alternate Names | DDI; drug-drug interaction; Contraindication; Alert
DetectedIssue.identifier
Definition | Business identifier associated with the detected issue record.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Requirements | Allows linking instances of the same detected issue found on different servers.
Summary | true
DetectedIssue.status
Definition | Indicates the status of the detected issue.
Control | 1..1
Terminology Binding | ObservationStatus (Required)
Type | code
Is Modifier | true
Alternate Names | status
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes cancelled and entered-in-error that mark the issue as not currently valid.
DetectedIssue.category
Definition | Identifies the general type of issue identified.
Control | 0..1
Terminology Binding | Detected Issue Category (Preferred)
Type | CodeableConcept
Alternate Names | type
Summary | true
DetectedIssue.severity
Definition | Indicates the degree of importance associated with the identified issue based on the potential impact on the patient.
Control | 0..1
Terminology Binding | DetectedIssueSeverity (Required)
Type | code
Alternate Names | severity
Summary | true
DetectedIssue.patient
Definition | Indicates the patient whose record the detected issue is associated with.
Control | 0..1
Type | Reference(Patient)
Requirements | While the subject could be inferred by tracing the subject of the implicated resources, it's useful to have a direct link for query purposes.
Summary | true
DetectedIssue.date
Definition | The date or date-time when the detected issue was initially identified.
Control | 0..1
Type | dateTime
Requirements | No-one can be responsible for mitigation prior to the issue being identified.
Summary | true
DetectedIssue.author
Definition | Individual or device responsible for the issue being raised. For example, a decision support application or a pharmacist conducting a medication review.
Control | 0..1
Type | Reference(Practitioner | Device)
Summary | true
DetectedIssue.implicated
Definition | Indicates the resource representing the current activity or proposed activity that is potentially problematic.
Control | 0..*
Type | Reference(Any)
Alternate Names | cause
Summary | true
Comments | There's an implicit constraint on the number of implicated resources based on DetectedIssue.type; e.g. For drug-drug, there would be more than one. For timing, there would typically only be one.
DetectedIssue.detail
Definition | A textual explanation of the detected issue.

Control | 0..1
Type | string
Comments | Should focus on information not covered elsewhere as discrete data - no need to duplicate the narrative.
DetectedIssue.reference
Definition | The literature, knowledge-base or similar reference that describes the propensity for the detected issue identified.
Control | 0..1
Type | uri
DetectedIssue.mitigation
Definition | Indicates an action that has been taken or is committed to to reduce or eliminate the likelihood of the risk identified by the detected issue from manifesting. Can also reflect an observation of known mitigating factors that may reduce/eliminate the need for any action.
Control | 0..*
DetectedIssue.mitigation.action
Definition | Describes the action that was taken or the observation that was made that reduces/eliminates the risk associated with the identified issue.
Control | 1..1
Terminology Binding | Detected Issue Mitigation Action (Preferred)
Type | CodeableConcept
Comments | The "text" component can be used for detail or when no appropriate code exists.
DetectedIssue.mitigation.date
Definition | Indicates when the mitigating action was documented.
Control | 0..1
Type | dateTime
Comments | This may not be the same as when the mitigating step was actually taken.
DetectedIssue.mitigation.author
Definition | Identifies the practitioner who determined the mitigation and takes responsibility for the mitigation step occurring.
Control | 0..1
Type | Reference(Practitioner)

9.10.11 Resource DetectedIssue - Mappings

URL: https://hl7.org/fhir/STU3/detectedissue-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.10.11 Resource DetectedIssue - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the detectedissue resource.

9.10.11.1 Mappings for RIM Mapping (http://hl7.org/v3)

DetectedIssue | Observation[classCode=ALRT, moodCode=EVN)
identifier | .id
status
category | .code
severity | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=SEV].value
patient | .participation[typeCode=SBJ].role[classCode=PAT]
date | .effectiveTime
author | .participation[typeCode=AUT].role[classCode=ASSIGNED]
implicated | .outboundRelationship[typeCode=SUBJ].target[moodCode=EVN, INT]
detail | .text
reference | .outboundRelationship[typeCode=INST].target[classCode=ALRT, moodCode=DEFN].text.uri
mitigation | .outboundRelationship[typeCode=MITG].target[classCode=ACT, moodCode=EVN]
action | .code
date | .effectiveTime
author | .participation[typeCode=AUT].role[classCode=ASSIGNED]

9.10.11.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

DetectedIssue | Partial mapping from messages containing the AL1 or IAM segment, such as the ADT^A01, ADT^A08, and ADT^A60.
identifier | IAM-7
status
category | AL1-2 / IAM-9
severity | AL1-4/ IAR-2
patient | PID-3
date | ALI-6 / IAM-11
author | IAM-14 / IAM-16
implicated
detail | AL1-5 / IAM-8

reference
mitigation
action
date
author

9.10.11.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

DetectedIssue | Event
identifier | Event.identifier
status | Event.status
category | Event.code
severity
patient | Event.subject
date | Event.occurrence
author | Event.performer.actor
implicated | Event.reasonReference
detail | Event.note
reference
mitigation
action
date
author

9.10.12 Resource DetectedIssue - Design Notes

URL: https://hl7.org/fhir/STU3/detectedissue-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.10.12 Resource DetectedIssue - Design Notes

to do

9.10.13 Resource DetectedIssue - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/detectedissue-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.10.13 Resource DetectedIssue - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the DetectedIssue resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.10.13.1 Profiles

No Profiles defined for this resource

9.10.13.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

operationoutcome-detectedIssue | detectedIssue | for Standard OperationOutcome Extensions

9.10.13.3 Search Extensions

No Search Extensions defined for this resource

9.10.14 Resource DetectedIssue - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/detectedissue-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.10.14 Resource DetectedIssue - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

R2 : R3 Converson maps for DetectedIssue.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (4 errors). (see documentation)

9.10.14.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DetectedIssue2to3" = "R2 to R3 Conversions for DetectedIssue"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DetectedIssue" alias DetectedIssueR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DetectedIssue" alias DetectedIssue as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DetectedIssue extends DomainResource
  input src : DetectedIssueR2 as source
  input tgt : DetectedIssue as target

  "DetectedIssue-patient" : for src.patient make tgt.patient
  "DetectedIssue-category" : for src.category make tgt.category
  "DetectedIssue-severity" : for src.severity make tgt.severity
  "DetectedIssue-implicated" : for src.implicated make tgt.implicated
  "DetectedIssue-detail" : for src.detail make tgt.detail
  "DetectedIssue-date" : for src.date make tgt.date
  "DetectedIssue-author" : for src.author make tgt.author
  "DetectedIssue-identifier" : for src.identifier make tgt.identifier
  "DetectedIssue-reference" : for src.reference make tgt.reference
  "DetectedIssue-mitigation" : for src.mitigation as vs0 make tgt.mitigation as vt0 then dtiMitigation(vs0, vt0)
endgroup

group dtiMitigation extends BackboneElement
  input src as source
  input tgt as target

  "DetectedIssue.mitigation-action" : for src.action make tgt.action
  "DetectedIssue.mitigation-date" : for src.date make tgt.date
  "DetectedIssue.mitigation-author" : for src.author make tgt.author
endgroup
```

9.10.14.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DetectedIssue3to2" = "R3 to R2 Conversion for DetectedIssue"

uses "http://hl7.org/fhir/StructureDefinition/DetectedIssue" alias DetectedIssue as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DetectedIssue" alias DetectedIssueR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DetectedIssue extends DomainResource
  input src : DetectedIssue as source
  input tgt : DetectedIssueR2 as target

  "DetectedIssue-patient" : for src.patient make tgt.patient
  "DetectedIssue-category" : for src.category make tgt.category
  "DetectedIssue-severity" : for src.severity make tgt.severity
  "DetectedIssue-implicated" : for src.implicated make tgt.implicated
  "DetectedIssue-detail" : for src.detail make tgt.detail
  "DetectedIssue-date" : for src.date make tgt.date
  "DetectedIssue-author" : for src.author make tgt.author
  "DetectedIssue-identifier" : for src.identifier make tgt.identifier
  "DetectedIssue-reference" : for src.reference make tgt.reference
  "DetectedIssue-mitigation" : for src.mitigation as vs0 make tgt.mitigation as vt0 then dtiMitigation(vs0, vt0)
endgroup

group dtiMitigation extends BackboneElement
  input src as source
  input tgt as target

  "DetectedIssue.mitigation-action" : for src.action make tgt.action
  "DetectedIssue.mitigation-date" : for src.date make tgt.date
  "DetectedIssue.mitigation-author" : for src.author make tgt.author
endgroup
```

9.10.14.3 R3 Validation Errors

ddi | ERROR @ DetectedIssue: Profile http://hl7.org/fhir/StructureDefinition/DetectedIssue, Element 'DetectedIssue.status': minimum required = 1, but only found 0 (src = InstanceValidator)

allergy | ERROR @ DetectedIssue: Profile http://hl7.org/fhir/StructureDefinition/DetectedIssue, Element 'DetectedIssue.status': minimum required = 1, but only found 0 (src = InstanceValidator)

duplicate | ERROR @ DetectedIssue: Profile http://hl7.org/fhir/StructureDefinition/DetectedIssue, Element 'DetectedIssue.status': minimum required = 1, but only found 0 (src = InstanceValidator)

lab | ERROR @ DetectedIssue: Profile http://hl7.org/fhir/StructureDefinition/DetectedIssue, Element 'DetectedIssue.status': minimum required = 1, but only found 0 (src = InstanceValidator)

9.11 Resource RiskAssessment - Content

URL: https://hl7.org/fhir/STU3/riskassessment.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

An assessment of the likely outcome(s) for a patient or other subject as well as the likelihood of each outcome.

9.11.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow, specifically Event.

This resource captures predicted outcomes for a patient or population on the basis of source information. Examples include:

* A prognosis statement for a particular condition

* Risk of health outcome (heart attack, particular type of cancer) on the basis of lifestyle factors and/or family history

* List of potential health risks based on a patient's genetic analysis

* A prediction of outbreak infection rates within a geography based on immunization rates

This resource can be used to represent the results of formal scoring/decision support tools that evaluate risk. It can also be used to capture a practitioner's subjective assessment of the patient's risk based on existing knowledge and previous experience.

9.11.2 Boundaries and Relationships

Risk assessments are a specialized type of observation. We use a specialized resource to provide a simpler mechanism to capture of a series of risks and to associate those risks with time-ranges, probabilities, etc. All risk assertions are captured at one time based on a single set of source inputs. Capture of a single risk MAY be done using the Observation class, particularly in circumstances where it's treated as a generic observation, but for consistency, all risk assessments and prognosis SHOULD be captured using RiskAssessment.

Risk assessments may be based on a variety of factors, including:

* Basic demographic information from the Patient or Group resources

* Various Observations including vital signs, lab information, assessments, genetic information, etc.

* FamilyMemberHistory

* Current, past and proposed therapies Immunization, Procedure, CarePlan, etc.

Because so many resources can potentially be used, no limit is placed on what resources can be sent in RiskAssessment.basis. However, some resources would be nonsensical for this use (e.g. MessageHeader).

AllergyIntolerance and RiskAssessment

AllergyIntolerance describes a specific type of risk - propensity to reaction to a substance while RiskAssessment describes general risks to a subject, not generally based on a reaction.

9.11.3 Background and Context

Risk Assessments can be determined manually by health-care providers based on their professional expertise, by using an algorithmic scoring system or through a combination. Risk assessments can also be computed by decision support systems. OperationDefinitions may be defined by servers to generate risk assessments based on particular combinations of source resources or other inputs.

This resource is referenced by ClinicalImpression and Goal

9.11.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name	Flags	Card.	Type	Description & Constraints
RiskAssessment			DomainResource	Potential outcomes for a subject with likelihood Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	?	0..1	Identifier	Unique identifier for the assessment
basedOn		0..1	Reference(Any)	Request fulfilled by this assessment
parent		0..1	Reference(Any)	Part of this occurrence
status		1..1	code	registered | preliminary | final | amended + ObservationStatus (Required)
method	?	0..1	CodeableConcept	Evaluation mechanism
code	?	0..1	CodeableConcept	Type of assessment
subject	?	0..1	Reference(Patient | Group)	Who/what does assessment apply to?
context	?	0..1	Reference(Encounter | EpisodeOfCare)	Where was assessment performed?
occurrence[x]	?	0..1		When was assessment made?
occurrenceDateTime			dateTime	
occurrencePeriod			Period	

condition | ? | 0..1 | Reference(Condition) | Condition assessed
performer | ? | 0..1 | Reference(Practitioner | Device) | Who did assessment?
reason[x] | | 0..1 | | Why the assessment was necessary?
reasonCodeableConcept | | | CodeableConcept
reasonReference | | | Reference(Any)
basis | | 0..* | Reference(Any) | Information used in assessment
prediction | I | 0..* | BackboneElement | Outcome predicted + Must be <= 100
outcome | | 1..1 | CodeableConcept | Possible outcome for the subject
probability[x] | I | 0..1 | | Likelihood of specified outcome + low and high must be percentages, if present
probabilityDecimal | | | decimal
probabilityRange | | | Range
qualitativeRisk | | 0..1 | CodeableConcept | Likelihood of specified outcome as a qualitative value Risk Probability (Example)
relativeRisk | | 0..1 | decimal | Relative likelihood
when[x] | | 0..1 | | Timeframe or age range
whenPeriod | | | Period
whenRange | | | Range
rationale | | 0..1 | string | Explanation of prediction
mitigation | | 0..1 | string | How to reduce risk
comment | | 0..1 | string | Comments on the risk assessment
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RiskAssessment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique identifier for the assessment --></identifier>
 <basedOn><!-- 0..1 Reference(Any) Request fulfilled by this assessment --></basedOn>
 <parent><!-- 0..1 Reference(Any) Part of this occurrence --></parent>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <method><!-- 0..1 CodeableConcept Evaluation mechanism --></method>
 <code><!-- 0..1 CodeableConcept Type of assessment --></code>
 <subject><!-- 0..1 Reference(Patient|Group) Who/what does assessment apply to? --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Where was assessment performed? --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period When was assessment made? --></occurrence[x]>
 <condition><!-- 0..1 Reference(Condition) Condition assessed --></condition>
 <performer><!-- 0..1 Reference(Practitioner|Device) Who did assessment? --></performer>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Why the assessment was necessary? --></reason[x]>
 <basis><!-- 0..* Reference(Any) Information used in assessment --></basis>
 <prediction>  <!-- 0..* Outcome predicted -->
  <outcome><!-- 1..1 CodeableConcept Possible outcome for the subject --></outcome>
  <probability[x]><!-- 0..1 decimal|Range Likelihood of specified outcome --></probability[x]>
  <qualitativeRisk><!-- 0..1 CodeableConcept Likelihood of specified outcome as a qualitative value
    --></qualitativeRisk>
  <relativeRisk value="[decimal]"/><!-- 0..1 Relative likelihood -->
  <when[x]><!-- 0..1 Period|Range Timeframe or age range --></when[x]>
  <rationale value="[string]"/><!-- 0..1 Explanation of prediction -->
 </prediction>
 <mitigation value="[string]"/><!-- 0..1 How to reduce risk -->
 <comment value="[string]"/><!-- 0..1 Comments on the risk assessment -->
</RiskAssessment>
```

JSON Template

```
{
  "resourceType" : "RiskAssessment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique identifier for the assessment
  "basedOn" : { Reference(Any) }, // Request fulfilled by this assessment
  "parent" : { Reference(Any) }, // Part of this occurrence
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "method" : { CodeableConcept }, // Evaluation mechanism
  "code" : { CodeableConcept }, // Type of assessment
  "subject" : { Reference(Patient|Group) }, // Who/what does assessment apply to?
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Where was assessment performed?
  // occurrence[x]: When was assessment made?. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "condition" : { Reference(Condition) }, // Condition assessed
  "performer" : { Reference(Practitioner|Device) }, // Who did assessment?
  // reason[x]: Why the assessment was necessary?. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "basis" : [{ Reference(Any) }], // Information used in assessment
  "prediction" : [{ // Outcome predicted
    "outcome" : { CodeableConcept }, // R!  Possible outcome for the subject
    // probability[x]: Likelihood of specified outcome. One of these 2:
    "probabilityDecimal" : <decimal>,
    "probabilityRange" : { Range },
    "qualitativeRisk" : { CodeableConcept }, // Likelihood of specified outcome as a qualitative value
    "relativeRisk" : <decimal>, // Relative likelihood
    // when[x]: Timeframe or age range. One of these 2:
    "whenPeriod" : { Period },
```

```
    "whenRange" : { Range },
    "rationale" : "<string>" // Explanation of prediction
  }],
  "mitigation" : "<string>", // How to reduce risk
  "comment" : "<string>" // Comments on the risk assessment
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RiskAssessment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RiskAssessment.identifier [ Identifier ]; # 0..1 Unique identifier for the assessment
  fhir:RiskAssessment.basedOn [ Reference(Any) ]; # 0..1 Request fulfilled by this assessment
  fhir:RiskAssessment.parent [ Reference(Any) ]; # 0..1 Part of this occurrence
  fhir:RiskAssessment.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:RiskAssessment.method [ CodeableConcept ]; # 0..1 Evaluation mechanism
  fhir:RiskAssessment.code [ CodeableConcept ]; # 0..1 Type of assessment
  fhir:RiskAssessment.subject [ Reference(Patient|Group) ]; # 0..1 Who/what does assessment apply to?
  fhir:RiskAssessment.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Where was assessment performed?
  # RiskAssessment.occurrence[x] : 0..1 When was assessment made?. One of these 2
    fhir:RiskAssessment.occurrenceDateTime [ dateTime ]
    fhir:RiskAssessment.occurrencePeriod [ Period ]
  fhir:RiskAssessment.condition [ Reference(Condition) ]; # 0..1 Condition assessed
  fhir:RiskAssessment.performer [ Reference(Practitioner|Device) ]; # 0..1 Who did assessment?
  # RiskAssessment.reason[x] : 0..1 Why the assessment was necessary?. One of these 2
    fhir:RiskAssessment.reasonCodeableConcept [ CodeableConcept ]
    fhir:RiskAssessment.reasonReference [ Reference(Any) ]
  fhir:RiskAssessment.basis [ Reference(Any) ], ... ; # 0..* Information used in assessment
  fhir:RiskAssessment.prediction [ # 0..* Outcome predicted
    fhir:RiskAssessment.prediction.outcome [ CodeableConcept ]; # 1..1 Possible outcome for the subject
    # RiskAssessment.prediction.probability[x] : 0..1 Likelihood of specified outcome. One of these 2
      fhir:RiskAssessment.prediction.probabilityDecimal [ decimal ]
      fhir:RiskAssessment.prediction.probabilityRange [ Range ]
    fhir:RiskAssessment.prediction.qualitativeRisk [ CodeableConcept ]; # 0..1 Likelihood of specified outcome as a
    qualitative value
    fhir:RiskAssessment.prediction.relativeRisk [ decimal ]; # 0..1 Relative likelihood
    # RiskAssessment.prediction.when[x] : 0..1 Timeframe or age range. One of these 2
      fhir:RiskAssessment.prediction.whenPeriod [ Period ]
      fhir:RiskAssessment.prediction.whenRange [ Range ]
    fhir:RiskAssessment.prediction.rationale [ string ]; # 0..1 Explanation of prediction
  ], ...;
  fhir:RiskAssessment.mitigation [ string ]; # 0..1 How to reduce risk
  fhir:RiskAssessment.comment [ string ]; # 0..1 Comments on the risk assessment
]
```

Changes since DSTU2

RiskAssessment
RiskAssessment.basedOn | Added Element
RiskAssessment.parent | Added Element
RiskAssessment.status | Added Element
RiskAssessment.code | Added Element
RiskAssessment.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
RiskAssessment.occurrence[x] | Added Element
RiskAssessment.reason[x] | Added Element
RiskAssessment.prediction.probability[x] | Remove CodeableConcept
RiskAssessment.prediction.qualitativeRisk | Added Element
RiskAssessment.comment | Added Element
RiskAssessment.date | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (4 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
RiskAssessment | | | DomainResource | Potential outcomes for a subject with likelihood Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Unique identifier for the assessment
basedOn | | 0..1 | Reference(Any) | Request fulfilled by this assessment
parent | | 0..1 | Reference(Any) | Part of this occurrence
status | | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
method | ? | 0..1 | CodeableConcept | Evaluation mechanism
code | ? | 0..1 | CodeableConcept | Type of assessment
subject | ? | 0..1 | Reference(Patient | Group) | Who/what does assessment apply to?
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Where was assessment performed?
occurrence[x] | ? | 0..1 | | When was assessment made?

occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
condition | ? | 0..1 | Reference(Condition) | Condition assessed
performer | ? | 0..1 | Reference(Practitioner | Device) | Who did assessment?
reason[x] | | 0..1 | | Why the assessment was necessary?
reasonCodeableConcept | | | CodeableConcept
reasonReference | | | Reference(Any)
basis | | 0..* | Reference(Any) | Information used in assessment
prediction | I | 0..* | BackboneElement | Outcome predicted + Must be <= 100
outcome | | 1..1 | CodeableConcept | Possible outcome for the subject
probability[x] | I | 0..1 | | Likelihood of specified outcome + low and high must be percentages, if present
probabilityDecimal | | | decimal
probabilityRange | | | Range
qualitativeRisk | | 0..1 | CodeableConcept | Likelihood of specified outcome as a qualitative value Risk Probability (Example)
relativeRisk | | 0..1 | decimal | Relative likelihood
when[x] | | 0..1 | | Timeframe or age range
whenPeriod | | | Period
whenRange | | | Range
rationale | | 0..1 | string | Explanation of prediction
mitigation | | 0..1 | string | How to reduce risk
comment | | 0..1 | string | Comments on the risk assessment
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RiskAssessment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique identifier for the assessment --></identifier>
 <basedOn><!-- 0..1 Reference(Any) Request fulfilled by this assessment --></basedOn>
 <parent><!-- 0..1 Reference(Any) Part of this occurrence --></parent>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <method><!-- 0..1 CodeableConcept Evaluation mechanism --></method>
 <code><!-- 0..1 CodeableConcept Type of assessment --></code>
 <subject><!-- 0..1 Reference(Patient|Group) Who/what does assessment apply to? --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Where was assessment performed? --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period When was assessment made? --></occurrence[x]>
 <condition><!-- 0..1 Reference(Condition) Condition assessed --></condition>
 <performer><!-- 0..1 Reference(Practitioner|Device) Who did assessment? --></performer>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Why the assessment was necessary? --></reason[x]>
 <basis><!-- 0..* Reference(Any) Information used in assessment --></basis>
 <prediction>  <!-- 0..* Outcome predicted -->
  <outcome><!-- 1..1 CodeableConcept Possible outcome for the subject --></outcome>
  <probability[x]><!-- 0..1 decimal|Range Likelihood of specified outcome --></probability[x]>
  <qualitativeRisk><!-- 0..1 CodeableConcept Likelihood of specified outcome as a qualitative value
    --></qualitativeRisk>
  <relativeRisk value="[decimal]"/><!-- 0..1 Relative likelihood -->
  <when[x]><!-- 0..1 Period|Range Timeframe or age range --></when[x]>
  <rationale value="[string]"/><!-- 0..1 Explanation of prediction -->
 </prediction>
 <mitigation value="[string]"/><!-- 0..1 How to reduce risk -->
 <comment value="[string]"/><!-- 0..1 Comments on the risk assessment -->
</RiskAssessment>
```

JSON Template

```
{
  "resourceType" : "RiskAssessment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique identifier for the assessment
  "basedOn" : { Reference(Any) }, // Request fulfilled by this assessment
  "parent" : { Reference(Any) }, // Part of this occurrence
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "method" : { CodeableConcept }, // Evaluation mechanism
  "code" : { CodeableConcept }, // Type of assessment
  "subject" : { Reference(Patient|Group) }, // Who/what does assessment apply to?
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Where was assessment performed?
  // occurrence[x]: When was assessment made?. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "condition" : { Reference(Condition) }, // Condition assessed
  "performer" : { Reference(Practitioner|Device) }, // Who did assessment?
  // reason[x]: Why the assessment was necessary?. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "basis" : [{ Reference(Any) }], // Information used in assessment
  "prediction" : [{ // Outcome predicted
    "outcome" : { CodeableConcept }, // R!  Possible outcome for the subject
    // probability[x]: Likelihood of specified outcome. One of these 2:
    "probabilityDecimal" : <decimal>,
    "probabilityRange" : { Range },
    "qualitativeRisk" : { CodeableConcept }, // Likelihood of specified outcome as a qualitative value
    "relativeRisk" : <decimal>, // Relative likelihood
```

```
    // when[x]: Timeframe or age range. One of these 2:
    "whenPeriod" : { Period },
    "whenRange" : { Range },
    "rationale" : "<string>" // Explanation of prediction
  }],
  "mitigation" : "<string>", // How to reduce risk
  "comment" : "<string>" // Comments on the risk assessment
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RiskAssessment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RiskAssessment.identifier [ Identifier ]; # 0..1 Unique identifier for the assessment
  fhir:RiskAssessment.basedOn [ Reference(Any) ]; # 0..1 Request fulfilled by this assessment
  fhir:RiskAssessment.parent [ Reference(Any) ]; # 0..1 Part of this occurrence
  fhir:RiskAssessment.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:RiskAssessment.method [ CodeableConcept ]; # 0..1 Evaluation mechanism
  fhir:RiskAssessment.code [ CodeableConcept ]; # 0..1 Type of assessment
  fhir:RiskAssessment.subject [ Reference(Patient|Group) ]; # 0..1 Who/what does assessment apply to?
  fhir:RiskAssessment.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Where was assessment performed?
  # RiskAssessment.occurrence[x] : 0..1 When was assessment made?. One of these 2
    fhir:RiskAssessment.occurrenceDateTime [ dateTime ]
    fhir:RiskAssessment.occurrencePeriod [ Period ]
  fhir:RiskAssessment.condition [ Reference(Condition) ]; # 0..1 Condition assessed
  fhir:RiskAssessment.performer [ Reference(Practitioner|Device) ]; # 0..1 Who did assessment?
  # RiskAssessment.reason[x] : 0..1 Why the assessment was necessary?. One of these 2
    fhir:RiskAssessment.reasonCodeableConcept [ CodeableConcept ]
    fhir:RiskAssessment.reasonReference [ Reference(Any) ]
  fhir:RiskAssessment.basis [ Reference(Any) ], ... ; # 0..* Information used in assessment
  fhir:RiskAssessment.prediction [ # 0..* Outcome predicted
    fhir:RiskAssessment.prediction.outcome [ CodeableConcept ]; # 1..1 Possible outcome for the subject
    # RiskAssessment.prediction.probability[x] : 0..1 Likelihood of specified outcome. One of these 2
      fhir:RiskAssessment.prediction.probabilityDecimal [ decimal ]
      fhir:RiskAssessment.prediction.probabilityRange [ Range ]
    fhir:RiskAssessment.prediction.qualitativeRisk [ CodeableConcept ]; # 0..1 Likelihood of specified outcome as a
    qualitative value
    fhir:RiskAssessment.prediction.relativeRisk [ decimal ]; # 0..1 Relative likelihood
    # RiskAssessment.prediction.when[x] : 0..1 Timeframe or age range. One of these 2
      fhir:RiskAssessment.prediction.whenPeriod [ Period ]
      fhir:RiskAssessment.prediction.whenRange [ Range ]
    fhir:RiskAssessment.prediction.rationale [ string ]; # 0..1 Explanation of prediction
  ], ...;
  fhir:RiskAssessment.mitigation [ string ]; # 0..1 How to reduce risk
  fhir:RiskAssessment.comment [ string ]; # 0..1 Comments on the risk assessment
]
```

Changes since DSTU2

RiskAssessment
RiskAssessment.basedOn | Added Element
RiskAssessment.parent | Added Element
RiskAssessment.status | Added Element
RiskAssessment.code | Added Element
RiskAssessment.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
RiskAssessment.occurrence[x] | Added Element
RiskAssessment.reason[x] | Added Element
RiskAssessment.prediction.probability[x] | Remove CodeableConcept
RiskAssessment.prediction.qualitativeRisk | Added Element
RiskAssessment.comment | Added Element
RiskAssessment.date | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (4 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

9.11.4.1 Terminology Bindings

Path | Definition | Type | Reference
RiskAssessment.status | The status of the risk assessment; e.g. preliminary, final, amended, etc. | Required | ObservationStatus
RiskAssessment.method | The mechanism or algorithm used to make the assessment; e.g. TIMI, PRISM, Cardiff Type 2 diabetes, etc. | Unknown | No details provided yet
RiskAssessment.prediction.outcome | The condition or other outcome; e.g. death, remission, amputation, infection, etc. | Unknown | No details provided yet
RiskAssessment.prediction.qualitativeRisk | The likelihood of the occurrence of a specified outcome | Example | Risk Probability

9.11.4.2 Constraints

* ras-1: On RiskAssessment.prediction.probability[x]: low and high must be percentages, if present (expression on RiskAssessment.prediction.probability[x]: (low.empty() or ((low.code = '%') and (low.system = %ucum))) and (high.empty() or ((high.code = '%') and (high.system = %ucum))))

* ras-2: On RiskAssessment.prediction: Must be <= 100 (expression on RiskAssessment.prediction: probability is decimal implies probability.as(decimal) <= 100)

9.11.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
condition | reference | Condition assessed | RiskAssessment.condition (Condition)
date | date | When was assessment made? | RiskAssessment.occurrence.as(DateTime) | 18 Resources
encounter | reference | Where was assessment performed? | RiskAssessment.context (Encounter) | 12 Resources
identifier | token | Unique identifier for the assessment | RiskAssessment.identifier | 26 Resources
method | token | Evaluation mechanism | RiskAssessment.method
patient | reference | Who/what does assessment apply to? | RiskAssessment.subject (Patient) | 31 Resources
performer | reference | Who did assessment? | RiskAssessment.performer (Practitioner, Device)
probability | number | Likelihood of specified outcome | RiskAssessment.prediction.probability
risk | token | Likelihood of specified outcome as a qualitative value | RiskAssessment.prediction.qualitativeRisk
subject | reference | Who/what does assessment apply to? | RiskAssessment.subject (Group, Patient)

9.11.6 Resource RiskAssessment - Examples

URL: https://hl7.org/fhir/STU3/riskassessment-examples.html
This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.11.6 Resource RiskAssessment - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Example List:

Genetic risk assessment | genetic | XML | JSON | Turtle
Patient prognosis | prognosis | XML | JSON | Turtle
Cardiac risk assessment | cardiac | XML | JSON | Turtle
Public health population risk assessment | population | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.11.7 Resource RiskAssessment - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/riskassessment-definitions.html
This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.11.7 Resource RiskAssessment - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the RiskAssessment resource.

RiskAssessment
Definition | An assessment of the likely outcome(s) for a patient or other subject as well as the likelihood of each outcome.
Control | 1..1
Alternate Names | Prognosis
RiskAssessment.identifier
Definition | Business identifier assigned to the risk assessment.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
RiskAssessment.basedOn
Definition | A reference to the request that is fulfilled by this risk assessment.
Control | 0..1
Type | Reference(Any)
RiskAssessment.parent
Definition | A reference to a resource that this risk assessment is part of, such as a Procedure.
Control | 0..1
Type | Reference(Any)
RiskAssessment.status

Definition | The status of the RiskAssessment, using the same statuses as an Observation.
Control | 1..1
Terminology Binding | ObservationStatus (Required)
Type | code
RiskAssessment.method
Definition | The algorithm, process or mechanism used to evaluate the risk.
Control | 0..1
Terminology Binding | RiskAssessmentMethod:
Type | CodeableConcept
Requirements | The method can influence the results of the assessment.
Summary | true
RiskAssessment.code
Definition | The type of the risk assessment performed.
Control | 0..1
Type | CodeableConcept
Summary | true
RiskAssessment.subject
Definition | The patient or group the risk assessment applies to.
Control | 0..1
Type | Reference(Patient | Group)
Summary | true
RiskAssessment.context
Definition | The encounter where the assessment was performed.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Summary | true
RiskAssessment.occurrence[x]
Definition | The date (and possibly time) the risk assessment was performed.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | The assessment results lose validity the more time elapses from when they are first made.
Summary | true
RiskAssessment.condition
Definition | For assessments or prognosis specific to a particular condition, indicates the condition being assessed.
Control | 0..1
Type | Reference(Condition)
Summary | true
RiskAssessment.performer
Definition | The provider or software application that performed the assessment.
Control | 0..1
Type | Reference(Practitioner | Device)
Summary | true
RiskAssessment.reason[x]
Definition | The reason the risk assessment was performed.
Control | 0..1
Type | CodeableConcept|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
RiskAssessment.basis
Definition | Indicates the source data considered as part of the assessment (FamilyHistory, Observations, Procedures, Conditions, etc.).
Control | 0..*
Type | Reference(Any)
RiskAssessment.prediction
Definition | Describes the expected outcome for the subject.
Control | 0..*
Comments | Multiple repetitions can be used to identify the same type of outcome in different timeframes as well as different types of outcomes.
Invariants | Defined on this element ras-2: Must be <= 100 (expression : probability is decimal implies probability.as(decimal) <= 100, xpath: not(f:probabilityDecimal) or f:probabilityDecimal/@value <= 100)
RiskAssessment.prediction.outcome
Definition | One of the potential outcomes for the patient (e.g. remission, death, a particular condition).
Control | 1..1
Terminology Binding | RiskAssessmentOutcome:
Type | CodeableConcept
RiskAssessment.prediction.probability[x]
Definition | How likely is the outcome (in the specified timeframe).
Control | 0..1
Type | decimal|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | If range is used, it represents the lower and upper bounds of certainty; e.g. 40-60% Decimal values are expressed as percentages as well (max = 100).
Invariants | Defined on this element ras-1: low and high must be percentages, if present (expression : (low.empty() or ((low.code = '%') and (low.system = %ucum))) and (high.empty() or ((high.code = '%') and (high.system = %ucum))), xpath: (not(f:low) or f:low[f:code/@value='%' and f:system/@value='http://unitsofmeasure.org']) and (not(f:high) or f:high[f:code/@value='%' and f:system/@value='http://unitsofmeasure.org']))

RiskAssessment.prediction.qualitativeRisk
Definition | How likely is the outcome (in the specified timeframe), expressed as a qualitative value (e.g. low, medium, high).
Control | 0..1
Terminology Binding | Risk Probability (Example)
Type | CodeableConcept
RiskAssessment.prediction.relativeRisk
Definition | Indicates the risk for this particular subject (with their specific characteristics) divided by the risk of the population in general. (Numbers greater than 1 = higher risk than the population, numbers less than 1 = lower risk.).
Control | 0..1
Type | decimal
Requirements | Absolute risk is less meaningful than relative risk.
RiskAssessment.prediction.when[x]
Definition | Indicates the period of time or age range of the subject to which the specified probability applies.
Control | 0..1
Type | Period|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | If not specified, the risk applies "over the subject's lifespan".
RiskAssessment.prediction.rationale
Definition | Additional information explaining the basis for the prediction.
Control | 0..1
Type | string
RiskAssessment.mitigation
Definition | A description of the steps that might be taken to reduce the identified risk(s).
Control | 0..1
Type | string
Requirements | One of the main reasons for assessing risks is to identify whether interventional steps are needed to reduce risk.
RiskAssessment.comment
Definition | Additional comments about the risk assessment.
Control | 0..1
Type | string

9.11.8 Resource RiskAssessment - Mappings

URL: https://hl7.org/fhir/STU3/riskassessment-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.11.8 Resource RiskAssessment - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Open EHR Archetype Mapping

Workflow Mapping

Mappings for the riskassessment resource.

9.11.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

RiskAssessment | Observation[classCode=OBS, moodCode=EVN, code="Risk Assessment"]
identifier | .id
basedOn
parent
status
method | .methodCode
code
subject | .participation[typeCode=SBJ].role
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
occurrence[x] | .effectiveTime
condition | .outboundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN]
performer | .participation[typeCode=PFM].role[classCode=ASSIGNED]
reason[x]
basis | .outboundRelationship[typeCode=DRIV].target
prediction | .outboundRelationship[typeCode=VALUE].target[classCod=OBS, moodCode=RISK]
outcome | .value
probability[x] | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="probability"].value
qualitativeRisk
relativeRisk | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="relative risk"].value
when[x] | .effectiveTime or .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="age"].value
rationale | .reasonCode.text
mitigation | .outboundRelationship[typeCode=MTGT].target[classCode=ACT, moodCode=EVN].text

comment

9.11.8.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

RiskAssessment | N/A

9.11.8.3 Mappings for Open EHR Archetype Mapping (http://openehr.org)

RiskAssessment | OpenEHR-EHR-EVALUATION.risk.v1
identifier | Workflow Identifier
basedOn
parent
status
method
code
subject | Subject of Care
context
occurrence[x]
condition
performer | Information Provider
reason[x]
basis | Risk Factor; Link to Evidence
prediction
outcome | Risk
probability[x] | Category
qualitativeRisk
relativeRisk
when[x]
rationale | Rationale
mitigation
comment

9.11.8.4 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

RiskAssessment | Event
identifier | Event.identifier
basedOn | Event.basedOn
parent | Event.partOf
status | Event.status
method
code | Event.code
subject | Event.subject
context | Event.context
occurrence[x] | Event.occurrence
condition
performer | Event.performer.actor
reason[x] | Event.reason
basis | Event.reasonReference
prediction
outcome
probability[x]
qualitativeRisk
relativeRisk
when[x]
rationale
mitigation
comment | Event.note

9.11.9 Resource RiskAssessment - Design Notes

URL: https://hl7.org/fhir/STU3/riskassessment-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.11.9 Resource RiskAssessment - Design Notes

to do

9.11.10 Resource RiskAssessment - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/riskassessment-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.11.10 Resource RiskAssessment - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the RiskAssessment resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

9.11.10.1 Profiles

No Profiles defined for this resource

9.11.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

9.11.10.3 Search Extensions

No Search Extensions defined for this resource

9.11.11 Resource RiskAssessment - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/riskassessment-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

9.11.11 Resource RiskAssessment - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

R2 : R3 Converson maps for RiskAssessment.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (4 errors). (see documentation)

9.11.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/RiskAssessment2to3" = "R2 to R3 Conversions for RiskAssessment"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/RiskAssessment" alias RiskAssessmentR2 as source
uses "http://hl7.org/fhir/StructureDefinition/RiskAssessment" alias RiskAssessment as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  RiskAssessment extends DomainResource
  input src : RiskAssessmentR2 as source
  input tgt : RiskAssessment as target

  "RiskAssessment-subject" : for src.subject make tgt.subject
  "RiskAssessment-date" : for src.date make tgt.occurrence
  "RiskAssessment-condition" : for src.condition make tgt.condition
  "RiskAssessment-context" : for src.encounter make tgt.context
  "RiskAssessment-performer" : for src.performer make tgt.performer
  "RiskAssessment-identifier" : for src.identifier make tgt.identifier
  "RiskAssessment-method" : for src.method make tgt.method
  "RiskAssessment-basis" : for src.basis make tgt.basis
  "RiskAssessment-prediction" : for src.prediction as vs0 make tgt.prediction as vt0 then rskPrediction(vs0, vt0)
  "RiskAssessment-mitigation" : for src.mitigation make tgt.mitigation
endgroup

group rskPrediction extends BackboneElement
  input src as source
  input tgt as target

  "RiskAssessment.prediction-outcome" : for src.outcome make tgt.outcome
  "RiskAssessment.prediction-probability" : for src.probability make tgt.probability
  "RiskAssessment.prediction-relativeRisk" : for src.relativeRisk make tgt.relativeRisk
  "RiskAssessment.prediction-when" : for src.when make tgt.when
  "RiskAssessment.prediction-rationale" : for src.rationale make tgt.rationale
endgroup
```

9.11.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/RiskAssessment3to2" = "R3 to R2 Conversion for RiskAssessment"

uses "http://hl7.org/fhir/StructureDefinition/RiskAssessment" alias RiskAssessment as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/RiskAssessment" alias RiskAssessmentR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  RiskAssessment extends DomainResource
  input src : RiskAssessment as source
  input tgt : RiskAssessmentR2 as target
```

```
  "RiskAssessment-subject" : for src.subject make tgt.subject
  "RiskAssessment-date" : for src.occurrence where $this is dateTime make tgt.date
  "RiskAssessment-condition" : for src.condition make tgt.condition
  "RiskAssessment-encounter" : for src.context make tgt.encounter
  "RiskAssessment-performer" : for src.performer make tgt.performer
  "RiskAssessment-identifier" : for src.identifier make tgt.identifier
  "RiskAssessment-method" : for src.method make tgt.method
  "RiskAssessment-basis" : for src.basis make tgt.basis
  "RiskAssessment-prediction" : for src.prediction as vs0 make tgt.prediction as vt0 then rskPrediction(vs0, vt0)
  "RiskAssessment-mitigation" : for src.mitigation make tgt.mitigation
endgroup

group rskPrediction extends BackboneElement
  input src as source
  input tgt as target

  "RiskAssessment.prediction-outcome" : for src.outcome make tgt.outcome
  "RiskAssessment.prediction-probability" : for src.probability make tgt.probability
  "RiskAssessment.prediction-relativeRisk" : for src.relativeRisk make tgt.relativeRisk
  "RiskAssessment.prediction-when" : for src.when make tgt.when
  "RiskAssessment.prediction-rationale" : for src.rationale make tgt.rationale
endgroup
```

9.11.11.3 R3 Validation Errors

genetic | ERROR @ RiskAssessment: Profile http://hl7.org/fhir/StructureDefinition/RiskAssessment, Element 'RiskAssessment.status': minimum required = 1, but only found 0 (src = InstanceValidator)

cardiac | ERROR @ RiskAssessment: Profile http://hl7.org/fhir/StructureDefinition/RiskAssessment, Element 'RiskAssessment.status': minimum required = 1, but only found 0 (src = InstanceValidator)

population | ERROR @ RiskAssessment: Profile http://hl7.org/fhir/StructureDefinition/RiskAssessment, Element 'RiskAssessment.status': minimum required = 1, but only found 0 (src = InstanceValidator)

prognosis | ERROR @ RiskAssessment: Profile http://hl7.org/fhir/StructureDefinition/RiskAssessment, Element 'RiskAssessment.status': minimum required = 1, but only found 0 (src = InstanceValidator)